

VT510 Video Terminal Programmer Information

This document is a reproduction of the first edition of this book, published by Digital in November 1993.

There are a lot of differences between the printed document and this online version. In particular, I have redrawn all graphics for the Set-Up screens in Chapter 2 and corrected a lot of errors in the original. A full list of errata for the printed version will appear here shortly.

Errata

- [DECTSR](#). The printed document says that the report starts:

DCS 1 \$ S D ... D ST

This is incorrect. The sequence contains a lowercase 's', not uppercase.

Part I VT510 Video Terminal

1 Overview

- 1.1 Introduction
- 1.2 Communications Features
- 1.3 Keyboard Features
- 1.4 Printer Port Features
- 1.5 Display and Text Capabilities
 - 1.5.1 ANSI
 - 1.5.2 ASCII
- 1.6 Enhanced Set-Up
- 1.7 Desktop Productivity Features
- 1.8 Character Set Support
- 1.9 Ergonomics (Human Factors) Features
- 1.10 Field-Upgradable Firmware
- 1.11 PCTerm Mode
- 1.12 Comparison with Other ANSI Products

2 Terminal Set-Up

- 2.1 Entering Set-Up
 - 2.1.1 Effects of Entering Set-Up
 - 2.1.2 Set-Up Languages
 - 2.1.3 Power-On Settings and Defaults
 - 2.1.4 Self-test Error Messages
 - 2.1.5 Context Sensitivity
 - 2.1.6 Set-Up Summary Line
 - 2.1.7 Set-Up Status Messages
 - 2.1.8 Status Line
 - 2.1.9 Keyboard Indicator Line
- 2.2 Set-Up Screen Text
- 2.3 Main Menu
 - 2.3.1 On-Line
 - 2.3.2 Save Settings
 - 2.3.3 Restore Settings
 - 2.3.4 Exit Set-Up
- 2.4 Actions Menu
 - 2.4.1 Clear Display
 - 2.4.2 Clear Communications
 - 2.4.3 Reset Terminal
 - 2.4.4 Restore Factory Defaults
 - 2.4.5 Clock, Calculator, Show Character Sets
 - 2.4.6 Banner Message
- 2.5 Display Menu
 - 2.5.1 Lines per Screen
 - 2.5.1.1 Auto Resize
 - 2.5.2 Lines per Page
 - 2.5.3 Columns per Page, Clear on Change
 - 2.5.4 Status Display

2.5.5	Scrolling Mode
2.5.6	Screen Background
2.5.7	Cursor Display
2.5.8	Cursor Direction
2.5.9	Copy Direction
2.5.10	Vertical Coupling
2.5.11	Page Coupling
2.5.12	Auto Wrap
2.5.13	New Line Mode
2.5.14	Lock User Preferences
2.5.15	Show Control Characters
2.5.16	CRT Saver
2.5.17	Overscan
2.5.18	Refresh Rate
2.5.19	Screen Alignment
2.6	Terminal Type Menu
2.6.1	Emulation Mode
2.6.2	Terminal ID to Host
2.6.3	VT Default Character Set
2.6.4	PCTerm Character Set
2.6.5	7-Bit NRCS Characters
2.6.6	Transmit 7-Bit Controls
2.7	ASCII Emulation Menu
2.8	Keyboard Menu
2.8.1	Keyboard Language
2.8.2	Define Key Editor
2.8.2.1	Copy of Key Default—Moving a Standard Function
2.8.2.2	Customization
2.8.2.3	Modifier Keys
2.8.2.4	Creating a New Function
2.8.2.5	Examples of Creating New Functions
2.8.2.6	Correcting a Mistake
2.8.2.7	Creating A New Alphanumeric Key or Keyboard Layout
2.8.2.8	Examples of Creating New Alphanumeric Keys
2.8.2.9	User-Defined Keys
2.8.2.10	Programming Notes
2.8.3	Save Key Definitions
2.8.4	Recall Key Definitions
2.8.5	Lock Key Definitions
2.8.6	Caps Lock Function
2.8.7	Keyclick, Warning Bell, and Margin Bell Volume
2.8.8	Keyboard Encoding
2.8.9	Auto Repeat
2.8.10	Data Processing Keys
2.8.11	Application Cursor Keys
2.8.12	Application Keypad Mode
2.9	Communication Menu
2.9.1	Port Select
2.9.2	Word Size
2.9.3	Parity
2.9.4	Stop Bits

2.9.5	Transmit Speed
2.9.6	Receive Speed
2.9.7	Transmit Flow Control
2.9.8	Receive Flow Control
2.9.9	Flow Control Threshold
2.9.10	Transmit Rate Limit, Fkey Rate Limit
2.9.11	Ignore Null Character
2.9.12	Local Echo
2.9.13	Half Duplex
2.9.14	Auto Answerback
2.9.15	Answerback Message
2.9.16	Answerback Concealed
2.10	Modem Menu
2.10.1	Enable Modem Control
2.10.2	Disconnect Delay
2.10.3	Modem High Speed
2.10.4	Modem Low Speed
2.11	Printer Menu
2.11.1	Port Select
2.11.2	Print Mode
2.11.3	Printer Type
2.11.4	DEC/ISO Character Sets
2.11.5	PC Character Sets
2.11.6	Print Extent
2.11.7	Print Terminator
2.11.8	Serial Print Speed
2.11.9	2-Way Communication
2.11.10	Transmit Flow Control
2.11.11	Receive Flow Control
2.11.12	Word Size
2.11.13	Parity
2.11.14	Stop Bits
2.12	Tabs . . .
2.13	Default All Modes

3 Desktop Features

3.1	Introduction
3.2	Calculator
3.3	Clock
3.4	Character Set Tables

Part II VT510 ANSI Mode

4 ANSI Control Functions Summary

4.1	Introduction
4.2	Control Characters
4.3	Control Functions
4.3.1	Sequence Format
4.3.2	Escape Sequences
4.3.3	Control Sequences
4.3.3.1	Numeric Parameters

- 4.3.3.2 Selective Parameters
- 4.3.4 Device Control Strings
- 4.3.5 Using Control Characters in Sequences
- 4.3.6 7-Bit Code Extension Technique
- 4.4 Working with 7-Bit and 8-Bit Environments
- 4.4.1 Conventions for Codes Received by the Terminal
- 4.4.2 Conventions for Codes Sent by the Terminal
- 4.5 Show Control Characters
- 4.6 ANSI Control Function Tables

5 ANSI Control Functions

ANSI Conformance Levels
 CBT—Cursor Backward Tabulation
 CHA—Cursor Horizontal Absolute
 CHT—Cursor Horizontal Forward Tabulation
 CNL—Cursor Next Line
 CPL—Cursor Previous Line
 CPR—Cursor Position Report
 CRM—Show Control Character Mode
 CUB—Cursor Backward
 CUD—Cursor Down
 CUF—Cursor Forward
 CUP—Cursor Position
 CUU—Cursor Up
 DA1—Primary Device Attributes
 DA2—Secondary Device Attributes
 DA3—Tertiary Device Attributes
 DCH—Delete Character
 DDD1, DDD2, DDD3
 DECAAM—Set/Reset Auto Answerback Mode
 DECALN—Screen Alignment Pattern
 DECANM—ANSI Mode
 DECARM—Autorepeat Mode
 DECARSM—Set/Reset Auto Resize Mode
 DECAUPSS—Assigning User-Preferred Supplemental Sets
 DECAWM—Autowrap Mode
 DECBI—Back Index
 DECBKM—Backarrow Key Mode
 DECCANSM—Conceal Answerback Message Mode
 DECCAPSLK—Caps Lock Mode
 DECCARA—Change Attributes in Rectangular Area
 DECCIR—Cursor Information Report (Terminal to Host)
 DECCKD—Copy Key Default
 DECCKM—Cursor Keys Mode
 DECCKSR—Memory Checksum Report
 DECCOLM—Select 80 or 132 Columns per Page
 DECCRA—Copy Rectangular Area
 DECCRTSM—Set/Reset CRT Save Mode
 DECDC—Delete Column
 DECDHL—Double-Width, Double-Height Line
 DECDLD—Dynamically Redefinable Character Sets

DECDMAC—Define Macro
DECDWL—Double-Width, Single-Height Line
DECEKBD—Extended Keyboard Report
DECELF—Enable Local Functions
DECERA—Erase Rectangular Area
DECESKM—Enable Secondary Keyboard Language Mode
DECFI—Forward Index
DECFNK—Function Key
DECFRA—Fill Rectangular Area
DECHDPXM—Set/Reset Half-Duplex Mode
DECHEBM—Hebrew/N-A Keyboard Mapping
DECHEM—Hebrew Encoding Mode
DECIC—Insert Column
DECINVM—Invoke Macro
DECIPEM—Enter/Return from IBM ProPrinter Emulation Mode
DECKBD—Keyboard Language Selection
DECKBUM—Typewriter or Data Processing Keys
DECKLHIM—Keyboard LED's Host Indicator Mode
DECKPAM—Keypad Application Mode
DECKPM—Key Position Mode
DECKPNM—Keypad Numeric Mode
DECLANS—Load Answerback Message
DECLBAN—Load Banner Message
DECLFKC—Local Function Key Control
DECLL—Load LEDs
DECLRMM—Left Right Margin Mode
DECLTOD—Load Time of Day
DECMCM—Set/Reset Modem Control Mode
DECMSR—Macro Space Report
DECNAKB—Greek/N-A Keyboard Mapping
DECNCSM—Set/Reset No Clearing Screen On Column Change
DECNKM—Numeric Keypad Mode
DECNRCM—National Replacement Character Set Mode
DECNULM—Set/Reset Ignoring Null Mode
DECNUMLK—Num Lock Mode
DECOM—Origin Mode
DECOSCNM—Set/Reset Overscan Mode
DECPAK—Program Alphanumeric Key
DECPCCM—Page Cursor-Coupling Mode
DECPCTERM—Enter/Exit PCTerm or Scancode Mode
DECPEX—Printer Extent Mode
DECPFF—Print Form Feed Mode
DECPFK—Program Function Key
DECPKA—Program Key Action
DECPKFMR—Program Key Free Memory Report
DECRRARA—Reverse Attributes in Rectangular Area
DECRC—Restore Cursor
DECRLCM—Right-to-Left Copy
DECRLM—Cursor Right to Left Mode
DECRPAK—Report All Modifiers/Alphanumeric Key State
DECRPDE—Report Displayed Extent

DECRPFK—Report Function Key Definition
DECRPKT—Report Key Type
DECRPM—Report Mode - Terminal To Host
DECRPSS—Report Selection or Setting
DECRPTUI—Report Terminal Unit ID
DECRQCRA—Request Checksum of Rectangular Area
DECRQDE—Request Displayed Extent
DECRQKD—Request Key Definition
DECRQKT—Key Type Inquiry
DECRQM—Request Mode - Host To Terminal
DECRQPKFM—Program Key Free Memory Inquiry
DECRQPSR—Request Presentation State Report
DECRQSS—Request Selection or Setting
DECRQTSR—Request Terminal State Report
DECRQUPSS—User-Preferred Supplemental Set
DECRSPS—Restore Presentation State
DECRSTS—Restore Terminal State
DECSACE—Select Attribute Change Extent
DECSASD—Select Active Status Display
DECSC—Save Cursor
DECSCA—Select Character Protection Attribute
DECSCCL—Select Conformance Level
DECSCCLM—Scrolling Mode
DECSCNM—Screen Mode: Light or Dark Screen
DECSCP—Select Communication Port
DECSCPP—Select 80 or 132 Columns per Page
DECSCS—Select Communication Speed
DECSCUSR—Set Cursor Style
DECSDDT—Select Disconnect Delay Time
DECSDPT—Select Digital Printed Data Type
DECSED—Selective Erase in Display
DECSEL—Selective Erase in Line
DECSERA—Selective Erase Rectangular Area
DECSFC—Select Flow Control
DECSKCV—Set Key Click Volume
DECSLCK—Set Lock Key Style
DECSLPP—Set Lines Per Page
DECSLRM—Set Left and Right Margins
DECSMBV—Set Margin Bell Volume
DECSMKR—Select Modifier Key Reporting
DECSNLS—Set Lines Per Screen
DECSPP—Set Port Parameter
DECSPPCS—Select ProPrinter Character Set
DECSPRTT—Select Printer Type
DECSR—Secure Reset
DECSRC—Secure Reset Confirmation
DECSRFR—Select Refresh Rate
DECSSCLS—Set Scroll Speed
DECSSDT—Select Status Display (Line) Type
DECSSL—Select Set-Up Language
DECST8C—Set Tab at Every 8 Columns

DECSTBM—Set Top and Bottom Margins
DECSTR—Soft Terminal Reset
DECSTRL—Set Transmit Rate Limit
DECSWBV—Set Warning Bell Volume
DECSWL—Single-Width, Single-Height Line
DECTABSR—Tab Stop Report
DECTCEM—Text Cursor Enable Mode
DECTID—Select Terminal ID
DECTME—Terminal Mode Emulation
DECTSR—Terminal State Report—Terminal to Host
DECTST—Invoke Confidence Test
DECUDK—User Defined Keys
DECVCCM—Vertical Cursor-Coupling Mode
DECXCPR—Extended Cursor Position
DECXRLM—Transmit Rate Limiting
DL—Delete Line
DSR—Device Status Reports
DSR—Cursor Position Report (CPR)
DSR—Data Integrity Report
DSR—Extended Cursor Position Report (DECXCPR)
DSR—Keyboard
DSR—Macro Space Report
DSR—Memory Checksum (DECCKSR)
DSR—Operating Status
DSR—Printer Port
DSR—User-Defined Keys (VT Level 4 Only)
ECH—Erase Character
ED—Erase in Display
EL—Erase in Line
HPA—Horizontal Position Absolute
HPR—Horizontal Position Relative
HTS—Horizontal Tab Set
HVP—Horizontal and Vertical Position
ICH—Insert Character
IL—Insert Line
IND—Index
IRM—Insert/Replace Mode
KAM—Keyboard Action Mode
LNM—Line Feed/New Line Mode
LS—Locking Shift
MC—Media Copy
NEL—Next Line
NP—Next Page
PP—Preceding Page
PPA—Page Position Absolute
PPB—Page Position Backward
PPR—Page Position Relative
RIS—Reset to Initial State
RM—Reset Mode
S7C1T—Send C1 Control Character to the Host
S8C1T—Send C1 Control Character to the Host

SCODFK—Define Function Key
SCORC—Restore Saved Cursor Position
SCOSC—Save Current Cursor Position
SCS—Select Character Set
SD—Pan Up
SGR—Select Graphic Rendition
SM—Set Mode
SRM—Local Echo: Send/Receive Mode
SS—Single Shifts
SU—Pan Down
TBC—Tab Clear
VPA—Vertical Line Position Absolute
VPR—Vertical Position Relative

6 SCO Console Emulation

- 6.1 SCO Console Emulation Mode
 - 6.1.1 Keyboard
 - 6.1.1.1 Local Function Keys in SCO Console Mode
 - 6.1.1.2 PC Keyboard
 - 6.1.1.3 VT Keyboard Key Mapping

7 Character Sets

- 7.1 Overview
- 7.2 Character Sets Supported
 - 7.2.1 VT Graphic Character Sets
 - 7.2.2 PCTerm Graphic Character Sets
 - 7.2.3 Control Character Sets
 - 7.2.4 Character Set Structure
 - 7.2.5 Characters Not in a Character Set.
- 7.3 VT Mode
 - 7.3.1 Selecting Character Sets
 - 7.3.2 Character Set Mappings
- 7.4 PCTerm Mode
 - 7.4.1 Selecting Character Sets
 - 7.4.2 Character Set Mappings
- 7.5 SCO Console Mode
- 7.6 Soft Character Sets

8 Keyboard Processing

- 8.1 Overview
- 8.2 PC Keyboard—VT Style vs. PC Style
 - 8.2.1 Differences—PC Style vs. VT Style
- 8.3 Top Row Function Keys
- 8.4 Main Key Array
 - 8.4.1 Character Sets
 - 8.4.2 Caps Lock
 - 8.4.3 Tab Key
 - 8.4.4 Left Alt
 - 8.4.5 Right Alt
 - 8.4.6 Alt Gr
 - 8.4.7 Modifier Keys

- 8.5 Editing Keypad Keys
- 8.6 Cursor Keypad Keys
- 8.7 Numeric Keypad Keys
- 8.8 Local Function Key Defaults
 - 8.8.1 Numeric Keypad Compose
 - 8.8.2 Accessibility Aids
- 8.9 Controlling Keyboard LEDs
- 8.10 Keyboard Languages
- 8.11 Switching Between Keyboard Languages
 - 8.11.1 Non-Spacing Accent Marks
 - 8.11.2 German
 - 8.11.3 Hebrew
- 8.12 Keystation Numbers
- 8.13 Scan Codes Sent in PCTerm Mode for EPC Keyboards
- 8.14 Scan Codes Sent in PCTerm Mode for VT Keyboards

9 Communications

- 9.1 Overview
- 9.2 Introduction to Communications
- 9.3 Physical Communications Link
 - 9.3.1 25-Pin Connectors
 - 9.3.2 DEC Corporate Modular Jacks (MMJ)
 - 9.3.3 Connector Pins Description
 - 9.3.3.1 Transmitted Data–TXD
 - 9.3.3.2 Received Data–RXD
 - 9.3.3.3 Request to Send–RTS
 - 9.3.3.4 Clear to Send–CTS
 - 9.3.3.5 Data Terminal Ready–DTR
 - 9.3.3.6 Data Set Ready–DSR
 - 9.3.3.6.1 Modem Initiated Disconnect
 - 9.3.3.6.2 Terminal Initiated Disconnect
 - 9.3.3.6.3 Modem off line
 - 9.3.3.7 Receive Line Signal Detector–RLSD
 - 9.3.3.8 Speed Indicator–SPDI
 - 9.3.3.9 Speed Select–SPDS
 - 9.3.4 Half-Duplex Mode
 - 9.3.5 Supported Modem Configurations
 - 9.3.5.1 CCITT Public Switched Telephone Network (2-wire only)
 - 9.3.5.2 CCITT Dedicated Line (2-Wire / 4-Wire) Data Network
 - 9.3.5.3 European Public Circuit Switched Data Network
 - 9.3.5.4 European Public Packet Switched Data Network (PSDN)
- 9.4 Data Link Layer
 - 9.4.1 7-Bit VT100 or 8-Bit VT510 Operation
 - 9.4.2 Flow Control
 - 9.4.2.1 Receive Buffer
 - 9.4.2.2 Flow Control Selection
 - 9.4.2.2.1 No Flow Control
 - 9.4.2.2.2 XOFF/XON or XPC Flow Control
 - 9.4.2.2.3 DSR/DTR Flow Control
 - 9.4.2.2.4 Software and Hardware Combined Flow Control
 - 9.4.2.2.5 Flow Control Threshold

- 9.4.2.3 Show Control Character Mode
- 9.4.2.4 Set-Up on Flow Control Handling
- 9.4.2.5 When Disconnect Occurs
- 9.5 Features that Affect Communications
- 9.5.1 Clear Communications
- 9.5.2 Connection
- 9.5.3 Line Disconnection
- 9.5.4 Break
- 9.5.4.1 Received Break

10 Printer Port

- 10.1 Overview
- 10.2 Serial Ports
 - 10.2.1 Serial Port Flow Control
- 10.3 Parallel Port
 - 10.3.1 Parallel Port Flow Control
- 10.4 Print Modes
 - 10.4.1 Printer Controller Mode
- 10.5 Print Operation
- 10.6 Transmitting Print Data
 - 10.6.1 DEC ANSI
 - 10.6.2 IBM ProPrinter or Compatible Printer
 - 10.6.2.1 Data Transmission
 - 10.6.3 DEC ANSI with IBM ProPrinter Emulation
 - 10.6.3.1 Data Transmission
 - 10.6.3.2 Printer Status
- 10.7 Suspending Print Operations
- 10.8 Aborting Print Operations

Part III VT510 ASCII Emulation

11 ASCII Emulations and Control Functions

- 11.1 Overview
- 11.2 ASCII Emulations
 - 11.2.1 Enhanced Mode
- 11.3 Coding Notation
 - 11.3.1 Communication Flow Control with PCTerm Mode
 - 11.3.2 Protecting Data

12 ASCII Escape Sequences

- 12.1 ASCII Communications
- 12.2 Terminal Management
- 12.3 Keyboard Processing
- 12.4 Screen Processing
- 12.5 Printing/Sending Data
 - 12.5.1 Sending Data
 - 12.5.2 Graphics Characters

13 Defining and Loading ASCII Character Fonts

- 13.1 Character Sets
 - 13.1.1 PCTerm on US Keyboard

- 13.1.2 WYSE and ADDS Enhanced Personalities
- 13.2 Loading, Defining, and Selecting Character Sets
- 13.3 Designing a Character Font

Figures

- 1-1 Sample VT Keyboard Layout
- 1-2 Sample PC Keyboard Layout
- 2-1 Set-Up
- 2-2 Set-Up Language Selection
- 2-3 Set-Up Summary Line
- 2-4 Main and Terminal Actions Menu
- 2-5 Display Menu
- 2-6 Lines per Screen Menu
- 2-7 Lines per Page Menu
- 2-8 Columns per Page Menu
- 2-9 Status Display Menu
- 2-10 Cursor Display Menu
- 2-11 Screen Alignment Display
- 2-12 Terminal Type and Emulation Mode Menu
- 2-13 Terminal ID to Host Menu
- 2-14 VT Default Character Set Menu
- 2-15 PCTerm Character Set Menu
- 2-16 ASCII Emulation Menu
- 2-17 Keyboard Menu
- 2-18 Define Key Editor, Select Function
- 2-19 Define Key Editor, Alphanumeric Alt Function
- 2-20 Communication Menu
- 2-21 Communication Transmit Speed Menu
- 2-22 Communication Fkey Rate Limit Menu
- 2-23 Communication Answerback Dialog Box
- 2-24 Modem Menu
- 2-25 Modem High Speed Menu
- 2-26 Printer Menu
- 2-27 DEC/ISO Character Sets Menu
- 2-28 PC Character Sets Menu
- 2-29 Tab Set-Up Dialog Box
- 3-1 Calculator with VT Keyboard Layout
- 3-2 Calculator with PC Keyboard Layout
- 3-3 Clock Function
- 4-1 Show Control Characters Font (Left Half)
- 4-2 Show Control Characters Font (Right Half)
- 8-1 VT Keyboard Layout
- 8-2 PC Keyboard Layout
- 8-3 Editing Keypad Legends on DEC and PC Layouts
- 8-4 VT Layout - Keystation Numbers
- 8-5 VT Layout - Typical North American Key Legends
- 8-6 Enhanced PC Layout - Map Numbers
- 8-7 Enhanced PC Layout - Typical North American Key Legends
- 9-1 25-Pin EIA 232 Port Pin Numbers
- 9-2 MMJ Modular Jack Pin Numbers
- 10-1 Parallel Printer Port

Tables

1-1	VT510 Model Differences
1-2	Configurations vs. Overscan Availability
1-3	VT510 Font Sizes
2-1	Error Messages
2-2	Context Sensitive Set-Up Features
2-3	Display Set-Up Features and Control Functions
2-4	Lines per Page vs. Lines per Screen When Auto Resize is Set
2-5	ASCII Emulation Features
2-6	Keyboard Features and Control Functions
2-7	Communication Features and Control Functions
2-8	Printer Features and Control Functions
2-9	Default All Modes
3-1	ASCII Emulation Character Sets
4-1	C0 (7-Bit) Control Characters Recognized
4-2	C1 (8-Bit) Control Characters Recognized
4-3	8-Bit Control Characters and Their 7-Bit Equivalents
4-4	Control Character Displayed on Screen
4-5	Text Processing Control Functions
4-6	Reports Control Functions
4-7	Terminal Management Control Functions
4-8	Keyboard Processing Control Functions
4-9	Communications Control Functions
5-1	VT52 Escape Sequences
5-2	DECDLD Parameter Characters
5-3	Valid DECDLD Parameter Combinations
5-4	VT510 Font Sizes
5-5	DECFNK Modifiers
5-6	Settings Supported by DECRQM, DECRPM, RM, and SM
5-7	ANSI Modes for DECRQM, DECRPM, SM, and RM
5-8	DEC Modes for DECRQM, DECRPM, SM, and RM
5-9	Terminal's Default Settings
5-10	Terminal Emulation Modes
5-11	VT Keyboards
5-12	PC Keyboards
5-13	VT and PC Keyboard Function Keys Compared
5-14	Keyboard Languages
5-15	Mapping Character Sets with Locking Shifts
5-16	Visual Character Attribute Values
6-1	SCO Console Mode with ASCII Keycodes and Scancodes
6-2	Local Functions on VT and Enhanced PC Layouts
6-3	SCO Console Mode Function Key Default Codes
6-4	SCO Console Mode Main Keypad Default Codes
6-5	SCO Console Mode Editing Keypad Default Codes
6-6	SCO Console Mode Cursor Keypad Default Codes
6-7	SCO Console Mode Numeric Keypad Default Codes, Num Lock Off
6-8	SCO Console Mode Numeric Keypad Default Codes, Num Lock On
8-1	VT Layout vs. Enhanced PC Layout Modifier Keys
8-2	Editing Keypad Sequences for PC Layout

8-3	PC Layout Numeric Keypad Sequences - VT Style
8-4	PC Layout Numeric Keypad Sequences - PC Style, Numeric Mode
8-5	Local Functions
8-6	Local Functions for ASCII emulations
8-7	VT Keyboard Layouts
8-8	Enhanced PC Keyboard Layouts
8-9	Default Character Set when changing Keyboard Language
8-10	Non-Spacing Accent Marks
9-1	EIA Interface Signals and Functions
9-2	MMJ Interface Signals
9-3	Flow Control Threshold Points
10-1	Parallel Connector Signals
11-1	Symbols Used in ASCII Terminal Escape Sequences
11-2	Commands that may Carry Protect Mode Features through Pages
11-3	Commands that Can Move the Cursor to Protected Area
11-4	Commands that Cannot Move Cursor into Protected Area
12-1	Serial Port Control Functions
12-2	Terminal Management Control Functions
12-3	Keyboard Processing Control Functions
12-4	Redefinable Keys
12-5	Screen Processing Control Functions
12-6	Data Protection Control Functions
12-7	Cursor Control Functions
12-8	ASCII Editing Control Functions
12-9	Local Edit Mode Function Keys
12-10	Tab Stop Control Functions
12-11	Page Edit Control Functions
12-12	Rectangle Area Control Functions
12-13	Send Data Delimiters
12-14	Delimiters for Other Personalities
12-15	Print/Send Control Functions
12-16	Printing Data Control Functions
12-17	Graphic Characters
13-1	Display and Load Character Sets
13-2	Loading Font Bank
13-3	Design a Character Font

Part I

VT510 Video Terminal

This part includes the following chapters:

- Chapter 1, [Overview](#)
- Chapter 2, [Terminal Set-Up](#)
- Chapter 3, [Desktop Features](#)

1.1 Introduction

The VT510 is a low-cost, single-session, text video terminal featuring enhanced keyboard and communications flexibility.

Three models of the product are available:

- North American
- International
- International MPR II compliant (low emissions)

All three versions have a 25-pin stacked male/female RS-232 connector (EIA-232), a 6-pin EIA-423 (MMJ) connector, and a Centronics parallel connector. These ports connect to a host or a printer.

Table 1–1 VT510 Model Differences

Feature	North American	International	Low Emission
FCC class (USA)	A	B	B
CSA class (Canada)	A	B	B
VDE compliant (Germany)	-	yes (B)	yes (B)
VCCI class (Japan)	-	1	1
MPR II compliant (Sweden)	no	no	yes
User-adjustable centering	yes	yes	yes
User-adjustable rotation	no	yes	yes
National keyboard languages	5	27	27
Power cord	fixed	detachable	detachable

1.2 Communications Features

The terminal provides:

- Two bidirectional, asynchronous serial communication ports for host and local printer
- Baud rates from 300 to 115.2K baud
- Centronics parallel printer port
- 25-pin serial connector
- Expanded flow control options:
 - Software flow control using XON/XOFF or XPC
 - Hardware flow control using DTR/DSR

1.3 Keyboard Features

The VT510 offers a choice of two keyboard layouts—a VT keyboard layout (LK450) or an enhanced PC keyboard layout (PCXAL). The VT keyboard ([Figure 1–1](#)) and the PC keyboard ([Figure 1–2](#)) differ in the placement of some of the keys, like the arrow keys. A VT keyboard has 20 function keys, **F1** through **F20**, above the main keypad, while a PC keyboard has 12 function keys, **F1** through **F12**.

Figure 1–1 Sample VT Keyboard Layout

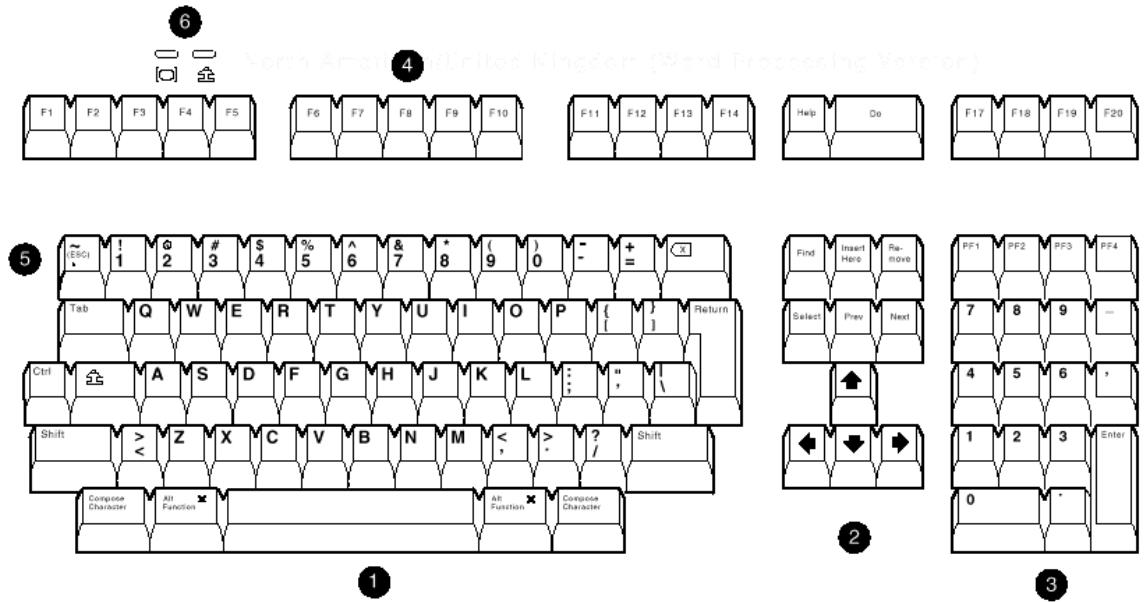
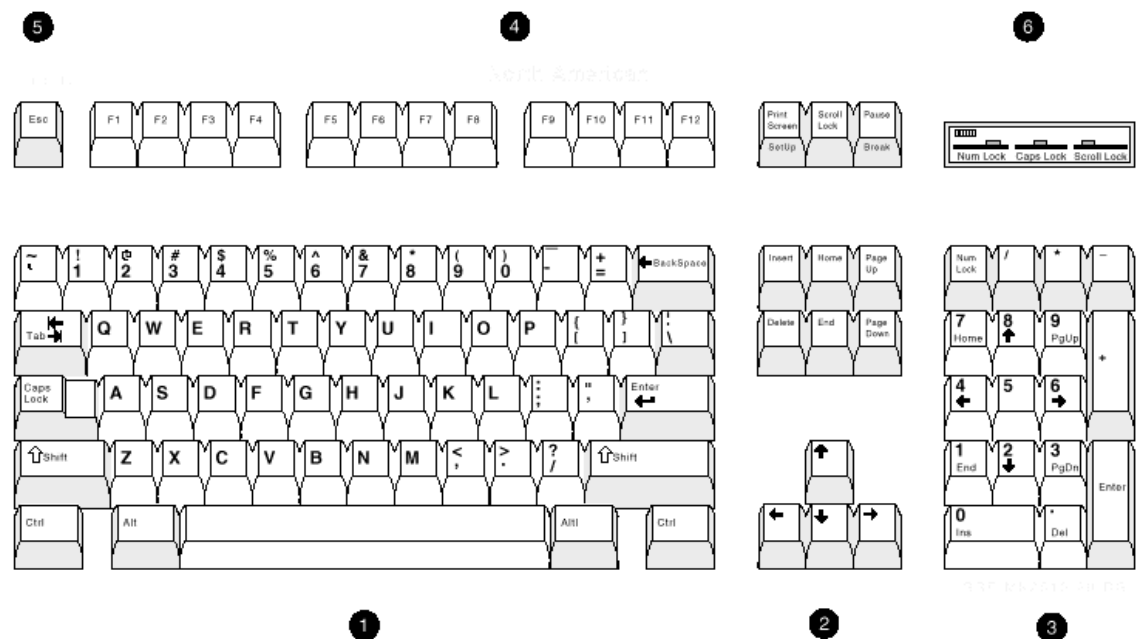


Figure 1–2 Sample PC Keyboard Layout



- (1) Main keypad, (2) Editing keypad, (3) Numeric keypad,
(4) Function keys, (5) Escape key, (6) Indicator lights.

The keyboard has the following features:

- An IBM PS/2 6-pin mini DIN connector.
- Keyboard keys that can be reprogrammed to send single characters, to send character sequences, or to invoke local functions (Hold, Copy and Paste, Remove, Insert, Print Page, Toggle Autoprint, Set-Up, Break, and so on).
- A local user-defined key editor. Key definitions can be saved in nonvolatile memory.
- Eight levels of user-defined function keys: Unshifted, Shifted, Control, Shift Control, Alt, Alt/Shift, Alt/Control and Alt/Shift/Control.
- 804 bytes of nonvolatile memory available for custom user-defined keys (UDKs). The maximum length for a UDK definition is 255 bytes.

1.4 Printer Port Features

The terminal has the following printer port features:

- A Centronics parallel printer port.
- IBM ProPrinter support.
- Bidirectional serial communication ports (male and female).
- Baud rates from 300 to 115.2K baud.
- Null characters that can pass as data to the printer port.

1.5 Display and Text Capabilities

The terminal has both ANSI and ASCII display and text processing capabilities. The features of each are listed below.

1.5.1 ANSI

- Macro capability
- Display: 26, 42, or 53 data lines
- Selectable page size: 24, 25, 36, 42, 48, 50, or 72 lines
- Full status line (occupies one data line)
- Separate keyboard indicator line
- Three pages of 24 lines
- SCO Console mode

1.5.2 ASCII

- Terminal emulations:
WYSE: 50+, 60 native, 160 text only, 120, 150, PCTerm
Teletype: TVI 950, 925, 910+
Applied Digital Data Systems: ADDS A2
- Dim and invisible character attributes

1.6 Enhanced Set-Up

The terminal provides an enhanced menu-based Set-Up, allowing common features to be accessed easily from a single screen. All Set-Up features are host controllable to allow remote configuration. The terminal also has a means to lock out the local setup.

Set-Up menus are available in English, French, German, Italian, or Spanish. For details, see Chapter 2, [Terminal Set-Up](#).

1.7 Desktop Productivity Features

The terminal has the following desktop productivity features:

- Local copy and paste (not available in PCTerm mode)
- Time of day clock (host or locally set, no battery back up)
- Desktop calculator with decimal, octal, and hex conversions
- Display character set tables

For details, see Chapter 3, [Desktop Features](#).

1.8 Character Set Support

In addition to the traditional Digital graphic character sets (DEC Multinational, NRCS, and so on), the VT510 supports Cyrillic, Greek, Hebrew, Turkish, most Eastern European languages, and PC character

sets. For details, see Chapter 7, [Character Sets](#).

1.9 Ergonomics (Human Factors) Features

The terminal is designed to provide a high-quality human interface and long-term reliability at low cost. The VT510 offers 72 Hz refresh, full overscan, or both at slightly reduced resolution. The terminal has the following features:

- 35.5 cm (14 in) flat face, antiglare CRT screen
- Improved resolution of 800 × 432 pixels
- 72 Hz refresh
- Overscan (see [Table 1–2](#))
- Additional fonts (see [Table 1–3](#))
- Tilt/swivel base
- Choice of keyboards
- Accessibility aids

Table 1–2 Configurations vs. Overscan Availability

Refresh Rate	Lines per Screen	Overscan Available
72 Hz	24/25/26	yes
72 Hz	36/42 or 48/50/53	no
60 Hz	all	yes

Table 1–3 VT510 Font Sizes

80 Columns	132 Columns	Maximum Number of Lines
10 × 16	6 × 16	26 lines + keyboard indicator line
10 × 10	6 × 10	42 lines + keyboard indicator line
10 × 8	6 × 8	53 lines + keyboard indicator line
10 × 13	6 × 13	26 lines + keyboard indicator line (72 Hz with overscan)

1.10 Field-Upgradable Firmware

All VT510 models include support for a 4 Mbit (512 KByte) ROM cartridge option. The base unit comes with a factory-installed ROM. The ROM cartridge connector is protected by a ROM cartridge cover. To install new code, the ROM cartridge cover is removed, and a ROM cartridge with attached cover is plugged into the ROM connector. The new code completely supersedes the factory-installed ROM code. It does not overlay or extend the factory code.

Note

If you service the terminal with a ROM cartridge installed, remove and save the ROM cartridge to put it on the new terminal.

1.11 PCTerm Mode

PCTerm mode is designed to allow the terminal to emulate the console of an industry-standard PC.

A separate character set selection is used for PCTerm mode. This selection is controlled by the **PCTerm character set** selection in Set-Up and the DECPCTERM control function. Changing this selection does not affect the VT character set selection.

1.12 Comparison with Other ANSI Products

The VT510 is functionally compatible with other ANSI products with the following exceptions:

- **Terminal ID**

The Primary Device Attributes can report additional extensions to cover new features. See [Chapter 4](#) for a summary of the control functions. The Secondary Device Attributes are different.

VT510 CSI > 61 ; Pv ; 0 c

Where Pv is the firmware version number.

- **Dynamically Redefinable Character Set (DRCS) Fonts**

The VT510 has four buffers, which can load up to two dynamically redefinable character sets, each with an 80-column font and a 132-column font.

- **Keyboards**

DEC LK401 and compatible keyboards cannot be used on the VT510; however, the VT510 can emulate other keyboards when configured accordingly.

- **Single Session**

The VT510 is a single-session terminal.

- **PC Character Sets**

PC character sets are available in SCO Console mode, and they do not require the keyboard to send scancodes, as PCTerm mode does.

Terminal Set-Up

This chapter describes the extensive VT510 Set-Up features, which include contemporary windowing techniques that are easy to access. These techniques make it easy to move about and select features. Set-Up is used to examine or change terminal operating features, such as transmit/receive speeds, type of cursor, or key definitions.

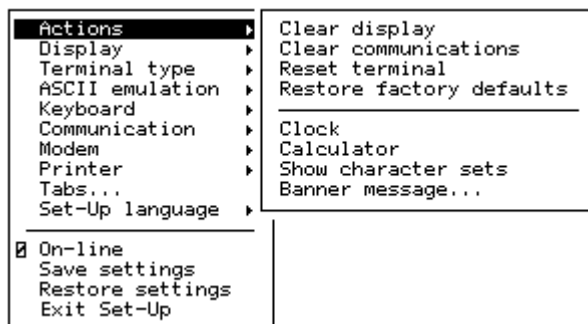
2.1 Entering Set-Up

You enter Set-Up by pressing the key designated as the Set-Up key. Pressing this key alternately places the terminal in Set-Up mode or returns it to the operating mode.

On . . .	The Set-Up key is . . .
a VT keyboard layout	F3 on the top row.
an Enhanced PC 101/102 keyboard	Caps Lock/Print Screen (or Alt/Print Screen) above the editing keypad.

The Set-Up function may be reassigned to other key combinations or can be disabled if desired.

Figure 2-1 Set-Up
Terminal Set-Up



The Set-Up display consists of pull-right menus. You can move back and forth with the arrow keys. Pressing **Enter**, **Return**, **Do**, or **Select** invokes the action or setting for the current menu item.

Note

Set-Up can be programmed to lock out setup; however, when you press **F3** as the first key after the terminal is powered on, you always enter Set-Up, regardless of which keyboard you use or how **F3** is defined.

2.1.1 Effects of Entering Set-Up

Placing the terminal in Set-Up mode causes no loss of data if a Flow Control protocol is in use on the communications port. When you enter Set-Up by pressing the Set-Up key, the text on the screen temporarily disappears and the first Set-Up menu is displayed on the screen. Upon exiting Set-Up, the previous text is restored.

Entering Set-Up has the following side effects:

- Any compose sequence in progress is aborted and the keyboard is unlocked.
- The Indicator Status Line is enabled and remains enabled while in Set-Up.
- Printer operations are suspended and are resumed upon exiting Set-Up.

2.1.2 Set-Up Languages

The VT510 Set-Up menus can be displayed in five languages using a Set-Up feature. A menu allows you to choose the language in which all subsequent Set-Up menus and displays are written. Selections from this Set-Up language menu (Figure 2–2) take effect immediately. This feature can also be invoked by DECSSL.

Figure 2–2 Set-Up Language Selection



2.1.3 Power-On Settings and Defaults

The VT510 terminal stores many of its feature settings in nonvolatile memory (NVM). Nonvolatile memory retains these settings even when power is shut off. In addition to storing operator-selected features, the terminal retains the factory default settings in permanent memory. By using Set-Up, you can modify individual terminal features, recall the feature settings stored in nonvolatile memory, or recall the factory default settings.

In this chapter, the defaults are shown with a solid bullet (•) in lists and in **boldface** in text.

2.1.4 Self-test Error Messages

At power-up, the VT510 terminal executes a series of self-tests, displays a message indicating whether the self-tests were successful, and displays a banner message. (To change the banner message, see [Section 2.4.6.](#))

If the self-tests detect an error, one of the following messages is displayed on the terminal in place of "Selftest OK."

Table 2–1 Error Messages

Message	Meaning
NVR Error - 1	A firmware update, or loss of power while writing to the NVR during a Save settings process may cause this error. Try Save settings again. If that does not work, Recall factory defaults and select Save settings again.
RS-232 Port Data Error - 2	Communications problem inside terminal. Call for service.
RS-232 Port Controls Error - 3	Communications problem inside terminal. Call for service.
Keyboard Error - 4	Turn the terminal off, and make sure the keyboard cable is plugged in. Turn the terminal on. If the problem continues, try another keyboard. If the new keyboard does not work, the problem is inside the terminal; in which case, call for service.
EIA-423 Port Error - 5	Communications problem inside terminal. Call for service.
Parallel Port Error - 6	Communications problem inside terminal. Call for service.
ROM Cartridge Error - 7	Turn the terminal off, and make sure the ROM cartridge is seated firmly on its connector pins. Turn the terminal on. If the problem continues, try another ROM cartridge. If the new ROM cartridge does not work, the problem is inside the terminal; in which case, call for service.

2.1.5 Context Sensitivity

On the screen, a check box ☐ or radio button ☒ indicates the current user selection and normally reflects the operating state of the terminal.

Certain feature selections overlap and could contradict each other, so they cannot be active at the same time. These are called *context sensitive* features, which are displayed with the dim video attribute in Set-Up. Example:

You cannot select **ASCII emulation** ▸ **Pages** menu item when you are in a VT emulation mode, because the **Display** ▸ **Lines per page** menu item performs the same function.

When a context sensitive feature is dimmed, it does not affect the user's selection, which may indicate the feature is enabled even though it is not currently selectable. Example:

You cannot have a 10 × 16 font with overscan at 72 Hz. If you select this font size at 72 Hz, *Overscan* is not available and is dimmed; however, its checkbox can still be checked.

Table 2–2 lists the set-up features that are context sensitive.

Table 2–2 Context Sensitive Set-Up Features

	Menu Item	Dimmed when . . .
Display ▸	Lines per screen	WYSE is selected; use ASCII emulation
	Lines per page	▸ Data lines and Pages .
	Cursor direction	Keyboard language is not Hebrew.
	Overscan	Display ▸ Refresh rate = 72 Hz and Lines per screen = 42 or 53.
Terminal type ▸	Terminal ID to host	Terminal type ▸ Emulation mode is other than a VT or SCO console.
	☒ 7-bit NRCS characters	Emulation mode is set to VT100/VT52; or Communication ▸ Word size is 7-bits; or no 8-bit NRC set exists for the keyboard language.
ASCII emulation ▸		Terminal type ▸ Emulation mode is set to VT or SCO console.
	TVI page-flip	Terminal type ▸ Emulation mode is other than TVI.
Keyboard ▸	Key encoding ▸ Character, Scan code and Key position	ASCII emulation is selected, or a PC keyboard is connected.
	Data processing keys	A PC keyboard is connected.
Communication ▸	Half duplex	The host Communication ▸ Port select is set to S1=comm2 .
	Transmit flow control ▸ DSR and Both	The Modem ▸ Enable modem control is selected.
	Receive flow control ▸ DTR and Both	The Modem ▸ Enable modem control is selected.
Modem ▸	Disconnect delay and Modem speed	The Modem ▸ Enable modem control is not selected.
Printer ▸	Serial print speed, 2-way communication, Transmit flow control, Receive flow control, Word size, Parity, and Stop bits	The Communication ▸ Port select is set to one of the print=parallel selections.
	DEC/ISO char sets	IBM ProPrinter is selected.
	PC character sets	DEC ANSI printer is selected.
	Print mode ▸ Controller	Display ▸ Show control characters is selected.

2.1.6 Set-Up Summary Line

The Set-Up Summary Line (Figure 2–3) shows the important set-up features that affect whether the terminal can successfully communicate with the host. If the terminal is not working, the Set-Up Summary Line allows you to quickly verify if the communication features are set appropriately. The summary line is displayed below the last data line, where the status line would normally appear.

Figure 2–3 Set-Up Summary Line

```
S1=comm1 9600N81 ISO Latin-1 North American VT510
```

The Set-Up Summary Line contains the following set-up information extracted from the Set-Up displays.

Information Displayed	Communication Feature
S1=comm1	Port selected: comm1 or comm2
9600	Transmit speed: 9600 baud
N	Parity: N=none, E=even, O=odd, M=mark, S=space
8	Word size: 7 or 8 bits
1	Stop bits: 1 or 2 bits
ISO Latin-1	Default character set or PC set in PCTerm mode
North American	Keyboard language
VT510	Emulation mode

The summary line is visible whenever the terminal is in Set-Up, except under the following conditions:

- You press a **Shift** key. Pressing either **Shift** key causes the normal Indicator Status Line to replace the Set-Up Summary Line.
- The status line is temporarily replaced by an error or status message from the terminal. In this case, the error or status message is shown in place of the status line until you type another key.

The Set-Up Summary Line appears in the currently selected set-up language.

2.1.7 Set-Up Status Messages

The Status or Set-Up Summary line is sometimes replaced by a Set-Up status message until the next user keystroke. Set-Up status messages are displayed whenever the effect of a set-up action would not otherwise be apparent. Example: "Done" is displayed after the Reset terminal action has completed.

The following are set-up status messages:

- Done
- NVR Error

2.1.8 Status Line

The VT510 terminal has a status display with the following features:

- The clock time appears on the status display if the local clock has been initialized since the terminal was last powered on.
- In the ASCII emulation modes, the host-writable status line becomes the WYSE Editing status line.
- The Accessibility Aids feature ("sticky keys") displays icons in the status line. For details, see Chapter 8, [Keyboard Processing](#).

The VT510 Indicator status line includes the Active Position Indicator, Printer Status Indicator, Modem Status Indicator, and local clock time in the desired format. Example:

1(24,008) Printer: None Modem: DSR 3:15 PM

2.1.9 Keyboard Indicator Line

The Keyboard Indicator Line has the following features:

- It is displayed in the Set-Up language.
- It has active keyboard layout. Nothing is displayed for Group 1. "G2" is displayed for Group 2, along with a small icon representing the right side of the top of the keycap. (To see this icon, press **Ctrl/Alt/F2**. To return from Group 2, press **Ctrl/Alt/F1**.)
- For Greek, Hebrew, and Russian, "G2" is replaced by "Grk", "Heb", and "Rus", respectively.
- When the Num Lock state is active on an EPC keyboard, "Num Lock" is displayed.

- When an EPC keyboard is operating in VT style, "VT" is displayed in place of the Num Lock indicator.
- In French and Spanish, an icon replaces the word "Lock" on the indicator line.

Example:

Heb Num Lock Copy Hold Lock Compose Wait

Using **Define key...** to redefine keys on the numeric keypad will affect only the Num Lock off state of these keys. When Num Lock is on, these keys will always send their factory default ASCII numerals and characters. The Num Lock behavior of these keys cannot be reprogrammed.

Scenario: Imagine the keyboard having two numeric keypads: one for Num Lock Off, and the other for Num Lock On. Only the Num Lock Off keypad can be reprogrammed. The Num Lock On keypad retains the factory defaults.

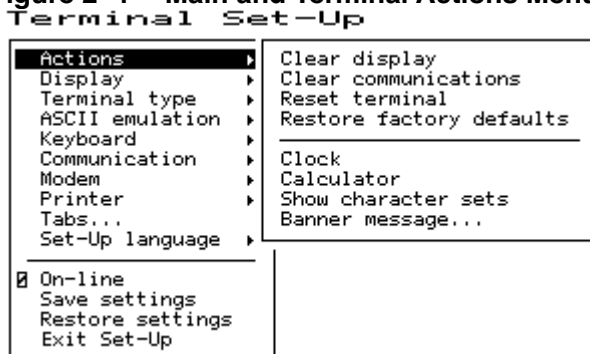
2.2 Set-Up Screen Text

The main menu, shown in [Figure 2-4](#), is the first menu displayed whenever you enter Set-Up. As the first menu, it holds the main directory of Set-Up features and those features which are commonly accessed and must be easy to find at all times. The Actions submenu is also displayed, since the Actions menu item is currently selected.

Unless stated otherwise, the following general descriptions apply:

- Changes to feature settings do not take effect until you exit Set-Up. Only net changes have an effect — a change to a new value and back to an old value has no effect.
- The setting choices for some features are shown in the corresponding Set-Up displays. Where all the choices could not be displayed, additional choices are listed immediately under the figure.

Figure 2-4 Main and Terminal Actions Menu



Pressing  moves the cursor to the next item in the main menu. Pressing  moves the cursor right to the first item of the submenu, making it the active menu.

2.3 Main Menu

The following descriptions refer to [Figure 2-4](#).

The items above the dividing line in the main menu form a directory to the major functional areas of set-up features. Each item has a corresponding submenu or dialog box. Below the dividing line are features that need to be easy to find from any menu.

2.3.1 On-Line

The **On-line** feature allows you to select whether the terminal directs keyboard input to its host port UART to communicate with a remote host (On-Line), or directs keyboard input directly to the screen (receive character parser) without sending to a remote host (off-line or local).

Note

The **On-line** selection is different from the Local echo mode in the Communications Set-Up submenu. When **On-line** is disabled (local mode), nothing is sent to the host. Any keyboard

input is directed to the receive character parser, which updates the screen.

In contrast, **Local echo** causes characters sent to the host port to also be directed back to the receive character parser, as if they had been received (echoed) from the host. Local echo is temporarily disabled when the terminal is in either Local mode or Local Controller mode, because keyboard input is already being redirected to the screen or printer port.

If an NVR error occurs during power-up, the terminal switches to Local mode.

Terminal reports, such as Answerback, DA, CPR, DECTSR, DECRPM, DECRPSS, DECAUPSS, DECCIR, DECTABSR and DSR, are not directed to the display. While in "local", host communications are put on "hold": All data received is buffered. If necessary, an XOFF is sent (or DTR is deasserted) to prevent buffer overflow. This buffered data is directed to the display when On-line mode is reentered. Note that the received character buffer (silo) is responsible for sending XON and XOFF when needed, not the On-line/Local state.

When in Local mode, modem signals, including DTR, are unaffected. No disconnect is performed. The functions of the printer port remain unchanged in Local mode. They function the same as in On-line mode.

When in Local mode and Printer To Host mode is selected, data from the printer port is displayed on the video terminal.

When both Local mode and Printer Controller mode are selected, the terminal is in Local Controller mode. Keyboard characters are sent to the printer port. Note that in "regular local" mode, any keyboard output is directed to the screen, but in "regular controller" mode, the keyboard output is directed at the host port. When Local Controller mode is exited to Local mode, the terminal redirects keyboard output from the printer to the screen (and parser). When you change from Local Controller mode to Printer Controller mode, the terminal transmits the characters accumulated in the receive silo to the printer.

2.3.2 Save Settings

The **Save settings** menu item, when invoked by pressing **[Return]** or **[Enter]**, causes the settings for most Set-Up controlled operating features to be saved in nonvolatile memory, where they become the power-on settings. These features include:

- Cursor direction
- Copy direction
- Auto wrap mode
- CRT saver
- Overscan
- Refresh rate
- Emulation mode
- VT default character set
- PCTerm mode and default character set
- Transmit 7-bit controls
- ASCII emulation features
- Keyboard language (VT or PC, not both)
- Up to 60 defined keys—Individual keys can have up to 255 characters. Limit = 804 bytes.
- All Communication submenu features (port select, word size...)
- All Modem submenu features
- All Printer submenu features
- Set-Up language

The **Keyboard encoding mode** is not saved.

2.3.3 Restore Settings

Restore settings is an action field that is used to replace all saved Set-Up parameters with the values stored in the NVR. As a side effect, this feature performs a disconnect sequence, clears the screen, reinitializes the user-definable keys (as saved), clears the soft character set, returns the cursor to the upper left corner, aborts any print operation in progress, aborts ESC, CSI, and DCS sequences (and control strings), and shows "Done" on the status line. This action can also be invoked by the RIS control function.

2.3.4 Exit Set-Up

The **Exit Set-Up** menu item, when invoked by pressing Enter, causes the terminal to exit Set-Up. The effect is identical to pressing the Set-Up key while in Set-Up.

2.4 Actions Menu

The following descriptions refer to [Figure 2-4](#). Certain Set-Up features instruct the terminal to perform an immediate action when they are invoked. These features are grouped in the terminal Actions menu.

2.4.1 Clear Display

The **Clear display** feature clears the active page (including the lines which have temporarily disappeared while the you are in Set-Up).

2.4.2 Clear Communications

This feature is used to clear communication buffers and flags. It does not affect the On-Line/Local state. It is not invoked by any control functions.

Clear communications:

- Aborts any print operation in progress
- Aborts any escape sequence, control sequence, control string, or character string processing (ESC, CSI, DCS, APC, OSC, PM, SOS)
- Clears the keyboard buffers
- Clears the receive silo
- Clears the transmit silo
- Takes the terminal out of Printer Controller mode
- Sends XON on to the host port
- Sends XON to the printer port if Printer To Host and XOFF are enabled
- Resets XOFF received flags on both ports
- Does not clear the screen
- Clears KAM locked condition
- Clears the "printer port has seen DSR since power up" flag

2.4.3 Reset Terminal

The **Reset terminal** Set-Up feature resets the terminal to a "known state." This function is like Soft Terminal Reset (DECSTR), except DECSTR resets the Character Set mode to "8-bit Characters" and **Reset terminal** does not.

Additions:

- Keyboard encoding is reset to "character."
- PCTerm mode is not selected.

Note

From Set-Up, a soft reset using **Reset terminal** always works regardless of the terminal mode; but using mode changing control functions (DECSCL, S8C1T, S7C1T, DECANM), the soft reset is sometimes not performed.

2.4.4 Restore Factory Defaults

Restore factory defaults is an action field that is used to replace all existing Set-Up parameters with the values stored in ROM (the factory defaults). As a side effect, it does the following:

- Performs a disconnect sequence
- Clears the screen
- Reinitializes the user-definable keys
- Clears the soft font
- Returns the cursor to upper-left corner
- Aborts print operations in progress
- Aborts ESC, CSI and DCS sequences
- Displays the "Done" message on the status line

2.4.5 Clock, Calculator, Show Character Sets

The **Clock**, **Calculator**, and **Show character sets** menu items are used to invoke the corresponding desktop productivity features. Refer to [Chapter 3](#).

2.4.6 Banner Message

The **Banner message** menu item invokes a dialog box with a 30-character limit. When this feature is selected, a dialog box is displayed with a reverse video text entry area 30 characters long. If a **Banner message** is currently defined, the message is displayed in the text entry area. The  or  key can then be used to delete backwards and enter a new **Banner message**.

All characters that can be generated by the keyboard are legal in this field (including primary and secondary keyboard language characters). As such, no errors are reported, except when trying to enter more than 30 characters. In this case, the bell rings (if enabled), and no more characters are accepted. The factory default message is no message at all (a blank field).

From the **Actions** menu, select **Banner message...**

1. Press  or  to display a dialog box.
2. Enter your banner message.
3. Press the  to select  button.
4. Press  or  to return to the Set-Up menu.
5. Use the **Save settings** menu item to save the Banner message.

2.5 Display Menu

Many of the features in [Figure 2-5](#) can be controlled by the host control sequences listed in [Table 2-3](#). The control functions are described in Chapter 5, [ANSI Control Functions](#).

Figure 2–5 Display Menu
Terminal Set-Up

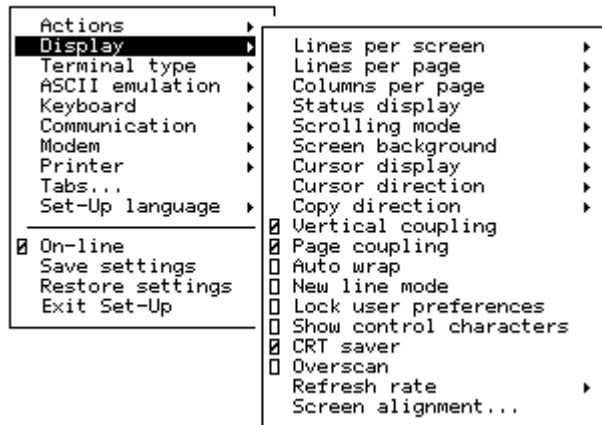
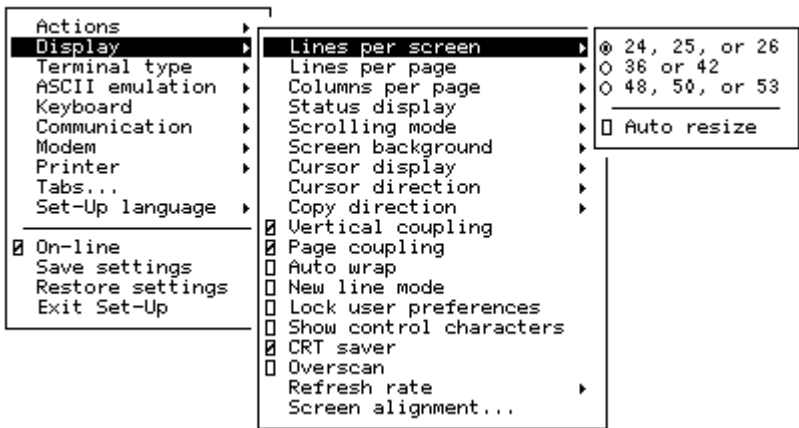


Table 2–3 Display Set-Up Features and Control Functions

Set-Up Feature	Control Function
Lines per screen	DECSNLS
Lines per page	DECSLPP
Columns per page	DECSCPP, DECCOLM
Clear on change	DECNCSM
Auto resize	DECARSM
Status display	DECSSDT
Scrolling mode	DECSCLM, DECSSCLS
Screen background	DECSCNM
Cursor display	DECTCEM, DECSCUSR
Cursor direction	DECRLM
Copy direction	DECRLCM
Vertical coupling	DECVCCM
Page coupling	DECPCCM
Auto wrap	DECAWM
New line mode	LNМ
Show control characters	CRM
CRT saver	DECCRTSM
Overscan	DECOSCNM
Refresh rate	DECSRFR

2.5.1 Lines per Screen

Figure 2–6 Lines per Screen Menu
Terminal Set-Up



This menu (Figure 2–6) chooses a font that enables the selected number of lines to be viewed. Note that you cannot view more lines than the number of lines on a page (**Lines per page**).

If the **Status display** is enabled, it is displayed using one of the same **Lines per screen** as the text for the corresponding session.

This is an operator preference feature and cannot be changed by control functions if the operator preference is set to **Lock user preferences**; however, if **Auto resize** is enabled at the same time, then the number of **Lines per screen** changes when the page size changes.

This feature can be changed from the host using the DECSNLS control function. Or, it may be changed indirectly from the host if **Auto resize** is selected and the page configuration is changed using DECSLPP.

2.5.1.1 Auto Resize

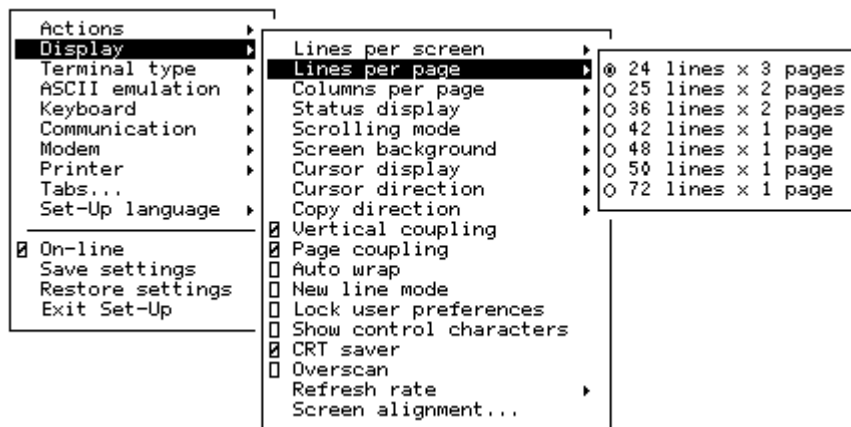
When **Auto resize** is enabled, the number of lines per screen changes automatically each time the page arrangement changes, either by the host or through Set-Up. Table 2–4 shows how the screen size changes whenever the page size changes when Auto Resize is set.

Table 2–4 Lines per Page vs.
Lines per Screen When Auto
Resize is Set

Lines per Page	Lines per Screen
24	26
25	26
36	43
42	43
43	43
48	52
52	52
72	52

2.5.2 Lines per Page

Figure 2–7 Lines per Page Menu
Terminal Set-Up

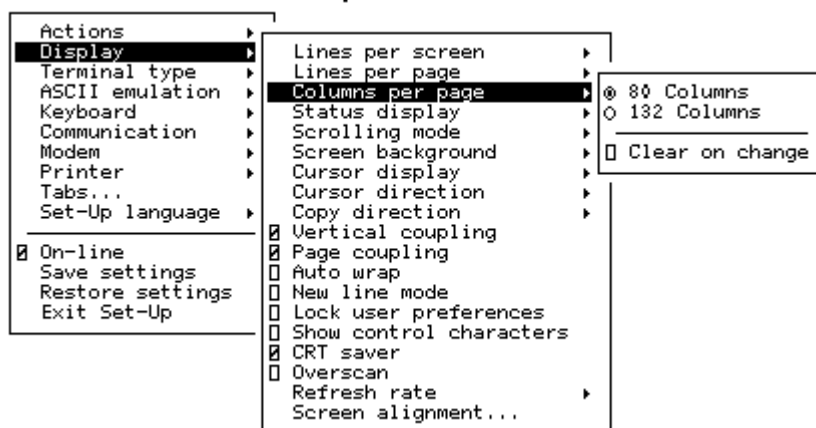


This menu (Figure 2–7) allows you to select the page size in lines and number of pages. The page size determines the addressing range for cursor positioning and scroll regions. The number of lines that can be displayed depends on the current setting of the **Lines per screen** feature.

This field is also invoked by the DECSLPP control function.

2.5.3 Columns per Page, Clear on Change

Figure 2–8 Columns per Page Menu
Terminal Set-Up



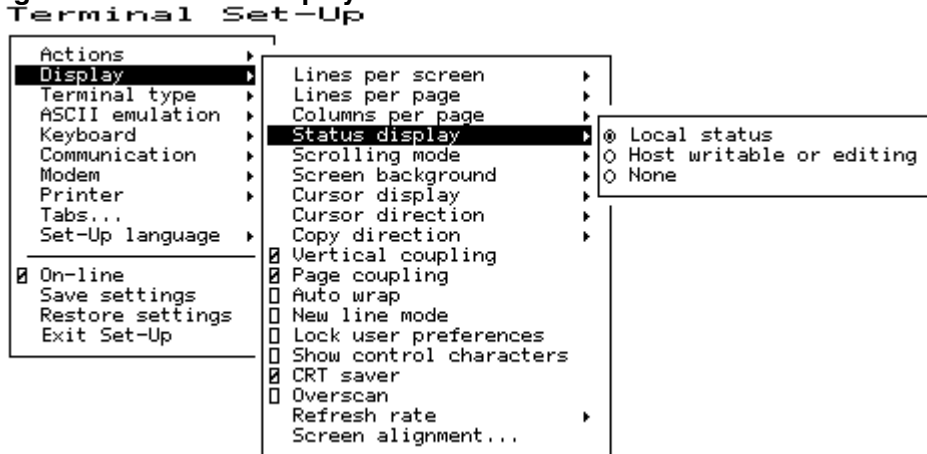
This menu (Figure 2–8) allows you to select an 80- or 132-column display for text. If **Clear on change** is disabled, changing this feature does not clear page memory, except when changing from 132 columns to 80 columns; then, columns 81 through 132 of each page are cleared. Changes to this field take effect immediately.

This field is also invoked by the DECSCPP and DECCOLM control functions. The **Clear on change** feature invokes the DECCOLM sequence only. *It does not erase page memory.*

2.5.4 Status Display

The **Status display** can be enabled in the selections shown in Figure 2–9.

Figure 2–9 Status Display Menu



2.5.5 Scrolling Mode

The **Scrolling mode** menu allows you to select how fast lines appear on the screen.

Scroll Selection	Scroll Rate
Slow smooth	Smooth steady scroll at approximately 9 lines per second (two scan lines per frame time).
Fast smooth	Smooth scrolls at approximately 18 lines per second.
Jump	Displays new lines as fast as they are received, causing a jump scroll on the screen.

This is a user-preference feature. If **Lock user preferences** is set, then this field is not invoked by any control functions.

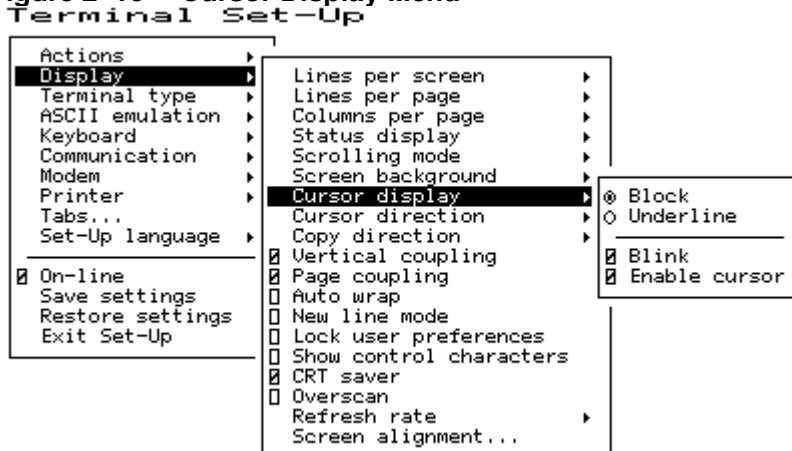
2.5.6 Screen Background

The **Screen background** can be either dark or light, with **dark** as the default.

2.5.7 Cursor Display

The **Cursor display** can be enabled in the selections shown in [Figure 2–10](#).

Figure 2–10 Cursor Display Menu



2.5.8 Cursor Direction

The **Cursor direction** default is **left to right**. If a Hebrew keyboard is connected to the terminal, a selection is available for right to left.

2.5.9 Copy Direction

The **Copy direction** default is **left to right**. If desired, you can change this selection to right to left. The cursor direction and the copy direction should be matched.

2.5.10 Vertical Coupling

Vertical coupling selects whether the user window automatically pans to follow the cursor when the cursor is moved vertically to a part of the page that is not in the currently displayed user window. Panning does not occur until the input buffer becomes empty and the cursor is displayed.

When **Vertical coupling** is disabled, the user window does not follow the cursor. If the cursor moves above or below the visible portion of the page, the cursor is no longer visible.

2.5.11 Page Coupling

Page coupling selects whether the user window automatically follows the cursor when the cursor is moved to a page that is not currently in the user window. When page coupling is enabled, moving the cursor to a page that is not currently displayed causes that page to be displayed. Panning does not occur until the input buffer becomes empty and the cursor is displayed.

When **Page coupling** is disabled, the user window does not follow the cursor. If the cursor moves to a page that is not currently displayed, the cursor is no longer visible.

2.5.12 Auto Wrap

Auto wrap selects where a received character is displayed when the cursor is at the right margin.

This field can be invoked by the DECAWM control function. The **Auto wrap** setting can be saved in NVM.

2.5.13 New Line Mode

When enabled, **New line mode** selects the characters transmitted by Return, CR, or CR+LF. It also determines the action taken by the terminal when receiving line feed, form feed, and vertical tab, LF, or CR+LF.

This feature can be invoked by the LNM (SM/RM) control function.

2.5.14 Lock User Preferences

The **Lock user preferences** feature allows you to prevent the host from modifying user-preference features. If set to lock, the user-preference features cannot be changed by host control functions.

The following user-preference features are locked and unlocked by this feature and when locked, cannot be changed by control functions:

- Auto repeat
- Scrolling mode
- Screen background
- Tab stops
- Keyboard definitions
- Lines per screen
- Auto resize
- Overscan
- Refresh rate

2.5.15 Show Control Characters

Show control characters allows you to select a normal display or a display called Control Representation mode or CRM. This "monitor and show all" display includes graphic representation of control characters.

When **Show control characters** is enabled, special CRM symbols are used for the C0 and C1 control areas of ISO 2022 conforming character sets. Other characters are displayed using the current user preference supplemental set. See Chapter 4, [ANSI Control Functions Summary](#).

2.5.16 CRT Saver

If the **CRT saver** feature is enabled and the terminal is inactive for 30 minutes (no keyboard activity or input from a host computer), the monitor screen goes blank to reduce wear on the CRT. No data is lost when the **CRT saver** feature is active. Any keyboard activity or input from the host computer reactivates the monitor. The factory default setting is **CRT saver** enabled.

When the monitor goes blank, a blinking hollow cursor is displayed in the lower-right corner of the screen.

2.5.17 Overscan

Overscan can be enabled for most character fonts. Overscan is disabled (dimmed) when the display **Refresh rate** is set to 72 Hz and the **Lines per screen** is set to either 36 or 48 lines.

2.5.18 Refresh Rate

The screen **Refresh rate** can be set to **72 Hz** (default) or to 60 Hz.

2.5.19 Screen Alignment

The **Screen alignment** menu item is used to invoke a screen alignment display ([Figure 2–11](#)). For international units only, screen rotation is provided to compensate for the Earth's magnetic field. Follow the instructions on the screen.

Figure 2–11 Screen Alignment Display

Hold down the Shift key while using the four arrow keys
to center the border on the screen.

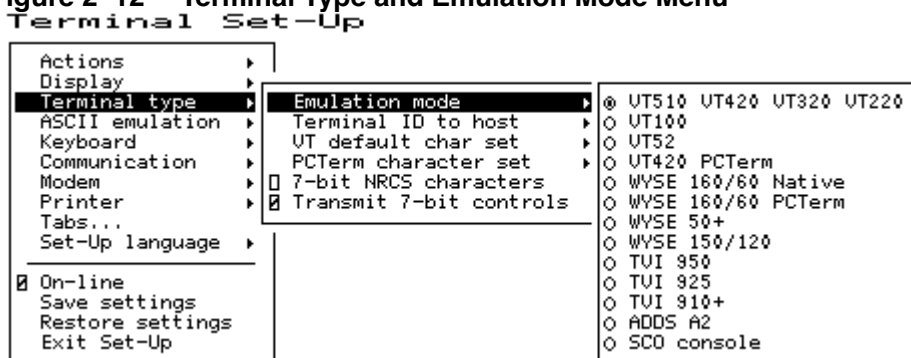
Hold down the Control key while using the left and right arrow keys
to rotate the border on the screen.

Press Enter to return to the Set-Up menu.

2.6 Terminal Type Menu

You can select terminal type and emulation modes using the menu shown in [Figure 2–12](#).

Figure 2–12 Terminal Type and Emulation Mode Menu



2.6.1 Emulation Mode

Emulation mode is the primary means of selecting different terminal modes or emulations. Changing terminal modes or emulations usually performs some initialization of the terminal state.

Example: Selecting a VT mode performs a soft terminal reset (DECSTR).

PCTerm mode can be selected or deselected independently from the other emulations. The ASCII modes use WYSE PCTerm emulation. The ANSI modes use VT420 PCTerm emulation.

This feature corresponds to the DECPCTERM, DECTME, DECANM, and DECSCL control functions described in Chapter 5, [ANSI Control Functions](#).

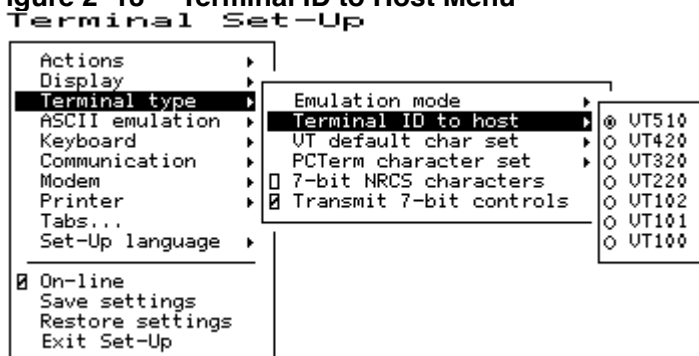
Changing the conformance level (DECSCL) does not change the operating level reported to identify the terminal to the host. It only changes the way extensions are reported.

Notes

- Printer operations are not affected or halted by a change in mode.
- A soft reset is always performed as a result of a mode change from Set-Up.
- Changes resulting from most, but not all, of the above sequences also cause a soft reset. Exception: Entering VT52 mode (via DECANM) does not cause a soft reset from VT100 mode, but does cause a soft reset from VT500 mode.

2.6.2 Terminal ID to Host

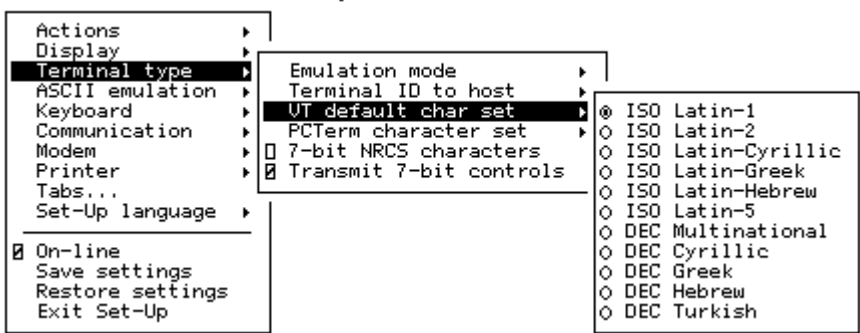
Figure 2–13 Terminal ID to Host Menu



Terminal ID to host (Figure 2–13) selects how the terminal identifies itself to host software, specifically the Primary Device Attributes response (DA). The default ID is "VT510." For a list of the responses, see the Device Attributes sections in Chapter 5. This field has no effect when the terminal is in VT52 mode. The feature corresponds to the DECTID control function described in Chapter 5.

2.6.3 VT Default Character Set

Figure 2–14 VT Default Character Set Menu

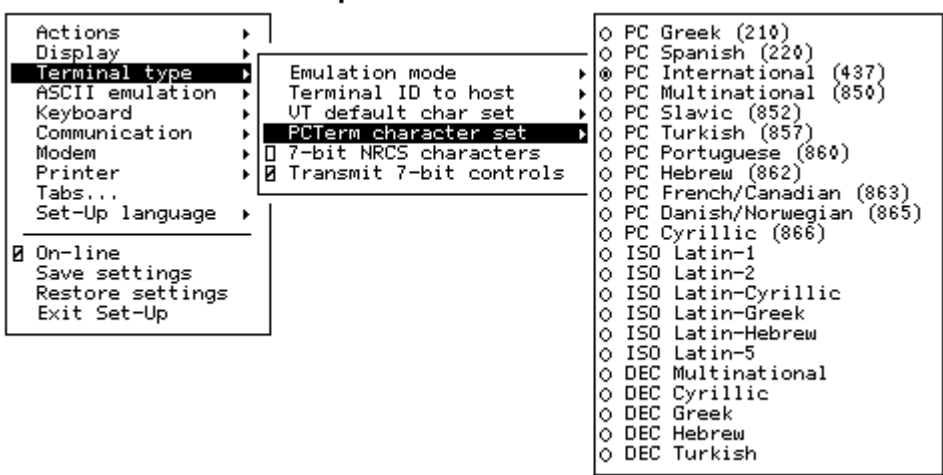


This feature (Figure 2–14) selects the default international character set to use for output as initialized in G0-G3 and GL/GR. It corresponds to the DECAUPSS (Assign User Preference Supplemental Set) control function described in Chapter 5.

Only ISO Latin-1 and DEC Multinational are available on the North American terminal.

2.6.4 PCTerm Character Set

Figure 2–15 PCTerm Character Set Menu



This feature (Figure 2–15) selects which PC code page will be used in PCTerm mode. This feature is also invoked by the DECPCTERM control function described in Chapter 5.

PC Greek, PC Spanish, PC International, PC Multilingual, and PC French/Canadian are available on the North American terminal.

2.6.5 7-Bit NRCS Characters

This feature enables National Replacement Character Set mode (DECNRCM), as described in Chapter 5. When 7-bit NRCS characters are selected, a corresponding 7-bit NRC set is used depending on the keyboard language. See Chapter 8, Keyboard Processing for details. Group 2 or secondary keyboard language characters may still be used if they are in the 7-bit character set. This feature is dimmed unless there is a corresponding NRC set available.

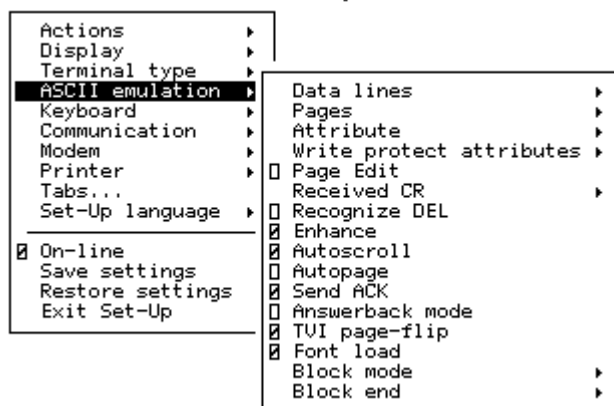
2.6.6 Transmit 7-Bit Controls

Transmit 7-bit controls selects whether C1 control codes are sent in their original 8-bit form (disabled) or converted to their 7-bit ESC Fe form.

This field is invoked by a parameter to the DECSCL control function and by the S7C1T and S8C1T control functions.

2.7 ASCII Emulation Menu

Figure 2–16 ASCII Emulation Menu
Terminal Set-Up



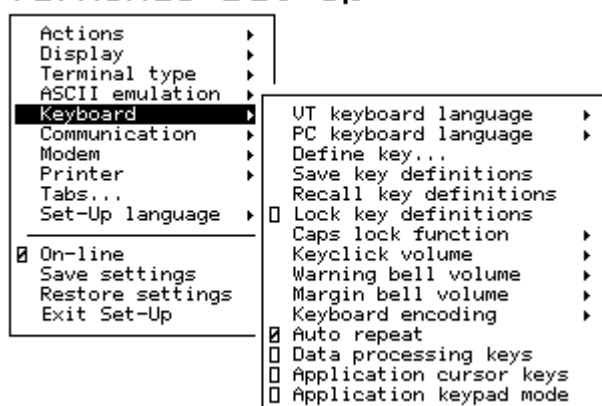
The features, shown in [Figure 2–16](#) and listed in [Table 2–5](#), are specific to the ASCII emulation modes supported in the VT510. The default is shown in **boldface** type. Refer to Part III for details.

Table 2–5 ASCII Emulation Features

Feature	Selections	Determine...
Data lines	24 , 25, 42, 43 lines	The number of data display lines visible, not counting any status lines.
Pages	1 × lines 2 × lines 4 × lines *	The page size and number of pages. 1 × lines selects each page to be the number of data lines visible on the screen. The * makes the display of a single page invisible.
Attribute	Character , line, or page	How visual attributes are applied per character, line, or page.
Write protect attributes	Dim , blank, reverse, underline, and/or blink.	The visual attributes used to highlight and specify write protected data.
Received CR	CR or CRLF	Carriage return or carriage return line feed.
Block Mode	Conversation or Block	Transfer method.
Block End	US/CR or CRLF/ETX	At the end of a block transfer, transmit a carriage return or an end-of-text character.

2.8 Keyboard Menu

Figure 2–17 Keyboard Menu
Terminal Set-Up



Several of the features shown in [Figure 2–17](#) are self-explanatory and correspond directly to control functions listed in [Table 2–6](#). For additional information, see [Chapters 4 and 5](#).

Table 2–6 Keyboard Features and Control Functions

Set-Up Feature	Control Function
Keyboard language	DECKBD
Switch between primary and secondary keyboard language	DECESKM (general), DECHEBM (Hebrew), DECNAKB (Greek), DECCYRM (Cyrillic)
Caps lock function	DECSLCK
Keyclick volume	DECSKCV
Warning bell volume	DECSWBV
Margin bell volume	DECSMBV
Keyboard encoding	DECKPM
Auto repeat	DECARM
Data processing keys	DECKBUM
Application cursor keys	DECCKM
Application keypad mode	DECKPNM, DECKPAM, DECNKM

2.8.1 Keyboard Language

Some keyboards allow you to select two different keyboard layouts and easily switch between them (English and Hebrew, for example). This feature allows the VT510 to support both existing conventions and emerging standards for extending the graphic input and/or switching between languages in dual language environments. (The North American keyboard has only one keyboard language, English.)

The primary keyboard language corresponds to Group 1 (per DIN 2137) and generally references the legends on the left half of the keytops.

The secondary keyboard language corresponds to Group 2 (Group Shift per DIN 2137) and generally references the legends on the right half of the keytops.

Unless overridden, **Ctrl/Alt/F1** makes the primary keyboard language active, and **Ctrl/Alt/F2** makes the secondary keyboard language active. (These are factory defaults, standard on PCs.)

Selecting a new keyboard language in Set-Up automatically initializes the keyboard character set, as described in [Chapter 8, Keyboard Processing](#).

Changes to this field take effect immediately. Changing this field to North American or Dutch and then exiting Set-Up or going to another menu resets the **7-bit NRCS characters** feature to disabled. There is no "7-bit characters" mode for the North American or Dutch keyboards. This field affects the language of the keyboard indicator line.

2.8.2 Define Key Editor

The Define Key Editor is an advanced function that allows the terminal to adapt to environments that do not match the preprogrammed settings.

Caution

This function is designed to be easy to use; however, unless you understand the consequences of remapping various key combinations, use caution when using this feature. When you press **F3** as the first key after the terminal is powered on, you always enter Set-Up, regardless of which keyboard you use or how **F3** is defined.

The **Define key** menu invokes the Define Key Editor.

To ensure consistent access to Set-Up features, the following rules apply for interpreting keystrokes.

- Within Set-Up and the Define Key Editor, the four arrow keys along with **Enter**, **Return**, **Do**, and **Select** are interpreted according to their standard definitions, regardless of how they may be redefined at other times. The one exception is when pressing the key to be defined.
- When copying one key definition to another, the standard key definition for a key is copied, not the current definition. This copy method ensures that keyboard functions are not lost within the Define Key Editor.

Figure 2–18 Define Key Editor, Select Function

2.8.2.1 Copy of Key Default—Moving a Standard Function

The simplest way to reprogram a key is to copy the behavior of another key. You can use **Copy of key default** to copy the factory default for a key to be defined to any position on the keyboard. You cannot, however, use this feature to edit the code transmitted by individual keys. To move factory default key functions:

1. From the **Keyboard** menu item, select the **Define key ...** function, and the **Define Key Editor** menu will appear.
2. Press the key for which you want to assign a new behavior. If the key is a function key, a screen similar to Figure 2–18 is displayed. Names are displayed in the Set-Up language selected (not according to keyboard language). The ± symbol indicates a toggle feature. Names are truncated to 12 characters in the definition field.
3. Press **⇒** to highlight the **Copy of key default** radio button ☒ and press **Enter**.
4. Press the key whose factory default behavior is what you want your defined key to do.
5. Press **⇩** to highlight the OK or Apply button and press **Enter**.

2.8.2.2 Customization

If you want to program a key to behave differently than one of the factory defined keys, then you will need to know about the following VT510 key categories:

- Function: Keys such as the arrow keys (**↑**, **↓**, **←**, **→**), the **Shift** modifier key, or the Set-Up key, used to transmit function key sequences or to perform local terminal functions.
- Alphanumeric: Keys used to transmit alphanumeric characters.

2.8.2.3 Modifier Keys

Modifier keys vary from within the Function and Alphanumeric categories. A modifier key is a key that modifies the behavior of other keys when it is pressed and held down. For example, pressing an alphanumeric key in combination with the **Shift** modifier key will normally send the shifted or uppercase characters for that key.

Modifier keys are treated as a special kind of local terminal function. The VT510 function modifier keys are: **Shift**, **Ctrl**, and **Alt**. VT510 alphanumeric keys can also be modified by pressing **Group Shift** (**Alt Gr** on enhanced PC keyboards) and **Alt/Shift** (Shift-2). Modifier keys themselves cannot normally be

modified by other keys. A key assigned to act as the **Shift** modifier, for example, cannot transmit a function sequence when pressed in combination with **Alt**. Defining a key as a modifier key makes all assignable combinations of that key act as a modifier.

2.8.2.4 Creating a New Function

To define a new function key:

1. From the **Keyboard** menu item, select the **Define key . . .** function, and the **Define Key Editor** menu will appear.
2. Press the key for which you want to assign a new behavior.
3. Press the  or  key to highlight the **Function** radio button  and press .
4. Press the  and  keys to highlight the modifier combination that you want to define (Unshifted, Shifted, Control, and so on) and press .
5. Press the  key to move to the **Select function** scroll box. Press the  and  keys to highlight the desired keystroke function from the list and press .
6. Press the  key to return to the **Modifiers**.
7. Repeat steps 4 through 6 to define other modifier combinations as desired.
8. Use the arrow keys (, , , ) to highlight the OK or Apply button and press .

2.8.2.5 Examples of Creating New Functions

Examples of when to create new functions include:

- Changing the  or  key to delete when unshifted and to backspace when shifted.
- Disabling the , , and Set-Up keys by assigning them to have no function.

2.8.2.6 Correcting a Mistake

If you make a mistake or want to start over, select the Cancel button or select the **Exit Set-Up** menu item. Your changes will not be recognized until you select the OK or Apply button. To save key definitions, select the **Save key definitions** menu item from the **Keyboard** menu.

The Define Key Editor is very powerful and you may make mistakes in learning how to use it. But no matter how you redefine the keys, you can always enter Set-Up by pressing  after powering on. Additionally, you can always select the **Restore factory defaults** menu item from the **Actions** menu.

2.8.2.7 Creating A New Alphanumeric Key or Keyboard Layout

The method for creating a new alphanumeric key is similar to that for function keys, except that you can define different modifier combinations and you enter alphanumeric values differently.

If the key was previously programmed as a function key, the function definition will be empty. Once any function definition is applied, all the alphanumeric definitions for that key are lost. *A single key cannot act as both a function key and an alphanumeric key simultaneously.*

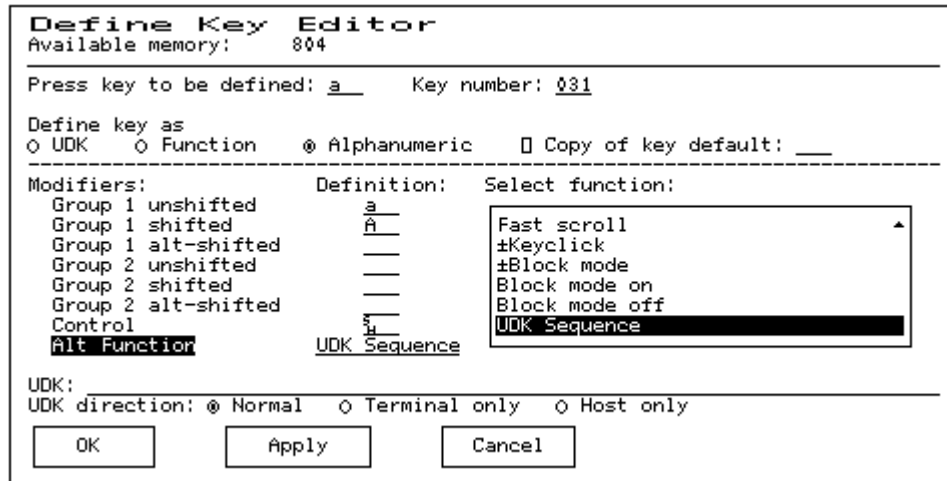
To define a new alphanumeric key:

1. From the **Keyboard** menu item, select the **Define key . . .** function, and the **Define Key Editor** menu will appear.
2. Press the key for which you want to assign a new behavior.
3. Press the  or  key to highlight the **Alphanumeric** radio button  and press . The character transmitted when this key is unshifted is highlighted.
4. To select a different character, press the corresponding key on the keyboard, or use the numeric compose key to generate a new character.
5. If desired, press the  key to move to the **Non-spacing accent** scroll box. Press the  and  keys to highlight any non-spacing accent from the list and press .
6. The code transmitted by the unshifted alphanumeric key when pressed in combination with the Control key is calculated automatically and displayed in the **Definition** column. If desired, you may

redefine this function.

7. If desired, select an Alt Function modifier combination for the alphanumeric key from the **Select Function** scroll box. You may choose a function, including a user-defined key (UDK) sequence. The default for the Alt Function is **None**.
8. Press the  key to return to the modifier selection.
9. Use the arrow keys (, , , ) to highlight the OK or Apply button and press .

Figure 2–19 Define Key Editor, Alphanumeric Alt Function



To enter alphanumeric values you can type the desired character or numeral, you can compose the desired character including numeric keypad compose, or you can select a nonspacing accent from the nonspacing accent scroll box.

2.8.2.8 Examples of Creating New Alphanumeric Keys

Examples of when to create new alphanumeric keys include:

- Making the comma and period keys send comma and period instead of angle brackets when shifted.
- Changing the keyboard from a QWERTY to QWERTZ layout.
- Defining an alternate key layout to match your own preference or local typing conventions (Dvorak, or Russian Cyrillic, instead of Bulgarian Cyrillic).

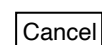
2.8.2.9 User-Defined Keys

The **UDK** radio button allows any key to be programmed with a user-defined sequence. UDKs are a subset of function keys. A separate UDK dialog box is provided for simplicity. Selecting **UDK** causes a **UDK:** text field to be displayed so you can enter a key sequence. This text field can scroll left or right as needed to allow longer strings to be entered.

Press  again to move the highlighting cursor to select one of the following UDK directions:

- Normal The sequence is sent to the host and/or to the screen depending on the communication settings (On-line, Local echo, Half duplex, and so on).
- Terminal only The sequence is sent only to the screen, as if it had just been received from the host.
- Host only The sequence is sent only to the host, regardless of the communication settings.

Press  again to select one of the following buttons:

When you select Cancel, you only cancel the changes to the currently selected modifier combination.

2.8.2.10 Programming Notes

- When you press  as the first key after the terminal is powered on, you always enter Set-Up, regardless of how  is defined.

- If a key is programmed to act as a modifier key, it operates as a modifier with any combination of **Shift**, **Alt**, or **Control**.
- If a key is programmed to be a modifier key and modifier key reporting is enabled using DECSMKR, the key sends the VT sequence for the left modifier key when there is more than one.
- **Caps Lock** combinations such as **Caps Lock/Print Screen** are local keyboard extensions and cannot be reprogrammed by DECPFK. The **Caps Lock** key acts as a modifier. The **Caps Lock** function can be moved to another key, but **Pause** must still send the Break.
- Use of the Define Key Editor to redefine keys on the numeric keypad affects only the Num Lock off state of these keys. When Num Lock is engaged, these keys will always send their factory default ASCII numerals (and related characters) intended for numeric input. The Num Lock behavior of these keys cannot be reprogrammed. (This Num Lock behavior does not apply to numeric keypad mode on the VT keyboard or PC keyboard in VT style.)
- The toggle Num Lock function can be assigned to any single key combination. Although the Num Lock state modifies other keys, the toggle Num Lock function is not a modifier key. This means modifier combinations of **Num Lock/Key** can be assigned to any other function or user-defined sequence.
- Use of the numeric keypad to compose characters by entering their decimal code in combination with **Alt Gr** or **Compose** (held down) should continue to work regardless of any reprogramming of keys on the numeric keypad, unless such reprogramming directly conflicts with use of the keypad in this manner. Example: A numeric keypad key has been defined as **Compose** or **Alt Gr**.
- Function keys also work in VT100 mode.

2.8.3 Save Key Definitions

To save your key definitions so they will be available the next time you turn the power on, select the **Save key definitions** menu item from the **Keyboard** menu.

This action field on the keyboard menu (Figure 2–17) causes the **Define key** modifications to be saved to nonvolatile memory independently from any other Set-Up features. Key definitions are saved on a first-come first-served basis and are limited by the total amount of nonvolatile memory available.

This feature cannot be invoked by a host control function.

2.8.4 Recall Key Definitions

This action field causes previously saved **Define key** modifications to be recalled from nonvolatile memory independently from any other Set-Up features.

2.8.5 Lock Key Definitions

The **Lock key definitions** menu item operates as a check box. When key definitions are locked, they cannot be reprogrammed from the host.

2.8.6 Caps Lock Function

This function allows you to enable the **Lock** or **Caps Lock** key to do the following:

- Caps lock Lock alpha keys on main keypad in uppercase setting.
- Shift lock Lock alpha and numeric keys on main keypad in shifted setting.
- Reverse lock Lock numeric keys in shifted setting, but lock alpha keys in lowercase setting.

2.8.7 Keyclick, Warning Bell, and Margin Bell Volume

The volume of these settings can be set to high, low, or off, with **high** as the default.

2.8.8 Keyboard Encoding

The **Keyboard encoding** menu allows you to select from the following:

- Character (ASCII) The keyboard uses normal ANSI/ISO character encoding.
- Scancode The keyboard transmits a scancode that represents the physical position of the key pressed within the keypad array.
- Key position The keyboard transmits key position codes for each down transition. This menu allows alternate keyboard layouts to be supported by application software.

This feature corresponds to the DECKPM (key position mode) control function described in Chapter 5, [ANSI Control Functions](#).

2.8.9 Auto Repeat

The **Auto repeat** feature selects whether keys begin auto repeating if still held down after a delay interval. The repeat rate is the number of "keystrokes" per second, not characters per second. Changes to this field take effect immediately.

This is a user-preference feature. When it is unlocked, this feature can be invoked by the DECARM control function.

2.8.10 Data Processing Keys

This feature corresponds to the DECKBUM control function described in Chapter 5, [ANSI Control Functions](#). Changes to this field take effect immediately.

2.8.11 Application Cursor Keys

This feature corresponds to the DECCKM control function described in Chapter 5, [ANSI Control Functions](#). This feature cannot be saved in NVM and is reset to the factory default setting by DECSTR. This feature cannot be locked. Changes to this field take effect immediately.

2.8.12 Application Keypad Mode

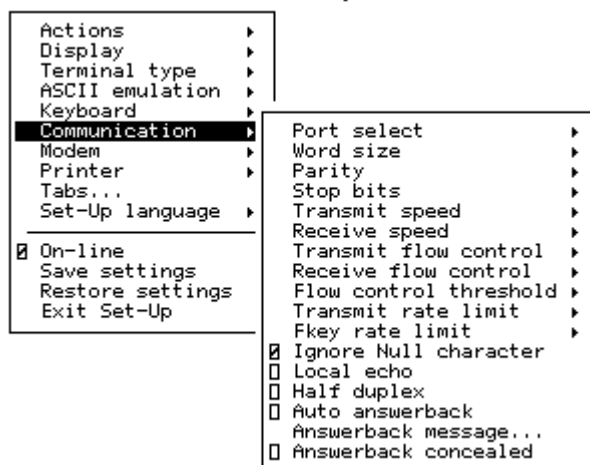
This feature selects whether the numeric keypad sends ASCII numerals or application function sequences. It corresponds to the DECKPNM, DECKPAM, and DECNKM control functions described in Chapter 5, [ANSI Control Functions](#).

This field is not stored in NVR. This field is reset to the power-up setting when a soft reset occurs (Reset Session or receipt of DECSTR).

This field is not a user-preference feature. It cannot be locked. Changes to this field take effect immediately so you can use the keypad to enter an Answerback message.

2.9 Communication Menu

Figure 2–20 Communication Menu
Terminal Set-Up



Several of the Communication features shown in [Figure 2–20](#) are self-explanatory and correspond directly to control functions listed in [Table 2–7](#). These functions are described in Chapter 5, [ANSI Control Functions](#).

Table 2–7 Communication Features and Control Functions

Set-Up Feature	Control Function
Comm port select	DECSCP
Comm word size	DECSP
Comm parity	DECSP
Comm stop bits	DECSP
Comm transmit speed	DECSCS
Comm receive speed	DECSCS
Comm transmit flow control	DECSFC
Comm receive flow control	DECSFC
Comm flow control threshold	DECSFC
Transmit rate limit	DECXRLM, DECSTR
Fkey rate limit	DECXRLM, DECSTR
Ignore Null character	DECNULM
Local echo	SRM
Half duplex	DECHDPXM
Auto answerback	DECAAM
Answerback message	DECLANS
Answerback concealed	DECCANSM

2.9.1 Port Select

This selection enables the cable configuration at the back of the terminal as follows:

- S1=comm1 print=comm2
 - S1=comm1 print=parallel
 - S1=comm2 print=comm1
 - S1=comm2 print=parallel

2.9.2 Word Size

The communication word size can be **8 bits** (default) or 7 bits.

2.9.3 Parity

You can select any of the following parity checks:

- None
 - Even
 - Odd
 - Even, unchecked
 - Odd, unchecked
 - Mark
 - Space

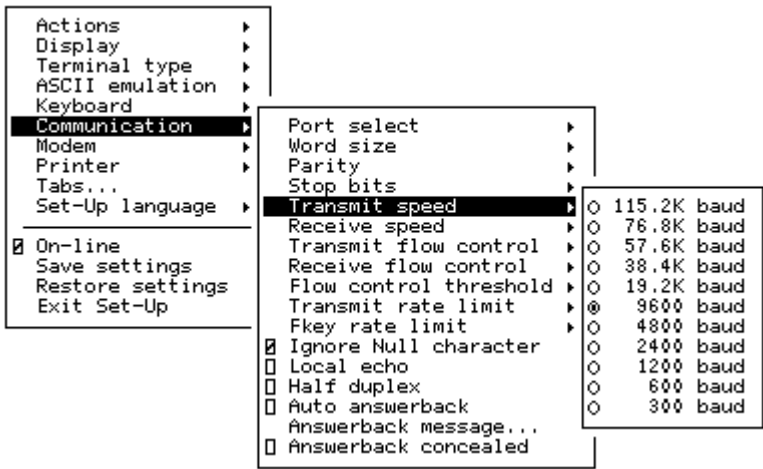
2.9.4 Stop Bits

For communication, **1** (default) or 2 stop bits can be enabled.

2.9.5 Transmit Speed

The communication **Transmit speed** is set to 9600 baud. You can select transmit speeds from the menu shown in [Figure 2–21](#).

Figure 2–21 Communication Transmit Speed Menu
Terminal Set-Up



2.9.6 Receive Speed

Like the communication **Transmit speed**, you can select the **Receive speed** from 300 to 115.2K baud. The default receive speed matches the transmit speed selection.

2.9.7 Transmit Flow Control

The **Transmit flow control** method can be one of the following:

- None
- XON/XOFF
- DSR (Data Send Ready)
- Both

Note

When an ASCII emulation is selected, the default **Transmit flow control** is **None**.

2.9.8 Receive Flow Control

The **Receive flow control** method can be one of the following:

- None
- XON/XOFF or XPC
- DTR (Data Transmit Ready)
- Both

2.9.9 Flow Control Threshold

You can set the **Flow control threshold** to Low (64 characters) or to High (768 characters). The default is **64**.

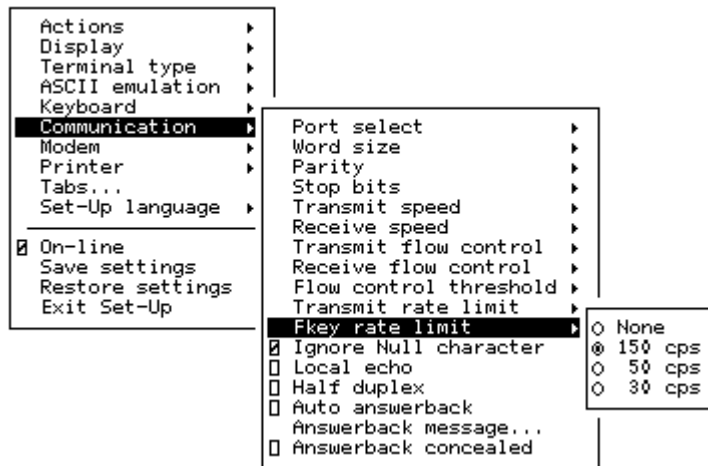
2.9.10 Transmit Rate Limit, Fkey Rate Limit

The **Transmit rate limit** choice limits the character rate from the keyboard to between 30 and 150 characters per second (selectable with fairly uniform separation), regardless of baud rate. This is fast enough to allow any keystroke to auto repeat at 30 Hz (baud rate permitting). The default is **150**.

This feature allows you to limit the transmit rate from the terminal so as to reduce the interrupt burden on the operating system.

You can select a different transmit rate for graphic keys and function keys using **Fkey rate limit**. The F keys transmit more than 1 byte per key press. This function may be selected through the DECXRLM and DECSTRL control function. Refer to Chapter 5, [ANSI Control Functions](#), for details.

Figure 2-22 Communication Fkey Rate Limit Menu
Terminal Set-Up



2.9.11 Ignore Null Character

When this menu item is selected, the terminal ignores received NUL control codes.

2.9.12 Local Echo

With **Local echo** enabled, most characters sent from the keyboard to the host are also displayed on the screen. The answerback message and XON/XOFFs are not echoed locally. Everything else is.

This feature corresponds to the SRM control function described in Chapter 5, [ANSI Control Functions](#).

Note

Local echo is different from the "Local" setting of **On-line** in the main menu of the Set-Up directory. **Local echo** causes keystrokes to be echoed on the screen, but it does not direct keyboard input to the ANSI parser, which also updates the screen. **Local echo** is temporarily disabled when the terminal is in either Local mode or Local Controller mode, because the keyboard input is already being redirected to the screen through the parser or to the printer.

2.9.13 Half Duplex

You can enable or disable **Half duplex** communication. The default is **disabled**. Half duplex communication is available only on the Comm1 communications port. If communication is set for the Comm2 port, this feature is disabled (dimmed).

2.9.14 Auto Answerback

This item is used to enable or disable **Auto answerback** on power-up or upon connection. The factory default is **disabled**.

2.9.15 Answerback Message

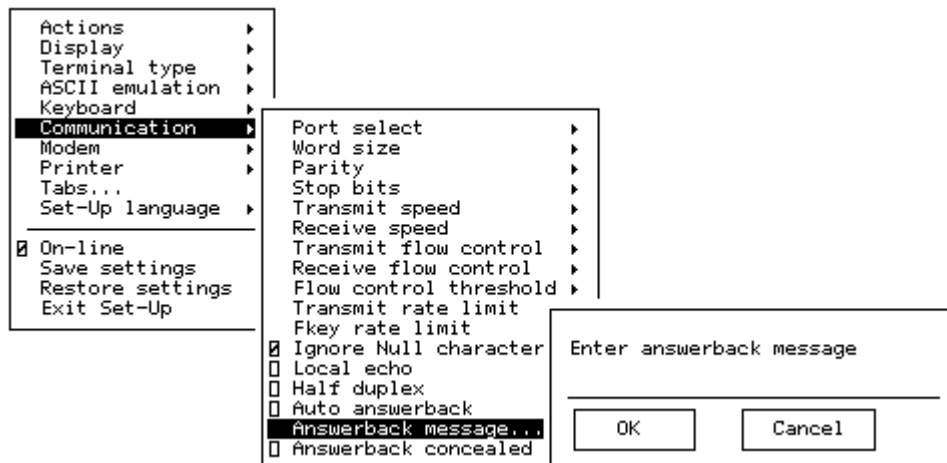
The **Answerback message . . .** menu item invokes a dialog box with a 30-character limit. All characters that can be generated by the keyboard are legal in this field (including primary and secondary keyboard language characters). As such, no errors are reported, except when trying to enter more than 30 characters. In this case, the bell rings (if enabled), and no more characters are accepted. The factory default message is no message at all (a blank field). Only a single Answerback message is saved in the

NVR. The last session from which a **Save settings** is performed overwrites this single Answerback message.

When this feature is selected, a dialog box is displayed with a reverse video text entry area 30 characters long. If an Answerback message is currently defined, the message is displayed in the text entry area. The ☒ or **Delete** key can then be used to delete backwards and enter a new Answerback message.

If the Answerback message is concealed, the cursor is displayed in the first position of the text entry area. In this case, the existing Answerback message does not need to be deleted before entering a new message.

Figure 2–23 Communication Answerback Dialog Box
Terminal Set-Up



The Answerback message can be up to 30 characters in length. Control characters are displayed using the CRM font. Pressing **Return** enters a return character in the Answerback message. Pressing **↓** with the highlighting cursor in the Answerback text field moves to the OK button.

The current Answerback message is displayed in the answerback dialog box unless **Answerback concealed** is selected. In this case, the previous Answerback message is not shown. Choosing the OK button to enter a new Answerback message clears the Answerback concealed check box.

2.9.16 Answerback Concealed

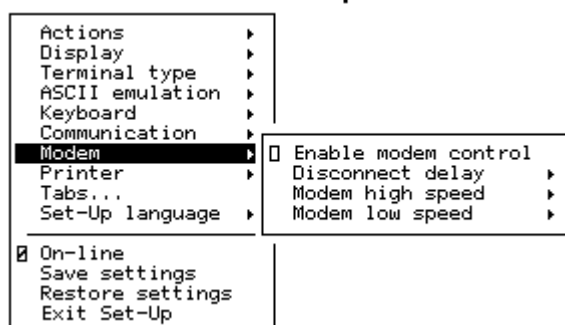
Changes to this field take effect immediately. The factory default is **disabled**.

If disabled, the Answerback message is visible in Set-Up. If enabled, the message is not visible and the text "<concealed>" replaces the message. The only way to return this field to disabled (visible) is to enter a new message.

2.10 Modem Menu

You can select several modem features using the menu shown in [Figure 2–24](#).

Figure 2–24 Modem Menu
Terminal Set-Up



2.10.1 Enable Modem Control

The field allows you to select whether the additional hardware control signals at the interface connector are used for modem control. Modem control disabled is also referred to as "Data leads only." Refer to Chapter 9, [Communications](#), for details on modem control signals.

This feature corresponds to DECMCM in Chapter 5, [ANSI Control Functions](#).

2.10.2 Disconnect Delay

When modem control is enabled, the **Disconnect delay** feature determines the time allowed before the terminal disconnects from the communications line when the received line signal detect (RLSD) is lost. Disconnect delay is in effect only when RLSD (CD) is lost. If DSR is lost, the terminal performs a disconnect immediately. The following selections are available:

- 2 seconds
 - 60 ms
 - No disconnect

All countries except the United Kingdom should use a delay of 2 seconds. The 60 ms delay is for use in the United Kingdom.

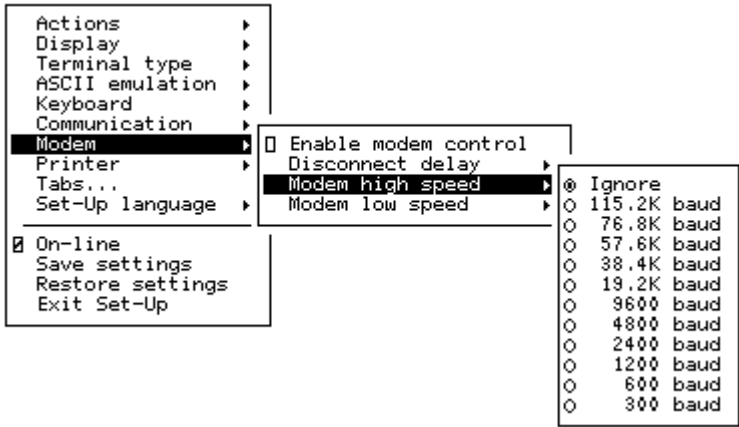
If the VT510 detects a loss of carrier and you selected No disconnect, the VT510 ignores RLSD (CD) after the beginning of the connection.

If you try to disconnect and reconnect the line, the VT510 checks if RLSD is asserted before granting the connection. Once it is connected, the terminal ignores the loss of carrier.

You can select disconnect delay through Set-Up or through control function DECSDDT. Refer to Chapter 9, [Communications](#), for details on the connect and disconnect process.

2.10.3 Modem High Speed

Figure 2–25 Modem High Speed Menu
Terminal Set-Up



When modem control is enabled, the speed indicator signal (SI) from the modem may be used to select the communication rate. This feature sets the communication rate to be used when the speed indicator line is "on." Selecting Ignore causes the terminal to use its regular transmit and receive speeds as it would when modem control is disabled.

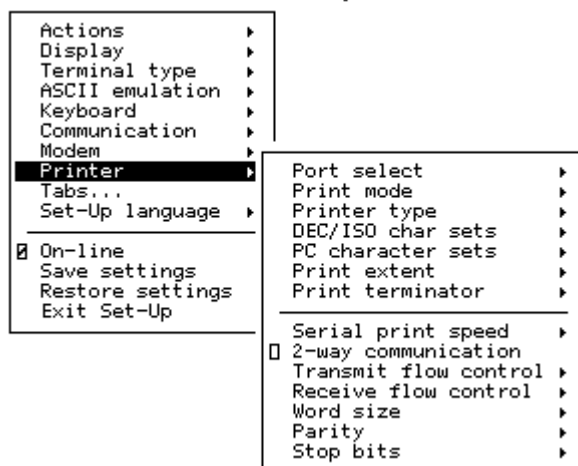
This feature can be invoked by DECSCS.

2.10.4 Modem Low Speed

These selections are the same as the modem high speed selections shown in [Figure 2–25](#). When modem control is enabled, the speed indicator (SI) signal from the modem may be used to select the communication rate. This feature sets the communication rate to be used when the speed indicator line is "off." Selecting Ignore causes the terminal to use its regular transmit and receive speed as it would when modem control is disabled. This feature can be invoked by DECSCS.

2.11 Printer Menu

Figure 2–26 Printer Menu
Terminal Set-Up



Printer features shown in [Figure 2–26](#) correspond to the control functions listed in [Table 2–8](#). These functions are described in Chapter 5, [ANSI Control Functions](#) and Chapter 10, [Printer Port](#).

2.11.1 Port Select

The **Port select** menu is the same one that appears in the **Communication** menu. You can select the cable connections from either menu.

- S1=comm1 print=comm2
 - S1=comm1 print=parallel
 - S1=comm2 print=comm1
 - S1=comm2 print=parallel

2.11.2 Print Mode

The **Print mode** menu allows you to select the printer operating mode as follows:

- Normal
 - Autoprint
 - Controller

Note

We recommend that you do not save the Controller mode selection in NVR because this may result in a "hung" terminal if the printer does not have DTR asserted.

Local echo is temporarily disabled when the terminal is in either Local mode or Local Controller mode, because the keyboard input is already being redirected to the screen through the parser or to the printer.

Table 2–8 Printer Features and Control Functions

Set-Up Feature	Control Function
Port select	DECSCP
Print mode	MC
Printer type	DECSPRTT
DEC/ISO char sets	DECSDPT
PC character sets	DECSPPCS
Print extent	DECPEX
Print terminator	DECPFF
Serial print speed	DECSCS
2-way communication	MC
Transmit flow control	DECSCFC
Receive flow control	DECSCFC
Word size	DECSP
Parity	DECSP
Stop bits	DECSP

2.11.3 Printer Type

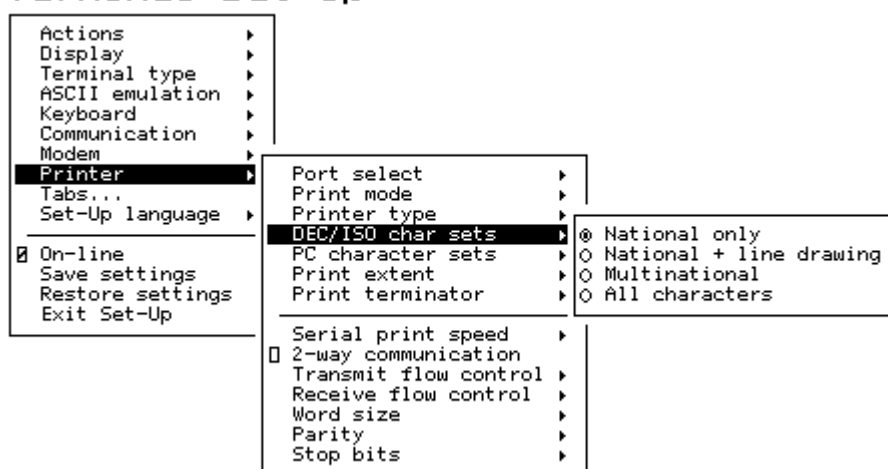
You can select the printer type as follows:

- DEC ANSI
 - IBM ProPrinter
 - DEC + IBM

2.11.4 DEC/ISO Character Sets

On the international versions of the terminal, you can enable the character set categories shown in [Figure 2–27](#).

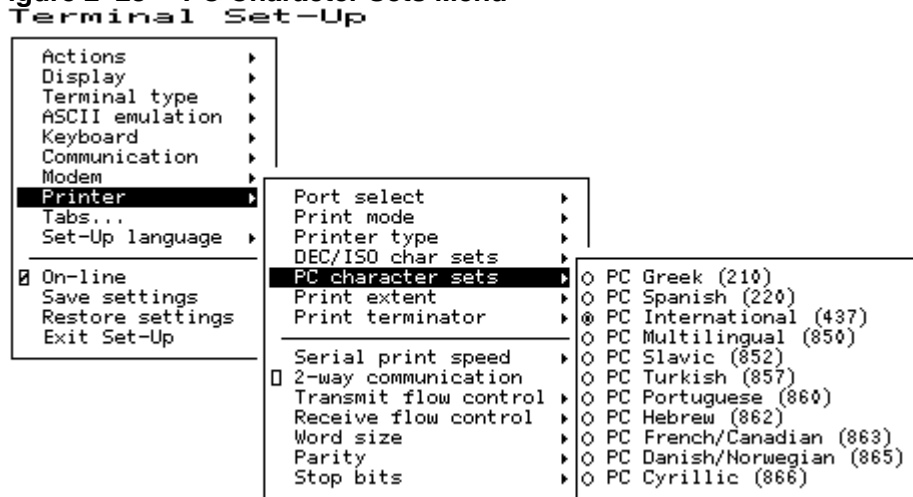
Figure 2–27 DEC/ISO Character Sets Menu



2.11.5 PC Character Sets

On the international versions of the terminal, you can enable the character set categories shown in [Figure 2–28](#). The numbers in parentheses refer to standard PC code pages.

Figure 2–28 PC Character Sets Menu



2.11.6 Print Extent

This selection allows you to print a **full page** or just the scroll region.

2.11.7 Print Terminator

The print terminator can be **None** or FF (form feed).

2.11.8 Serial Print Speed

Like the **Communication** menu speed selection (see [Figure 2–21](#)), you can select the serial printer speed from 300 to 115.2K baud. The default is **4800** baud.

2.11.9 2-Way Communication

This item allows you to enable and disable bidirectional communications on the printer port. The default is 2-way communication **disabled**.

2.11.10 Transmit Flow Control

The printer transmit flow control method can be one of the following:

- None
- XON/XOFF
- DSR (Data Send Ready)
- Both

2.11.11 Receive Flow Control

The printer receive flow control method can be one of the following:

- None
- XON/XOFF
- DTR (Data Terminal Ready)
- Both

2.11.12 Word Size

The printer word size can be **8 bits** (default) or 7 bits.

2.11.13 Parity

You can select any of the following parity checks to the printer:

- None
 - Even
 - Odd
 - Mark
 - Space

2.11.14 Stop Bits

For the printer, **1** (default) or 2 stop bits can be enabled.

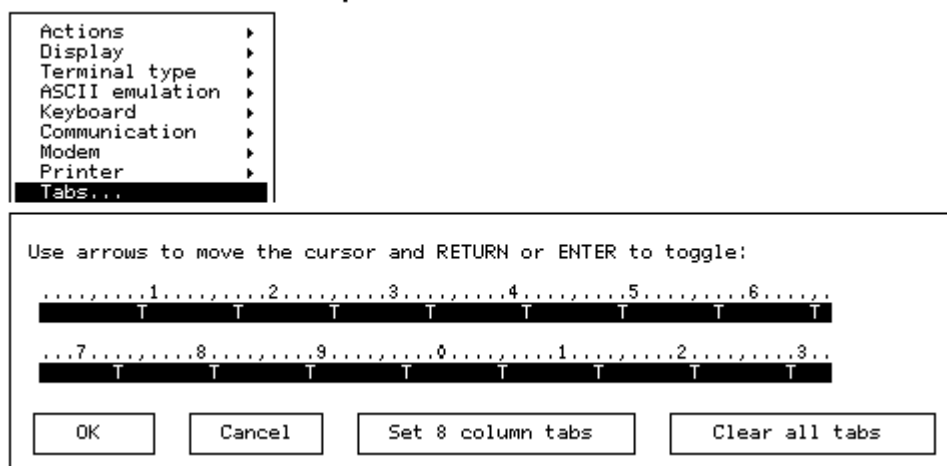
2.12 Tabs . . .

The **Tabs . . .** menu item is used to invoke the Tab Set-Up dialog box, displaying a 132-column tab ruler. [Figure 2–29](#) shows that tab stops are indicated by the letter T.

The highlighting cursor is initially displayed over the entire tab field. The normal character cursor appears in column 1 of the tab field.

The ← and → keys move the cursor within the tab field. Pressing Tab advances the cursor to the next tab stop in the tab field (if any). The character cursor automatically wraps between the end of the first row and the beginning of the second row in the tab field.

Figure 2–29 Tab Set-Up Dialog Box
Terminal Set-Up



The labeled buttons allow you to clear all tabs or to set 8-column tabs directly. To set individual tabs:

1. Position the cursor in the desired column.
2. Press Enter to set or clear a tab in that column.

Tab setting can also be invoked by the HTS and DECST8C control functions.

2.13 Default All Modes

[Table 2–9](#) lists the default for each feature in the Set-up Menu.

Table 2–9 Default All Modes

From Set-up Menu . . .	Set-Up Feature	Default
Display ▸	Columns per page	80
	Clear on change	On
	Screen background	Dark
	Cursor display	Block, Blink, On
	Auto wrap	Off
	New line mode	Off
	Status display	Local status
	Scrolling mode	Slow smooth
	CRT Saver	On
	Show control characters	Off
Terminal Type ▸	Emulation mode	WYSE 160/60 Native
ASCII Emulation ▸	Data lines	24
	Character cell	10 × 16
	Pages	1 × lines
	Attribute	Char
	Write protect attributes	Dim
	Page edit	Off
	Received CR	CR
	Recognize DEL	Off
	Enhance	On
	Autoscroll	On
	Autopage	Off
	Send ACK	On
	Auto Answerback	Off
	Font load	On
	Block mode	Off
	Block end	US/CR
	Keyboard language	North American
	Caps lock function	Caps lock
Keyboard ▸	Keyclick volume	High
	Warning bell volume	High
	Margin bell	Off
	Keyboard encoding	Character (ASCII)
	Auto repeat	On
	Port select	S1=comm1, print=parallel
Communication ▸	Word size	8 bits
	Parity	None
	Stop bits	1

From Set-up Menu . . .	Set-Up Feature	Default
	Transmit speed	9600 baud
	Receive speed	Transmit speed
	Transmit flow control	None
	Receive flow control	XON/XOFF or XPC
	Flow control threshold	Low (64)
	Transmit rate limit	150 cps
	Fkey rate limit	150 cps
	Auto answerback	Off
	Answerback concealed	Off
Printer ▸	Port select	S1=comm1, print=parallel
	Print mode	Normal
Set-Up Language ▸		English
Tabs . . . ▸		Set 8 column tabs

3.1 Introduction

To enhance productivity, the VT510 provides three local functions that can be used at any time the terminal is powered up, even when the terminal is not connected to a host. These functions are:

- Calculator
- Clock
- Character Set Tables

This chapter describes these programs and their interaction with the terminal. The local functions can be accessed through Set-Up or through a user-programmable key combination. Chapter 2, [Terminal Set-Up](#), describes how to invoke these programs and how to assign a key combination to a program. When one of these programs is invoked, the screen is partially obscured by that program. Pressing `Ctrl/Z`, `F10` (`Exit`), or `ESC` exits the current program and reveals the entire screen contents.

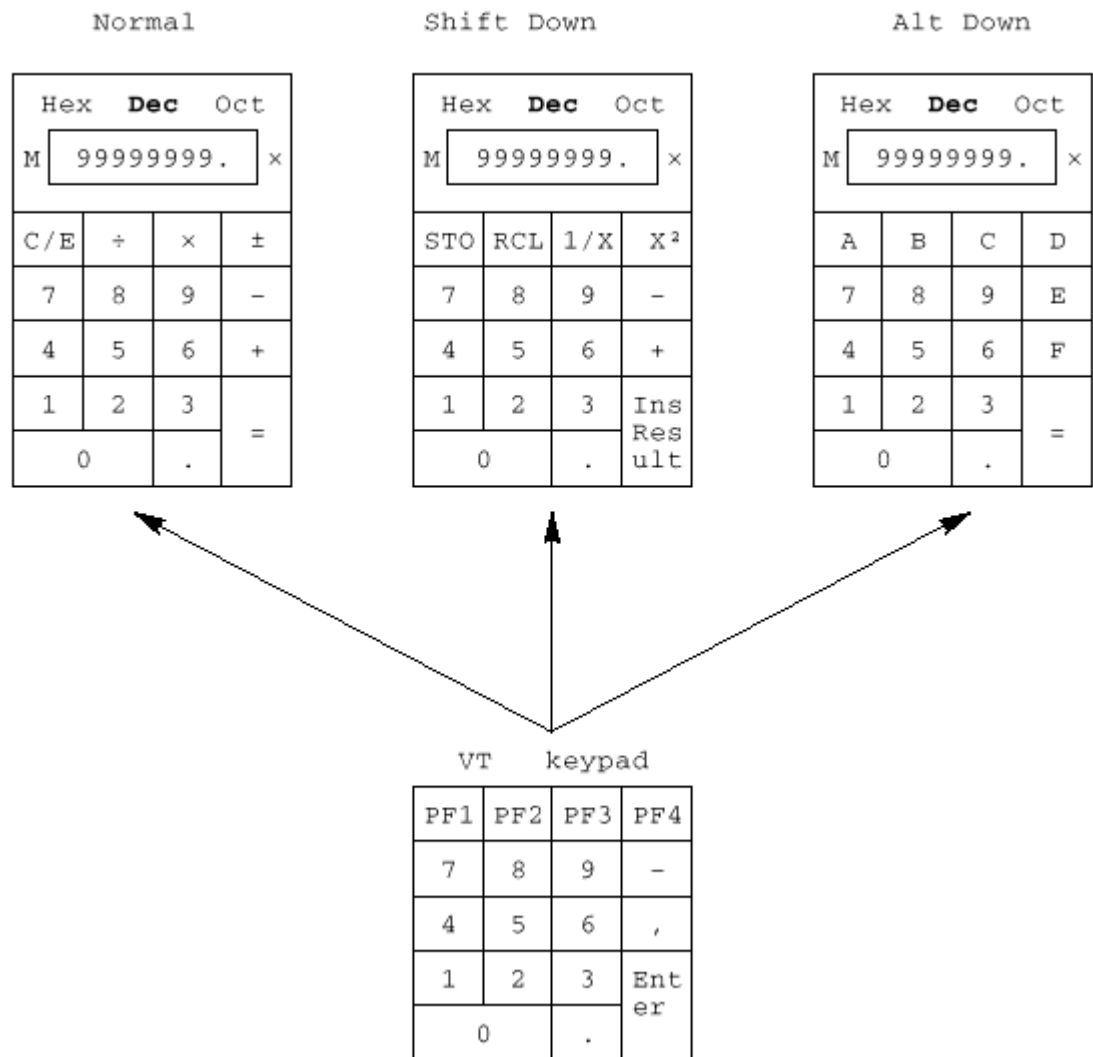
3.2 Calculator

You can enable the **Calculator** function in Set-Up by pressing `Caps Lock` `Alt/F12` if you are not in an ASCII emulation mode.

When you select the Calculator function, the terminal displays a simple calculator. See the leftmost layout in Figures 3-1 and 3-2. The keypad keys map directly to the calculator keys. The layout depends on whether a VT or Enhanced PC keyboard is attached, and whether a modifier key (`Shift` or `Alt`) is currently held down.

When a keypad key is pressed, the corresponding key is highlighted (shown in reverse video) on the screen. If a modifier key is pressed down, the display changes to show the new keypad assignment. See the two rightmost layouts in Figures 3-1 and 3-2. All keys not associated with the calculator are ignored.

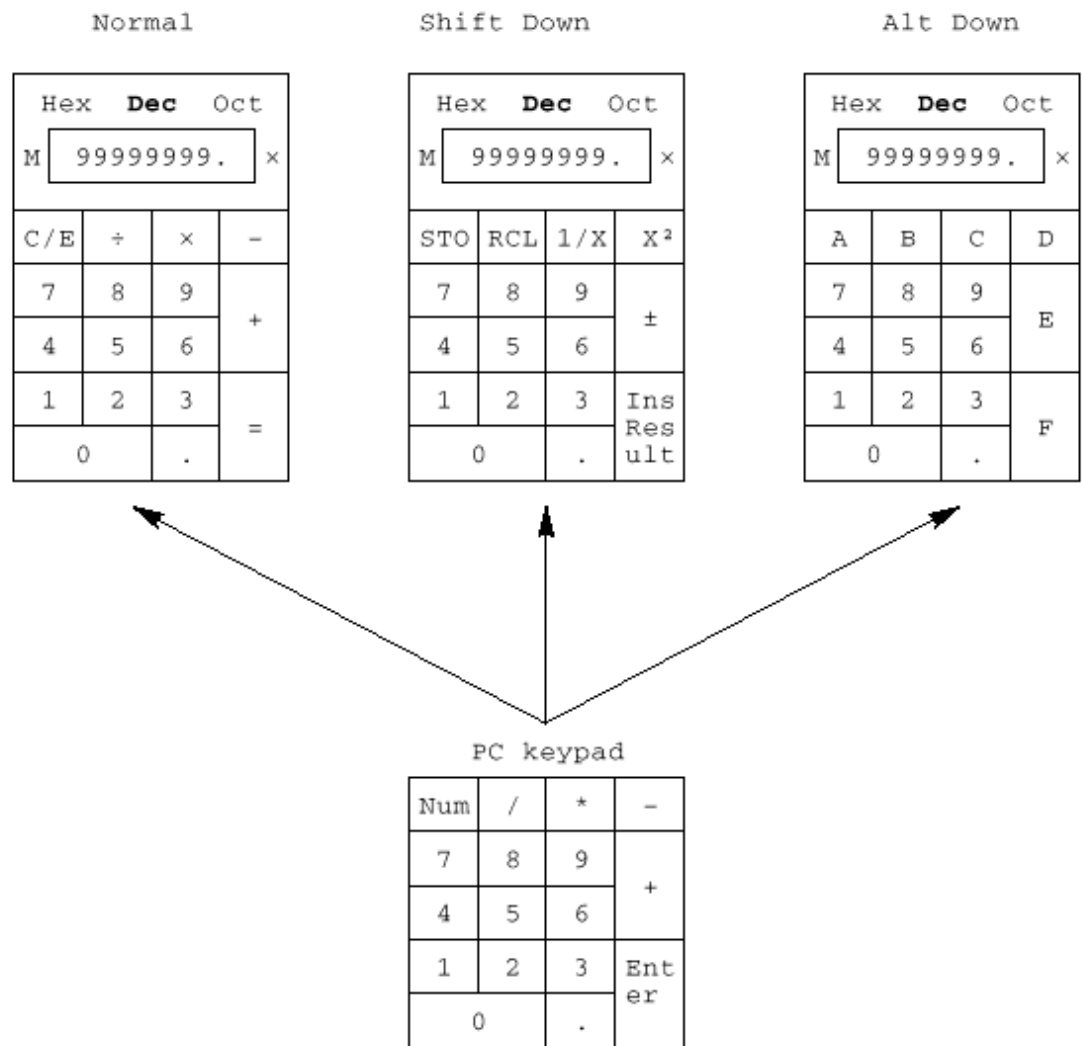
Figure 3–1 Calculator with VT Keyboard Layout



In addition to the numbers on the numeric keypad, you can use the following keys with the calculator:

Key	Function
[H], [O], or [D]	Select hexadecimal, octal, or decimal modes, respectively.
Arrow keys	Move the position of the calculator on the screen.
[Shift]	Changes the keypad display to allow you to select [STO], [RCL], [1/x], [X²], and Insert Result ([Shift/Enter]).
[Alt]	Changes the keypad display to hexadecimal and allows you to select keys [A] through [F] on the numeric keypad.
[C/E]	Clears the entry.
[STO]	Stores the number from the display in memory.
[RCL]	Recalls the number from memory and places it in the display.
[Shift/Enter] (Insert Results)	Inserts the result at the current cursor position after exiting the calculator feature.

Figure 3–2 Calculator with PC Keyboard Layout



Some keys on the main keypad can also be used for calculator functions:

Main Key	Key	Calculator Function
+	[.] (VT keyboard); [+] (PC keyboard)	
−	−	
*	×	
/	÷	
=	same as [Enter]	
.	.	
<X	[C/E]	

While the calculator is on the screen, all other terminal operations are suspended. The first time the calculator is invoked after the terminal is powered on, the display and memory registers contain 0. The display is eleven characters wide: eight characters are used to display the number's digits, one is used for the decimal point, one for the leading sign bit, and the one in the rightmost position is left blank.

The calculator allows the user to add, subtract, multiply, and divide real numbers in the range [−99,999,999. through +99,999,999.]. While one of these operations is in progress, the calculator displays the corresponding symbol [+ − × ÷] to the right of the display. (See the × in the figures.) You can also compute the reciprocal or the square of a number by pressing the [1/X] or [X²] keys, respectively.

All calculator math operations have equal priority except 1/X and X². If a result is wider than the display, then a rounded number will be displayed. The nonrounded result will continue to be used in subsequent

calculations.

The calculator has a memory register in which an intermediate result can be stored. When you press **STO**, you replace the content of this register with the number in the display. When you press **RCL**, you place the number from memory in the display. If any value is stored in memory, the calculator displays the letter M to the left of the display.

The number in the display (usually the last result) can be inserted at the current cursor position by pressing **Shift/Enter** (Insert Result), which exits the calculator and transmits the data to the host.

The calculator can also be used with hexadecimal and octal numbers in addition to the default (decimal). The **H**, **O**, or **D** keys put the calculator in hexadecimal, octal, or decimal modes, respectively. The current mode is in plain text and the other two modes are displayed with the dim attribute. In hexadecimal mode, the numbers A through F are entered by holding down **Alt** and typing the corresponding keypad key. See Figures 3-1 and 3-2. The decimal point cannot be used in the hexadecimal or octal modes, and any fractional part is truncated in these two modes.

3.3 Clock

The current time is displayed in the status line, if this feature is enabled. If the 12-hour format is selected, an A or P is displayed after the minutes.

The **Clock** function allows the current time to be set, the hourly chime to be enabled or disabled, and the alarms to be set and armed or disarmed. The current time can also be set from the host through an escape sequence. When the clock function is selected, the window shown in Figure 3-3 is displayed on the screen.

Figure 3-3 Clock Function

The screenshot shows a terminal window titled "Clock Function". It contains the following elements:

- A "Time:" label followed by a text field containing "12:34P" and a checkbox labeled "24 hour format".
- A checked checkbox labeled "Alarm1" followed by a time field "12:00A" and a message field "Message 1".
- An unchecked checkbox labeled "Alarm2" followed by a time field "12:00A" and a message field "Message 2".
- An unchecked checkbox labeled "Hourly chime".
- An unchecked checkbox labeled "Request time at Power-up".

The following keys have these functions:

Key(s)	Function
↓ and Tab	Move the cursor to the next text field or check box.
↑ and Shift/Tab	Move the cursor to the previous text field or check box.
← and →	Move the cursor inside a text field.
Return and Enter	Toggle the state of the check box.
← and →	Move the cursor inside the current input field. To change the contents of a field, type in the new information.
A or P	If the 24-hour time format is selected, these keys are ignored. If the 12-hour time format is selected, sets the time to either AM or PM.

You do not need to type the colon (:) character in the time field; the cursor skips over that character position. The check boxes next to the alarms are used to enable or disable the alarms. When an alarm time comes due and the alarm is enabled, the terminal sounds the alarm for 5 seconds or until you type any key.

The terminal also flashes "Alarm1" or "Alarm2" in the error line and the corresponding message associated with the alarm. The maximum size of the alarm messages is 20 characters. When the alarm is cleared (either by typing a key or after the 5 second timeout), the error line also disappears, revealing the status line if it was enabled.

If the hourly chime check box is enabled, the terminal emits a double beep every hour on the hour.

The state of the alarms, the alarm times, and the alarm messages are saved in NVM if the user chooses **Save settings** from Set-Up.

3.4 Character Set Tables

When you select **Show character sets**, the current character set is shown on the screen. Nonprinting characters are shown in control representation mode wherever possible. The following keys have these functions:

Key(s)	Function
Arrow keys	Highlight (reverse video) any character in the table. The corresponding 8-bit code is displayed in hexadecimal, decimal, octal and binary modes.
Shift/Enter	Inserts the highlighted character at the current cursor position. Note that this only works with the current character set.
Next Screen and Prev Screen or Page Up and Page Down	Cycle through the available character sets.
Shift/L	Displays the line drawing character set.
Shift/T	Displays the technical character set.

ASCII terminal emulations support the character sets listed in [Table 3–1](#). They do not support any 8-bit or ISO standard character sets.

Table 3–1 ASCII Emulation Character Sets

Emulation Mode	Character Set
WYSE 160/60 Native	Native/WY-50+
WYSE 160/60 PCTerm	PC Multilingual (Code Page 850)
WYSE 50+, 150/120	Native/WY-50+
TVI 950, 925, 910+	Native/WY-50+
ADDS A2	Native/WY-50+

Part II

VT510 ANSI Mode

This part includes the following chapters:

- Chapter 4, [ANSI Control Functions Summary](#)
- Chapter 5, [ANSI Control Functions](#)
- Chapter 6, [SCO Console Emulation](#)
- Chapter 7, [Character Sets](#)
- Chapter 8, [Keyboard Processing](#)
- Chapter 9, [Communications](#)
- Chapter 10, [Printer Port](#)

ANSI Control Functions Summary

4.1 Introduction

This chapter provides a general description of ANSI escape and control sequences, defining control characters, control functions, escape sequences, and device control strings. This chapter provides information on working with 7- and 8-bit environments and the conventions used in each, and it describes the **Show control characters** Set-Up feature. In addition, this chapter provides summary tables of the ANSI control functions.

4.2 Control Characters

The purpose of a control character is to control an action such as line spacing, paging, or data flow. The terminal does not display control characters unless you select **Show control characters** from the **Display** Set-Up menu. There are two groups of control characters.

- C0 7-bit control characters, in columns 0 and 1 of the 8-bit code table

- C1 8-bit control characters, in columns 8 and 9 of the 8-bit code table

[Table 4–1](#) lists the C0 control characters the VT510 recognizes. [Table 4–2](#) lists the C1 control characters the VT510 recognizes. You can also code C1 control characters as 7-bit escape sequences. [Table 4–3](#) lists the equivalent 7-bit sequences for 8-bit control characters. All three tables give column/row locations to help you find the characters in the character sets.

Table 4–1 C0 (7-Bit) Control Characters Recognized

Name	Mnemonic Column/Row	Function
Null	NUL 0/0	NUL has no function (ignored by the terminal).
Enquiry	ENQ 0/5	Sends the answerback message. (Communications Set-Up)
Bell	BEL 0/7	Sounds the bell tone if the bell is enabled in Keyboard Set-Up.
Backspace	BS 0/8	Moves the cursor one character position to the left. If the cursor is at the left margin, no action occurs.
Horizontal tab	HT 0/9	Moves the cursor to the next tab stop. If there are no more tab stops, the cursor moves to the right margin. HT does not cause text to auto wrap.
Line feed	LF 0/10	Causes a line feed or a new line operation, depending on the setting of line feed/new line mode.
Vertical tab	VT 0/11	Treated as LF.
Form feed	FF 0/12	Treated as LF.
Carriage return	CR 0/13	Moves the cursor to the left margin on the current line.
Shift out (locking shift 1)	SO (LS1) 0/14	Maps the G1 character set into GL. You designate G1 by using a select character set (SCS) sequence.
Shift in (locking shift 0)	SI (LS0) 0/15	Maps the G0 character set into GL. You designate G0 by using a select character set (SCS) sequence.
Device control 1 (XON)	DC1 1/1	Also known as XON. If XON/XOFF flow control is enabled in Communications Set-Up, DC1 clears DC3 (XOFF). This action causes the VT510 to continue sending characters.
Device control 3 (XOFF)	DC3 1/3	Also known as XOFF. If XON/XOFF flow control is enabled in Communications Set-Up, DC3 causes the VT510 to stop sending characters. The terminal cannot resume sending characters until it receives a DC1 control character.
Cancel	CAN 1/8	Immediately cancels an escape sequence, control sequence, or device control string in progress. In this case, the VT510 does not display any error character.
Substitute	SUB 1/10	Immediately cancels an escape sequence, control sequence, or device control string in progress, and displays a reverse question mark as an error character.
Escape	ESC 1/11	Introduces an escape sequence. ESC also cancels any escape sequence, control sequence, or device control string in progress.
Delete	DEL 7/15	Ignored when received, unless a 96- character set is mapped into GL. DEL is not used as a fill character. Digital does not recommend using DEL as a fill character. Use NUL instead.

Table 4–2 C1 (8-Bit) Control Characters Recognized

Name	Mnemonic Column/Row	Function
Index	IND 8/4	Moves the cursor down one line in the same column. If the cursor is at the bottom margin, the page scrolls up.
Next line	NEL 8/5	Moves the cursor to the first position on the next line. If the cursor is at the bottom margin, the page scrolls up.
Horizontal tab set	HTS 8/8	Sets a horizontal tab stop at the column where the cursor is.
Reverse index	RI 8/13	Moves the cursor up one line in the same column. If the cursor is at the top margin, the page scrolls down.
Single shift 2	SS2 8/14	Temporarily maps the G2 character set into GL, for the next graphic character. You designate the G2 set by using a select character set (SCS) sequence.
Single shift 3	SS3 8/15	Temporarily maps the G3 character set into GL, for the next graphic character. You designate the G3 set by using a select character set (SCS) sequence.
Device control string	DCS 9/0	Introduces a device control string that uses 8-bit characters. A DCS control string is used for loading function keys or a soft character set.
Start of string	SOS 9/8	Ignored.
VT identification	DECID 9/10	Makes the terminal send its device attributes response to the host (same as an ANSI device attributes (DA) sequence). Programs should use the ANSI DA sequence instead. If the printer is in controller mode, the terminal sends the sequence to the printer.
Control sequence introducer	CSI 9/11	Introduces a control sequence that uses 8-bit characters.
String terminator	ST 9/12	Ends a device control string. You use ST in combination with DCS.
Operating system command	OSC 9/13	Introduces an operating system command.*
Privacy message	PM 9/14	Introduces a privacy message string.*
Application program command	APC 9/15	Introduces an application program command.*

*The VT510 ignores all following characters until it receives a SUB, ST, or any other C1 control character.

Table 4–3 8-Bit Control Characters and Their 7-Bit Equivalents

Name	8-Bit Character	7-Bit Sequence
Index	IND 8/4	ESC D 1/11 4/4
Next line	NEL 8/5	ESC E 1/11 4/5
Horizontal tab set	HTS 8/8	ESC H 1/11 4/8
Reverse index	RI 8/13	ESC M 1/11 4/13
Single shift 2	SS2 8/14	ESC N 1/11 4/14
Single shift 3	SS3 8/15	ESC O 1/11 4/15
Device control string	DCS 9/0	ESC P 1/11 5/0
Start of string	SOS 9/8	ESC X 1/11 5/8
VT identification	DECID 9/10	ESC Z 1/11 5/10
Control sequence introducer	CSI 9/11	ESC [1/11 5/11
String terminator	ST 9/12	ESC \ 1/11 5/12
Operating system command	OSC 9/13	ESC] 1/11 5/13
Privacy message	PM 9/14	ESC ^ 1/11 5/14
Application program	APC 9/15	ESC _ 1/11 5/15

4.3 Control Functions

You use control functions to make the terminal perform special actions in your applications. Examples:

- Move the cursor.
- Delete a line of text.
- Select bold or underlined text.
- Change character sets.
- Make the terminal emulate a VT52 or VT100 terminal.

There are single-character and multiple-character control functions.

The single-character functions are the C0 and C1 control characters. You can use C0 characters in a 7-bit or 8-bit environment. C1 characters provide a few more functions than C0 characters, but you can only use C1 characters directly in an 8-bit environment.

Multiple-character functions provide many more functions than the C0 and C1 characters. Multiple-character functions can use control characters and graphic characters. There are three basic types of multiple-character functions.

- Escape sequences
- Control sequences
- Device control strings

Many sequences are based on ANSI and ISO standards and are used throughout the industry. Others are private sequences, created by some manufacturers, for specific families of products. ANSI sequences and private sequences follow ANSI and ISO standards for control functions.

In this manual, control functions created for the VT have the prefix DEC in their mnemonic name. For example, column mode has the mnemonic DECCOLM. All other control functions are standardized.

The following sections describe the format for escape sequences, control sequences, and device control strings.

Programming Tip

When you use control functions, remember that the binary codes define a function—not the graphic characters. This manual uses graphic characters from a Multinational character set to show control functions. If you use another character set, the graphic characters for control functions may change, but the code is always the same.

4.3.1 Sequence Format

This manual shows escape and control sequences in their 8-bit format. You can also use equivalent 7-bit sequences ([Table 4–3](#)).

The 8-bit format uses the C0 and C1 control characters and ASCII characters from the Multinational character set. The sequences also show each character's column/row position in the character set table, below the character. The column /row code eliminates confusion over similar-looking characters such as 0 (3/0) and O (4/15).

Note

Spaces appear between characters in a sequence for clarity. These spaces are not part of the sequence. If a space is part of the sequence, the SP (2/0) character appears.

4.3.2 Escape Sequences

An escape sequence uses two or more bytes to define a specific control function. Escape sequences do not include variable parameters, but may include intermediate characters. Here is the format for an escape sequence.

ESC	I	F
1/11	2/0 to 2/15	3/0 to 7/14
Escape character	Intermediate characters (zero or more characters)	Final character (one character)

ESC introduces escape sequences. After receiving the ESC control character, the terminal interprets the next received characters as part of the sequence.

I represents zero or more *intermediate characters* that can follow the ESC character. Intermediate characters come from the 2/0 through 2/15 range of the code table.

F is the *final character*. This character indicates the end of the sequence. The final character comes from the 3/0 through 7/14 range of the code table. The intermediate and final characters together define a single control function.

For example, the following escape sequence changes the current line of text to double-width, single-height characters:

ESC # 6
1/11 2/3 3/6

4.3.3 Control Sequences

A control sequence uses two or more bytes to define a specific control function. Control sequences usually include variable parameters. Here is the format for a control sequence.

CSI	P...P	I...I	F
9/11	3/0 to 3/15	2/0 to 2/15	4/0 to 7/14
Control sequence introducer	Parameter (zero or more characters)	Intermediate (zero or more characters)	Final (one character)

CSI is the *control sequence introducer*. You can also use the equivalent 7-bit sequence, ESC (1/11) [(5/11), as a substitute for CSI. After receiving CSI, the terminal interprets the next received characters as part of the sequence.

P...P are *parameter characters* received after CSI. These characters are in the 3/0 to 3/15 range in the code table. Parameter characters modify the action or interpretation of the sequence. You can use up to 16 parameters per sequence. You must use the ; (3/11) character to separate parameters.

All parameters are unsigned, positive decimal integers, with the most significant digit sent first. Any parameter greater than 9999 (decimal) is set to 9999 (decimal). If you do not specify a value, a 0 value is assumed. A 0 value or omitted parameter indicates a *default value* for the sequence. For most sequences, the default value is 1.

Note

All parameters must be positive decimal integers. Do not use a decimal point in a parameter —the terminal will ignore the command.

If the first character in a parameter string is the ? (3/15) character, it indicates that VT parameters follow. The terminal interprets VT parameters according to ANSI X3.64 and ISO 6429.

The VT510 processes two types of parameters, numeric and selective.

4.3.3.1 Numeric Parameters

A numeric parameter indicates a number value such as a margin location. In this manual, numeric parameters appear as actual values or as *Pn*, *Pn1*, *Pn2*, and so on.

The following is an example of a control sequence with numeric parameters:

CSI	5	;	2	0	r
9/11	3/5	3/11	3/2	3/0	7/2
Control sequence introducer	First numeric parameter	Delimiter	Second numeric parameter		Final character

This sequence sets the top and bottom margins of the current page. The top margin is at line 5, the bottom is at line 20. The ; (3/11) separates the two parameters.

4.3.3.2 Selective Parameters

A selective parameter selects an action associated with the specific parameter. In this manual, selective parameters usually appear as *Ps*, *Ps1*, *Ps2*, and so on.

The following is an example of a control sequence using selective parameters:

CSI	1	;	4	m
9/11	3/1	3/11	3/4	6/13
Control sequence introducer	First selective parameter	Delimiter	Second selective parameter	Final character

This control sequence turns on the bold and underline attribute at the cursor position. The parameters are 1 (indicating the bold attribute) and 4 (indicating the underline attribute). The ; (3/11) delimiter separates the two parameters.

I...I are zero or more *intermediate characters* received after CSI. These characters are in the 2/0 to 2/15 range.

F is the *final character* from the 4/0 to 7/14 range. The final character indicates the end of the sequence. The intermediate and final characters together define a control function. If there are no intermediate characters, the final character defines the function.

4.3.4 Device Control Strings

Device control strings (DCS), like control sequences, use two or more bytes to define specific control functions. However, a DCS also includes a data string. Here is the format for a device control string.

DCS	P...P	I...I	F	Data string	ST
9/0	3/0	2/0	4/0	*****	9/12
	to	to	to		
	3/15	2/15	7/14		
Device control string introducer	Zero or more parameter characters	Zero or more intermediate characters	Final	String	String terminator

DCS is the *device control string introducer*. DCS is the C1 control character at position 9/0. You can also use the equivalent 7-bit sequence, ESC (1/11) P (5/0). After receiving DCS, the terminal processes the next received characters as part of the string function.

P...P are *parameter characters* received after DCS. The use of parameter characters in a device control string is a Digital extension to the ANSI syntax. According to ANSI standards, any elements included after DCS are part of the data string.

Parameter characters are in the 3/0 to 3/15 range. They modify the action or interpretation of the device control string. You can use up to 16 parameters per string. Each parameter is separated with a ; (3/11) character. These characters follow the same rules as in a control sequence. See the "[Section 4.3.3](#)" section in this chapter.

I...I are zero or more intermediate characters received after CSI. These characters are in the 2/0 to 2/15 range.

F is the final character in the 4/0 to 7/14 range. The final character indicates the end of the string. The intermediate and final characters define the string. If there are no intermediates, the final character defines the string.

Data string follows the final character and usually includes several definition strings. Each definition string can be several characters in length. Individual strings are separated by the ; (3/11) delimiter.

ST is the *string terminator*. ST (9/12) indicates the end of a string. You can also use the equivalent 7-bit sequence, ESC (1/11) \ (5/12).

The following is an example of a device control string:

DCS	0	!	u	%	5	ST
9/0	3/0	2/1	7/5	2/5	3/5	9/12
Device control string introducer	Parameter	Intermediate	Final	Data string		String terminator

This device control string assigns a Supplemental Graphic set as the user-preferred supplemental set.

4.3.5 Using Control Characters in Sequences

You can use control characters—ESC, CAN, and SUB—to interrupt or recover from errors in escape sequences, control sequences, and device control strings.

- You can send ESC (1/11) to cancel a sequence in progress and begin a new sequence.
- You can send CAN (1/8) to indicate the present data is in error or to cancel a sequence in progress. The VT510 interprets the characters following CAN as usual.
- You can send SUB (1/10) to cancel a sequence in progress. The VT510 interprets the characters following SUB as usual.

The VT510 does not lose data when errors occur in escape or control sequences and device control strings. The terminal ignores unrecognized sequences and strings, unless they end a current escape

sequence.

4.3.6 7-Bit Code Extension Technique

You can represent all C1 control characters as 7-bit escape sequences. You can use the C1 characters indirectly, by representing them as 2-character escape sequences. ANSI calls this technique a 7-bit code extension. The 7-bit code extension provides a way of using C1 characters in applications written for a 7-bit environment. Here are some examples.

8-Bit C1 Character	7-Bit Code Extension Escape Sequence
CSI	ESC [
9/11	1/11 5/11
SS3	ESC O
8/15	1/11 4/15
IND	ESC D
8/4	1/11 4/4
DCS	ESC P
9/0	1/11 5/0

In general, you can use the 7-bit code extension technique in two ways.

- You can represent any C1 control character as a 2-character escape sequence. The second character of the sequence has a code that is 40 16 or 64 10 less than the code of the C1 character.
- You can make any escape sequence whose second character is in the range of 4/0 through 5/15 one byte shorter by removing the ESC character and adding 40 16 to the code of the second character. This generates an 8-bit control character. For example, you can change **ESC [** to **CSI** with this method.

4.4 Working with 7-Bit and 8-Bit Environments

There are three requirements for using one of the terminal's 8-bit character sets.

- Your program and communication environment must be 8-bit compatible.
- The terminal cannot be in national replacement character set mode (DECNRCM).
- The terminal must operate in VT level 4 or PC TERM mode. When the terminal operates in VT level 1 mode or VT52 mode, you are limited to working in a 7-bit environment.)

The following sections describe conventions that apply in VT level 4.

4.4.1 Conventions for Codes Received by the Terminal

The terminal expects to receive character codes in a form compatible with 8-bit coding. Your application can use the C0 and C1 control characters, as well as the 7-bit C1 code extensions, if necessary. The terminal always interprets these codes correctly.

When your program sends GL or GR codes, the terminal interprets the character codes according to the graphic character sets in use. When you turn on or reset the terminal, you automatically select the ASCII character set in GL and the current user-preferred character set in GR. You select the user-preferred set in the **Terminal type, VT default character set** Set-Up menu. This mapping assumes the current terminal mode is VT level 4.

4.4.2 Conventions for Codes Sent by the Terminal

The terminal can send data to an application in two ways.

- Directly from the keyboard
- In response to commands from the host (application or operating system)

Most function keys on the keyboard send multiple-character control functions. Many of these functions start with CSI (9/11) or SS3 (8/15), which are C1 characters. If your application cannot handle 8-bit characters, you can make the terminal automatically convert all C1 characters to their equivalent 7-bit code extensions before sending them to the application. To convert C1 characters, you use the DECSCL commands.)

By default, the terminal is set to automatically convert all C1 characters sent to the application to 7-bit code extensions. However, to ensure the correct mode of operation, always use the appropriate DECSCl commands.

Note

In VT level 4, the terminal can send GR graphic characters to an application, even if the application cannot handle 8-bit codes. However, in a 7-bit environment, the terminal sends C1 controls as 7-bit escape sequences and does not send 8-bit graphic characters.

New programs should accept both 7-bit and 8-bit forms of the C1 control characters.

4.5 Show Control Characters

The VT510 lets you display control characters as graphic characters, when you want to debug your applications. With this mode enabled, the terminal does not perform all control functions.

To enable this mode, use the **Display** Set-Up menu and enable the checkbox for **Show control characters**; or you can use the **control representation mode (CRM)** control sequence. (You cannot select this mode with an escape sequence.)

In VT level 4 mode

The terminal temporarily loads a special graphic character set into C0, GL, C1, and GR. Figures 4–1 and 4–2 shows this special set, called the display controls font. The terminal uses this font to display control characters on the screen.

In VT level 1 or VT52 mode

The terminal temporarily loads the left half of the display controls font into C0 and GL. The terminal uses this half of the font to display all C0 and GL characters. (C1 and GR are meaningless in VT52 or VT100 emulations.)

When displaying 36 or more lines on the screen

When you display 36 or more lines on the screen (DECSNLS), the terminal uses a smaller font to display control characters. The smaller font represents each control character as a two-character symbol instead of a three-character symbol. Figures 4–1 and 4–2 show what the control characters look like when displaying 24 or 25 lines on the screen. Table 4–4 shows the abbreviations for the control characters in the smaller font.

Table 4–4 Control Character Displayed on Screen

Control Character in Large Font	Control Character in Small Font	Name
NUL	NL	Null
SOH	SH	Start of heading
STX	SX	Start of text
ETX	EX	End of text
EOT	ET	End of transmission
ENQ	EN	Enquire
ACK	AK	Acknowledge
BEL	BL	Bell
BS	BS	Backspace
HT	HT	Horizontal tab
LF	LF	Line feed
VT	VT	Vertical tab
FF	FF	Form feed
CR	CR	Carriage return
SO	SO	Shift out
SI	SI	Shift in
DLE	DE	Data link escape
DC1	D1	Device control 1 (XON)
DC2	D2	Device control 2
DC3	D3	Device control 3 (XOFF)
DC4	D4	Device control 4
NAK	NK	Negative acknowledge
SYN	SY	Synchronous idle
ETB	EB	End of transmission block
CAN	CA	Cancel
EM	EM	End of medium
SUB	SB	Substitute
ESC	EC	Escape
FS	FS	Field separator
GS	GS	Group separator
RS	RS	Record separator
US	US	Unit separator
IND	IN	Index
NEL	NE	Next line
SSA	SA	Start selected area
ESA	EA	End selected area
HTS	HS	Horizontal tab set

Control Character in Large Font	Control Character in Small Font	Name
HTJ	HJ	Horizontal tab with justify
VTs	VS	Vertical tab set
PLD	PD	Partial line down
PLU	PU	Partial line up
RI	RI	Reverse index
SS2	S2	Single shift 2
SS3	S3	Single shift 3
DCS	DC	Device control string
PU1	P1	Private use 1
PU2	P2	Private use 2
STS	SS	Set transmit state
CCH	CC	Cancel character
MW	MW	Message waiting
SPA	SP	Start protected area
EPA	EP	End protected area
CSI	CS	Control sequence introducer
ST	ST	String terminator
OSC	OS	Operating system command
PM	PM	Private message
APC	AP	Application program command
NS	NS	No-break space

Exceptions

Some control functions still work in this mode.

- LF, FF, and VT cause a carriage return and line feed (CR LF) that move the cursor to a new line. The terminal displays the LF, FF, or VT character before performing the new line function.
- XOFF (DC3) and XON (DC1) maintain flow control, if enabled in set-up. The terminal displays the DC1 or DC3 character after performing the control function.
- The terminal does not display SSU session management commands.

Figure 4-1 Show Control Characters Font (Left Half)

Column		0		1		2		3		4		5		6		7	
b8 Bits		0		0		0		0		0		0		0		0	
b7		0		0		0		0		0		0		0		0	
b6		0		0		1		1		0		1		1		1	
b5		0		1		0		1		0		1		0		1	
Row	b4 b3 b2 b1																
0	0 0 0 0	NUL	00	DLO	20	SP	40	0	60	@	100	P	120	'	140	p	160
1	0 0 0 1	SOH	01	DC1	21	!	41	1	61	A	101	Q	121	a	141	q	161
2	0 0 1 0	STX	02	DC2	22	"	42	2	62	B	102	R	122	b	142	r	162
3	0 0 1 1	ETX	03	DC3	23	#	43	3	63	C	103	S	123	c	143	s	163
4	0 1 0 0	EOT	04	DC4	24	\$	44	4	64	D	104	T	124	d	144	t	164
5	0 1 0 1	ENQ	05	NAK	25	%	45	5	65	E	105	U	125	e	145	u	165
6	0 1 1 0	ACK	06	SYN	26	&	46	6	66	F	106	V	126	f	146	v	166
7	0 1 1 1	BEL	07	ETB	27	'	47	7	67	G	107	W	127	g	147	w	167
8	1 0 0 0	BS	10	CAN	30	(	50	8	70	H	110	X	130	h	150	x	170
9	1 0 0 1	HT	11	EM	31	)	51	9	71	I	111	Y	131	i	151	y	171
10	1 0 1 0	LF	12	?	32	*	52	:	72	J	112	Z	132	j	152	z	172
11	1 0 1 1	VT	13	ESC	33	+	53	;	73	K	113	[	133	k	153	{	173
12	1 1 0 0	FF	14	FS	34	,	54	<	74	L	114	\	134	l	154		174
13	1 1 0 1	CR	15	GS	35	-	55	=	75	M	115	]	135	m	155	}	175
14	1 1 1 0	SO	16	RS	36	.	56	>	76	N	116	^	136	n	156	~	176
15	1 1 1 1	SI	17	US	37	/	57	?	77	O	117	_	137	o	157	D _T	177

← C0 Codes → GL Codes (ASCII Graphic) →

Key		
Character	ESC	33 27 1B
		Octal Decimal Hex

Figure 4–2 Show Control Characters Font (Right Half)

8		9		10		11		12		13		14		15		Column	
1 0 0 0		1 0 0 1		1 0 1 0		1 0 1 1		1 1 0 0		1 1 0 1		1 1 1 0		1 1 1 1		b8 Bits b7 b6 b5 b4 b3 b2 b1	
8 0	200 128 80	D C S	220 144 90	N S P	240 160 A0	•	260 176 B0	À	300 192 C0	Ð	320 208 D0	à	340 224 E0	ø	360 240 F0	0 0 0 0	0
8 1	201 129 81	P U 1	221 145 91	i	241 161 A1	+	261 177 B1	Á	301 193 C1	Ñ	321 209 D1	á	341 225 E1	ñ	361 241 F1	0 0 0 1	1
8 2	202 130 82	P U 2	222 146 92	¢	242 162 A2	2	262 178 B2	Â	302 194 C2	Ò	322 210 D2	â	342 226 E2	ò	362 242 F2	0 0 1 0	2
8 3	203 131 83	S T S	223 147 93	£	243 163 A3	3	263 179 B3	Ã	303 195 C3	Ó	323 211 D3	ã	343 227 E3	ó	363 243 F3	0 0 1 1	3
I N D	204 132 84	C R H	224 148 94	¤	244 164 A4	'	264 180 B4	Ä	304 196 C4	Ô	324 212 D4	ä	344 228 E4	ô	364 244 F4	0 1 0 0	4
N E L	205 133 85	M W	225 149 95	¥	245 165 A5	µ	265 181 B5	Å	305 197 C5	Õ	325 213 D5	å	345 229 E5	õ	365 245 F5	0 1 0 1	5
S S A	206 134 86	S P A	226 150 96	¦	246 166 A6	¶	266 182 B6	Æ	306 198 C6	Ö	326 214 D6	æ	346 230 E6	ö	366 246 F6	0 1 1 0	6
E S A	207 135 87	E P A	227 151 97	§	247 167 A7	•	267 183 B7	Ç	307 199 C7	×	327 215 D7	ç	347 231 E7	÷	367 247 F7	0 1 1 1	7
H T S	210 136 88	9 8	230 152 98	"	250 168 A8	,	270 184 B8	È	310 200 C8	Ø	330 216 D8	é	350 232 E8	ø	370 248 F8	1 0 0 0	8
H T J	211 137 89	9 9	231 153 99	©	251 169 A9	1	271 185 B9	É	311 201 C9	Ù	331 217 D9	é	351 233 E9	ù	371 249 F9	1 0 0 1	9
V T S	212 138 9A	9 A	232 154 9A	ª	252 170 AA	0	272 186 BA	Ê	312 202 CA	Ú	332 218 DA	ê	352 234 EA	ú	372 250 FA	1 0 1 0	10
P L D	213 139 9B	C S I	233 155 9B	«	253 171 AB	»	273 187 BB	Ë	313 203 CB	Û	333 219 DB	ë	353 235 EB	û	373 251 FB	1 0 1 1	11
P L U	214 140 9C	S T	234 156 9C	¬	254 172 AC	1/4	274 188 BC	Ì	314 204 CC	Ü	334 220 DC	ì	354 236 EC	ü	374 252 FC	1 1 0 0	12
R I	215 141 9D	O S C	235 157 9D	—	255 173 AD	1/2	275 189 BD	Í	315 205 CD	Ý	335 221 DD	í	355 237 ED	ý	375 253 FD	1 1 0 1	13
S S 2	216 142 9E	P M	236 158 9E	®	256 174 AE	3/4	276 190 BE	Î	316 206 CE	Þ	336 222 DE	î	356 238 EE	þ	376 254 FE	1 1 1 0	14
S S 3	217 143 9F	A P C	237 159 9F	—	257 175 AF	¿	277 191 BF	Ï	317 207 CF	ß	337 223 DF	ï	357 239 EF	ÿ	377 255 FF	1 1 1 1	15

← C1 Codes → GR Codes (ISO Latin-1 Supplemental Graphic) →

4.6 ANSI Control Function Tables

The tables in this section summarize the ANSI Control Functions for ANSI-compatible VT mode and PCTerm mode. Default conditions are in boldface type. [Chapter 5](#) describes all the ANSI control functions in alphabetical order by mnemonic.

[Table 4–5](#) lists the text processing control functions; [Table 4–6](#), the reports control functions; [Table 4–7](#), the terminal management control functions; [Table 4–8](#), the keyboard processing control functions; and [Table 4–9](#), the communications control functions. In these tables, the word "same" signifies the same control sequences as in the VT510 column.

Table 4–5 Text Processing Control Functions

Active Position and Cursor	Control Function			
	VT510	VT420	VT100	Mnemonic
Auto Resize Mode		n/a	n/a	DECARSM
Set (enable):	CSI ? 98 h			
Reset (disable):	CSI ? 98 l			
Backspace	BS	same	same	BS
Carriage Return	CR	same	same	CR
Cursor Backward	CSI <i>Pc</i> D	same	same	CUB
Cursor Backward Tabulation	CSI <i>Pn</i> Z	n/a	n/a	CBT
Cursor Down	CSI <i>Pl</i> B	same	same	CUD
Cursor Forward	CSI <i>Pc</i> C	same	same	CUF
Cursor Horizontal Absolute	CSI <i>Pn</i> G	n/a	n/a	CHA
Cursor Horizontal Forward Tab	CSI <i>Pn</i> I	n/a	n/a	CHT
Cursor Information Report	DCS 1\$u D.....D ST	same	n/a	DECCIR
Cursor Next Line	CSI <i>Pn</i> E	n/a	n/a	CNL
Cursor Previous Line	CSI <i>Pn</i> F	n/a	n/a	CPL
Cursor Position	CSI <i>Pl</i> ; <i>Pc</i> H	same	same	CUP
Cursor Position Report	CSI 6 n	same	same	CPR
(Also see DECCIR, DECXCPR, and DSR.)				
Response:	CSI <i>Pl</i> ; <i>Pc</i> R			
<i>Pl</i> No. of lines				
<i>Pc</i> No. of columns				
Cursor Up	CSI <i>Pl</i> A	same	same	CUU
Extended Cursor Position Report	CSI ? 6 n	same	n/a	DECXCPR
Response:	CSI <i>Pl</i> ; <i>Pc</i> ; <i>Pp</i> R			
<i>Pl</i> Line				
<i>Pc</i> Column				
<i>Pp</i> Page				
Form Feed	FF	same	same	FF
Horizontal Tab	HT	same	same	HT
Horizontal/Vertical Position	CSI <i>Pl</i> ; <i>Pc</i> f	same	same	HVP
<i>Pl</i> No. of lines to move to				
<i>Pc</i> No. of columns to move to				

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor	Control Function			
	VT510	VT420	VT100	Mnemonic
Horizontal Position Absolute	CSI <i>Pn</i> `	same	same	HPA
Horizontal Position Relative	CSI <i>Pn</i> a	same	same	HPR
Line Feed	LF	same	same	LF
Line Feed/New Line Mode		same	same	LNМ
Set (NL):	CSI 20 h			
Reset (LF):	CSI 20 I			
Next Line	ESC E	same	same	NEL
No clear screen on column change		n/a	n/a	DECNCМ
Do not clear:	CSI ? 95 h			
Clear screen:	CSI ? 95 I			
Restore Cursor	ESC 8	same	same	DECRC
Restore Saved Cursor (SCO)	CSI u	n/a	n/a	SCORC
Right-to-Left Copy		n/a	n/a	DECRLCM
Set:	CSI ? 96 h			
Reset:	CSI ? 96 I			
Right-to-Left Mode		n/a	n/a	DECRLM
Set:	CSI ? 34 h			
Reset:	CSI ? 34 I			
Save Cursor	ESC 7	same	same	DECSC
Save Cursor Position (SCO)	CSI s	n/a	n/a	SCOSC
Set Cursor Style	CSI <i>Ps</i> SP q	n/a	n/a	DECSCUSR
<u><i>Ps</i></u>				
none , 0, 1	Blinking block			
2	Steady block			
3	Blinking underline			
4	Steady underline			
Select Set-Up Language	CSI <i>Ps</i> p	n/a	n/a	DECSSL
<u><i>Ps</i></u>				
none , 0, 1	English			
2	French			
3	German			
4	Spanish			
5	Italian			
Show Control Character Mode		same	same	CRM
Set (show):	CSI 3 h			
Reset (interpret):	CSI 3 I			

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor	Control Function			
	VT510	VT420	VT100	Mnemonic
Text Cursor Enable Mode		same	n/a	DECTCEM
Set (visible):	CSI ? 25 h			
Reset (invisible):	CSI ? 25 l			
Vertical Line Position Absolute	CSI <i>P</i> n d	n/a	n/a	VPA
Vertical Line Position Relative	CSI <i>P</i> n e	n/a	n/a	VPA
Vertical Tab	VT	same	same	VT
Vertical Autowrap		same	n/a	DECAWM
Set (enable):	CSI ? 7 h			
Reset (disable):	CSI ? 7 l			

Horizontal Tabs—Set/Clear	Control Function			Mnemonic
	VT510	VT420	VT100	
Set Horizontal Tab	ESC H	same	same	HTS
Set Tab at every 8 columns	CSI ? 5 W	n/a	n/a	DECST8C
Clear Tabs	CSI <i>P</i> s g	same	same	TBC
<i>P</i> s <u>Clear . . .</u>				
none , 0 tab at cursor				
3 all tabs				

Margins and Scrolling	Control Function			
	VT510	VT420	VT100	Mnemonic
Back Index	ESC 6	same	n/a	DECBI
Forward Index	ESC 9	same	n/a	DECFI
Index	ESC D	same	same	IND
Origin Mode		same	same	DECOM
Set (in margins):	CSI ? 6 h			
Reset (upper-left):	CSI ? 6 l			
Reverse Index	ESC M	same	same	RI
Scrolling Mode		same	same	DECSCLM
Set (smooth):	CSI ? 4 h			
Reset (jump):	CSI ? 4 l			
Set Left and Right Margin	CSI <i>Pl</i> ; <i>Pr</i> s	same	same	DECSLRM
<i>Pl</i> Left column margin (default = 1)				
<i>Pr</i> Right column margin (default = 80 or 132 , depending on page width)				
Set Scroll Speed	CSI <i>P</i> s SP p	n/a	n/a	DECSSCLS

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor		Control Function			Mnemonic
		VT510	VT420	VT100	
<u>Ps</u>					
none, 0, 1, 2, 3	Smooth 2				
4, 5, 6, 7, 8	Smooth 4				
9	Jump				
Set Top and Bottom Margin		CSI <i>Pt</i> ; <i>Pb</i> r	same	same	DECSTBM
<i>Pt</i>	Top margin (default = 1)				
<i>Pb</i>	Bottom margin (default = lines per screen)				
Page Arrangement and Size ¹		Control Function			Mnemonic
		VT510	VT420	VT100	
Column Mode			same	same	DECCOLM
Set (132 columns):		CSI ? 3 h			
Reset (80 columns):		CSI ? 3 l			
Set Columns per Page		CSI <i>Ps</i> \$ l	same	n/a	DECSCPP
<u>Ps</u>	<u>Set each page to . . .</u>				
0 or 80	80 columns				
132	132 columns				
Set Lines Per Page		CSI <i>Pn</i> t	same	same	DECSLPP
Vertical Split Screen			same	n/a	DECVSSM
Set (enable):		CSI 69 h			
Reset (disable):		CSI 69 l			
Page Movement		Control Function			Mnemonic
		VT510	VT420	VT100	
Next Page		CSI <i>Pn</i> U	same	same	NP
<i>Pn</i>	Number of pages to move (0 or 1 = 1 page)				
Preceding Page		CSI <i>Pn</i> V	same	same	PP
<i>Pn</i>	Number of pages to move (0 or 1 = 1 page)				
Page Position Absolute		CSI <i>Pn</i> SP P	same	same	PPA
Page Position Backward		CSI <i>Pn</i> SP R	same	same	PPB
Page Position Relative		CSI <i>Pn</i> SP Q	same	same	PPR
WINDOW MANAGEMENT					
Coupling Mode					
Page Cursor Coupling			same	n/a	DECPCCM
Set:		CSI ? 64 h			
Reset:		CSI ? 64 l			
Vertical Cursor Coupling			same	n/a	DECVCCM

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

		Control Function			
Active Position and Cursor		VT510	VT420	VT100	Mnemonic
Set:		CSI ? 61 h			
Reset:		CSI ? 61 l			
Display Extent		Control Function			Mnemonic
		VT510	VT420	VT100	
Request Displayed Extent		CSI " v	same	n/a	DECRQDE
Report Displayed Extent		CSI Ph ; Pw ; Pml ; Pmt ; Pmp " w			
			same	n/a	DECRPDE
Select number of lines per screen		CSI Pn * l	same	same	DECSNLS
Pan Direction					
Pan Down		CSI Pl S	same	same	SD
Pan Up		CSI Pl T	same	same	SU
Audible and Visual Attributes					
Bell		BEL	same	same	BEL
Double Height Line Top		ESC # 3	same	same	DECDHLT
Double Height Line Bottom		ESC # 4	same	same	DECDHLB
Single Width Line		ESC # 5	same	same	DECSWL
Double Width Line		ESC # 6	same	same	DECDWL
Screen Mode – Light/Dark			same	same	DECSCNM
Set	(light background):	CSI ? 5 h			
Reset	(dark background):	CSI ? 5 l			
Select Graphic Rendition ²		CSI Ps ; Ps m	same	same	SGR
<u>Ps</u>	<u>Attribute</u>				
0	All attributes off				
1	Bold				
4	Underline				
5	Blinking				
7	Negative image				
8	Invisible image				
10	ASCII char set (SCO only) ³				
11	Map 00-7F (SCO only)				
12	Map 80-FF (SCO only)				
22	Bold off				
24	Underline off				
25	Blinking off				
27	Negative image off				
28	Invisible image off				
GRAPHIC CHARACTER SETS					
¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.					
² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.					
³ See SGR—Select Graphic Rendition in Chapter 5 .					
⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.					

Active Position and Cursor		Control Function			
		VT510	VT420	VT100	Mnemonic
Assign User-Preference Supp Set		DCS <i>Ps ! u Dscs</i> ST	same	n/a	DECAUPSS
<u><i>Ps</i></u>					
0	94-Character Set				
1	96-Character Set				
<u><i>Dscs</i></u>	<u>Default 94-Character Set</u>	<u><i>Dscs</i></u>	<u>Default 96-Character Set</u>		
% 5	DEC Supplemental	A	ISO Latin-1 Supplemental		
" ?	DEC Greek	B	ISO Latin-2 Supplemental		
" 4	DEC Hebrew	F	ISO Greek Supplemental		
% 0	DEC Turkish	H	ISO Hebrew Supplemental		
& 4	DEC Cyrillic	L	ISO Latin-Cyrillic		
<	User-preferred supplemental	M	ISO Latin-5 Supplemental		
		<	User-preferred supplemental		
Display PC Characters (SCO)		ESC xx	n/a	n/a	
Select Character Set		ESC <i>I Dscs</i>	same	n/a	SCS
<u><i>I</i></u>	<u>94-Character G-set</u>	<u><i>I</i></u>	<u>96-Character G-set</u>		
(	G0	-	G1		
)	G1	.	G2		
*	G2	/	G3		
+	G3				
<u><i>Dscs</i></u>	<u>Default 94-Character Set</u>	<u><i>Dscs</i></u>	<u>Default 96-Character Set</u>		
& 4	Cyrillic (DEC)	A	ISO Latin-1 Supplemental		
A	British/U.K. NRCS	B	ISO Latin-2 Supplemental		
5 or C	Finnish NRCS	F	ISO Greek Supplemental		
R	French NRCS	H	ISO Hebrew Supplemental		
9 or Q	French Canadian NRCS	M	ISO Latin-5 Supplemental		
K	German NRCS	L	ISO Latin-Cyrillic		
" ?	Greek (DEC)	<	User-preferred Supplemental		
" >	Greek NRCS				
" 4	Hebrew (DEC)				
% =	Hebrew NRCS				
Y	Italian NRCS				
~, E, or 6	Norwegian/Danish NRCS				
% 6	Portuguese NRCS				
& 5	Russian NRCS				
% 3	SCS NRCS				
Z	Spanish NRCS				
7 or H	Swedish NRCS				
=	Swiss NRCS				
% 0	Turkish (DEC)				
% 2	Turkish NRCS				
0	DEC Special Graphic				
% 5	DEC Supplemental				
>	DEC Technical Character Set				
<	User-preferred Supplemental				
Supplemental Character Set		Control Function		Mnemonic	

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor	Control Function			
	VT510	VT420	VT100	Mnemonic
	VT510	VT420	VT100	
National Replacement Character Set Mode		same	n/a	DECNRCM
Set (National): (7-bit characters)	CSI ? 42 h			
Reset (Multinational): (8-bit characters)	CSI ? 42 l			
Request User-Preference Supplemental Set	CSI & u	same	n/a	DECRQUPSS
Response:	DEC <i>P_s</i> ! u <i>D_{scs}</i> ST			See DECAUPSS
Parameters:				
Send Control Character to Host				
7-bit controls	ESC SP F	n/a	n/a	S7C1T
8-bit controls	ESC SP G	n/a	n/a	S8C1T
Single Shift 2	ESC N	same	same	SS2
Single Shift 3	ESC O	same	same	SS3

EDITING AND TEXT PROCESSING	Control Function			Mnemonic
	VT510	VT420	VT100	
Delete Character	CSI <i>P_n</i> P	same	same	DCH
<i>P_n</i> Number of characters to delete (default = 1).				
Delete Column	CSI <i>P_n</i> ' ~	same	n/a	DECDC
<i>P_n</i> Number of columns to delete (default = 1).				
Delete Line	CSI <i>P_n</i> M	same	same	DL
<i>P_n</i> Number of lines to delete (default = 1).				
Erase Character	CSI <i>P_n</i> X	same	n/a	ECH
<i>P_n</i> Number of characters to erase (<i>P_n</i> = 0 or 1 (default) = 1 character).				
Erase in Display	CSI <i>P_s</i> J	same	same	ED
<i>P_s</i> Erase from . . .				
0 Cursor to end of display				
1 Top of display through cursor				
2 Top to bottom of display				
Erase in Line	CSI <i>P_s</i> K	same	same	EL
<i>P_s</i> Erase from . . .				
0 Cursor to end of line				
1 Start of line through cursor				
2 Start to end of line				
Insert Character	CSI <i>P_n</i> @	same	n/a	ICH

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor	Control Function			
	VT510	VT420	VT100	Mnemonic
Insert Column	CSI <i>Pn</i> ' }	same	n/a	DECIC
<i>Pn</i> Number of columns to insert (default = 1).				
Insert Line	CSI <i>Pn</i> L	same	same	IL
<i>Pn</i> Number of lines to insert (default = 1).				
Insert/Replacement Mode		same	same	IRM
Set (insert):	CSI 4 h			
Reset (replace):	CSI 4 l			
Select Character Attribute ⁴	CSI <i>Ps</i> " q	same	n/a	DECSCA
<i>Ps</i> <u>Meaning</u>				
0 DECSER and DECSEL can erase characters.				
1 DECSER and DECSEL cannot erase characters.				
2 Same as 0.				
Selective Erase in Display	CSI ? <i>Ps</i> J	same	n/a	DECSER
<i>Ps</i> <u>Erase from . . .</u>				
0 Cursor to end of display				
1 Top of display through cursor				
2 Top to bottom of display				
Selective Erase in Line	CSI ? <i>Ps</i> K	same	n/a	DECSEL
<i>Ps</i> <u>Erase from . . .</u>				
0 Cursor to end of line				
1 Start of line through cursor				
2 Start to end of line				

Rectangle Area Processing	Control Function			Mnemonic
	VT510	VT420	VT100	
Change Attribute in Rectangle	CSI <i>Pt;Pl;Pb;Pr; Ps l;...Psn</i> \$ r			
		same	n/a	DECCARA
<i>Ps</i> <u>Attribute</u>				
0 Off				
1 Bold	<i>Pt</i> Top line			
4 Underline	<i>Pl</i> Left column			
5 Blink	<i>Pb</i> Bottom line			
7 Negative image	<i>Pr</i> Right column			
22 No bold				
24 No underline				
25 No blink				
27 Positive image				
Copy Rectangular Area	CSI <i>Pts;Pls;Pbs;Prs;Pps;Ptd;Pld;Ppd</i> \$ v			
		same	n/a	DECCRA

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Active Position and Cursor		Control Function			
		VT510	VT420	VT100	Mnemonic
	<u>Source</u>				
<i>Pts</i>	Top line border	<i>Ptd</i>	Top line border		
<i>Pls</i>	Left column border	<i>Pld</i>	Left column border		
<i>Pbs</i>	Bottom line border	<i>Ppd</i>	Page number		
<i>Prs</i>	Right column border				
<i>Pps</i>	Page number				
Erase Rectangular Area		CSI <i>Pt;Pl;Pb;Pr</i> \$ <i>z</i>	same	n/a	DECERA
Fill Rectangular Area		CSI <i>Pch; Pt;Pl;Pb;Pr</i> \$ <i>x</i>			
			same	n/a	DECFRA
<i>Pch</i>	Character				
Reverse Attribute in Rectangle		CSI <i>Pt;Pl;Pb;Pr Psl;...Psn</i> \$ <i>t</i>			
			same	n/a	DECRARA
<i>Ps</i>	<u>Reverse Attributes</u>				
0	All				
1	Bold				
4	Underline				
5	Blink				
7	Negative image				
Select Attribute and Change Extent Mode		CSI <i>Ps</i> * <i>x</i>	same	n/a	DECSACE
Selective Erase Rectangular Area		CSI <i>Pt;Pl;Pb;Pr</i> \$ {	same	n/a	DECSERA
Memory Checksum Report or DSR-Memory Checksum Response:		CSI ? 63; <i>Pid</i> n DCS <i>Pid</i> ! ~ <i>D...D</i> ST	same	n/a	DECCKSR or DSR-Mem
Request Checksum of Rect Area Response:		CSI <i>Pid;Pp;Pt;Pl;Pb;Pr</i> * <i>y</i> DCS <i>Pid</i> ! ~ <i>D...D</i> ST	same	n/a	DECRCRA

¹ The page size can be 24, 25, 36, 42, 48, 52, and 72 lines with 80 or 132 columns.

² The ANSI mode supports Bold, Blink, Underline, and Reverse graphic renditions.

³ See [SGR—Select Graphic Rendition](#) in [Chapter 5](#).

⁴ DECSCA does not affect visual character attributes set by the select graphic rendition (SGR) function.

Table 4–6 Reports Control Functions

Macro	Control Function			
	VT510	VT420	VT100	Mnemonic
Define Macro	DCS <i>Pid;Pdt;Pen</i> ! z <i>D...D</i> ST			
		same	n/a	DECDMAC
DSR - Macro Space	CSI ? 62 n	same	n/a	DSR-MSR
Invoke Macro	CSI <i>Pid</i> * z	same	n/a	DECINVM
Macro Space Report	CSI <i>Pn</i> * {	same	n/a	DECMSR, DSR-MSR
$Pn = \frac{\text{number of bytes}}{16}$				
Reports				
DSR - Data Integrity Report	CSI ? 75 n	same	n/a	DSR-Data
Response:	CSI ? <i>Ps</i> n			
<u><i>Ps</i></u>	<u>Meaning</u>			
70	Ready, no power loss or comm errors.			
71	Malfunction, comm error.			
73	No reported data loss since last power up.			
Presentation State Report (See DECRQPSR.)		n/a	n/a	DECPSR
Report Mode (DECRPM) (See Report Mode (ANSI) below.)		same	n/a	DECRPM
Request Mode (ANSI)	CSI <i>Pa</i> \$ p	same	n/a	DECRQM ¹
Request Mode (DEC)	CSI ? <i>Pd</i> \$ p	same	n/a	DECRQM
Request Presentation State Report	CSI <i>Ps</i> \$ w	same	n/a	DECRQPSR
<u><i>Ps</i></u>	<u>Report Requested</u>			
0	Error, restore ignored.			
1	Cursor information report (DECCIR)			
2	Tab stop report (DECTABSR)			
Report Selection or Setting (See DECRQSS.)		same	n/a	DECRPSS
Request Selection or Setting		same	n/a	DECRQSS
Request (DECRQSS):	DCS \$ q <i>D...D</i> ST			
Report (DECRPSS):	DCS <i>Ps</i> \$ r <i>D...D</i> ST			

¹A DECRQM sequence can only ask about one mode at a time. When the terminal is set to VT52 mode, DECRQM provides no report.

²The * means this mode is always Reset in the terminal and cannot be changed.

		Control Function			
Macro		VT510	VT420	VT100	Mnemonic
<u>Setting Control Sequence</u>	<u>Final Characters</u>	<u>Mnemonic</u>			
Select Active Status Display	\$ g	DECSASD			
Select Attribute Change Extent	* x	DECSACE			
Set Character Attribute	" q	DECSCA			
Set Conformance Level	" p	DECSCCL			
Set Columns Per Page	\$ l	DECSCPP			
Set Lines Per Page	t	DECSLPP			
Set Number of Lines per Screen	* l	DECSNLS			
Set Status Line Type	\$ ~	DECSSDT			
Set Left and Right Margins	s	DECSLRM			
Set Top and Bottom Margins	r	DECSTBM			
Set Graphic Rendition	m	SGR			
Select Set-Up Language	p	DECSSL			
Select Printer Type	\$ s	DECSPRTT			
Select Refresh Rate	" t	DECSRFR			
Select Digital Printed Data Type	) p	DECSDP			
Select ProPrinter Character Set	* p	DECSPPCS			
Select Communication Speed	* r	DECSCS			
Select Communication Port	* u	DECSCP			
Set Scroll Speed	SP p	DECSSCLS			
Set Cursor Style	SP q	DECSCUSR			
Set Key Click Volume	SP r	DECSKCV			
Set Warning Bell Volume	SP t	DECSWBV			
Set Margin Bell Volume	SP u	DECSMBV			
Set Lock Key Style	SP v	DECSLCK			
Select Flow Control Type	* s	DECSFC			
Select Disconnect Delay Time	\$ q	DECSDDT			
Set Transmit Rate Limit	" u	DECSTRL			
Set Port Parameter	+ w	DECSPP			

Reports (Cont.)		Control Function			Mnemonic
		VT510	VT420	VT100	
Request Terminal State Report			same	n/a	DECRQTSR
Request:		CSI 1 \$ u			
Reply to Host:		DCS 1 \$ s <i>D...D</i> ST			
<u>Effects</u>					
DDD1	DECRLM, DECHEBM, and DEHCCEM are set.				DDD1
DDD2	DECRLM is reset. ASCII is designated in G1.				DDD2
DDD3	ASCII is designated in G0.				DDD3
Restore Presentation State		DCS <i>P</i> s \$ t <i>D...D</i> ST	same	n/a	DECRSPS
<u><i>P</i>s</u>	<u>Data String Format</u>				
0	Error, restore ignored.				
1	Selects the format of the cursor information report (DECCIR).				
2	Selects the format of the tab stop report (DECTABSR).				

¹A DECRQM sequence can only ask about one mode at a time. When the terminal is set to VT52 mode, DECRQM provides no report.

²The * means this mode is always Reset in the terminal and cannot be changed.

Macro	Control Function			
	VT510	VT420	VT100	Mnemonic
<i>D...D</i> Data string of tab stops or cursor position.				
Restore Terminal State	DCS <i>Ps</i> \$ p <i>D...D</i> ST	same	n/a	DECRSTS
<i>Ps</i> <u>Data String Format</u>				
0 Error, restore ignored.				
1 Selects the format of the terminal state report (DECTSR).				
<i>D...D</i> Data string of restored information.				
Report Mode (ANSI)	CSI <i>Pa</i> ; <i>Ps</i> \$ y	same	n/a	
Report Mode (DEC)	CSI ? <i>Pd</i> ; <i>Ps</i> \$ y	same	n/a	
Reset Mode	CSI <i>Pl</i> <i>Pn</i> l	same	same	RM
Set Mode	CSI <i>Pl</i> <i>Pn</i> h	same	same	SM

¹A DECRQM sequence can only ask about one mode at a time. When the terminal is set to VT52 mode, DECRQM provides no report.

²The * means this mode is always Reset in the terminal and cannot be changed.

Macro	Control Function				
	VT510		VT420	VT100	Mnemonic
Mode Description	ANSI Mode		DEC Mode		
	<i>Pa</i>	<u>Mnemonic</u> ²	<i>Pd</i>	<u>Mnemonic</u>	<u>Mode</u>
Guarded area transfer	1	GATM *	?1	DECCKM	Cursor keys
Keyboard action	2	KAM	?2	DECANM	ANSI/VT52
Control representation	3	CRM	?3	DECCOLM	Column
Insert/replace	4	IRM	?4	DECSCLM	Scrolling
Status reporting transfer	5	SRTM *	?5	DECSCNM	Screen
			?6	DECOM	Origin
Vertical editing	7	VEM *	?7	DECAWM	Autowrap
Horiz editing	10	HEM *	?8	DECARM	Autorepeat
Positioning unit	11	PUM *			
Send/receive	12	SRM			
Format effector action	13	FEAM *			
Format effector transfer	14	FETM *			
Multiple area transfer	15	MATM *			
Transfer termination	16	TTM *			
Selected area transfer	17	SATM *			
Tabulation stop	18	TSM *	?18	DECPFF	Print form
Editing boundary	19	EBM *	?19	DECPEX	Print extent
Linefeed/New line	20	LNМ			
Text cursor enable			?25	DECTCEM	
Right-to-left			?34	DECRLM	
Hebrew keyboard map			?35	DECHEBM	
Hebrew encoding			?36	DECHEM	
NRCS characters			?42	DECNRCM	
Greek/N.A.			?57	DECNAKB	
Horiz cursor coupling			?60	DECHCCM *	
Vertical cursor coupling			?61	DECVCCM	
Page cursor coupling			?64	DECPCCM	
Numeric keypad			?66	DECNKM	
Backarrow key			?67	DECBKM	
Keyboard usage			?68	DECKBUM	
Vertical split screen			?69	DECVSSM	
Transmit rate limiting			?73	DECXRLMM	
No clear screen			?95	DECNCSM	
Right-to-left copy			?96	DECRLCM	
CRT save			?97	DECCRTSM	
Auto resize			?98	DECARSM	
Modem control			?99	DECMCM	
Auto answerback			?100	DECAAM	
Conceal answerback			?101	DECCANSМ	
Null			?102	DECNULM	
Half-duplex			?103	DECHDPXM	
Enable secondary keyboard language			?104	DECESKM	
Overscan			?106	DECOSCM	

¹A DECRQM sequence can only ask about one mode at a time. When the terminal is set to VT52 mode, DECRQM provides no report.

²The * means this mode is always Reset in the terminal and cannot be changed.

Macro	Control Function			
	VT510	VT420	VT100	Mnemonic
Set/Reset Parameters				
	ANSI Mode	DEC Mode		
	<i>P_s</i>			
Not recognized	0	same		
Set	1	same		
Reset	2	same		
Set permanently	3	same		
Reset permanently	4	same		

Reports (Cont.)	Control Function			Mnemonic
	VT510	VT420	VT100	
Set Mode/Reset Mode (See also DECRQM.)				SM/RM
Send/Receive Mode (Local Echo)		same	same	SRM
Set (Local echo off):	CSI 12 h			
Reset (Local echo on):	CSI 12 l			
Tabulation Stop Report	DCS 2 \$ u <i>D...D</i> ST	same	n/a	DECTABSR
<i>D...D</i> Column numbers of tab stops.				
Terminal State Report (See DECRQTSR.)		same	n/a	DECTSR

¹A DECRQM sequence can only ask about one mode at a time. When the terminal is set to VT52 mode, DECRQM provides no report.

²The * means this mode is always Reset in the terminal and cannot be changed.

Table 4–7 Terminal Management Control Functions

Purpose	Control Function			
	VT510	VT420	VT100	Mnemonic
CRT Save Mode		n/a	n/a	DECCRTSM
Set (enable CRT saver):	CSI ? 97 h			
Reset (disable CRT saver):	CSI ? 97 l			
Device Attributes	CSI c or CSI 0 c	same	same	DA1
Response:	CSI ? Psc ; PSl ; ... Psn c			
(See Primary Device Attributes (DA1).)				
Secondary Device Attributes	CSI > c or CSI > 0 c	same	same	DA2
VT Response:	CSI > 61; Pv ; 0 c			
PC Response:	CSI > 61; Pv ; 1 c			
(See Secondary Device Attributes (DA2).)				
Tertiary Device Attributes	CSI = c or CSI = 0 c	same	same	DA3
Device Status Report – ANSI:	CSI Ps n	same	same	DSR
Device Status Report – DEC:	CSI ? Ps n	same	same	DSR
Enter/Exit PCTerm Mode or Select ASCII characters or scancodes	CSI ? Psl ; Ps2 r	same	n/a	DECPCTERM
<u>Ps1</u> none, 0 1	<u>VT Emulation Mode</u> VT Mode PCTerm Mode	<u>SCO Console Mode</u> ASCII characters Scancodes		
<u>Ps2</u> 210 220 437 850 852 857 860 862 863 865 866	<u>Character Set</u> PC Greek PC Spanish PC International PC Multilingual PC Slavic (Latin 2) PC Turkish PC Portuguese PC Hebrew PC French-Canadian PC Danish PC Cyrillic	<u>Ps2</u> none, 0 1 2 3 4 5 6 7	<u>Character Set</u> No Change PC Multilingual PC International PC Danish/Norwegian PC Spanish PC Portuguese DEC Supplemental ISO Latin 1	
Identify Device	ESC Z	n/a	same	DECID ¹
Invoke Confidence Test ²	CSI 4 ; Ps ; ... ; Ps y	same	n/a	DECTST

¹DECID may not be supported; use primary device attributes (DA1) instead.

²"ESC # 8" invokes the Screen Alignment test. DECTST causes a disconnect; therefore, it should not be used in conjunction with a modem.

³Overscan and 72 Hz refresh rate cannot co-exist when the terminal is set to 42 or 53 lines per screen. Refer to DECSNLS —Set Lines Per Screen for details.

⁴DECTID has no effect when the terminal is in VT52 mode.

Purpose		Control Function			
		VT510	VT420	VT100	Mnemonic
<u>Ps</u>	<u>Test</u>				
0	"All Tests" (1, 2, 3, 6)				
1	Power-Up Self Test				
2	RS-232 Port Data Loopback				
3	Printer Port Loopback				
4	Speed Select and Indicator				
5	Reserved - No action				
6	Modem Loopback Test				
7	EIA-423 Port Loopback				
8	Parallel Port Loopback				
9	Repeat other tests				
Overscan Mode			n/a	n/a	DECOSCNM
Set (enable overscan): ³		CSI ? 106 h			
Reset (disable overscan):		CSI ? 106 I			
Primary Device Attributes		CSI c or CSI 0 c	same	same	DA1
<u>Model</u>	<u>Response to DA1</u>				
North American	CSI ? 64;1;2;7;8;9;15;18;21;44;45;46 c				
World-Wide	CSI ? 64;1;2;7;8;9;12;15;18;21;23;24;42;44;45;46 c				
<u>Ext</u>	<u>Description</u>	<u>Ext</u>	<u>Description</u>		
1	132 columns	18	Windowing capability		
2	Printer port extension	21	Horizontal scrolling		
4	Sixel extension	23	Greek extension		
6	Selective erase	24	Turkish extension		
7	DRCS	42	ISO Latin 2 character set		
8	UDK	44	PCTerm		
9	NRCS	45	Soft key map		
12	SCS extension	46	ASCII emulation		
15	Technical character set				
Report Terminal Unit ID		DCS !! D...D ST	same	n/a	DECRPTUI
Reset to Initial state		ESC c	same	same	RIS
Secondary Device Attributes		CSI > c or CSI > 0 c	same	n/a	DA2
<u>Keyboard</u>	<u>Response</u>				
VT keyboard	CSI > 61; Pv; 0 c				
PC keyboard	CSI > 61; Pv; 1 c				
Pv	Firmware version				
Secure Reset		ESC [Pr + p	same	n/a	DECSR
Secure Reset Confirmation		ESC [Pr * q	same	same	DECSRC
Select Conformance (Operating) Level		CSI 6 PsI ; Ps2 ; " p	same	same	DECSCL

¹DECID may not be supported; use primary device attributes (DA1) instead.

²"ESC # 8" invokes the Screen Alignment test. DECTST causes a disconnect; therefore, it should not be used in conjunction with a modem.

³Overscan and 72 Hz refresh rate cannot co-exist when the terminal is set to 42 or 53 lines per screen. Refer to DECSNLS — Set Lines Per Screen for details.

⁴DECTID has no effect when the terminal is in VT52 mode.

				Control Function			Mnemonic
Purpose				VT510	VT420	VT100	
<u>Ps1</u>	<u>Level Selected</u>	<u>Ps2</u>	<u>Data Bits</u>				
none,0,1	1 - VT100	none, 0, 2	8-bit controls				
2,3,4	4 - VT510	1	7-bit controls				
Select Refresh Rate ³				CSI Ps " t	n/a	n/a	DECSRFR
<u>Ps</u>	<u>Cursor Style</u>						
1	50 Hz						
2	60 Hz						
none, 0 or 3	≥ 70 Hz						
Select Terminal ID ⁴				CSI Ps , q	same	same	DECTID
<u>Ps</u>	<u>ID Type</u>	<u>Response to DECTID</u>	<u>Terminal Type</u>				
0	VT100	ESC [? 1; 2 c	VT100				
1	VT101	ESC [? 1; 0 c	VT101				
2	VT102	ESC [? 6 c	VT102				
5	VT220	CSI ? 62; 1; 2; 7; 8 c	VT220 North American				
		CSI ? 62; 1; 2; 7; 8; 9 c	VT220 International				
7	VT320	CSI ? 63; 1; 2; 7; 8 c	VT320 North American				
		CSI ? 63; 1; 2; 7; 8; 9 c	VT320 International				
9	VT420	CSI ? 64; 1; 2; 7; 8; 9; 15; 18; 21 c	VT420 North American				
		CSI ? 64; 1; 2; 7; 8; 9; 15; 18; 21 c	VT420 International				
10	VT510	Refer to DA1 for response.					
Soft Terminal Reset				CSI ! p	same	n/a	DECSTR
Terminal Mode Emulation				CSI Ps SP ~	n/a	n/a	DECTME
<u>Ps</u>	<u>Terminal Mode</u>						
none, 0 or 1	VT510 (VT Level 4)						
2	VT100						
3	VT52						
4	VT420 PCTerm						
5	WYSE 60/160						
6	WYSE 60/160 PCTerm						
7	WYSE 50+						
8	WYSE 150/120						
9	TVI 950						
10	TVI 925						
11	TVI 910+						
12	ADDS A2						
13	SCO Console						
Exit ANSI (VT52) Mode				ESC <	same	same	DECANM
Status Display Extension Functions				Control Function			Mnemonic
				VT510	VT420	VT100	
Select Active Status Display				CSI Ps \$ }	same	n/a	DECSASD

¹DECID may not be supported; use primary device attributes (DA1) instead.

²"ESC # 8" invokes the Screen Alignment test. DECTST causes a disconnect; therefore, it should not be used in conjunction with a modem.

³Overscan and 72 Hz refresh rate cannot co-exist when the terminal is set to 42 or 53 lines per screen. Refer to DECSNLS —Set Lines Per Screen for details.

⁴DECTID has no effect when the terminal is in VT52 mode.

Purpose		Control Function			
		VT510	VT420	VT100	Mnemonic
<i>Ps</i>	Send data to . . .				
0	Main display only				
1	Status line only				
Select Status Display (Line) Type		CSI <i>Ps</i> \$ ~	n/a	n/a	DECSSDT
<i>Ps</i>	<u>Status Line Selected</u>				
0	No status line				
1	Indicator status line				
2	Host-writable status line				

Desktop Features Control Functions		Control Function			Mnemonic
		VT510	VT420	VT100	
Load Time of Day		CSI <i>Ps1</i> ; <i>Ps2</i> , p	n/a	n/a	DECLTOD
<i>Ps1</i>	Hour (decimal number between 00 and 23)				
<i>Ps2</i>	Minutes (decimal number between 00 and 59)				

¹DECID may not be supported; use primary device attributes (DA1) instead.

²"ESC # 8" invokes the Screen Alignment test. DECTST causes a disconnect; therefore, it should not be used in conjunction with a modem.

³Overscan and 72 Hz refresh rate cannot co-exist when the terminal is set to 42 or 53 lines per screen. Refer to DECSNLS —Set Lines Per Screen for details.

⁴DECTID has no effect when the terminal is in VT52 mode.

Table 4–8 Keyboard Processing Control Functions

Purpose		Control Function			Mnemonic
		VT510	VT420	VT100	
Autorepeat Mode			same	same	DECARM
	Set (repeat):	CSI ? 8 h			
	Reset (disabled):	CSI ? 8 l			
Backarrow Key Mode			same	same	DECBKM
	Set (backspace):	CSI ? 67 h			
	Reset (delete):	CSI ? 67 l			
Copy Key Default		DCS " z D...D ST	n/a	n/a	DECCKD
<i>D...D</i>	<i>Key-s/Key-d;Key-s/Key-d...;</i>				
<i>Key-s</i>	Source key				
<i>Key-d</i>	Destination key				
Cursor Keys Mode			same	same	DECCKM
	Set (application):	CSI ? 1 h			
	Reset (cursor):	CSI ? 1 l			
Define Function Key (SCO)		ESC Q <i>Fn string</i>	n/a	n/a	SCODFK
Device Status Report (keyboard status)					DSR
	Request Keyboard Report:	CSI ? 26 n	same	same	
	Reply to Keyboard Request:	CSI ? 27 ; <i>Pn; Pst; Ptyp</i> n			
			same	same	
<i>Pn</i>	<u>Keyboard Language (VT/PC).</u>	<i>Pn</i>	<u>Keyboard Language (VT or PC).</u>		
0	Unknown	15	Spanish		
1	North American	16	Portuguese		
2	British	19	Hebrew		
3	Belgian (Flemish)	22	Greek		
4	French Canadian	28	Canadian English		
5	Danish	29	Turkish Q/Turkish		
6	Finnish	30	Turkish F/Turkish		
7	German/German	31	Hungarian		
8	Dutch	33	Slovak		
9	Italian	34	Czech		
10	Swiss French	35	Polish		
11	Swiss German	36	Romanian		
12	Swedish	38	SCS		
13	Norwegian	39	Russian		
14	French	40	Latin American		
<i>Pst</i>	<u>Keyboard Status</u>	<i>Ptyp</i>	<u>Keyboard Type</u>		
0	Keyboard Ready	4	LK450		
3	No Keyboard	5	PCXAL		
8	Keyboard Busy				

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRRPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Purpose	Control Function			
	VT510	VT420	VT100	Mnemonic
Down-line Load	DCS <i>Pfn; Pcn; Pe; Pcmw; Pss; Pu; Pcmh; Pcsc</i> { <Dscs> <sixel-font-patterns> ST			
		same	n/a	DECDLD
Extended Keyboard Report	APC : <i>ppp mm</i> ST	same	n/a	DECEKBD
Enable Local Functions	CSI <i>Pfl; PcI; ...Pfn; Pcn</i> + q		n/a	DECELF
Enable Secondary Keyboard Language Mode		same	n/a	DECESKM
Set (secondary):	CSI ? 104 h			
Reset (primary):	CSI ? 104 l			
Function Key	CSI <i>PsI; Ps2</i> ~	same	n/a	DECFNK
<i>PsI</i> Function key				
<i>Ps2</i> Modifier				
Hebrew Encoding Mode ¹		same	n/a	DECHEM
Set (8-bit char):	CSI ? 36 h			
Reset (7-bit char):	CSI ? 36 l			
Hebrew Keyboard Map Mode		same	n/a	DECHEBM
Set (Hebrew keyboard):	CSI ? 35 h			
Reset (North American):	CSI ? 35 l			
Keyboard Action Mode		same	same	KAM
Set (lock):	CSI 2 h			
Reset (unlock):	CSI 2 l			
Keypad Application Mode	ESC =	same	same	DECKPAM
Keyboard Language Selection	CSI <i>PsI; Ps2</i> SP }	same	n/a	DECKBD
<i>PsI</i> Keyboard Type				
1,0 VT keyboard layout				
2 Enhanced PC keyboard layout				

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRQPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Purpose		Control Function			
		VT510	VT420	VT100	Mnemonic
<u>Ps2</u>	<u>Keyboard Language (VT/PC).</u>	<u>Ps2</u>	<u>Keyboard Language (VT or PC).</u>		
none, 0, 1	North American	16	Portuguese		
2	British	19	Hebrew		
3	Belgian	22	Greek		
4	French Canadian	28	Canadian English		
5	Danish	29	Turkish Q/Turkish		
6	Finnish	30	Turkish F/Turkish		
7	German	31	Hungarian		
8	Dutch	33	Slovak		
9	Italian	34	Czech		
10	Swiss French	35	Polish		
11	Swiss German	36	Romanian		
12	Swedish	38	SCS		
13	Norwegian	39	Russian		
14	French	40	Latin American		
15	Spanish				
Keypad Numeric Mode		ESC >	same	same	DECKPNM
Key Position Mode			same	n/a	DECKPM
	Set (key position):	CSI ? 81 h			
	Reset (char codes):	CSI ? 81 I			
Keyboard Usage Mode			same	n/a	DECKBUM
	Set (data processing):	CSI ? 68 h			
	Reset (typewriter):	CSI ? 68 I			
Local Function Key Control		CSI <i>Pk1</i> ; <i>Pf1</i> ; <i>Pk2</i> ; <i>Pf2</i> ... * g			
			same	n/a	DECLFKC
<u>Pk</u>	<u>Function</u>	<u>VT Keyboard</u>	<u>Enhanced PC Keyboard</u>		
1	Hold	F1	Scroll Lock		
2	Print	F2	Print Screen		
3	Set-Up	F3	Cap Lock/Print Screen		
Locking Shift					LS
LS0	Map G0 into GL	SI	same	same	
LS1	Map G1 into GL	SO	same	same	
LS1R	Map G1 into GR	ESC ~	same	n/a	
LS2	Map G2 into GL	ESC n	same	n/a	
LS2R	Map G2 into GR	ESC }	same	n/a	
LS3	Map G3 into GL	ESC o	same	n/a	
LS3R	Map G3 into GR	ESC l	same	n/a	
North American/Greek Selection			same	n/a	DECNAKB
	Set (North American):	CSI ? 57 h			
	Reset (Greek keyboard):	CSI ? 57 I			

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRQPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Purpose	Control Function			
	VT510	VT420	VT100	Mnemonic
Numeric Keypad Mode		same	n/a	DECNKM
Set (application):	CSI ? 66 h			
Reset (numeric):	CSI ? 66 I			
Program Alphanumeric Key	DCS " y <i>D...D</i> ST	n/a	n/a	DECPAK
Program Function Key	DCS " x <i>D...D</i> ST	n/a	n/a	DECPFK
Program Key Action	CSI <i>Ps</i> + z	n/a	n/a	DECPKA
<i>Ps</i> 1	<u>Program Key Action</u>			
none , 0	None			
1	Lock Keys and Modifiers			
2	Restore Factory Defaults			
3	Recall Key Definitions			
Program Key Free Memory Report ²	CSI <i>Ps</i> 1 ; <i>Ps</i> 2 + y	n/a	n/a	DECPKFMR
<i>Ps</i> 1	Total memory (bytes)			
<i>Ps</i> 2	Free memory			
Program Key Free Memory Inquiry	CSI + x	n/a	n/a	DECRQPKFM
Response: DECPKFMR				
Request Key Type	CSI <i>Ps</i> , u	n/a	n/a	DECRQKT
Report Key Type	CSI <i>Ps</i> 1 ; <i>Ps</i> 2 , v	n/a	n/a	DECRPKT
Report Function Key Definition	DCS " } <i>D...D</i> ST	n/a	n/a	DECRPFK
Report Modifiers/Key State	DCS " ~ <i>D...D</i> ST	n/a	n/a	DECRPAK
Set Key Click Volume	CSI <i>Ps</i> SP r	same	n/a	DECSKCV
<i>Ps</i>	<u>Volume</u>			
1	Off			
2, 3, 4	Low			
none , 0, 5, 6, 7, 8	High			
Set Lock Key Style	CSI <i>Ps</i> SP v	n/a	n/a	DECSLCK
<i>Ps</i>	<u>Lock Style</u>			
none , 0, 1	Caps Lock			
2	Shift Lock			
3	Reverse Lock			
Num Lock Mode		same	same	DECNUMLK
Set:	CSI ? 108 h			
Reset:	CSI ? 108 I			
Caps Lock Mode		same	same	DECCAPSLK
Set:	CSI ? 109 h			
Reset:	CSI ? 109 I			
Keyboard LEDs Host Indicator Mode		same	same	DECKLHIM
Set:	CSI ? 110 h			

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRQPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Purpose	Control Function			
	VT510	VT420	VT100	Mnemonic
Reset:	CSI ? 110 I			
Load LEDs	CSI <i>Ps</i> q	n/a	n/a	DECLL
<u><i>Ps</i></u> <u>Volume</u>				
0 Clear all LEDs				
1 Light Num Lock				
2 Light Caps Lock				
3 Light Scroll Lock				
21 Extinguish Num Lock				
22 Extinguish Caps Lock				
23 Extinguish Scroll Lock				
Set Margin Bell Volume	CSI <i>Ps</i> SP u	same	n/a	DECSMBV
<u><i>Ps</i></u> <u>Volume</u>				
none , 0, 1 Off				
2, 3, 4 Low				
5, 6, 7, 8 High				
Select Modifier Key Reporting (See reference.)	CSI <i>Pm</i> l; <i>Pf</i> l... <i>Pmn</i> ; <i>Pfn</i> + r			
		same	n/a	DECSMKR
Set Warning Bell Volume	CSI <i>Ps</i> SP t	n/a	n/a	DECSWBV
<u><i>Ps</i></u> <u>Volume</u>				
1 Off				
2, 3, 4 Low				
none , 0, 5, 6, 7, 8 High				

User-Defined Keys	Control Function			Mnemonic
	VT510	VT420	VT100	
User Defined Keys	DCS <i>Ps</i> l; <i>Ps</i> 2; <i>Ps</i> 3 <i>D</i> ... <i>D</i> ST			
		same	n/a	DECUDK
<u><i>Ps</i> l</u> <u>Clear . . .</u>				
none, 0 All UDKs before loading new values				
1 Old UDKs only when redefined; load new UDK values.				
<u><i>Ps</i> 2</u> <u>Lock . . .</u>				
none, 0 UDKs against future redefinition				
1 Don't lock the UDKs against future redefinition.				
<u><i>Ps</i> 3</u> <u>Define . . .</u>				
none, 0 Shifted function key				
1 Unshifted function key				
2 Shifted function key				
3 Alt unshifted function key				
4 Alt shifted function key				
<i>D</i> ... <i>D</i> <i>Key</i> 1/ <i>St</i> 1/ <i>Dir</i> ; <i>Key</i> 2/ <i>St</i> 2/ <i>Dir</i> ; <i>Key</i> <i>n</i> / <i>St</i> <i>n</i> / <i>Dir</i> ;				
<i>St</i> <i>n</i> Sequence user-defined key (UDK)				

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRQPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Purpose		Control Function			
		VT510	VT420	VT100	Mnemonic
<u>Keyn</u>	<u>VT Keyboard</u>	<u>PC Keyboard</u>			
11	F1	F1			
12	F2	F2			
13	F3	F3			
14	F4	F4			
15	F5	F5			
17	F6	F6			
18	F7	F7			
19	F8	F8			
20	F9	F9			
21	F10	F10			
23	F11	F11			
24	F12	F12			
25	F13				
26	F14				
28	F15				
29	F16				
31	F17				
32	F18				
33	F19				
34	F20				
<u>Dir</u>	<u>Direction³</u>				
0, none	Normal				
1	Local				
2	Remote				
DSR - User-Defined Key Request			same	n/a	DSR-UDK
Request UDK lock:		CSI ? 25 n			
Reply to UDK locked status:		CSI ? 20 n			
Reply to UDK unlocked status:		CSI ? 21 n			

¹DECHEM is opposite in action to DECNRCM; that is, setting DECHEM has the same result as resetting DECNRCM.

²The VT510 reports this sequence upon receiving DECRQPKFM from the host. See reference for limitations.

³See DECPFK for direction definitions.

Table 4–9 Communications Control Functions

Printer Port Function	Control Function			
	VT510	VT420	VT100	Mnemonic
DSR - Printer Port Report		same	same	DSR-Print
Request:	CSI ? 15 n			
Response:	CSI ? Ps n			
<u>Ps</u> <u>Printer . . .</u>				
10 Ready				
11 Not ready				
13 Off line				
18 Busy				
19 On another session.				
Media Copy - Send Screen Data through Host Port	CSI 2 i	n/a	n/a	MC
Media Copy - Start/Stop Printer-to-Host Session				MC
ANSI:	CSI <i>Pa</i> i	same	same	
DEC:	CSI ? <i>Pd</i> i	same	CSI ?8i	
ANSI Mode		DEC Mode		
<u><i>Pa</i></u> <u>Function</u>		<u><i>Pd</i></u> <u>Function</u>		
0, none Print page		1 Print cursor line		
4 Reset printer controller mode		4 Reset autoprint mode		
5 Set printer controller mode		5 Set autoprint mode		
6 Start printer to host session		8 Stop printer to host session		
7 Stop printer to host session		9 Start printer to host session		
		10 Print composed main display		
		11 Print all pages		
		18 Assign printer to active host session		
		19 Accept printer commands from sessions		
Print Extent Mode		same	same	DECPEX
Set (entire page):	CSI ? 19 h			
Reset (scroll region):	CSI ? 19 l			
Print Form Feed Mode		same	same	DECPFF
Set (FF):	CSI ? 18 h			
Reset (no FF):	CSI ? 18 l			
Select Digital Printed Data Type ¹	CSI <i>Ps</i>) p	n/a	n/a	DECSDPPT
<u><i>Ps</i></u> <u>Printed Data Type</u>				
none, 0, 1 Print National Only				
2 National and Line Drawing				
3 Print Multinational				
4 Print All Characters				
Select Printer Type	CSI <i>Ps</i> \$ s	n/a	n/a	DECSPPRTT

¹This sequence takes effect only when a ProPrinter or Digital printer with ProPrinter emulation mode is attached.

²The answerback data string may consist of from 0 to 30, 7- or 8-bit characters, which are hex pairs in the 3/0 through 3/9 (0 through 9) range, 4/1 through 4/6 (A through F) range, and 6/1 through 6/6 (A through F) range. Pressing Ctrl/Break or receiving an ENQ code causes the VT510 to transmit the answerback message.

† Not available for the printer port. If specified in the sequence to configure the printer port, the Ps3 value is ignored.

Printer Port Function		Control Function			Mnemonic
		VT510	VT420	VT100	
<u>Ps</u>	<u>Printer Type</u>				
none, 0, 1	DEC ANSI				
2	IBM ProPrinter				
3	DEC + IBM				
Select ProPrinter Character Set		CSI <i>Pn</i> * p	n/a	n/a	DECSPPCS
<u>Pn</u>	<u>Printer</u>				
210	PC Greek				
220	PC Spanish				
437	PC International				
850	PC Multilingual				
852	PC Slavic (Latin 2)				
857	PC Turkish				
860	PC Portuguese				
862	PC Hebrew				
863	PC French-Canadian				
865	PC Danish				
866	PC Cyrillic				
Start/stop printer; see Media Copy.					
DEC Printers with ProPrinter Emulation		Control Function			Mnemonic
		VT510	VT420	VT100	
Enter/Return from IBM ProPrinter Emulation Mode			n/a	n/a	DECIPEM
Enter:		CSI ? 58 h			
Return:		CSI ? 58 l			
Terminal Synchronization					
Auto Answerback Mode			same	same	DECAAM
Set:		CSI ? 100 h			
Reset:		CSI ? 100 l			
Conceal Answerback Message Mode			n/a	n/a	DECCANSM
Set:		CSI ? 101 h			
Reset:		CSI ? 101 l			
Communication Port		CSI <i>Ps1</i> ; <i>Ps2</i> * u	n/a	n/a	DECSCP
<u>Ps1</u>	<u>Printer Port</u>				
1	Parallel port				
2	Comm1				
3	Comm2				
<u>Ps2</u>	<u>Session 1</u>				
1	Comm1				
2	Comm2				
Communication Speed		CSI <i>Ps1</i> ; <i>Ps2</i> * r	n/a	n/a	DECSCS

¹This sequence takes effect only when a ProPrinter or Digital printer with ProPrinter emulation mode is attached.

²The answerback data string may consist of from 0 to 30, 7- or 8-bit characters, which are hex pairs in the 3/0 through 3/9 (0 through 9) range, 4/1 through 4/6 (A through F) range, and 6/1 through 6/6 (A through F) range. Pressing Ctrl/Break or receiving an ENQ code causes the VT510 to transmit the answerback message.

† Not available for the printer port. If specified in the sequence to configure the printer port, the Ps3 value is ignored.

Printer Port Function		Control Function			Mnemonic
		VT510	VT420	VT100	
<u>Ps1</u>	<u>Comm Line</u>	<u>Ps2</u>	<u>Communication Speed</u>		
none, 0, 1	Host Transmit	none, 0	Use default speed.		
2	Host Receive	1	300		
3	Printer	2	600		
4	Modem Hi	3	1200		
5	Modem Lo	4	2400		
		5	4800		
		6	9600		
		7	19200		
		8	38400		
		9	57600		
		10	76800		
		11	115200		
<u>Communication Line</u>		<u>Default Communication Speed</u>			
Host Transmit		9600			
Host Receive		Receive=transmit			
Printer		4800			
Modem Hi		Ignore			
Modem Lo		Ignore			
Device Control 1 (XON)		DC1	same	same	
Device Control 3 (XOFF)		DC3	same	same	
Disconnect Delay Time		CSI Ps \$ q	same	same	DECSDDT
<u>Ps</u>	<u>Delay Time</u>				
1	None				
2	60 ms				
none, 0, 3	2 s				
Flow Control		CSI Ps1; Ps2; Ps3; Ps4 * s			
			n/a	n/a	DECSFC
<u>Ps1</u>	<u>Port Type</u>	<u>Ps3</u>	<u>Flow Control Type</u>		
none, 0, 1	Comm Port	none, 0 or 1	XON/XOFF or XPC (for receive only)		
2	Printer Port	2	DTR		
		3	Both		
		4	None		
<u>Ps2</u>	<u>Transmit or Receive</u>	<u>Ps4</u>	<u>Flow Control Threshold</u>		
1, 0 or none	Transmit	none, 0 or 1	Low (64)		
2	Receive	2	High (768)		
3	Transmit and Receive				
Half-Duplex Mode			n/a	n/a	DECHDPXM
Set:		CSI ? 103 h			
Reset:		CSI ? 103 l			
Load Answerback Message ²		DCS Ps v D...D ST	n/a	n/a	DECLANS
Load Banner Message		DCS Ps r D...D ST	n/a	n/a	DECLBAN

¹This sequence takes effect only when a ProPrinter or Digital printer with ProPrinter emulation mode is attached.

²The answerback data string may consist of from 0 to 30, 7- or 8-bit characters, which are hex pairs in the 3/0 through 3/9 (0 through 9) range, 4/1 through 4/6 (A through F) range, and 6/1 through 6/6 (A through F) range. Pressing **Ctrl/Break** or receiving an ENQ code causes the VT510 to transmit the answerback message.

† Not available for the printer port. If specified in the sequence to configure the printer port, the Ps3 value is ignored.

Printer Port Function	Control Function			
	VT510	VT420	VT100	Mnemonic
Modem Control Mode		n/a	n/a	DECMCM
Set:	CSI ? 99 h			
Reset:	CSI ? 99 l			
Null Mode		n/a	n/a	DECNULM
Set (ignore NUL):	CSI ? 102 h			
Reset (accept NUL):	CSI ? 102 l			
Set Port Parameters	CSI <i>Ps1</i> ; <i>Ps2</i> ; <i>Ps3</i> ; <i>Ps4</i> + w			
		n/a	n/a	DECSPP
<u><i>Ps1</i></u>	<u>Port</u>	<u><i>Ps3</i></u>	<u>Parity</u>	
none , 0 or 1	Comm Port	none , 0 or 1	No parity	
2	Printer Port	2	Even	
		3	Odd	
<u><i>Ps2</i></u>	<u>Data</u>	4†	Even	
1, 0 or none	8 bits	5†	Odd, unchecked	
2	7 bits	6	Mark	
		7	Space	
<u><i>Ps4</i></u>	<u>Stop Bit</u>			
none , 0 or 1	1 bit			
2	2 bits			
Set Transmit Rate Limit	CSI <i>Ps1</i> ; <i>Ps2</i> " u	n/a	n/a	DECSTRL
<u><i>Ps1</i></u>	<u>Key Type</u>			
none , 0 or 1	All keys			
2	Graphic key			
3	Function key			
<u><i>Ps2</i></u>	<u>Characters Per Second</u>			
none , 0, 1	150 cps			
2	50 cps			
3	30 cps			
Transmit Rate Limiting Mode		same	n/a	DECXRLM
Set (limited):	CSI ? 73 h			
Reset (unlimited):	CSI ? 73 l			

¹This sequence takes effect only when a ProPrinter or Digital printer with ProPrinter emulation mode is attached.

²The answerback data string may consist of from 0 to 30, 7- or 8-bit characters, which are hex pairs in the 3/0 through 3/9 (0 through 9) range, 4/1 through 4/6 (A through F) range, and 6/1 through 6/6 (A through F) range. Pressing Ctrl/Break or receiving an ENQ code causes the VT510 to transmit the answerback message.

† Not available for the printer port. If specified in the sequence to configure the printer port, the Ps3 value is ignored.

ANSI Control Functions

This chapter describes the ANSI escape and control sequences in detail, whereas the previous chapter provides only a brief summary of these sequences. Each command in this chapter is formatted in such a way that you can refer to the command by either the mnemonic, name, or control function.

ANSI Conformance Levels

This control function lets an application map certain character sets into the terminal's in-use table as default sets. The character sets are based on ANSI conformance levels.

Format

ESC SP *Final*
1/11 2/0 ***

Parameters

Final

indicates the ANSI conformance level for the subsequent data exchange.

Final ANSI Conformance Level	
L	Level 1
M	Level 2
N	Level 3

Description

ANSI conformance levels represent an agreement between the sender and receiver for compatible data exchange. The control function:

1. Acts as an *announcer* for the data exchange that follows between the terminal and application software.
2. Selects which character sets the terminal uses by default in the data exchange.

The VT510 supports three ANSI conformance levels.

ANSI Levels 1 and 2

- ASCII designated as G0.
- ISO Latin-1 supplemental designated as G1.
- G0 mapped into GL.
- G1 mapped into GR.

ANSI Level 3

- ASCII designated as G0.
- G0 mapped into GL.

Notes

- If the terminal is reset, turned off, or changed with a set conformance level (DECSCL) sequence, then the software must send another announcer sequence to the terminal. Otherwise, the terminal uses the default character sets (ASCII in GL, DEC Supplemental or ISO Latin-1 supplemental in GR).

CBT—Cursor Backward Tabulation

Move the active position *n* tabs backward.

Default: 1.

Format

CSI	<i>Pn</i>	Z
9/11	3/ <i>n</i>	5/10

Parameters

Pn

is the number of active position tabs to move backward.

Description

The active position is moved to the character position corresponding to the *n*-th preceding horizontal tabulation stop. If an attempt is made to move the active position past the first character position on the line, then the active position stays at column one.

CHA—Cursor Horizontal Absolute

Move the active position to the n-th character of the active line.

Default: 1.

Format

CSI	<i>Pn</i>	G
9/11	3/ <i>n</i>	4/7

Parameters

Pn

is the number of active positions to the n-th character of the active line.

Description

The active position is moved to the n-th character position of the active line.

CHT—Cursor Horizontal Forward Tabulation

Move the active position n tabs forward.

Default: 1.

Format

CSI	Pn	I
9/11	3/ n	4/9

Parameters

Pn

is the number of active position tabs to move forward.

Description

The active position is moved to the character position corresponding to the following n -th horizontal tabulation stop.

CNL—Cursor Next Line

Move the cursor to the next line.

Default: 1.

Format

CSI	<i>Pn</i>	E
9/11	3/ <i>n</i>	4/5

Parameters

Pn

is the active position to the first character of the n-th following line.

Description

The active position is moved to the first character of the n-th following line.

CPL—Cursor Previous Line

Move the cursor to the preceding line.

Default: 1.

Format

CSI	<i>Pn</i>	F
9/11	3/ <i>n</i>	4/6

Parameters

Pn

is the number of active position moved to the first character of the n-th preceding line.

Description

The active position is moved to the first character of the n-th preceding line.

CPR—Cursor Position Report

The host asks the terminal for a cursor position report.

Format

CSI	6	n			Request cursor position.
9/11	3/6	6/14			
CSI	<i>PI</i>	;	<i>Pc</i>	R	Report cursor position.
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	5/2	

Parameters

PI

indicates what line the cursor is on.

Pc

indicates what column the cursor is at.

CRM—Show Control Character Mode

Enable or disable control characters to be displayed.

Default: Interpret

Format

CSI	3	h	Set: show control character.
9/11	3/3	6/8	
CSI	3	I	Reset: interpret control character.
9/11	3/3	6/12	

Description

When CRM mode is set, all characters (graphic or control) are displayed in the CRM font. All control functions are displayed as graphic characters. No control functions are executed, except LF, FF, and VT, which are represented in the CRM font before a CRLF (new line) is executed. XOFF (DC3) and XON (DC1) are used for flow control (if enabled). DC1 and DC3 are displayed after execution.

This function affects data received in the data stream following the change to CRM mode. Characters already displayed are not affected. Entering Control Representation mode temporarily causes G0 and GL to be loaded with ASCII. G2 and GR are loaded with user-selected Supplemental Graphic characters if in 8-bit Multinational Character mode.

Notes on CRM

- CRM mode works in all VT terminal modes.
- NUL is displayed and is not useful as a fill character.
- CRM mode overrides Printer Controller mode. When in CRM, Printer Controller mode is temporarily off.
- In local mode and CRM (or local echo and CRM), CSI sequences may be represented as either ESC Fe sequences (such as ESC I) or as C1 controls. Conversely, C1 controls may be represented as either C1 controls or as ESC Fe sequences.
- In CRM mode, auto-wrap always occurs at the right margin. Sending a sequence or disabling auto-wrap in Set-Up is ignored.
- Terminal Device Protocol commands are not affected if Control Representation mode is entered.

CUB—Cursor Backward

This control function moves the cursor to the left by a specified number of columns. The cursor stops at the left border of the page.

Format

CSI	<i>Pn</i>	D
9/11	3/ <i>n</i>	4/4

Parameters

Pn

is the number of columns to move the cursor to the left.

CUD—Cursor Down

This control function moves the cursor down a specified number of lines in the same column. The cursor stops at the bottom margin. If the cursor is already below the bottom margin, then the cursor stops at the bottom line.

Format

CSI	<i>Pn</i>	B
9/11	3/ <i>n</i>	4/2

Parameters

Pn
is the number of lines to move the cursor down.

CUF—Cursor Forward

This control function moves the cursor to the right by a specified number of columns. The cursor stops at the right border of the page.

Format

CSI	<i>Pn</i>	C
9/11	3/ <i>n</i>	4/3

Parameters

Pn

is the number of columns to move the cursor to the right.

CUP—Cursor Position

This control function moves the cursor to the specified line and column. The starting point for lines and columns depends on the setting of origin mode (DECOM). CUP applies only to the current page.

Format

CSI	<i>Pl</i>	;	<i>Pc</i>	H
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	4/8

Parameters

Pl

is the number of the line to move to. If Pl is 0 or 1, then the cursor moves to line 1.

Pc

is the number of the column to move to. If Pc is 0 or 1, then the cursor moves to column 1.

CUU—Cursor Up

Moves the cursor up a specified number of lines in the same column. The cursor stops at the top margin. If the cursor is already above the top margin, then the cursor stops at the top line.

Format

CSI	<i>Pn</i>	A
9/11	3/ <i>n</i>	4/1

Parameters

Pn

is the number of lines to move the cursor up.

DA1 — Primary Device Attributes

In this DA exchange, the host asks for the terminal's architectural class and basic attributes.

Host Request

The host uses the following sequence to send this request:

```
CSI  c   or  CSI  0   c
9/11 6/3    9/11 3/0 6/3
```

Terminal Response

The terminal responds by sending its architectural class and basic attributes to the host. This response depends on the terminal's current operating VT level.

Response from North American Terminal

```
CSI  ?   6   4   ;   Ps1 ;   ...  Psn  c
9/11 3/15 3/6 3/4 3/11 3/n 3/11 ... 3/n 6/3
```

Response from International Terminal

```
CSI  ?   6   4   ;   Ps1 ;   ...  Psn  c
9/11 3/15 3/6 3/4 3/11 3/n 3/11 ... 3/n 6/3
```

Parameters

Ps1 ; . . . Psn

reports different device attributes between the North American terminal and the international terminal.

The value of the first parameter is encoded so a simple range check can be performed to determine the basic service class and level of the device. The VT510 is a level 4 terminal so its service class code is 64. The following extensions to level 4 are provided:

Ps	Meaning
----	---------

1	132 columns
---	-------------

2	Printer port
---	--------------

4	Sixel
---	-------

6	Selective erase
---	-----------------

7	Soft character set (DRCS)
---	---------------------------

8	User-defined keys (UDKs)
---	--------------------------

9	National replacement character sets (NRCS) (International terminal only)
---	---

12	Yugoslavian (SCS)
----	-------------------

15	Technical character set
----	-------------------------

18	Windowing capability
----	----------------------

21	Horizontal scrolling
----	----------------------

23	Greek
----	-------

24	Turkish
----	---------

42	ISO Latin-2 character set
----	---------------------------

44	PCTerm
----	--------

45	Soft key map
----	--------------

46	ASCII emulation
----	-----------------

Primary DA Example

Here is a typical primary DA exchange.

Exchange	Sequence	Meaning
Request (Host to VT510)	CSI c or CSI 0 c	The host asks for the terminal's architectural class code and supported extensions.
Response—N.A. (VT510 to host)	CSI ? 64; 1; 2; 7; 8; 9; 15; 18; 21; 44; 45; 46 c	The terminal is a class 4 device (64) and supports the Ps parameters listed above.
Response— International (VT510 to host)	CSI ? 64; 1; 2; 7; 8; 9; 12; 15; 18; 21; 23; 24; 42; 44; 45; 46 c	The terminal is a class 4 device (64) and supports the Ps parameters listed above.

DA2—Secondary Device Attributes

In this DA exchange, the host requests the terminal's identification code, firmware version level, and hardware options.

Host Request

The host uses the following sequence to send this request:

CSI > c or CSI > 0 c
9/11 3/14 6/3 9/11 3/14 3/0 6/3

Terminal Response

The terminal with a VT keyboard uses the following sequence to respond:

CSI > 6 1 ; Pv ; 0 c DA2R for terminal with STD keyboard.
9/11 3/14 3/6 3/1 3/11 3/n 3/11 3/0 6/3
CSI > 6 1 ; Pv ; 1 c DA2R for terminal with PC keyboard.
9/11 3/14 3/6 3/1 3/11 3/n 3/11 3/1 6/3

Parameters

61
indicates the identification code of the terminal for the secondary device attributes command.

Pv
indicates the version level of the firmware implementing the terminal management functions, for example, editing, as shown in the following table.

Pv Version	
10	V1.0 (released version 1.0)
20	V2.0 (released version 2.0)

Secondary DA Example

The following is a typical secondary DA exchange:

Exchange	Sequence	Meaning
Request (Host to VT510)	CSI > c or CSI > 0 c	The host asks for the terminal's identification, firmware version, current hardware options.
Response (VT510 to host)	CSI > 61; 20; 1 c	The terminal identifies itself as a VT510 that uses version 2.0 firmware, and has a PC keyboard option.

DA3—Tertiary Device Attributes

In this DA exchange, the host asks for the terminal unit identification code. This ID code serves as a way to identify each terminal in a system. The unit ID code is preset at the factory.

Host Request

The host uses the following sequence to send this request:

CSI = c or CSI = 0 c
9/11 3/13 6/3 9/11 3/13 3/0 6/3

Terminal Response

The terminal responds by sending a report terminal unit ID (DECRPTUI) control string to the host. DECRPTUI is available in VT Level 4 mode only.

DCS ! | D...D ST
9/0 2/1 7/12 ... 9/12

Parameters

D...D
is the unit ID of the terminal, consisting of four hexadecimal pairs. The first pair represents the manufacturing site code. This code can be any hexadecimal value from 00 through FF.

The last three hexadecimal pairs are the terminal ID number. This number is unique for each terminal manufactured at that site.

Tertiary DA Example

Here is a typical tertiary DA exchange.

Exchange	Sequence	Meaning
Request (Host to Terminal)	CSI = c or CSI = 0 c	The host asks for the terminal unit ID.
DECRPTUI Response (Terminal to host)	DCS ! 00 01 02 05 ST	The terminal was manufactured at site 00 and has a unique ID number of 125.

DCH—Delete Character

This control function deletes one or more characters from the cursor position to the right.

Format

CSI	<i>Pn</i>	P
9/11	3/ <i>n</i>	5/0

Parameters

Pn

is the number of characters to delete. If *Pn* is greater than the number of characters between the cursor and the right margin, then DCH only deletes the remaining characters.

Default: *Pn* = 1.

Description

As characters are deleted, the remaining characters between the cursor and right margin move to the left. Character attributes move with the characters. The terminal adds blank spaces with no visual character attributes at the right margin. DCH has no effect outside the scrolling margins.

DDD1, DDD2, DDD3

DDD1, DDD2, and DDD3 are responses to the host for a Request Terminal State Report (DECRQTSR) control function. These responses only work for a Hebrew keyboard.

Format

ESC) 1 DDD1
1/11 2/9 3/1
ESC) B DDD2
1/11 2/9 4/2
ESC (1 DDD3
1/11 2/8 3/1

Description

These escape sequences have the following effects:

Escape Sequence	Meaning
DDD1	DECRLM is set. DECHEBM is set. DECHEM is set
DDD2	DECRLM is reset. ASCII is designated in G1.
DDD3	ASCII is designated in G0.

DECAAM—Set/Reset Auto Answerback Mode

Selects whether the terminal automatically sends (enables) or does not send (disables) the answerback message to the host computer after a communication line connection.

Default: Auto answerback disabled.

Format

CSI	?	1	0	0	h	Set: enables auto answerback.
9/11	3/15	3/1	3/0	3/0	6/8	
CSI	?	1	0	0	l	Reset: disables auto answerback.
9/11	3/15	3/1	3/0	3/0	6/12	

Description

When modem control is enabled, the answerback message is sent 500 ms after the connection is made.

DECALN—Screen Alignment Pattern

This control function fills the complete screen area with a test pattern used for adjusting screen alignment. Normally, only manufacturing and service personnel would use DECALN.

Format

ESC	#	8
1/11	2/3	3/8

Notes on DECALN

DECALN sets the margins to the extremes of the page, and moves the cursor to the home position.

Also see the screen alignment in [Chapter 2](#).

DECANM—ANSI Mode

DECANM changes the terminal to the VT52 mode of operation. In VT52 mode, the terminal acts like a VT52 terminal. This mode lets you use applications designed for the VT52 terminal.

Format

CSI ? 2 I
9/11 3/15 3/2 6/12

Description

Table 5–1 VT52 Escape Sequences

Sequence	Action
ESC A	Cursor up.
ESC B	Cursor down.
ESC C	Cursor right.
ESC D	Cursor left.
ESC F	Enter graphics mode.
ESC G	Exit graphics mode.
ESC H	Cursor to home position.
ESC I	Reverse line feed.
ESC J	Erase from cursor to end of screen.
ESC K	Erase from cursor to end of line.
ESC Y Pn	Move cursor to column Pn.
ESC Z	Identify (host to terminal).
ESC /Z	Report (terminal to host).
ESC =	Enter alternate keypad mode.
ESC >	Exit alternate keypad mode.
ESC <	Exit VT52 mode. Enter VT100 mode.
ESC ^	Enter autoprint mode.
ESC _	Exit autoprint mode.
ESC W	Enter printer controller mode.
ESC X	Exit printer controller mode.
ESC]	Print screen.
ESC V	Print the line with the cursor.

DECARM—Autorepeat Mode

This control function determines whether or not keys automatically repeat their character when held down. If DECARM is set, then most keys you press for more than 0.5 seconds send a character repeatedly until you release the key.

Default: Repeat (set)

Format

CSI ? 8 h Set: keys autorepeat when pressed for more than 0.5 seconds.
9/11 3/15 3/8 6/8
CSI ? 8 l Reset: keys do not autorepeat.
9/11 3/15 3/8 6/12

Notes on DECARM

- The following keys on the VT keyboard do not repeat:

F1 (Hold)	F5 (Break)	Return
F2 (Print)	Compose Character	Lock
F3 (Set-Up)	Shift	Ctrl
F4 (Session)	Alt Function	Extend

- The following keys on the PC keyboard do not repeat:

Alt	AltGr
Caps Lock	Pause
Ctrl	Print Screen
Enter	Scroll Lock
Num Lock	Shift

DECARSM—Set/Reset Auto Resize Mode

Enable or disable changing the lines per screen automatically when the page arrangement changes.
Default: Auto resize disabled

Format

CSI	?	9	8	h	Set: enable auto resize.
9/11	3/15	3/9	3/8	6/8	
CSI	?	9	8	l	Reset: disable auto resize.
9/11	3/15	3/9	3/8	6/12	

Description

When Auto Resize is enabled, the number of Lines/Screen changes automatically each time the page arrangement changes.

Auto Resize and Number of lines per screen are user features that can be locked from being redefined by the host if the User Feature Lock in Set-Up is enabled.

The following table indicates how the lines/screen changes when the page size changes, assuming Auto Resize is set.

Page size	Lines/Screen
24	26 (24 data lines, 1 status line, 1 blank line)
25	26 (25 data lines, 1 status line)
36	42 (36 data lines, 1 status line, 5 blank lines)
41	42 (41 data lines, 1 status line)
42	42 (42 data lines, no status line)
48	53 (48 data lines, 1 status line, 4 blank lines)
52	53 (52 data lines, 1 status line)
53	53 (53 data lines, 0 status line)
72	53 (53 data lines, 0 status line)

DECAUPSS—Assigning User-Preferred Supplemental Sets

You can assign the supplemental character set you use most often as a special standby set. This standby set is called the user-preferred supplemental set. This feature provides applications with an easy way to access the user's preferred supplemental set.

Default: DEC Supplemental Graphic

Format

DCS Pn ! u Dscs ST
9/0 3/n 2/1 7/5 3/n 9/12

Parameters

Pn
represents a character set designator.

Pn	Meaning
0	DEC, ISO Latin-1, ISO Latin-2
1	ISO Latin-5, ISO Latin-7, ISO Cyrillic, ISO Hebrew

Dscs represents a character set.

Dscs	Character Set
% 5	DEC Supplemental
" ?	DEC Greek
" 4	DEC Hebrew
% 0	DEC Turkish
& 4	DEC Cyrillic
<	User-preferred Supplemental
A	ISO Latin-1 Supplemental
B	ISO Latin-2 Supplemental
F	ISO Greek Supplemental
H	ISO Hebrew Supplemental
M	ISO Latin-5 Supplemental
L	ISO Latin-Cyrillic

Description

You can assign any supplemental or ISO Latin supplemental set as the standby set. See SCS—Select Character Set for more information.

After you assign a set:

1. Designate the set as G1, G2, or G3.
2. Map the set into GR.

DECAWM—Autowrap Mode

This control function determines whether or not received characters automatically wrap to the next line when the cursor reaches the right border of a page in page memory.

Default: No autowrap

Format

CSI	?	7	h	Set: autowrap.
9/11	3/15	3/7	6/8	
CSI	?	7	l	Reset: no autowrap.
9/11	3/15	3/7	6/12	

Description

If the DECAWM function is set, then graphic characters received when the cursor is at the right border of the page appear at the beginning of the next line. Any text on the page scrolls up if the cursor is at the end of the scrolling region.

If the DECAWM function is reset, then graphic characters received when the cursor is at the right border of the page replace characters already on the page.

DECBI—Back Index

This control function moves the cursor backward one column. If the cursor is at the left margin, then all screen data within the margin moves one column to the right. The column that shifted past the right margin is lost.

Available in: VT Level 4 mode only

Format

ESC	6
1/11	3/6

Description

DECBI adds a new column at the left margin with no visual attributes. DECBI is not affected by the margins. If the cursor is at the left border of the page when the terminal receives DECBI, then the terminal ignores DECBI.

DECBKM—Backarrow Key Mode

This control function determines whether the  key works as a backspace key or delete key.

Note

To be compatible with some software, DECBKM should be reset (delete key setting).

Default: Delete key

Format

CSI	?	6	7	h	Set: backspace key.
9/11	3/15	3/6	3/7	6/8	
CSI	?	6	7	l	Reset: delete key.
9/11	3/15	3/6	3/7	6/12	

Description

If DECBKM is set,  works as a backspace key. When you press , the terminal sends a BS character to the host.

If DECBKM is reset,  works as a delete key. When you press , the terminal sends a DEL character to the host.

DECCANSM—Conceal Answerback Message Mode

Conceal the answerback message.
Default: Answerback message is not concealed

Format

CSI	?	1	0	1	h	Conceal answerback message
9/11	3/15	3/1	3/0	3/1	6/8	
CSI	?	1	0	1	l	Answerback message is not concealed
9/11	3/15	3/1	3/0	3/1	6/12	

Description

Setting	Function
Not concealed	The answerback message is visible in Set-Up.
Concealed	The text "concealed" replaces the answerback message. The answerback message cannot be set back to "not concealed" either through a host control function or Set-Up. Only entering a new message returns this field to "not concealed." If the user or host requests the terminal to reset to the default setting (answerback message not concealed (CSI ? 101 l)) when the answerback message is set to "concealed," then the terminal will not reset the answerback message to "not concealed."

DECCAPSLK—Caps Lock Mode

DECCAPSLK controls the state of the Caps Lock key on the PC keyboard.

Format

CSI	?	1	0	9	h	Set: caps lock mode.
9/11	3/15	3/1	3/0	3/9	6/8	
CSI	?	1	0	9	l	Reset: caps lock mode.
9/11	3/15	3/1	3/0	3/9	6/12	

DECCARA—Change Attributes in Rectangular Area

This control function lets you change the visual character attributes (bold, blink, reverse video, and underline) of a specified rectangular area in page memory. The select attribute change extent (DECSACE) control function determines whether all or some of the character positions in the rectangle are affected. DECCARA does not change the values of characters just the visual attributes of those characters.

Available in: VT Level 4 mode only

Format

CSI	<i>Pt; Pl; Pb; Pr;</i>	<i>Ps1; . . . Psn</i>	\$	<i>r</i>
9/11	area to be changed	attributes to change	2/4	7/2

Parameters

Pt; Pl; Pb; Pr;

define the rectangular area to be changed. A semicolon (3/11) separates parameters.

Pt

is the top-line border. *Pt* must be less than or equal to *Pb*.

Default: *Pt* = 1.

Pl

is the left-column border. *Pl* must be less than or equal to *Pr*.

Default: *Pl* = 1.

Pb

is the bottom-line border.

Default: *Pb* = last line of the active page.

Pr

is the right-column border.

Default: *Pr* = last column of the active page.

Ps1; . . . Psn

select the visual character attributes to change. These values correspond to the values used in the select graphic rendition.

Psn	Meaning
0 (default)	Attributes off (no bold, no underline, no blink, positive image)
1	Bold
4	Underline
5	Blink
7	Negative image
22	No bold
24	No underline
25	No blink
27	Positive image

DECCARA ignores all other parameter values. When you use more than one parameter in a command, DECCARA executes them cumulatively in sequence.

Examples

The following sequence changes the video attributes of the complete screen to blink and underscore:

CSI ; ; ; ; 0; 4; 5 \$ r

The following sequence turns off the blinking character attribute from position line 10, column 2 to position line 14, column 45 on the current page:

CSI 10; 2; 14; 45; 25 \$ r

Notes on DECCARA

- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECCARA is not affected by the page margins.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, then the value is treated as the width or height of that page.
- DECCARA does not change the active cursor position.
- DECCARA does not change the current rendition set by the select graphic rendition (SGR) function.
- The exact character positions affected by DECCARA depend on the current setting of the select attribute change extent (DECSACE) function.

DECCIR—Cursor Information Report (Terminal to Host)

The terminal sends this sequence in response to a request presentation state report (DECRQPSR) sequence, CSI 1 \$ w. DECCIR reports the status of the cursor position, including visual attributes and character protection attributes. DECCIR also reports the status of origin mode (DECOM) and the current active character sets.

Format

DCS 1 \$ u D ... D ST
9/0 3/1 2/4 7/5 D ... D 9/12

Description

DECCIR reports all the character sets if they are assigned to one of the G0, G1, G2, or G3 sets. Refer to SCS—Select Character Set for all character set designators.

Parameters

D...D
is the data string containing the cursor information. The format for this data string is as follows:
Pr; Pc; Pp; Srend; Satt; Sflag; Pgl; Pgr; Scss; Sdesig

Pr
is the number of the line the cursor is on.

Pc
is the number of the column the cursor is at.

Pp
is the number of the current page.

Srend
is one or more characters indicating the visual attributes, such as bold or blinking, currently in use for writing. To find out what attributes are set, you must convert the character to an 8-bit binary number. The bits are used as follows.

Bit	Attribute	Bit Value
8	–	Always 0 (off).
7	–	Always 1 (on).
6	Extension indicator	1 = another character (byte) of visual attribute data follows this one. 0 = no more attribute data.
5	–	Always 0 (off).
4	Reverse video	0 = off. 1 = on.
3	Blinking	0 = off. 1 = on.
2	Underline	0 = off. 1 = on.
1	Bold	0 = off. 1 = on.

Example
If the bold and underline attributes are currently set for writing, *Srend* is the ASCII uppercase C character (01000011₂).

Satt

is one or more characters indicating any selective erase attributes currently set for writing.

To find what attributes are set, you must convert each character to an 8-bit binary number. Use the following table to find the meaning of the 8-bit binary number:

Bit	Attribute	Bit Value
8	–	Always 0 (off).
7	–	Always 1 (on).
6	Extension indicator	1 = another character (byte) of selective erase data follows this one. 0 = no more protection data.
5	–	0 Reserved for future use.
4	–	0 Reserved for future use.
3	–	0 Reserved for future use.
2	–	0 Reserved for future use.
1	Selective erase (DECSCA)	0 = off. 1 = on.

Example

If the selective erase protection attribute is currently set for writing, then *Satt* is the ASCII uppercase A character (01000001₂).

Sflag

is one or more characters that indicate several flags and modes the terminal must save.

To see the current state of the flags and modes, you must convert each character to an 8-bit binary number. Use the following table to find the meaning of the 8-bit binary number:

Bit	Attribute	Bit Value
8	–	Always 0 (off).
7	–	Always 1 (on).
6	Extension indicator	1 = another character (byte) of flag data follows this one. 0 = no more flag data.
5	–	0 Reserved for future use.
4	Autowrap	1 = autowrap pending. 0 = autowrap not pending.
3	Single shift 3 (SS3) setting	1 = G3 is mapped into GL for the next typed character only. 0 = single shift 3 is off.
2	Single shift 2 (SS2) setting	1 = G2 is mapped into GL for the next typed character only. 0 = single shift 2 is off.
1	Origin mode	1 = origin mode set. 0 = origin mode reset.

Example

If origin mode is set, autowrap is pending, and a single shift 3 has been received, then *Sflag* is the ASCII uppercase M character (01001101₂).

Pgl

indicates the number of the logical character set (G0 through G3) mapped into GL.

0 = G0 is in GL. 2 = G2 is in GL.

1 = G1 is in GL. 3 = G3 is in GL.

Pgr

indicates the number of the logical character set (G0 through G3) mapped into GR.

0 = G0 is in GR. 2 = G2 is in GR.

1 = G1 is in GR. 3 = G3 is in GR.

Scss

is a character indicating the size of the character sets in G0 through G3.

To find out what the character means, you must convert it to an 8-bit binary number. Use the following table to find the meaning of the 8-bit binary number:

Bit	Attribute	Bit Value
8	–	Always 0 (off).
7	–	Always 1 (on).
6	Extension indicator	1 = another character (byte) of character size data follows this one. 0 = no more size data.
5	–	0 Reserved for future use.
4	G3 set size	0 = 94 characters. 1 = 96 characters.
3	G2 set size	0 = 94 characters. 1 = 96 characters.
2	G1 set size	0 = 94 characters. 1 = 96 characters.
1	G0 set size	0 = 94 characters. 1 = 96 characters.

Example

Suppose the following conditions exist:

- The ISO Latin-1 supplemental set is designated as G2 and G3.
- The ASCII set is designated as G0 and G1.
- Single shift 2 (SS2) is set.

Then Scss is the ASCII backslash \ character (01011100₂).

Sdesig

is a string of intermediate and final characters indicating the character sets designated as G0 through G3. These final characters are the same as those used in select character set (SCS) sequences.

Example

Suppose the ASCII character set is designated as G0, DEC Special Graphic as G1, and DEC Supplemental Graphic as G2 and G3. The *Sdesig* string would be B0%5%5. Each character corresponds to a final character in an SCS sequence, as follows:

G0	G1	G2	G3
B	0	%5	%5
ASCII set	DEC Special Graphic	DEC Supplemental Graphic	DEC Supplemental Graphic

Example

The following is an example of a cursor information report:

DCS 1 \$ u 1; 1; 1; @; @; @; 0; 2; @; BB%5%5 ST

1; 1; 1; indicates that the cursor is at row 1, column 1, on the first page.

@; @; @; indicates that (1) no visual character attributes or selective erase attributes are set for writing, (2) DECOM is reset, and (3) there is no SS2, SS3, or autowrap pending.

0; 2; indicates that G0 is mapped into GL, and G2 is in GR.

@; indicates that all character sets have 94 characters.

BB%5%5 indicates that ASCII is in G0 and G1, and DEC Supplemental Graphic is in G2 and G3.

Notes on DECCIR

- **The cursor information in a DECCIR sequence is the same information saved through a save cursor (DECSC) command.**

DECCKD—Copy Key Default

Copies the default states of a key and its key/modifier combination to another key and its key/modifier combination.

Format

```
DCS " z D...D ST
9/0 2/2 7/10 D...D 9/12
```

Description

This control function copies the default states of a key and all its modifier combinations to another key and its modifier combinations. The source key may have been reprogrammed to perform a non-default function when copy operation is executed. Only the default states of the source key are copied.

The source key and the destination key can be two different keys or the same key. If they are different keys, then the copy operation occupies the memory space that is reserved for the operations of control functions, DECPFK, DECPAK, and DECUDK and their equivalent functions in Set-Up. (Refer to DECPFK—Program Function Key for details.) If they are the same key, then the function of the copy operation is to restore the key's default. In this state, the terminal does not allocate new memory space. The terminal frees up the space by deleting the non-default functions of the destination key and its modifier combinations.

A function key can copy the default state from a function key or a graphic key. When a graphic key's default is copied, the function key operates as a graphic key. Only the modifiers that are suitable for a graphic key operate with the key. This logic is used when a graphic key copies the default from a function key.

Multiple keys can be copied in one sequence.

The data string D...D format is:

Key-s/Key-d;Key-s/Key-d.....;

Where:

- Key-s is the keystation number of the key to be copied.
- Key-d is the keystation number of the destination key.

Once programmed, the key states are not affected by the terminal mode changes. Refer to DECPFK—Program Function Key for the list of terminal mode changes.

This function works in all the VT modes except in VT52 mode.

DECCKM—Cursor Keys Mode

This control function selects the sequences the arrow keys send. You can use the four arrow keys to move the cursor through the current page or to send special application commands.

Default: Cursor

Format

CSI	?	1	h	Set: application sequences.
9/11	3/15	3/1	6/8	
CSI	?	1	l	Reset: cursor sequences.
9/11	3/15	3/1	6/12	

Description

If the DECCKM function is set, then the arrow keys send application sequences to the host.

If the DECCKM function is reset, then the arrow keys send ANSI cursor sequences to the host.

DECCKSR—Memory Checksum Report

The host asks for a memory checksum report of current text macro definitions.

Programming Tip

Enter your macro definitions, then request and save a memory checksum. After this procedure, you can request another checksum to determine if the checksum has changed.

Format

CSI	?	6	3	;	Pid	n	Request memory checksum.
9/11	3/15	3/6	3/3	3/11	3/n	6/14	
DCS	Pid	!	~	D...D	ST		Response with checksum.
9/0	3/n	2/1	7/14	...		9/12	

Parameters

Pid (request)

is an optional numeric parameter that provides a label to identify the particular checksum request. The checksum report returns the label. This label lets you distinguish between multiple checksum reports.

Pid (response)

is a label indicating which DSR request the report is for.

D ... D

is the data string consisting of four hexadecimal digits indicating the checksum. The digits can be in the range of 3/0 through 3/9 and 4/0 through 4/6.

DECCOLM—Select 80 or 132 Columns per Page

There are two control functions that can set the page width to 80 or 132 columns, DECSCPP (DECSCPP—Select 80 or 132 Columns per Page) and DECCOLM.

Note

It is recommended that new applications use DECSCPP rather than DECCOLM. DECSCPP does not clear page memory or reset the scrolling regions, as does DECCOLM. DECCOLM is provided mainly for compatibility with previous products.

Default: 80 columns

Format

CSI	?	3	h	Set: 132-column font.
9/11	3/15	3/3	6/8	
CSI	?	3	l	Reset: 80-column font.
9/11	3/15	3/3	6/12	

Notes on DECCOLM

- DECCOLM sets the number of columns on the page to 80 or 132 and selects the corresponding 80- or 132-column font.
- If you change the DECCOLM setting, the terminal:
 - Sets the left, right, top and bottom scrolling margins to their default positions.
 - Erases all data in page memory.
- DECCOLM resets vertical split screen mode (DECLRMM) to unavailable.
- DECCOLM clears data from the status line if the status line is set to host-writable.

DECCRA—Copy Rectangular Area

This control function copies a rectangular area of characters from one section to another in page memory. The copied text retains its character values and attributes.

Available in: VT Level 4 mode only

Format

CSI	<i>Pts; Pls; Pbs; Prs; Pps;</i>	<i>Ptd; Pld; Ppd</i>	\$	v
9/11	<i>area to be copied</i>	<i>destination</i>	2/4	7/6

Parameters

Pts; Pls; Pbs; Prs; Pps;

define the rectangular area to be copied (the source). A semicolon (3/11) separates parameters.

Pts

is the top-line border. *Pts* must be less than or equal to *Pbs*. Default: *Pts* = 1.

Pls

is the left-column border. *Pls* must be less than or equal to *Prs*. Default: *Pls* = 1.

Pbs

is the bottom-line border. Default: *Pbs* = the last line of the page.

Prs

is the right-column border. Default: *Prs* = the last column of the page.

Pps

is the number of the page where the rectangular area is located. Default: *Pps* = 1.

Ptd; Pld; Ppd;

define the destination of the copied rectangular area.

Ptd

is the top-line border. Default: *Ptd* = 1.

Pld

is the left-column border. Default: *Pld* = 1.

Ppd

is the number of the page. Default: *Ppd* = 1.

Notes on DECCRA

- If *Pbs* is greater than *Pts*, or *Pls* is greater than *Prs*, the terminal ignores DECCRA.
- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECCRA is not affected by the page margins.
- The copied text takes on the line attributes of the destination area.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, then the value is treated as the width or height of that page.
- If a page value exceeds the number of pages available in the current page arrangement, then the value is treated as the last available page number.
- If the destination area is partially off the page, then DECCRA clips the off-page data.
- DECCRA does not change the active cursor position.

DECCRTSM—Set/Reset CRT Save Mode

Enable or disable CRT saver.

Default: Enable CRT saver.

Format

CSI	?	9	7	h	Set: enable CRT saver.
9/11	3/15	3/9	3/7	6/8	
CSI	?	9	7	l	Reset: disable CRT saver.
9/11	3/15	3/9	3/7	6/12	

Description

If CRT saver is enabled and the terminal is inactive for 30 minutes (no keyboard activity or input from a host computer), then the monitor screen goes blank to prolong the life of the CRT. No data is lost when CRT saver is active. Keyboard activity or input from the host computer re-activates the monitor.

When the monitor goes blank, the keyboard LEDs flash.

DECDC—Delete Column

This control function deletes one or more columns in the scrolling region, starting with the column that has the cursor.

Available in: VT Level 4 mode only

Format

```
CSI  Pn  '  ~  
9/11 3/n 2/7 7/14
```

Parameters

Pn

is the number of columns to delete.

Default: *Pn* = 1.

Description

As columns are deleted, the remaining columns between the cursor and the right margin move to the left. The terminal adds blank columns with no visual character attributes at the right margin. DECDC has no effect outside the scrolling margins.

DECDHL—Double-Width, Double-Height Line

These two control functions make the line with the cursor the top or bottom half of a double-height, double-width line. You must use these sequences in pairs on adjacent lines. In other words, the same display characters must appear in the same positions on both lines to form double-height characters. If the line was single width and single height, then all characters to the right of the screen center are lost.

Format

ESC	#	3	Top Half
1/11	2/3	3/3	
ESC	#	4	Bottom Half
1/11	2/3	3/4	

Description

The following sequences make the phrase "VT510 Video Terminal" a double-height, double-width line.

ESC#3 VT510 Video Terminal

ESC#4 VT510 Video Terminal

DECDLD—Dynamically Redefinable Character Sets

DECDLD loads your soft character set into dynamically redefinable character set (DRCS) font buffers. Although the terminal has six font buffers, it only supports four buffers for soft character sets.

The terminal arranges the DRCS font buffers into sets. Each DRCS set has two font buffers: one for 80-column fonts and one for 132-column fonts. These buffers also allow you to load two different font heights or character sets in both 80- and 132-column widths.

Although the terminal allows you to use two different soft character sets at one time, it only allows you to use one DRCS set at one time.

Format

`DCS Pfn ; Pcn; Pe; Pcmw; Pss; Pt; Pcmh; Pcsc f Dscs Sxbp1 ; Sxbp2 ; . . . ; Sxbpn ST`

Description

DCS is the *device control string introducer*. DCS is an 8-bit C1 character. You can use the equivalent 7-bit sequence, ESC P (1/11, 5/0), when coding for a 7-bit environment.

ST is the *string terminator*. ST is an 8-bit C1 character. You can use the equivalent 7-bit sequence, ESC \ (1/11, 5/12), when coding for a 7-bit environment.

Parameters

Pfn; Pcn; Pe; Pcmw; Pss; Pt; Pcmh; Pcsc

are *parameter characters*, separated by semicolons (3/11). [Table 5–2](#) describes these parameters and lists their possible values. [Table 5–3](#) describes the combinations of *Pcmw*, *Pt*, and *Pcmh* that you can use for 80- and 132-column fonts. If you use any other combinations, then the terminal ignores the DECDLD string.

{

is the *final character*. It marks the end of the parameter characters and indicates that this string is a DECDLD function.

Dscs

defines the name for the soft character set. You use this name in the select character set (SCS) escape sequence. You use the following format for the *Dscs* name:

I F

I is 0, 1, or 2 intermediate characters from the 2/0 to 2/15 range in the ASCII character set.

F is a final character in the 3/0 to 7/14 range.

Notes

- If two downline loaded soft character sets have the same *Dscs*, then the most recently loaded set is used. The terminal looks for a matching *Dscs* in reverse load order (last loaded first).
- If the terminal receives a DRCS control string for a given font number with a different *Dscs* (designating sequence), *Pcmh* (height) or *Pcsc* (character set size), then the terminal erases the previously downline loaded set (if any) in both 80- and 132-column mode.
- If there is no DRCS available that is compatible with the current number of lines per screen (font height), then the terminal displays a reverse question mark in place of any characters.

Examples of *Dscs* Names

Name	Function
SP @ 2/0 4/0	Defines the character set as an unregistered soft set. This value is the recommended default for user-defined sets. The value of <i>Pcss</i> defines whether this set has 94 or 96 characters.
A 4/2	If <i>Pcss</i> is 0, then A defines the soft character set as the U.K. national replacement character set. If <i>Pcss</i> is 1, then A defines the soft character set as the ISO Latin-1 supplemental set.
& % C 2/6 2/5 4/3	Defines the soft character set as % C, which is currently an unregistered set. The value of <i>Pcss</i> defines whether this set has 94 or 96 characters. Sxbp1 ; Sxbp2 ; . . . ; Sxbpn are the sixel bit patterns for individual characters, separated by semicolons (3/11). Your character set can have 1 to 94 patterns or 1 to 96 patterns, depending on the setting of the character set size parameter (<i>Pcss</i>). Each sixel bit pattern is in the following format: <i>S...S/S...S</i> the first <i>S...S</i> represents the sixels in Group A of the soft character. <i>/ (2/5)</i> advances the sixel pattern to Group B of the soft character. the second <i>S...S</i> represents the sixels in Group B of the soft character.

Table 5–2 DECDLD Parameter Characters

Parameter	Name	Description
<i>Pfn</i>	Font number	Selects the DRCS (soft character set) to load. The previous set (if any), that you load with that font number, is erased or overwritten as specified by the control sequence. You can assign a value of 0, 1, and 2 to <i>Pfn</i> . Other values cause the terminal to ignore the entire sequence. <i>Pfn</i> = 0 defaults to the first empty font buffer, or, if none are empty, to font buffer 1.
<i>Pcn</i>	Starting character	Selects where to load the first character in the DRCS font buffer. The location corresponds to a location in the ASCII code table. For example, a <i>Pcn</i> value of 0 means that the first soft character is loaded into position 2/0 of the character table. A <i>Pcn</i> value of 1 means position 2/1 in the table, and so on, up to <i>Pcn</i> = 95 (position 7/15). <i>Pcn</i> is affected by the character set size. See <i>Pcss</i> below.
<i>Pe</i>	Erase control	Selects which characters to erase from the DRCS buffer before loading the new font. 0 = Erases all characters in the DRCS buffer with this number, width, and rendition. 1 = Erases only characters in locations being reloaded. 2 = Erases all renditions of the soft character set (normal, bold, 80-column, 132-column).
Note Erased characters are undefined (not blank). The terminal displays these characters as the error character (reverse question mark).		
<i>Pcmw</i>	Character matrix width	Selects the maximum character cell width. When <i>Pss</i> specifies 80 columns, <i>Pt</i> specifies font usage full cell as follows: 0 = 10 pixels wide for 80 columns, 6 pixels wide for 132 columns. (default). 2 = 5 × 10 pixel cell (width × height). 3 = 6 × 10 pixel cell (width × height). 4 = 7 × 10 pixel cell (width × height). 5 = 5 pixels wide. 6 = 6 pixels wide. . . . 10 = 10 pixels wide. When <i>Pss</i> specifies 80 columns, <i>Pt</i> specifies font usage text as follows:

Parameter	Name	Description
		0 = 10 pixels wide for 80 columns, 6 pixels wide for 132 columns. (default) 2 = 5 × 10 pixel cell (width × height). 3 = 6 × 10 pixel cell (width × height). 4 = 7 × 10 pixel cell (width × height). 5 = 5 pixels wide. . . . 9 = 9 pixels wide. When <i>Pss</i> specifies 132 columns, <i>Pt</i> specifies font usage full cell as follows: 0 = 10 pixels wide for 80 columns, 6 pixels wide for 132 columns. (default) 2 = 5 × 10 pixel cell (width × height). 3 = 6 × 10 pixel cell (width × height). 4 = 7 × 10 pixel cell (width × height). 5 = 5 pixels wide. 6 = 6 pixels wide. When <i>Pss</i> specifies 132 columns, <i>Pt</i> specifies font usage text as follows: 0 = 10 pixels wide for 80 columns, 6 pixels wide for 132 columns. (default) 2 = 5 × 10 pixel cell (width × height). 3 = 6 × 10 pixel cell (width × height). 4 = 7 × 10 pixel cell (width × height). 5 = 5 pixels wide. If you omit a <i>Pcmw</i> value, the terminal uses the default character width. If the <i>Pe</i> parameter is 0, <i>Pcmw</i> must be less than 10. Any <i>Pcmw</i> value over 10 is illegal. For illegal values of <i>Pcmw</i>, the terminal ignores the rest of the load sequence.
For VT200 compatible software: Use a <i>Pcmw</i> value of 2, 3, or 4. When you use these values, the VT510 doubles the height of the font definitions in memory. This makes the soft set the same height as the hard sets. If you use a <i>Pcmw</i> value of 0, then the VT510 does not double the height and the soft set appears smaller than the hard sets. Remember that the VT200 fonts may appear different on the VT510.		
For fonts designed for the VT510: Use values 5 through 10.		
For compatibility between VT200 and VT510 terminals: Use a <i>Pcmw</i> value of 5, 6, or 7.		
<i>Pss</i>	Font set size	Defines the screen width and screen height for this font.

Parameter	Name	Description
		0,1 = 80 columns, 24 lines. (default) 2 = 132 columns, 24 lines 11 = 80 columns, 36 lines 12 = 132 columns, 36 lines 21 = 80 columns, 48 lines 22 = 132 columns, 48 lines For illegal values of <i>Pss</i>, the terminal ignores the rest of the load sequence. If the number of columns or lines/screen changes, then the terminal uses the appropriate variation of the soft set. If you try to display a DRCS character when there is no soft set defined for the current number of lines and columns, then the terminal displays the error character (reverse question mark).
Note For VT220 compatibility, you must use a soft font of 80 or 132 columns (not both). The VT220 allows a single font to be used for 80 and 132 columns.		
		You can load a font for any of the six sizes at any time. When a new soft set is loaded, any characters from the previous soft set are lost. Characters currently displayed from the old set become reverse question marks.
<i>Pt</i>	Text or full cell	Defines the font as a text font or <i>full-cell font</i>. 0 = text. (default) 1 = text. 2 = full cell. Full-cell fonts can individually address all pixels in a cell. Text fonts cannot individually address all pixels. If you specify a text cell, then the terminal automatically performs spacing and centering of the characters.
<i>Pcmh</i>	Character matrix height	Selects the maximum character cell height. 0 or omitted = 16 pixels high. (default) 1 = 1 pixel high. 2 = 2 pixels high. 3 = 3 pixels high. . . . 16 = 16 pixels high. <i>Pcmh</i> values over 16 are illegal. If the value of <i>Pcmw</i> is 2, 3, or 4, then <i>Pcmh</i> is ignored.
<i>Pcss</i>	Character set size	Defines the character set as a 94- or 96- character graphic set.

Parameter	Name	Description
		0 = 94-character set. (default)
		1 = 96-character set.
		The value of <i>Pcss</i> changes the meaning of the <i>Pcn</i> (starting character) parameter above.
		If Pcss = 0 (94-character set)
		The terminal ignores any attempt to load characters into the 2/0 or 7/15 table positions.
	Pcn	Specifies
	1	column 2/row 1
		.
		.
		.
	94	column 7/row 14
		If Pcss = 1 (96-character set)
	Pcn	Specifies
	0	column 2/row 0
		.
		.
		.
	95	column 7/row 15

Table 5–3 Valid DECDLD Parameter Combinations

Pt	Pcmw	Pcmh	Pss†
<i>80-column, 24 lines</i>			
0 or 1	0 to 8	0 to 16	0 or 1
2	0 to 10	0 to 16	0, 1
<i>132-column, 24 lines</i>			
0 or 1	0 to 5	0 to 16	2
2	0 to 6	1 to 16	2
<i>80-column, 36 lines</i>			
0 or 1	0 to 8	0 to 10	11
2	0 to 10	0 to 10	11
<i>132-column, 36 lines</i>			
0 or 1	0 to 5	0 to 10	12
2	0 to 6	0 to 10	12
<i>80-column, 48 lines</i>			
0 or 1	0 to 8	0 to 8	21
2	0 to 10	0 to 8	21
<i>132-column, 48 lines</i>			
0 or 1	0 to 5	0 to 8	22
2	0 to 6	0 to 8	22
†The default values are the maximum legal values in each case.			

Note

For illegal combinations of the *Pss*, *Pt*, *Pcmw*, and *Pcmh* parameters, the terminal ignores the rest of the sequence.

Table 5–4 VT510 Font Sizes

80 Columns	132 Columns	Maximum Number of Lines
10 × 16	6 × 16	26 lines + keyboard indicator line
10 × 10	6 × 10	42 lines + keyboard indicator line
10 × 8	6 × 8	53 lines + keyboard indicator line
10 × 13	6 × 13	26 lines + keyboard indicator line (72 Hz with overscan)

DECDMAC—Define Macro

This device control string lets you define a macro consisting of a string of ANSI text and control functions. After you define the macro, you can execute it by using the invoke macro control function (DECINVM). The terminal processes the macro as normal input.

Available in: VT Level 4 mode only.

Programming Tip

You can find how much memory space you have available for macros by using a device status report sequence.

Format

```
DCS  Pid ; Pdt ; Pen ! z D...D ST
9/0  3/? 3/11 3/? 3/11 3/? 2/1 7/10 ... 9/12
```

Parameters

Pid

is the macro ID number. *Pid* can be any number between 0 and 63, otherwise the terminal ignores the command. If you give a macro an ID that already exists, then DECDMAC deletes the old macro definition before using the new one.

Pdt

defines how the terminal treats new macro definitions.

Pdt	Meaning
0 or omitted	DECDMAC deletes the old macro with the same ID number before defining this new macro.
1	DECDMAC deletes all current macro definitions before defining this macro.
Other	The terminal ignores the macro.

Pen

selects the encoding format for the text of the macro definition.

Pen	Meaning
0 or omitted	Uses standard ASCII characters in the macro.
1	Uses hex pairs in the macro. Each pair of characters in the macro is the hex value for a single ASCII character.
Other	The terminal ignores the macro.

D...D

is the data of the control string. This data is the string of text and control functions performed when an application invokes the macro. The data can consist of ASCII text or hex pairs.

When you use ASCII text (Pen = 0 or omitted), the macro consists of the graphic characters that appear in the device control string. Only characters from positions 2/0 through 7/14 and 10/00 through 15/15 in the code table are valid.

When you use hex pairs (Pen = 1), the macro consists of pairs of hex numbers, each of which represents an ASCII character. When using hex pairs, you can also use a repeat introducer in the data string. The repeat introducer is the ! (2/1) character. The repeat introducer lets you repeat any hex pair within the definition string any number of times. You embed repeat sequences within the data string of the macro definition.

Note

If ASCII text is specified as the encoding format, then you cannot use the repeat sequence.

Notes on DECDMAC

- **!** is the repeat sequence introducer. The format of the hex string with the repeat introducer is as follows:
! Pn ; D...D ;
Pn is a numeric parameter that specifies the number of times to repeat the sequence. If *Pn* is omitted, then the macro repeats the sequence once.
D...D is the sequence of hex pairs to repeat *Pn* times.
A semicolon ; (3/11) separates the repeat count from the sequence of hex pairs. Another semicolon ends the sequence. If any characters are included before the first semicolon, then the terminal ignores the macro. If a string terminator is placed before the final semicolon, then the repeat string ends normally.
- The VT510 has 6 Kbytes of memory available for the storage of macros. The terminal ignores any macro definitions that cannot fit into this space.
- A reset to initial state (RIS) or secure reset (DECSR) operation clears all macro definitions. A soft terminal reset (DECSTR) has no effect on stored macro definitions.
- You can include characters from positions 0/8 through 0/13 to format the device control string, but these characters cannot be part of the macro definition.

DECDWL—Double-Width, Single-Height Line

This control function makes the line with the cursor a double-width, single-height line. If the line was single width and single height, then all characters to the right of the screen's center are lost.

Format

ESC	#	6
1/11	2/3	3/6

DECEKBD—Extended Keyboard Report

This control function is a report that provides applications with more complete keystroke information. Each report consists of an ISO key position code and the current state of the eight modifier keys—pressed or released.

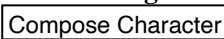
Format

APC : *ppp* *mm* ST
9/15 3/10 *** ** 9/12

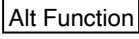
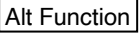
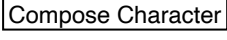
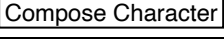
Parameters

ppp
is a three-character ISO key position name, for example C01. For keys that occupy more than one position in the ISO key position map, the VT510 uses the largest lower-left position. The position for the space bar is A02;  is C13.

mm
are two ASCII hexadecimal digits that represent the state of the modifier keys.
The terminal represents the state of each modifier key as a bit in an 8-bit binary number. A bit value of 1 means the key is pressed, and a bit value of 0 means the key is not pressed. *mm* is the hexadecimal equivalent of the resulting 8-bit binary number.

The following table shows the order in which keys are represented in the binary number, and their corresponding hexadecimal values when pressed. The least significant bit represents the left  key. The most significant bit represents the right  key.

For example, 00000010₂ means that the right  key is pressed. The hexadecimal value for 00000010₂ is 02₁₆.

mm Key	
01	left  (Least significant bit)
02	right 
04	
08	
10	left 
20	right 
40	left 
80	right  (Most significant bit)

Description

The terminal only sends extended key position reports when key position mode (DECKPM) is set.
The terminal uses the following rules when sending extended key reports:

- A single report may contain more than one keystroke. In this case, each keystroke is separated by a slash (/). The terminal sends the control string (APC ... ST) when a modifier key changes, or to separate keystroke data from other terminal reports.
- The terminal only sends the modifier key state when it changes in the first keystroke after an APC character.
- The terminal sends repeated key position reports as a single repeat character, period (.)

Examples

The following sequence reports an  key with no modifier keys, four  keys, one  key, and a  key:

APC : C01 00 / B18.../ B16 / B05 ST

The following sequence reports a left Shift key and an A key:

APC : B99 01 / C01 ST

DECELF—Enable Local Functions

This control function lets you enable or disable the keys used to perform several local terminal functions: cutting and pasting text, panning, and resizing windows. DECELF is useful for:

- Simplifying the keyboard for novice users
- Letting the corresponding keys send position codes to the host when key position mode (DECKPM) is set
- Disabling features that are not compatible with the application

Format

```
CSI  Pfn ; Pc1 ; ... Pfn ; Pcn + q
9/11 3/n 3/11 3/n 3/11 ... 3/n 3/11 3/n 2/11 7/1
```

Parameters

Pfn
is a number that specifies the local function affected.

Pfn	Local Functions Affected	Default Setting
0 (default)	All local functions listed for 1, 2, and 3.	
1	Copy and paste keys, including F1 (Hold), Select , Remove , and Insert Here .	The keys are enabled.
2	Panning keys, including Ctrl with ↑ , ↓ , ← , or → .	The keys are enabled.
3	Window resizing keys, including: Ctrl and Shift with ↑ or ↓ keys.	The keys are enabled.

Pcn
controls the action of the local function specified by the corresponding *Pfn* value.

Pcn	Meaning
0	Selects the default setting of the local function. In VT mode, the default setting is enabled. In PC Term mode, the default setting is disabled.
1	Enables the local function. The associated keys perform the local function and cannot send position codes to the host.
2	Disables the local function. The associated keys do not perform the local function and may send position codes to the host.

Description

DECELF can accept any number of parameter pairs. The terminal executes the parameters in the order in which it receives them.

DECERA—Erase Rectangular Area

This control function erases characters from the specified rectangular area in page memory. When an area is erased, DECERA replaces all character positions with the space character (2/0). DECERA erases character values and visual attributes from the specified area. DECERA does not erase line attributes.

Available in: VT Level 4 mode only

Format

CSI	<i>Pt</i>	;	<i>Pl</i>	;	<i>Pb</i>	;	<i>Pr</i>	\$	<i>z</i>
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	3/11	3/ <i>n</i>	3/11	3/ <i>n</i>	2/4	7/10

Parameters

Pt, Pl, Pb, and Pr

define the rectangular area to be erased:

Pt is the top-line border. *Pt* must be less than or equal to *Pb*.

Default: *Pt* = 1.

Pl is the left-column border. *Pl* must be less than or equal to *Pr*.

Default: *Pl* = 1.

Pb is the bottom-line border.

Default: *Pb* = the last line of the active page.

Pr is the right-column border.

Default: *Pr* = the last column of the active page.

Notes on DECERA

- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECERA is not affected by the page margins.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, then the value is treated as the width or height of that page.
- DECERA does not change the active cursor position.

DECESKM—Enable Secondary Keyboard Language Mode

Select a secondary or primary keyboard language.

Format

CSI	?	1	0	4	h	Secondary keyboard mapping.
9/11	3/15	3/1	3/0	3/4	3/8	
CSI	?	1	0	4	l	Primary keyboard mapping.
9/11	3/15	3/1	3/0	3/4	6/12	

Description

When enabled, a secondary keyboard mapping replaces the primary keyboard mapping. To use the secondary keyboard mapping, a secondary keyboard language has to be defined. See [Chapter 8](#) for information on secondary keyboard languages.

DECBI—Forward Index

This control function moves the cursor forward one column. If the cursor is at the right margin, then all screen data within the margins moves one column to the left. The column shifted past the left margin is lost.

Available in: VT Level 4 mode only

Format

ESC 9
1/11 3/6

Description

DECBI adds a new column at the right margin, with no visual attributes. DECBI is not affected by the margins. If the cursor is at the right border of the page when the terminal receives DECBI, then the terminal ignores DECBI.

DECFNK—Function Key

The DECFNK sequences encode the top row function keys, the six editing keypad keys, and the four cursor keys. The top row function keys, (F1 through F20), can be encoded unmodified or in combination with Shift, Ctrl, and Alt key combinations. The six editing keypad keys (Find, Insert Here, Remove, Select, Prev, Next), can be encoded unmodified or in combinations with the Ctrl and Alt keys. The four cursor keys (←, ↓, ↑, ⇒) can be encoded with the Alt key.

Format

CSI Ps1 ; Ps2 ~
9/11 3/? 3/11 3/? 7/14

Parameters

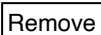
Ps1
indicates the function keystroke.

Ps2
indicates the modifier key state.

Description

All combinations of the listed modifiers are coded as shown in [Table 5–5](#).

Table 5–5 DECFNK Modifiers

Ps1	Function Keystroke	Modifiers Coded by Default
1		Unmodified, Control, Alt
2		Unmodified, Control, Alt
3		Unmodified, Control, Alt
4		Unmodified, Control, Alt
5		Unmodified, Alt
6		Unmodified, Alt
7		Alt
8		Alt
9		Alt
10		Alt
11		Unmodified, Shift, Control, Alt
12		Unmodified, Shift, Control, Alt
13		Unmodified, Shift, Control, Alt
14		Unmodified, Shift, Control, Alt
15		Unmodified, Shift, Control, Alt
17		Unmodified, Shift, Control, Alt
18		Unmodified, Shift, Control, Alt
19		Unmodified, Shift, Control, Alt
20		Unmodified, Shift, Control, Alt
21		Unmodified, Shift, Control, Alt
23		Unmodified, Shift, Control, Alt
24		Unmodified, Shift, Control, Alt
25		Unmodified, Shift, Control, Alt
26		Unmodified, Shift, Control, Alt
28		Unmodified, Shift, Control, Alt
29		Unmodified, Shift, Control, Alt
31		Unmodified, Shift, Control, Alt
32		Unmodified, Shift, Control, Alt
33		Unmodified, Shift, Control, Alt
34		Unmodified, Shift, Control, Alt

Ps2	Modifiers in effect
0 or omitted	none
1	none
2	Shift
3	Alt
4	Alt + Shift
5	Control
6	Control + Shift

Ps1	Function Keystroke	Modifiers Coded by Default
7	Alt + Control	
8	Alt + Control + Shift	

Notes

- **Shift** is not distinguished for the editing and numeric keypad keys by default so it does not interfere with touch typing patterns. This means that shifted keys send the same DECFNK sequence as the unshifted keys.
- The cursor keys are not available unmodified since they send standard cursor key sequences, not DECFNK sequences.
- **Ctrl** with **↑**, **↓**, **Next**, and **Prev** are used for local panning by default.
- Top row function keys **F1** through **F5** are only available when they are not used for local terminal functions.
- Shifted top row function keys **F6** through **F20** are still reserved for user-defined key sequences. The DECFNK sequences become the factory default UDK sequences for shifted **F6** through **F20**.

DECFRA—Fill Rectangular Area

This control function fills a rectangular area in page memory with a specified character. DECFRA replaces the rectangular area's character positions and attributes with the specified fill character. The fill character assumes the visual character attributes set by the last select graphic rendition (SGR) command. DECFRA does not change the current line attributes.

Available in: VT Level 4 mode only

Format

CSI	<i>Pch</i>	;	<i>Pt; Pl; Pb; Pr</i>	\$	<i>x</i>
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	2/4	7/8

Parameters

Pch

is the decimal value of the fill character. *Pch* can be any value from 32 to 126 or from 160 to 255. If *Pch* is not in this range, then the terminal ignores the DECFRA command. The decimal value refers to the character in the current GL or GR in-use table.

Pt; Pl; Pb; Pr

define the rectangular area to be filled:

Pt is the top-line border. *Pt* must be less than or equal to *Pb*.

Default: *Pt* = 1.

Pl is the left-column border. *Pl* must be less than or equal to *Pr*.

Default: *Pl* = 1.

Pb is the bottom-line border.

Default: *Pb* = the last line of the active page.

Pr is the right-column border.

Default: *Pr* = the last column of the active page.

Notes on DECFRA

- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECFRA is not affected by the page margins.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, the value is treated as the width or height of that page.
- DECFRA does not change the active cursor position.

DECHDPXM—Set/Reset Half-Duplex Mode

Enables half-/full-duplex communication.

Format

CSI	?	1	0	3	h	Set to half-duplex mode.
9/11	3/15	3/1	3/0	3/3	6/8	
CSI	?	1	0	3	l	Set to full-duplex mode.
9/11	3/15	3/1	3/0	3/3	6/12	

Description

Full-duplex mode allows simultaneously two-way communication; half-duplex mode is limited to one-way communication at any given time. Half-duplex mode is available only in an RS232 line. Refer to [Chapter 9](#) for details about half-duplex mode communication protocol.

DECHEBM—Hebrew/N-A Keyboard Mapping

This control function sets the keyboard mapping to Hebrew/N-A and resets the mapping to North American.

Format

CSI	?	3	5	h	Set
9/11	3/15	3/3	3/5	6/8	
CSI	?	3	5	l	Reset
9/11	3/15	3/3	3/5	6/12	

Description

When 8-bit characters are selected, DECHEBM switches between the Group 1 and the Group 2 character set mappings. It does not affect the keyboard character set selection or the processing of received characters.

When 7-bit characters are selected or the terminal is set to VT100 mode, setting DECHEBM does the following:

- Switches keyboard map to Group 2.
- Makes DEC 7-bit Hebrew the keyboard character set.
- Causes received characters in the range of 60 to 7A (hex) to be displayed using the DEC 7-bit Hebrew when ASCII is designated or invoked and keyboard Group 2 is active.

Resetting DECHEBM does the following:

- Switches keyboard map to Group 1.
- Makes ASCII the keyboard character set.
- Causes received characters to be displayed using the graphic left (GL) table of characters in use at the time.

Note

DECHEBM is only valid when a Hebrew keyboard language is selected. Also it does not convert ASCII or Hebrew characters already on the display.

DECHEM—Hebrew Encoding Mode

This control function sets the terminal's character set mode to DEC (8-bit) multinational mode. On a reset, the sequence resets the terminal's character set mode to (7-bit) national mode (NRC sets).

The terminal maintains DECHEM for the main display and status line separately in the same way as it maintains National Replacement Character Set Mode (DECNRCM).

Format

CSI	?	3	6	h	Set
9/11	3/15	3/3	3/6	6/8	
CSI	?	3	6	l	Reset
9/11	3/15	3/3	3/6	6/12	

DECIC—Insert Column

This control function inserts one or more columns into the scrolling region, starting with the column that has the cursor.

Available in: VT Level 4 mode only

Format

```
CSI  Pn  '  }  
9/11 3/? 2/7 7/13
```

Parameters

Pn

is the number of columns to insert.

Default: *Pn* = 1.

Description

As columns are inserted, the columns between the cursor and the right margin move to the right. DECIC inserts blank columns with no visual character attributes. DECIC has no effect outside the scrolling margins.

DECINVM—Invoke Macro

This control function lets you execute a stored macro. You store macros with the define macro (DECDMAC) control string. The terminal substitutes the DECINVM sequence with the contents of the macro definition and executes the macro.

Available in: VT level 4 mode only.

Format

CSI	<i>Pid</i>	*	<i>z</i>
9/11	3/ <i>n</i>	2/10	7/10

Parameters

Pid

is the macro ID number. If the *Pid* is not associated with a particular macro, the terminal ignores the command.

Notes on DECINVM

Any functions executed with a macro invocation remain in affect after the terminal executes the macro.

Examples

Suppose you want to create a macro that clears all pages in page memory, writes the word "top" at the top of each page, and leaves the cursor on page 1.

First, you would write the string with ASCII characters as follows:

```
DCS 1 ; 0 ; 1 ! z CSI 6 SP P ! 6 ; CSI 1 ; 1 H CSI 2 J TOP CSI 1 V ; ST
```

Then, you must convert the string into hexadecimal code because the string uses the repeat (!) character. You can only use the repeat character with hexadecimal sequences. The following example is the same string written in hex:

```
DCS 1 ; 0 ; 1 ! z 9B 36 20 50 ! 6 ; 9B 31 3B 48 9B 32 49 54 4F 50 9B 31 56 ; ST
```

To invoke the macro, use the following sequence:

```
CSI 1 * z
```


DECIPEM—Enter/Return from IBM ProPrinter Emulation Mode

Exit from DEC protocol and interpret subsequent data according to the IBM ProPrinter protocol syntax. This sequence also resets IBM ProPrinter mode to its initial state.

Format

CSI	?	5	8	h	Set: Enter ProPrinter Emulation mode.
9/11	3/15	3/5	3/8	6/8	
CSI	?	5	8	l	Reset: Return from ProPrinter Emulation mode.
9/11	3/15	3/5	3/8	6/12	

DECKBD—Keyboard Language Selection

Select a national keyboard language on either a VT keyboard or an Enhanced PC keyboard.

Format

CSI *Ps1* ; *Ps2* SP }
9/11 3/*n* 3/11 3/*n* 2/0 7/13

Parameters

Ps1
denotes a keyboard type (layout): VT or enhanced PC.

Ps1	Keyboard Layout
0, 1, or None	VT
2	Enhanced PC

Ps2
indicates the national keyboard language.

Ps2	Keyboard Language (VT/PC)
0, 1 or None	North American
2	British
3	Belgian (Flemish)
4	French Canadian
5	Danish
6	Finnish
7	German
8	Dutch
9	Italian
10	Swiss French
11	Swiss German
12	Swedish
13	Norwegian
14	French
15	Spanish
16	Portuguese
19	Hebrew
22	Greek
28	Canadian (English)
29	Turkish Q/Turkish
30	Turkish F
31	Hungarian
33	Slovak
34	Czech
35	Polish
36	Romanian
38	SCS
39	Russian
40	Latin American

Description

The first selective parameter, *Ps1*, denotes a keyboard type, VT layout or Enhanced PC keyboard layout. The second parameter, *Ps2*, indicates the national keyboard language.

Selecting a new keyboard language automatically initializes the keyboard character set. Refer to [Chapter 8](#) for more details.

DECKBUM—Typewriter or Data Processing Keys

You can select whether the main keyboard keys act as data processing keys or typewriter keys by using the keyboard usage mode (DECKBUM) function.

Default: Typewriter keys

Keyboard Usage Mode (DECKBUM)

CSI	?	6	8	h	Set: data processing keys.
9/11	3/15	3/6	3/8	6/8	
CSI	?	6	8	l	Reset: typewriter keys.
9/11	3/15	3/6	3/8	6/12	

Description

If DECKBUM is set, then data processing keys send the characters on the right half of their keycaps.

If DECKBUM is reset, then data processing keys send the characters on the left half of their keycaps.

Notes on DECKBUM

- DECKBUM changes the characters that the main keyboard keys send. Make sure you are aware of this change if you use DECKBUM in an application.
- If you use the North American language, then DECKBUM should always be reset (typewriter). For all other languages, you can use either mode.

DECKLHIM—Keyboard LED's Host Indicator Mode

DECKLHIM controls the state of the keyboard LED's host indicator mode.

Default: Reset.

Format

CSI	?	1	1	0	h	Set: keyboard LED's host indicator mode.
9/11	3/15	3/1	3/1	3/0	6/8	
CSI	?	1	1	0	l	Reset: keyboard LED's host indicator mode.
9/11	3/15	3/1	3/1	3/0	6/12	

Description

DECLL controls keyboard LEDs independently of any keyboard state. The use of LEDs for this purpose conflicts with their use as keyboard state indicators. DECKLHIM selects a mode of how the keyboard LEDs are to be used: as keyboard indicators; or host indicators. If host indicators is selected, then the DECLL sequence can be used to control the keyboard LEDs.

For DECLL to function, DECKLHIM must be set. See DECLL for the implications of using DECLL to control the keyboard LEDs independently of any keyboard state.

DECKPAM—Keypad Application Mode

DECKPAM enables the numeric keypad to send application sequences to the host. DECKPNM enables the numeric keypad to send numeric characters.

DECKPAM and DECKPNM function the same as numeric keypad mode (DECNKM).

Format

ESC = Send application sequences.
1/11 3/13

Note on DECKPAM

The setting is not saved in NVM. When you turn on or reset the terminal, it automatically selects numeric keypad mode.

DECKPM—Key Position Mode

This control function selects whether the keyboard sends character codes or key position reports to the host. DECKPM lets new applications take full control of the keyboard including single shifts, locking shifts, and compose character processing.

Default: Send character codes

Format

CSI	?	8	1	h	Set: send key position reports.
9/11	3/15	3/8	3/1	6/8	
CSI	?	8	1	l	Reset: send character codes.
9/11	3/15	3/8	3/1	6/12	

Description

If the DECKPM function is set, then all keyboard keys send extended reports that include the key position and the state of modifier keys when pressed. A *modifier key* is pressed in combination with another key to modify the code sent by that key. The Ctrl key is a modifier key.

If the DECKPM function is reset, then the keyboard keys send character codes.

Note on DECKPM

DECKPM only affects keyboard input; it does not affect how the terminal interprets data from the host.

DECKPNM—Keypad Numeric Mode

DECKPNM enables the keypad to send numeric characters to the host. DECKPAM enables the keypad to send application sequences.

DECKPNM and DECKPAM function the same as numeric keypad mode (DECNKM).

Default: Send numeric keypad characters.

Format

ESC > Send numeric keypad characters.
1/11 3/14

Description

DECKPNM enables the numeric keypad to send the characters shown on each key—number, comma, period, or minus sign. Keys PF1 to PF4 send application sequences. See DECKPAM—Keypad Application Mode for more information.

Note on DECKPNM

The setting is not saved in NVM. When you turn on or reset the terminal, it automatically selects numeric keypad mode.

DECLANS—Load Answerback Message

This sequence loads an answerback message into the terminal's memory.

Format

DCS	<i>Ps</i>	<i>v</i>	<i>D . . . D</i>	ST
9/0	3/ <i>n</i>	7/6	...	9/12

Description

The answerback data string may consist of from 0 to 30, 7- or 8-bit characters, which are hex pairs in the range 3/0 - 3/9 (0 through 9), 4/1 - 4/6 (A through F), and 6/1 - 6/6 (A through F).

Once defined, the message may be stored in the non-volatile memory if Save settings is executed in the Set-Up.

The selective parameter denotes the encoding of the answerback message. A "1" is always used for the VT510 indicating answerback is encoded as hex pairs. The sequence is ignored if any number other than "1" is specified in the *Ps* field.

Pressing Ctrl/Break or receiving ENQ code causes the VT510 to transmit the answerback message. In on-line mode, the answerback message is transmitted to the appropriate host port. The answerback is not echoed in local echo and is not "sent" in local mode.

The answerback message is transmitted as keyboard data. Any speed restrictions (as set in limited transmit) apply.

DECLBAN—Load Banner Message

DECLBAN loads the banner message to be displayed on the terminal screen at power up.

Format

DCS *Ps* *r* *D . . . D* ST
9/0 3/s 7/2 ... 9/12

Parameters

Ps
denotes the encoding of the message.

Ps	Description
0	Zero or omitted, default to Ps=2.
1	Interprets as ASCII hexadecimal pairs. See DECLANS.
2	Loads as text in the VT default character set.

D ... D
is a 30-character string. If a matching *Ps* value is selected as an encoding method, then you can select to define the string by the characters in the default character set or by hexadecimal pairs.

Description

This command only loads the banner message into dynamic memory. It is necessary to do a Save settings in the Set-Up menu to store the banner message in NVM so that it will be displayed at power up.

The terminal displays the banner message in double size characters above the Selftest OK message.

DECLFKC—Local Function Key Control

This control function lets you select the action performed by some of the local function keys on the top row of the keyboard:

Pk	Function	VT Keyboard	Enhanced PC Keyboard
1	Hold	F1	Scroll Lock
2	Print	F2	Print Screen
3	Set-Up	F3	Caps Lock/Print Screen

You can have the key perform its local function, send a function key sequence to the host, or do nothing.

This control function is useful for:

- Simplifying the keyboard for novice users
- Disabling features that are not compatible with the application, such as local printing

Format

```
CSI  Pk1  ;   Pf1  ... Pkn  ;   Pfn  *   }  
9/11 3/n  3/11 3/n  ... 3/n  3/11 3/n  2/10 7/13
```

Parameters

Pkn

is a number from 0 to 4 that corresponds to the local function key affected (F1 to F4). A *Pkn* value of 0 means that all four keys (F1 through F4) are affected.

Pfn

controls the action of the local function key specified by the corresponding *Pkn* value.

Pfn	Meaning	VT Mode	PCTerm Mode
0	Selects the default function of the key.	Local print	Scan code
1	The key performs its local function either shifted, unshifted, or with the Ctrl key.	Print	Local Print
2	The unshifted key sends its function key sequence to the host. The key sends nothing when pressed with Shift or Ctrl .	Function Key	Scan Code
3	The key is disabled.	Disabled	Does not exist

Description

DECLFKC can accept any number of parameter pairs. The terminal executes the parameters in the order in which it receives them.

Note on DECLFKC

Applications cannot disable the F5 (Break) key using this command. This restriction is required for secure systems, which ensures that the operator has control of the communication line.

DECLL—Load LEDs

Load the programmable LEDs on the keyboard according to its parameters.

Format

CSI *Ps* *q*
9/11 3/*n* 7/1

Parameters

Ps
selects the state of the Num Lock, Caps Lock, and Scroll Lock LEDs as follows:

Ps	LED	Light Condition
0	Num Lock	Off
	Caps Lock	Off
	Scroll Lock	Off
1	Num Lock	On
2	Caps Lock	On
3	Scroll Lock	On
21	Num Lock	Off
22	Caps Lock	Off
23	Scroll Lock	Off

Description

DECLL controls keyboard LEDs independently of any keyboard state. The use of LEDs for this purpose conflicts with their use as keyboard state indicators. DECKLHIM selects a mode of how the keyboard LEDs are to be used: as keyboard indicators; or host indicators. If host indicators is selected, then the DECLL sequence can be used to control the keyboard LEDs.

For DECLL to function, DECKLHIM must be set.

DECLRMM—Left Right Margin Mode

This control function defines whether or not the set left and right margins (DECSLRM) control function can set margins.

Available in: VT Level 4 mode only.
Default: DECSLRM cannot set margins.

Format

CSI	?	6	9	h	Set: DECSLRM can set margins.
9/11	3/15	3/6	3/9	6/8	
CSI	?	6	9	l	Reset: DECSLRM cannot set margins.
9/11	3/15	3/6	3/9	6/12	

Description

When DECLRMM is set, DECSLRM can set the left and right margins. All line attributes currently in page memory for the session are set to single width, single height. The terminal ignores any sequences to change the line attributes to double width or double height (DECDWL or DECDHL).

When DECLRMM is reset, DECSLRM cannot set the left and right margins. The margins are set to the page borders for every page. The terminal can process sequences to change the line attributes to double width or double height (DECDWL or DECDHL).

DECLTOD—Load Time of Day

This sequence loads the time of day into the terminal's memory. When loaded, the time of day is displayed in the Status line if the Status line is enabled.

Format

CSI	<i>Ps1</i>	;	<i>Ps2</i>	,	p
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	2/12	7/0

Parameters

Ps1

specifies the hour, which is a 2-digit decimal number between 00 and 23.

The default value is 8, indicating 8:00 a.m.

Ps2

specifies the minutes, which is a 2-digit decimal number between 00 and 59.

The default value is 00, indicating 8:00 a.m.

DECMCM—Set/Reset Modem Control Mode

Enables (set) or disables (reset) modem control.

Format

CSI	?	9	9	h	Enable Modem Control
9/11	3/15	3/6	3/6	6/8	
CSI	?	9	9	l	Disable modem control - default
9/11	3/15	3/6	3/6	6/12	

Description

When enabled for the Comm2 connector, no data is transmitted or received unless DSR is asserted. Assertion of DSR alone is sufficient to establish a connection. Loss of DSR causes a disconnect.

When DSR is not asserted, no data is transmitted, and incoming receive data is ignored. When DSR is again asserted, the communications state is re-initialized to prepare for the new connection.

When enabled for the Comm1 (25-pin EIA) connector, the terminal does not transmit unless CTS is asserted. If a connection has not been established (DSR is deasserted), RLSD is ignored.

Transmission and reception may still occur if CTS is asserted. Assertion of CTS with DSR and RLSD asserted indicates a connection has been established. (CTS is checked last.)

When Modem Control is disabled, transmission and reception occur regardless of the state of the modem control signal. The terminal communicates using data leads only. DTR and RTS are always asserted except during an operator initiated disconnect sequence, or the input buffer reaches the 64-byte threshold when DTR flow control scheme is selected.

DECMSR—Macro Space Report

DECMSR is the terminal's response to a DSR from the host. DECMSR indicates the number of bytes available for macro definitions.

Format

CSI	<i>Pn</i>	*	{
9/11	3/ <i>n</i>	3/10	7/11

Parameters

Pn

$$Pn = \frac{\text{number of bytes}}{16} \text{ rounded down.}$$

DECNAKB—Greek/N-A Keyboard Mapping

This control function sets the keyboard mapping to North American and resets the mapping to Greek/N-A.

Format

CSI	?	5	7	h	Set.
9/11	3/15	3/5	3/7	6/8	
CSI	?	5	7	l	Reset.
9/11	3/15	3/5	3/7	6/12	

DECNCSCM—Set/Reset No Clearing Screen On Column Change

Set/reset the following condition: do not clear the screen whenever the column mode changes.

Format

CSI	?	9	5	h	Set: do not clear screen when column mode changes.
9/11	3/15	3/9	3/5	6/8	
CSI	?	9	5	l	Reset: clear the screen when column mode changes.
9/11	3/15	3/9	3/5	6/12	

Description

When enabled, a column mode change (either through Set-Up or by the escape sequence DECCOLM) does not clear the screen. When disabled, the column mode change clears the screen as a side effect.

This sequence does not affect the column mode change caused by the sequence, DECSCPP.

DECNKM—Numeric Keypad Mode

This control function works like the DECKPAM and DECKPNM functions. DECNKM is provided mainly for use with the request and report mode (DECRQM/DECRPM) control functions.

Available in: VT Level 4 mode only

Default: Numeric

Format

CSI	?	6	6	h	Set: application sequences.
9/11	3/15	3/6	3/6	6/8	
CSI	?	6	6	l	Reset: keypad characters.
9/11	3/15	3/6	3/6	6/12	

DECNRCM—National Replacement Character Set Mode

Enable operation in 7-bit or 8-bit character mode.

Format

CSI	?	4	2	h	Set: 7-bit characters.
9/11	3/15	3/4	3/2	6/8	
CSI	?	4	2	l	Reset: 8-bit characters (default)
9/11	3/15	3/4	3/2	6/12	

Description

The character set mode is reset to 8-bit characters. In 8-bit mode, the terminal operates as a normal level 4 terminal allowing 8-bit controls and using the default character set, which is user-selectable for transmission and reception.

DECNULM—Set/Reset Ignoring Null Mode

Allow Null (NUL) characters to be ignored or buffered at printer port only.

Format

CSI	?	1	0	2	h	Set: ignore NUL (default).
9/11	3/15	3/1	3/0	3/2	6/8	
CSI	?	1	0	2	l	Reset: accept NUL.
9/11	3/15	3/1	3/0	3/2	6/12	

Description

When set, the NUL character is discarded on receipt from the host. When reset, the NUL character is passed through to the printer.

DECNUMLK—Num Lock Mode

DECNUMLK controls the state of the keyboard's Num Lock key.

Format

CSI	?	1	0	8	h	Set: num lock mode.
9/11	3/15	3/1	3/0	3/8	6/8	
CSI	?	1	0	8	l	Reset: num lock mode.
9/11	3/15	3/1	3/0	3/8	6/12	

DECOM—Origin Mode

This control function sets the origin for the cursor. DECOM determines if the cursor position is restricted to inside the page margins. When you power up or reset the terminal, you reset origin mode.

Default: Origin is at the upper-left of the screen, independent of margins.

Format

CSI	?	6	h	Set: within margins.
9/11	3/15	3/6	6/8	
CSI	?	6	l	Reset: upper-left corner.
9/11	3/15	3/6	6/12	

Description

When DECOM is set, the home cursor position is at the upper-left corner of the screen, within the margins. The starting point for line numbers depends on the current top margin setting. The cursor *cannot* move outside of the margins.

When DECOM is reset, the home cursor position is at the upper-left corner of the screen. The starting point for line numbers is independent of the margins. The cursor *can* move outside of the margins.

DECOSCNM—Set/Reset Overscan Mode

Sets or resets overscan mode.

Default: Overscan disabled (reset).

Format

CSI	?	1	0	6	h	Enable overscan
9/11	3/15	3/1	3/0	3/6	6/8	
CSI	?	1	0	6	l	Disable overscan (Default)
9/11	3/15	3/1	3/0	3/6	6/12	

Description

When set, overscan is enabled.

Overscan and 72 Hz refresh rate cannot co-exist when the terminal is set to 42 or 53 lines per screen. Refer to DECSNLS—Set Lines Per Screen for details.

DECPAK—Program Alphanumeric Key

Allow users to reprogram keys to transmit user-preferred alphanumeric codes or to perform user-preferred functions.

Format

DCS	"	y	D . . . D	ST
9/0	2/2	7/9	...	9/12

Description

This sequence programs one or more keys (and their modifiers) to:

- Transmit desired alphanumeric codes
- Perform a local function or send user-defined sequence

This function works in all the VT modes except in VT52 mode.

The memory allocation and deallocation processes (program key lock and unlock process) on the DECPAK are the same as on the DECPFK. Refer to DECPFK—Program Function Key for details.

The data string D...D format is:

Key1/HEX Code String/Function #/UDS/UDS Direction;Key2/...;

Parameters

Keyn

is the key station number and slash "/" is a delimiter.

HEX code string

is the hex character code to be transmitted with each of the seven modifier states. The hex code represents a valid code in the current character set (either 7-bit or 8-bit). Use a period "." as a place holder if a modifier combination is not to be defined. Use a minus "-" preceding the hex representation of a diacritical sign if a diacritical mark is to be defined.

The terminal always allows the ISO character set to be used for specifying accent marks to DECPAK even when:

1. 7-bit NRCs characters are selected
2. The ISO character set is not the user-preferred supplemental character set

The corresponding ISO character set would be based on the keyboard language. Refer to [Chapter 8](#) for keyboard information.

The position of the diacritical marks within the ISO Latin character sets is as follows:

Diacritical Sign	Position	Available ISO Character Set(s)
Circumflex	5/14	ASCII
Grave	6/0	ASCII
Tilde	7/14	ASCII
Dieresis (umlaut)	10/8	Latin 1, Latin 2, Latin Greek, Latin Turkish
Degree (ring)	11/0	Latin 1, Latin 2, Latin Greek, Latin Turkish
Acute	11/4	Latin 1, Latin 2, Latin Greek, Latin Turkish
Cedilla	11/8	Latin 1, Latin 2, Latin Turkish
Macron	10/15	Latin 1
Breve	10/2	Latin 2
Ogonek	11/2	Latin 2
Caron	11/7	Latin 2
Double Acute Accent	11/13	Latin 2
Dot Above	15/15	Latin 2

You should program the Hex string according in the following order:

1. Group 1 unshifted
2. Group 1 shifted
3. Group 1 alternate shifted (Shift 2)
4. Group 2 (Group Shift or Alt Gr) unshifted
5. Group 2 shifted
6. Group 2 alternate shifted
7. Control (if omitted, use default)

If the control state is not redefined in this sequence, then the Key/Ctrl combination generates the control code based on the "Group 1 unshifted" state.

A graphic key transmits alphanumeric codes only when pressed with the above modifiers. When pressed with Alt key, the graphic key performs either:

- Local terminal function
- User-defined sequence (same as UDK)

Function #

corresponds to a local function to be performed when the key is pressed in combination with the Alt function. Example: Function number "0" is used to make the Key/Alt combination inoperative. Function number "100" indicates a user-defined sequence (UDS), and a UDS direction is defined for the Key/Alt combination following the "/" delimiter.

UDS and UDS direction

is a user-defined sequence. These parameters are the same as for DECPFK. Refer to DECPFK—Program Function Key for the details.

Once programmed, the key states are not affected by the terminal mode changes. Refer to DECPFK—Program Function Key for the list of terminal mode changes.

DECPCCM—Page Cursor-Coupling Mode

This control function determines if a new page appears in the display when the cursor moves to a new page. DECPCCM is only useful with a multiple-page format.

Format

CSI	?	6	4	h	Set: couples the cursor to the display.
9/11	3/15	3/6	4/4	6/8	
CSI	?	6	4	l	Reset: uncouples the cursor.
9/11	3/15	3/6	4/4	6/12	

Description

When DECPCCM is set, the cursor is coupled to the display when the cursor moves to a new page. The new page appears in the display to keep the cursor in view.

When DECPCCM is reset, the cursor is uncoupled from the display. If the cursor moves to a new page, then the cursor disappears.

DECPCTERM—Enter/Exit PCTerm or Scancode Mode

DECPCTERM enables you to switch the terminal between VT mode and PCTerm mode, while the terminal is in VT mode. It also enables you to switch the terminal between ASCII character and scancode keycode mode, while the terminal is in SCO Console mode.

Format

```
CSI ? Ps1 ; Ps2 r
9/11 3/15 3/n 3/11 3/n 7/2
```

Parameters

Ps1
selects VT mode or PCTerm mode. In SCO console mode, this parameter selects ASCII characters or scancode keycode mode. When the mode changes, some operating features are changed to reflect the characteristics of the selected emulation mode.

Ps1	VT Emulation Mode	SCO Console Mode
0 or none	VT Mode	ASCII characters
1	PCTerm Mode	Scancodes

Ps2
indicates the character set selection. See SCS—Select Character Set for the character set designators.

Description

The character sets selectable with single-digit Ps2 parameters are available for both VT mode and PCTerm mode. If you select one of these character sets for use in PCTerm mode, then you can use the SCS—Select Character Set command to modify the PCTerm character set.

The PC character sets selectable with three-digit Ps2 parameters are available only in PCTerm mode or SCO console emulation. (SCO console uses PC character sets.) If you select one of these PC character sets, then the SCS command to invoke a character set is ignored.

Ps2	Character Set
0 or none	No Change
1	PC Multilingual
2	PC International
3	PC Danish/Norwegian
4	PC Spanish
5	PC Portuguese
6	DEC Supplemental
7	ISO Latin 1
8	VT character set
210	PC Greek
220	PC Spanish
437	PC International
850	PC Multilingual
852	PC Slavic (Latin 2)
857	PC Turkish
860	PC Portuguese
862	PC Hebrew
863	PC French Canadian
865	PC Danish
866	PC Cyrillic

DECPEX—Printer Extent Mode

This control function selects how much data you can print when you use the print page function.

Default: Scrolling region

Format

CSI	?	1	9	h	The print function prints the complete page.
9/11	3/15	3/1	3/9	6/8	
CSI	?	1	9	l	The print function prints the scrolling region only (data inside the
9/11	3/15	3/1	3/9	6/12	margins).

DECPFF—Print Form Feed Mode

This control function selects whether or not the terminal sends a form feed (FF) character to the printer at the end of a printing function. DECPFF does not affect the print cursor line function described in the next section.

Default: No form feed

Format

CSI	?	1	8	h	The terminal sends a form feed (FF) to the printer at the end of a printing
9/11	3/15	3/1	3/8	6/8	function.
CSI	?	1	8	l	The terminal sends nothing to the printer at the end of a printing function.
9/11	3/15	3/1	3/8	6/12	

DECPFK—Program Function Key

Used to reprogram function keys.

Format

DCS	"	x	D . . . D	ST
9/0	2/2	7/8	...	9/12

Description

This sequence programs one or more keys (and their modifier combinations) from the current state to perform user-desired functions. The type of the function programmed must be either:

- a local terminal function
- a user-defined sequence (same as UDK)

This sequence is ignored when the program key function is locked through DECPKA, DECUDK, or Set-Up.

There are 804 bytes reserved for programmable key operations. This memory is shared by the following four functions:

- Program keys through DECPFK or through Set-Up
- Program keys through DECPAK or through Set-Up
- Copy other key default through DECCKD or Set-Up
- Program user-defined keys through DECUDK

Space is supplied on a first-come, first-serve basis. Once the 804 bytes are used, no more keys may be redefined unless space is made available. Space may be made available by restoring keys back to their default states. There are four ways to do this:

- Execute a Restore factory defaults from the Set-Up main menu. All the terminal parameters, including the keys, return to their default states.
- Execute a "Restore factory defaults of all keys and key definitions" either through DECPKA (Program Key Action) or Recall key definitions in Set-Up. This action also releases the 804 bytes.
- Execute a Copy key default to copy a key's own default either through DECCKD or in Set-Up to free the space occupied by the destination key.
- Clear UDKs by issuing a clear function in DECUDK.

The free bytes left for key programming can be seen using the Define Key Editor in Set-Up. *D ... D* are the *key definition strings* such as:

Key1/Mod1/Function #/UDS/UDS Direction;
Key2/Mod2/Function #/UDS/UDS Direction;.....;

Parameters

Keyn

is the key station number and a slash "/" is a delimiter. Refer to [Chapter 8](#) for the key station number map.

Modn

denotes the modifier key that is to be pressed at the same time as the defined key.

Modn	Modifier Key
0 or None	Normal
1	Normal
2	Shift
3	Alt
4	Alt + Shift
5	Control
6	Shift + Control
7	Alt + Control
8	Alt + Control + Shift

Function

represents a number associated with a local function. For example, function number "0" is used to make the key or key/modifier inoperative. Function number "100" indicates a user-defined sequence (UDS) and UDS direction is defined following the slash "/" delimiter. Refer to [Chapter 8](#).

UDS

is a user-defined sequence with a maximum length of 255 bytes, which must be written in Hex format for DEC/ANSI terminals. Each hex code in this string represents a value transmitted by the defined key combined with the modifiers.

- UDS in DECPFK is functionally equivalent to the string Stn in the DECUDK control function.
- See DECUDK—User Defined Keys.

Dir (UDS Direction)

the user-defined sequence (UDS) direction of the transmission.

Dir	Direction	UDS is sent to . . .
0, none	Normal	Host and/or the terminal, depending on the setting of the COMM mode parameter. The UDS is directed to: • The host for full-duplex mode. • The screen for half-duplex mode.
1	Local	Terminal screen only.
2	Remote	Host only.

Default: None (Normal)

Description

Any of the following four methods may be used to define a key to transmit a user-defined sequence (either single byte or multiple bytes):

- Control function DECPFK
- Control function DECPAK
- Control function DECUDK
- In Set-Up, Define key ...

Once programmed, the key state is not effected by the terminal mode changes listed below:

- Keyboard mode change, such as switching between Data Processing mode and Typewriter mode
- Character Mode change, such as switching between 7-bit character and 8-bit character
- Keyboard Layout change, such as switching between PC layout and VT layout
- Keyboard Style change, such as switching between PC-style and VT-style

- **Terminal Mode change, such as switching between VT modes**

This function works in all the ANSI VT modes except VT52 mode.

DECPKA—Program Key Action

Allow changes to programmable keys, which affects all the keys on the keyboard in all the VT modes except in VT52 mode.

Format

CSI *Ps* + *z*
9/11 3/*n* 2/11 7/10

Parameters

Ps

This sequence has two functions:

- 1. Lock the keys from redefinition.
- 2. Restore the definitions of the keys either from the factory default or the NVR.

If set to "1," the parameter determines whether the keys will be locked. Any attempt to program keys will be invalid.

If the keys are not locked, then the keys are available for further definition either through Set-Up or control functions DECPFK, DECPAK, DECCKD, and DECUDK. But if "Lock UDK" is enabled by DECUDK, then the attempt to program the keys does not take effect. (Refer to DECUDK—User Defined Keys for details.)

Once locked, keys can only be unlocked through Set-Up. There is no host control function that can unlock the keys.

If the parameter Ps is set to 2, then all the keys and keys/modifiers combination restore to their factory default states. Refer to DECCKD—Copy Key Default if restoring defaults for a single key and key/modifier combination.

If the parameter Ps is set to 3, all saved settings of the keys and keys/modifiers combination are recalled from the NVR.

If the keys are locked for further definition, then restore and recall operations are not valid.

Ps1	Program Key Action
0 or none	None
1	Locks keys and keys/modifiers from further definition
2	Restores factory defaults of all the keys and their keys/modifiers
3	Recalls saved keys definition

DECPKFMR—Program Key Free Memory Report

Report the approximate free memory space for program key operations.

Format

```
CSI  Ps1  ;   Ps2  +   y
9/11 3/n  3/11 3/n  2/11 7/9
```

Parameters

Ps1

denotes the total memory space in bytes that is reserved for the programmable keys. The total memory available is 804 bytes, hence the Ps1 value is "804."

Ps2

denotes the memory space that is available at the time DECRQPKFM is received, depending on the number of bytes left in decimal format. A "0" indicates that no memory is left for any key programming operation; in which case, any attempt to program the keys will fail.

Description

The VT510 reports this sequence upon receiving DECRQPKFM from the host.

Note on DECPKFMR

The free bytes reported by Ps2 in this command represents only the approximate amount of memory available, because additional memory is used by the firmware data structure to program the key definitions.

DECRARA—Reverse Attributes in Rectangular Area

This control function lets you reverse the visual character attributes (bold, blink, reverse video, and underline) of a specified rectangular area in page memory. The select attribute change extent (DECSACE) control function determines whether all or some of the character positions in the rectangle are affected.

Reversing a visual attribute means changing the attribute to its opposite setting, on or off. For example, DECRARA can change characters from bold and not underlined to characters that are underlined and not bold. DECRARA does not change the values of characters, just the visual attributes of those characters.

Available in: VT Level 4 mode only

Format

```
CSI  Pt; Pl; Pb; Pr;      Ps1; . . . Psn      $  t
9/11  area to be reversed  attributes to reverse  2/4  7/4
```

Parameters

Pt, Pl, Pb, and Pr

define the rectangular area to be reversed.

Pt is the top-line border. *Pt* must be less than or equal to *Pb*.

Default: *Pt* = 1.

Pl is the left-column border. *Pl* must be less than or equal to *Pr*.

Default: *Pr* = 1.

Pb is the bottom-line border.

Default: *Pb* = last line of the active page.

Pr is the right-column border.

Default: *Pr* = last column of the active page.

Ps1; . . . Psn

select the visual character attributes to reverse. These values correspond to the values used in the select graphic rendition (SGR) function.

Ps	Meaning
0	Reverse all attributes (default).
1	Reverse the bold attribute.
4	Reverse the underline attribute.
5	Reverse the blink attribute.
7	Reverse the negative-image attribute.

DECRARA ignores all other parameter values.

When you use more than one parameter in a command, DECRARA executes them cumulatively in sequence.

Examples

The following sequence reverses the blink and underscore attributes of the complete screen:

```
CSI ; ; ; 0; 4; 5; $t
```

The following sequence reverses all attributes except the blink attribute, from position line 10, column 2 to position line 14, column 45 on the current page:

```
CSI 10; 2; 14; 45; 1; 4; 7 $ t
```

Notes on DECRARA

- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECRARA is not affected by the page margins.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, then the value is treated as the width or height of that page.
- DECRARA does not change the active cursor position.
- DECRARA does not change the current rendition set by the select graphic rendition (SGR) function.
- The exact character positions affected by DECRARA depend on the current setting of the select attribute change extent (DECSACE) function.

DECRC—Restore Cursor

Restores the terminal to the state saved by the save cursor (DECSC) function.

Format

ESC 8
1/11 3/8

Description

If nothing was saved by DECSC, then DECRC performs the following actions:

- Moves the cursor to the home position (upper left of screen).
- Resets origin mode (DECOM).
- Turns all character attributes off (normal setting).
- Maps the ASCII character set into GL, and the DEC Supplemental Graphic set into GR.

Notes on DECSC and DECRC

The terminal maintains a separate DECSC buffer for the main display and the status line. This feature lets you save a separate operating state for the main display and the status line.

DECRLCM—Right-to-Left Copy

Perform a copy/paste from right-to-left or from left-to-right.

Default: Disable right-to-left-copy.

Format

CSI	?	9	6	h	Enable right-to-left copy.
9/11	3/15	3/9	3/6	6/8	
CSI	?	9	6	l	Disable right-to-left copy (default).
9/11	3/15	3/9	3/6	6/12	

Description

When operating in Copy/Paste, you can paste the copied text according to the direction assigned in this command sequence.

If right-to-left copy is selected, then the text is stored in the paste buffer starting from right-to-left. For example if "ABCD" is to be copied right-to-left, when a Paste key is pressed, the string is saved in the paste buffer as "DCBA." Thus, the text is displayed as "DCBA".

DECRLM—Cursor Right to Left Mode

This control function starts/stops the right-to-left cursor direction.

Format

CSI	?	3	4	h	Set.
9/11	3/15	3/3	4/4	6/8	
CSI	?	3	4	l	Reset.
9/11	3/15	3/3	4/4	6/12	

Description

DECRLM state is common to both the main display and the status line display. The status line is effected only when you set it to Host Writable Status Display in the Display Set-Up screen. DECRLM does not affect column numbering within the line. The first line column is always at the left side of the display window. The last column is always at the right side of the display window.

The next character position is defined as the current cursor position minus 1.

If the terminal is in Replace mode (IRM is reset), then:

- The new character is inserted in the current cursor position.
- The cursor moves to the next character position.

If terminal is in Insert mode (IRM is set), then:

- All characters from the current cursor position to the left margin inclusively are shifted one position towards the left.
- The character previously displayed at the left margin disappears.
- The new character is inserted in the current cursor position.
- The cursor moves to the next character position.

The following escape sequences are not affected by the DECRLM setting:

ICH - Insert Character	DECSER - Selective Erase in Display
DCH - Delete Character	DECSEL - Selective Erase in Line
ED - Erase in Display	ECH - Erase Character
EL - Erase in Line	

The following escape sequences are not affected by the DECRLM setting:

CUP - Cursor Position
HVP - Horizontal and Vertical Position
CUF - Cursor Forward
CUB - Cursor Backward

Notes on DECRLM

- The BS control character increases the current cursor position by one, thus moving the cursor on the screen one position to the right.
- When new line mode (LNM) is set and the terminal receives one of the control codes LF, VT, or FF, the cursor moves to the last right-most column of the next line.
- When control code CR is received, the cursor moves to the last right-most column of the current line.
- When control code SUB is received, a number of cancellation activities result within the terminal and the terminal displays an error character, according to the current DECRLM setting.
- When control code NEL is received, the cursor moves to the last right-most column of the next line. If the cursor is at the bottom margin, then the page scrolls up.
- If auto-wrap is enabled (DECAWM is set), then a graphic character received when the cursor is at column number one appears at the right margin column of the next line. If the cursor is

at the end of the scrolling region, then any text already on the page scrolls up.

DECRPAK—Report All Modifiers/Alphanumeric Key State

This control string is in response to a Request Key Definition (DECRQKD).

Format

DCS	"	~	<i>D . . . D</i>	ST
9/0	2/2	7/14	...	9/12

Description

String *D...D* has a format same as DECPAK.

Parameters

D ... D
See DECPAK for a description of parameters.

DECRPDE—Report Displayed Extent

The terminal sends this control function in response to a request displayed extent (DECRQDE) sequence. DECRPDE indicates how much of the current page is displayed on the screen.

DECRPDE indicates how much of the screen has to display the current page.

Format

CSI	<i>Ph</i>	;	<i>Pw</i>	;	<i>Pml</i>	;	<i>Pmt</i>	;	<i>Pmp</i>	;	"	<i>w</i>
9/11	3/ <i>n</i>		3/11		3/ <i>n</i>		3/11		3/ <i>n</i>		3/11	2/2 7/7

Parameters

Ph

is the number of lines of the current page displayed excluding the status line.

Pw

is the number of columns of the current page displayed.

Pml

is the column number displayed in the left-most column.

Pmt

is the line number displayed in the top line.

Pmp

is the page number displayed.

DECRPFK—Report Function Key Definition

The terminal reports a function key's definition when it receives a DECRQKD. DECRPFK denotes a local function number or a defined sequence, that the target key, which is combined with a specific modifier key, performs or transmits.

Format

DCS	"	}	<i>D ... D</i>	ST
9/0	2/2	7/13	...	9/12

Parameters

D ... D

uses the same format as DECPFK, **Key/Modifier/Function#/UDS/UDS Direction**. **Function#** is skipped in the report, if the function key has not been redefined and is transmitting the default escape sequence as its UDS.

DECRPKT—Report Key Type

DECRPKT (from the terminal) is a response to a key type inquiry (DECRQKT).

Format

CSI *Ps1* ; *Ps2* , v
9/11 3/*n* 3/11 3/*n* 2/12 7/6

Parameters

Ps1
indicates is the target key's station number.

Ps2
indicates

Ps2 Key Type	
0	Alphanumeric
1	Function

DECRPM—Report Mode - Terminal To Host

Reports the terminal mode in response to a request mode (DECRCQM) function. In the response DECRPM informs the host if a certain mode is set (SM—Set Mode) or reset (RM—Reset Mode).

Programming Tip

Applications can use the information in the DECRPM report to save the current mode settings. Later, the application can restore the mode settings that were saved.

This operation is useful for applications that need to temporarily change some of the terminal's mode settings. When the application is finished, it can restore the mode settings that were in effect before the application changed them. You use the set mode (SM) and reset mode (RM) functions to restore mode settings. SM and RM are described later in this chapter.

There are two versions of DECRPM, for ANSI and DEC modes.

Reporting ANSI Modes

CSI	<i>Pa</i>	;	<i>Ps</i>	\$	<i>y</i>
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	2/4	7/9

Reporting DEC Modes

CSI	?	<i>Pd</i>	;	<i>Ps</i>	\$	<i>y</i>
9/11	3/15	3/ <i>n</i>	3/11	3/ <i>n</i>	2/4	7/9

Parameters

Pa

Table 5–6 lists the values for *Pa*.

Pd

indicates which DEC mode the terminal is reporting on. Table 5–6 lists the values for *Pd*.

Ps

indicates the setting of the mode. The *Ps* values are the same for the ANSI and DEC versions.

Ps Mode Setting	
0	Mode not recognized
1	Set
2	Reset
3	Permanently set
4	Permanently reset

Examples

The following sequence reports the setting of a DEC mode:

VT510	Report Meaning
CSI ? 57; h	(Greek) Keyboard mapping is currently set. (57 = DECNAKB, h = set)

The following sequences report the setting for some modes:

- CSI Ps \$ p (or CSI ? Ps \$ p (DEC)) (Request Mode)
- CSI Ps1; Ps2 \$ y (or CSI ? Ps1; Ps2 \$ y) (Report Mode)
- CSI P1.....Pn l (Reset Mode)
- CSI P1.....Pn h (Set Mode)

Notes on DECRPM

The terminal can only report on one mode at a time.

Description

DECRQM, DECRPM, SM, and RM support all the mode settings as follows:

Table 5–6 Settings Supported by DECRQM, DECRPM, RM, and SM

ANSI Mode		DEC Mode	
Pa	Mnemonic	Pd	Mnemonic
1	GATM*	?1	DECCKM
2	KAM	?2	DECANM†
3	CRM	?3	DECOLM
4	IRM	?4	DECSCLM
5	SRTM*	?5	DECSCNM
		?6	DECOM
7	VEM*	?7	DECAWM
		?8	DECARM
10	HEM*	–	–
11	PUM*	–	–
12	SRM	–	–
13	FEAM*	–	–
14	FETM*	–	–
15	MATM*	–	–
16	TTM*	–	–
17	SATM*	–	–
18	TSM*	?18	DECPFF
19	EBM*	?19	DECPEX
20	LNM	–	–
–	–	?25	DECTCEM
–	–	?34	DECRLM
–	–	?35	DECHEBM
–	–	?36	DECHEM
–	–	?42	DECNRCM
–	–	?57	DECNAKB
–	–	?60	DECHCCM *
–	–	?61	DECVCCM
–	–	?64	DECPCCM
–	–	?66	DECNKM
–	–	?67	DECBKM
–	–	?68	DECKBUM
–	–	?69	DECLRMM (DECVSSM)

*This mode is always reset in the terminal and cannot be changed.

†When the terminal is set to VT52 mode, DECRQM provides no report.

ANSI Mode		DEC Mode	
Pa	Mnemonic	Pd	Mnemonic
–	–	?73	DECXRLMM
–	–	?95	DECNCSM
–	–	?96	DECRLCM
–	–	?97	DECCRTSM
–	–	?98	DECARSM
–	–	?99	DECMCM
–	–	?100	DECAAM
–	–	?101	DECCANSM
–	–	?102	DECNULM
–	–	?103	DECHDPXM
–	–	?104	DECESKM
–	–	?106	DECOSCNM

*This mode is always reset in the terminal and cannot be changed.

†When the terminal is set to VT52 mode, DECRQM provides no report.

DECRPSS—Report Selection or Setting

The terminal sends the host this sequence in response to a request selection or setting (DECQRSS) sequence. The terminal sends DECRPSS to report the setting of a particular control function.

Programming Tip

Applications can use the information in the DECRPSS report to save the current selections or settings of some control functions. Later, the application can restore the control functions to their saved state.

This operation is useful for applications that need to temporarily change the settings of some of the terminal's control functions. When the application is finished, it can restore the control functions to their previous state.

Format

DCS	<i>Ps</i>	\$	<i>r</i>	<i>D ... D</i>	ST
9/0	3/ <i>n</i>	2/4	7/2	...	9/12

Parameters

Ps

indicates whether or not the request from the host is valid.

Ps Host Request

0 Host's request is valid.

1 Host's request is invalid.

D ... D

indicates the current setting of a valid control function that the host asked about. *D...D* consists of all the characters in the control function, except the CSI (9/11) or ESC [(1/11, 5/11) introducer characters.

Examples

The host requests the setting of the select graphic rendition (SGR) function. If the current graphic rendition is underline, blinking, and reverse, then the terminal responds with the following DECRPSS sequence:

DCS 0 \$ r 0 ; 4 ; 5 ; 7 m ST

0 ; 4 ; 5 ; 7 m are all the characters in the SGR sequence, except CSI.

The host requests the setting of the set top and bottom margin function (DECSTBM). If the current top and bottom margins are set to include the complete screen area (24 lines/screen), then the terminal responds with the following DECRPSS sequence:

DCS 0 \$ r 1 ; 24 r ST

1 ; 24 r are all the characters in the DECSTBM sequence, except CSI.

The host requests the setting of a function that the terminal does not recognize. The terminal responds with the following DECRPSS sequence:

DCS 1 \$ r ST

The terminal does not send a data string (*D...D*) to the host when the terminal receives an invalid request.

DECRPSS (and DECQRSS) support the following settings or selections:

Setting	Mnemonic	Final Character(s)
Select Active Status Display	DECSASD	\$ }
Select Attribute Change Extent	DECSACE	* x
Set Character Attribute	DECSCA	" q
Set Conformance Level	DECSCL	" p
Set Columns Per Page	DECSCPP	\$ l
Set Lines Per Page	DECSLPP	t
Set Number of Lines per Screen	DECSNLS	* l
Set Status Line Type	DECSSDT	\$ ~
Set Left and Right Margins	DECSLRM	s
Set Top and Bottom Margins	DECSTBM	r
Set Graphic Rendition	SGR	m
Select Set-Up Language	DECSSL	p
Select Printer Type	DECSPRTT	\$ s
Select Refresh Rate	DECSRFR	" t
Select Digital Printed Data Type	DECSDPT	(p
Select ProPrinter Character Set	DECSPPCS	* p
Select Communication Speed	DECSCS	* r
Select Communication Port	DECSCP	* u
Set Scroll Speed	DECSSCLS	SP p
Set Cursor Style	DECSCUSR	SP q
Set Key Click Volume	DECSKCV	SP r
Set Warning Bell Volume	DECSWBV	SP t
Set Margin Bell Volume	DECSMBV	SP u
Set Lock Key Style	DECSLCK	SP v
Select Flow Control Type	DECSFC	* s
Select Disconnect Delay Time	DECSDDT	\$ q
Set Transmit Rate Limit	DECSTR	" u
Set Port Parameter	DECSPP	+ w

DECRPTUI—Report Terminal Unit ID

DECRPTUI is a response from the terminal to a tertiary DA (DA3) request from the host. DECRPTUI is available in VT Level 4 mode only. See DA3 for more information.

Format

DCS	!		<i>D . . . D</i>	ST
9/0	2/1	7/12	...	9/12

Parameters

D...D
is the unit ID of the terminal, consisting of four hexadecimal pairs. The first pair represents the manufacturing site code. This code can be any hexadecimal value from 00 through FF.
The last three pairs are the terminal ID number. This number is unique for each terminal manufactured at that site.

DECRQCRA—Request Checksum of Rectangular Area

This control function requests a checksum of the specified rectangular area in the specified page. The terminal returns a checksum report (DECCKSR) in response to this request.

Format

```
CSI  Pid ;    Pp ;    Pt;Pl;Pb;Pr *    y
9/11 3/n  3/11 3/n  3/11 3/n...3/n 2/10 7/9
```

Parameters

Pid

is a numeric label you can provide to identify the checksum request. The checksum report returns this number. The number serves to differentiate between multiple checksum reports.

Pp

is the page number that has the rectangular area. If *Pp* is 0 or omitted, then the terminal ignores the following parameters and reports a checksum for all pages in page memory. If *Pp* is a higher number than the number of pages available, then the terminal reports on the last page.

Pt;Pl;Pb;Pr

are the top, left, bottom, and right borders of the rectangular area. *Pt* and *Pb* are line numbers. *Pt* must be less than or equal to *Pb*. *Pl* and *Pr* are column numbers. *Pl* must be less than or equal to *Pr*.

Defaults are *Pt* = 1, *Pb* = current page length, *Pr* = current page width. If these parameters are omitted, then the terminal returns a checksum of page *Pp*.

Note on DECRQCRA

The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).

DECRQDE—Request Displayed Extent

The host sends a DECRQDE to the terminal requesting how much of the current page is displayed on the screen for the active session.

Format

CSI	"	v
9/11	2/2	7/6

DECQRKD—Request Key Definition

DECQRKD allows you to inquire about the terminal's definition of:

- A function key or a function key combined with a single modifier key.
- An alphanumeric key combined with all of its available modifier keys.

The terminal reports the function key definition with DECPRFK, and the alphanumeric key definition with DECRLAK.

Format

```
CSI  Ps1  ;   Ps2  ,   w
9/11 3/n  3/11 3/n  2/12 7/7
```

Parameters

Ps1

indicates is the target key's station number. When inquiring for an alphanumeric key definition, *Ps2* is ignored by the terminal. DECRLAK always reports a hexadecimal string in which each hexadecimal code represents a value transmitted by pressing a combination of the target key and a modifier key. The whole sequence is ignored if you enter an invalid key/modifier. See DECRLAK.

Ps2

When inquiring for a function key definition, *Ps2* represents:

Ps2	Description
0, 1 or none	Normal
2	Shift
3	Alt
4	Alt + Shift
5	Control
6	Shift + Control
7	Alt + Control
8	Alt + Control + Shift

DECRQKT—Key Type Inquiry

The host issues a DECRQKT to the terminal for the key definition of a function or alphanumeric key. Upon receiving the request, the terminal reports using DECRPKT.

Format

CSI	<i>Ps</i>	,	u
9/11	3/ <i>n</i>	2/12	7/5

Parameters

Ps

is a decimal number representing the target key's station number.

DECRQM—Request Mode - Host To Terminal

The host sends this control function to find out if a particular mode is set or reset. The terminal responds with a report mode function (DECRPM—Report Mode - Terminal To Host).

DECRQM functions in ANSI and DEC modes.

Requesting ANSI Modes

CSI	<i>Pa</i>	\$	p
9/11	3/ <i>n</i>	2/4	7/0

Requesting DEC Modes

CSI	?	<i>Pd</i>	\$	p
9/11	3/15	3/ <i>n</i>	2/4	7/0

Parameters

Pa

indicates the ANSI mode that the host is asking about. Table 5–7 lists the values for *Pa*.

Pd

indicates the DEC mode the host is asking about. Table 5–8 lists the values for *Pd*.

Examples

The following sequences request the setting of some ANSI modes:

Host	Meaning
CSI 2 \$ p	What is the current state of keyboard action mode (KAM)? (KAM = 2)
CSI 4 \$ p	What is the current state of insert/replace mode (IRM)? (IRM = 4)

The following sequences request the setting of some DEC modes:

Host	Meaning
CSI ? 36 \$ p	What is the current state of Hebrew encoding mode (DECHEM)? (DECHEM = 36)
CSI ? 6 \$ p	What is the current state of origin mode (DECOM)? (DECOM = 6)

Notes on DECRQM

- The terminal does not respond to a DECRQM sequence when in VT52 mode.
- A DECRQM sequence can only ask about one mode at a time.

Table 5–7 ANSI Modes for DECRQM, DECRPM, SM, and RM

Mode	Mnemonic	Pa
Guarded area transfer	GATM*	1
Keyboard action	KAM	2
Control representation	CRM†	3
Insert/replace	IRM	4
Status reporting transfer	SRTM*	5
Vertical editing	VEM*	7
Horizontal editing	HEM*	10
Positioning unit	PUM*	11
Send/receive	SRM	12
Format effector action	FEAM*	13
Format effector transfer	FETM*	14
Multiple area transfer	MATM*	15
Transfer termination	TTM*	16
Selected area transfer	SATM*	17
Tabulation stop	TSM*	18
Editing boundary	EBM*	19
Line feed/new line	LNLM	20

*This control function is permanently reset.

†The host cannot change the setting of CRM. You can only change CRM from Set-Up. If CRM is set, then the terminal ignores DECRQM and most other control functions.

Table 5–8 DEC Modes for DECRQM, DECRPM, SM, and RM

Mode	Mnemonic	Pd
Cursor keys	DECCKM	1
ANSI	DECANM	2
Column	DECCOLM	3
Scrolling	DECSCLM	4
Screen	DECSCNM	5
Origin	DECOM	6
Autowrap	DECAWM	7
Autorepeat	DECARM	8
Print form feed	DECPFF	18
Printer extent	DECPEX	19
Text cursor enable	DECTCEM	25
Cursor direction, right to left	DECRLM	34
Hebrew keyboard mapping	DECHEBM	35
Hebrew encoding mode	DECHEM	36
National replacement character set	DECNRCM	42
Greek keyboard mapping	DECNAKB	57
Horizontal cursor coupling	DECHCCM*	60
Vertical cursor coupling	DECVCCM	61
Page cursor coupling	DECPCCM	64
Numeric keypad	DECNKM	66
Backarrow key	DECBKM	67
Keyboard usage	DECKBUM	68
Vertical split screen	DECVSSM / DECLRMM	69
Transmit rate limiting	DECXRLM	73
Key position	DECKPM	81
No clearing screen on column change	DECNCSM	95
Cursor right to left	DECRLCM	96
CRT save	DECCRTSM	97
Auto resize	DECARSM	98
Modem control	DECMCM	99
Auto answerback	DECAAM	100
Conceal answerback message	DECCANSM	101
Ignoring null	DECNULM	102
Half-duplex	DECHDPXM	103
Secondary keyboard language	DECESKM	104
Overscan	DECOSCNM	106

*This control function is permanently reset.

DECRQPKFM—Program Key Free Memory Inquiry

Inquire as to the amount of free memory for programmable key operations.

Format

CSI	+	x
9/11	2/11	7/8

Description

Upon receiving of this sequence, the terminal responds with DECPKFMR.

DECRQPSR—Request Presentation State Report

The host sends this sequence to request a cursor information report (DECCIR) or tabulation stop report (DECTABSR).

Format

CSI	Ps	\$	w
9/11	3/n	2/4	7/7

Parameters

Ps
indicates which report the host requests.

Ps Report Requested	
0	Error. Request ignored
1	Cursor information report (DECCIR)
2	Tab stop report (DECTABSR)

DECRQSS—Request Selection or Setting

The host requests the terminal setting. See DECRPSS for the terminal's response.

Format

DCS	\$	q	<i>D . . . D</i>	ST
9/0	2/4	7/1	...	9/12

Parameters

D ... D

indicates the current setting of a valid control function that the host asked about. *D . . . D* consists of all the characters in the control function, except the CSI (9/11) or ESC [(1/11, 5/11) introducer characters.

Description

DECRQSS (and DECRPSS) support the following settings or selections:

Setting	Mnemonic	Final Character(s)
Select Active Status Display	DECSASD	\$ }
Select Attribute Change Extent	DECSACE	* x
Set Character Attribute	DECSCA	" q
Set Conformance Level	DECSCL	" p
Set Columns Per Page	DECSCPP	\$ l
Set Lines Per Page	DECSLPP	t
Set Number of Lines per Screen	DECSNLS	* l
Set Status Line Type	DECSSDT	\$ ~
Set Left and Right Margins	DECSLRM	s
Set Top and Bottom Margins	DECSTBM	r
Set Graphic Rendition	SGR	m
Select Set-Up Language	DECSSL	p
Select Printer Type	DECSPRTT	\$ s
Select Refresh Rate	DECSRFR	" t
Select Digital Printed Data Type	DECSDPT	(p
Select ProPrinter Character Set	DECSPPCS	* p
Select Communication Speed	DECSCS	* r
Select Communication Port	DECSCP	* u
Set Scroll Speed	DECSSCLS	SP p
Set Cursor Style	DECSCUSR	SP q
Set Key Click Volume	DECSKCV	SP r
Set Warning Bell Volume	DECSWBV	SP t
Set Margin Bell Volume	DECSMBV	SP u
Set Lock Key Style	DECSLCK	SP v
Select Flow Control Type	DECSFC	* s
Select Disconnect Delay Time	DECSDDT	\$ q
Set Transmit Rate Limit	DECSTRL	" u
Set Port Parameter	DECSP	+ w

DECRQTSR—Request Terminal State Report

The host requests the terminal to report its state. The terminal reports (DECTSR) by indicating the setting of many device attributes. See DECTSR.

Format

CSI *Ps* \$ u
9/11 3/*n* 2/4 7/5

Parameters

Ps
indicates the type of report the host requests from the terminal.

Ps	Report Requested
0 or none	Ignored. No report sent.
1	Terminal state report (DECTSR)

Response

DCS 1 \$ S D . . . D ST
D ... D
See Device Attributes (DA1, DA2, DA3).

DECRQUPSS—User-Preferred Supplemental Set

Applications can ask for the current user-preferred supplemental set. The terminal responds with the user-preferred supplemental set (DECAUPSS) sequence.

Host Request (DECRQUPSS)

The host requests the current user-preferred supplemental set by sending the following sequence:

Format

CSI & u
9/11 2/6 7/5

Terminal Responses

The terminal uses the DECAUPSS device control string to report the current user-preferred supplemental set (UPSS). The terminal sends DECAUPSS in response to a DECRQUPSS sequence. The terminal can send one of the following reports:

Control Sequence UPSS	
DCS 0 ! u % 5 ST	The user-preferred supplemental set is DEC Supplemental Graphic.
DCS 1 ! u A ST	The user-preferred supplemental set is ISO Latin-1 supplemental.

DECRSPS—Restore Presentation State

This control function restores the terminal to a previous state based on one of the presentation state reports. There are two presentation state reports.

- Cursor information report (DECCIR)
- Tab stop report (DECTABSR)

A DECRSPS sequence can only restore the information from one report at a time, cursor information, or tab stop.

Programming Tip

Applications can use DECRSPS to restore the terminal to a previous state specified in a presentation state report. See the previous "Cursor Information Report (DECCIR)" section in this chapter.

Format

DCS	<i>Ps</i>	\$	<i>t</i>	<i>D ... D</i>	ST
9/0	3/ <i>n</i>	2/4	7/4	...	9/12

Parameters

Ps

indicates the format of the data string (*D . . . D*). You can use one of the two following formats for the data string. These formats correspond to the formats used in the two presentation state reports (DECPSPR). Make sure you use the format of the report you are restoring.

Ps Data String Format

- | | |
|----------|---|
| 0 | Error, restore ignored. |
| 1 | Selects the format of the cursor information report (DECCIR). |
| 2 | Selects the format of the tab stop report (DECTABSR). |
-

D ... D

is a data string that contains the restored information. This string is identical to the data string used in the report you are restoring— Cursor Information Report (DECCIR) or Tab Stop Report (DECTABSR).

Examples

The following DECRSPS sequence restores tab stops according to the tab stop report (DECTABSR):

DCS 2 \$ *t* 9/ 17/ 25/ 33/ 41/ 49/ 57/ 65/ 73 ST

Note that the data string format above is exactly the same as the format for the tab stop report (DECTABSR). If there is an invalid value in the DECRSPS sequence, then the terminal ignores the rest of the sequence. This action may leave the terminal in a partially restored state.

DECRSTS—Restore Terminal State

This sequence restores the terminal to a previous state specified in a terminal state report (DECTSR).

Programming Tip

Applications can use DECRSTS to restore the terminal to a previous operating state specified in a terminal state report.

Format

DCS	<i>Ps</i>	\$	p	<i>D . . . D</i>	ST
9/0	3/ <i>n</i>	2/4	7/0	...	9/12

Parameters

Ps
indicates the format of the data string (*D . . . D*).

Ps Data String Format	
0	Error, restore ignored
1	Selects the format of the terminal state report (DECTSR)

D ... D
is a data string that contains the restored information. This string is identical to the data string used by the report you are restoring.

Notes on DECRSTS

- If there is an invalid value in the DECRSTS sequence, then the terminal ignores the rest of the sequence. This action may leave the terminal in a partially restored state.
- Software should not expect the format of the terminal state report (DECTSR) to be the same for all terminals.

DECSACE—Select Attribute Change Extent

This control function lets you select which character positions in a rectangle can have their attributes changed or reversed. DECSACE controls the effect of two other functions—change attributes in rectangular area (DECCARA) and reverse attributes in rectangular area (DECRARA).

Available in: VT Level 4 mode only

Format

CSI *Ps* * *x*
9/11 3/*n* 2/10 7/8

Parameters

Ps
selects the area of character positions affected.

Ps	Area Effected
0 (default)	DECCARA or DECRARA affect the stream of character positions that begins with the first position specified in the DECCARA or DECRARA command, and ends with the second character position specified.
1	Same as 0.
2	DECCARA and DECRARA affect all character positions in the rectangular area. The DECCARA or DECRARA command specifies the top-left and bottom-right corners.

DECSASD—Select Active Status Display

DECSASD selects whether the terminal sends data to the main display or the status line.

Available in: VT Level 4 mode only

Default: Main display

Format

CSI *Ps* \$ }
9/11 3/*n* 2/4 7/13

Parameters

Ps indicates the display to which the terminal sends data.	
Ps	Action
0 (default)	Selects the main display. The terminal sends data to the main display only.
1	Selects the status line. The terminal sends data to the status line only.

DECSC—Save Cursor

Format

ESC 7
1/11 3/7

Description

Saves the following items in the terminal's memory:

- Cursor position
- Character attributes set by the SGR command
- Character sets (G0, G1, G2, or G3) currently in GL and GR
- Wrap flag (autowrap or no autowrap)
- State of origin mode (DECOM)
- Selective erase attribute
- Any single shift 2 (SS2) or single shift 3 (SS3) functions sent

DECSCA—Select Character Protection Attribute

DECSCA defines the characters that come after it as erasable or not erasable from the screen. The selective erase control functions (DECSED and DECSEL) can only erase characters defined as erasable.

Available in: VT Level 4 mode only

Format

CSI *Ps* " q
9/11 3/*n* 2/2 7/1

Parameters

Ps

defines all characters that follow the DECSCA function as erasable or not erasable.

Ps	Meaning
0 (default)	DECSED and DECSEL can erase characters.
1	DECSED and DECSEL cannot erase characters.
2	Same as 0.

Note on DECSCA

DECSCA does *not* effect visual character attributes set by the select graphic rendition (SGR) function.

DECSCL—Select Conformance Level

You select the terminal's operating level by using the following select conformance level (DECSCL) control sequences. The factory default is level 4 (VT Level 4 mode, 7-bit controls).

Note

When you change the conformance level, the terminal performs a hard reset (RIS).

Format

Sequence	Level Selected
	<i>Level 1</i>
CSI 6 1 " p 9/11 3/6 3/1 2/2 7/0	VT100 mode
	<i>Level 4†</i>
CSI 6 n " p 9/11 3/6 3/n 2/2 7/0	VT Level 4 mode, 8-bit controls
CSI 6 n ; 0 " p 9/11 3/6 3/n 3/11 3/0 2/2 7/0	VT Level 4 mode, 8-bit controls
CSI 6 n ; 1 " p 9/11 3/6 3/n 3/11 3/1 2/2 7/0	VT Level 4 mode, 7-bit controls
CSI 6 n ; 2 " p 9/11 3/6 3/n 3/11 3/2 2/2 7/0	VT Level 4 mode, 8-bit controls
†Level 4 includes levels 2 and 3. In these sequences, <i>n</i> can be 2, 3, or 4	

DECSCLM—Scrolling Mode

This control function selects the way the terminal scrolls lines. You can select one of two scroll settings, smooth or jump.

Default: Smooth scroll.

Format

CSI	?	4	h	Set: smooth scroll.
9/11	3/15	3/4	6/8	
CSI	?	4	l	Reset: jump scroll.
9/11	3/15	3/4	6/12	

Description

When DECSCLM is set, the terminal adds lines to the screen at a moderate, smooth rate. You can select a slow rate or fast rate in Display Set-Up.

When DECSCLM is reset, the terminal can add lines to the screen as fast as it receives them.

DECSCNM—Screen Mode: Light or Dark Screen

This control function selects a dark or light background on the screen.

Default: Dark background.

Format

CSI	?	5	h	Set: reverse video.
9/11	3/15	3/5	6/8	
CSI	?	5	l	Reset: normal display.
9/11	3/15	3/5	6/12	

Description

When DECSCNM is set, the screen displays dark characters on a light background.

When DECSCNM is reset, the screen displays light characters on a dark background.

Note on DECSCNM

Screen mode only effects how the data appears on the screen. DECSCNM does not change the data in page memory.

DECSCP—Select Communication Port

Select the port to be used for printer or host connection.

Format

```
CSI  Ps1  ;    Ps2  *    u
9/11 3/n  3/11 3/n  2/10 7/5
```

Parameters

Ps1

indicates the port selection for a printer connection. The VT510 enables you to select from a Centronics parallel port.

Ps1	Printer Port
-----	--------------

1	Parallel port
---	---------------

2	Comm1
---	-------

3	Comm2
---	-------

Ps2

indicates the port selection for a host connection. Since the VT510 is a single session terminal, the host session can connect to the Comm1 port or the Comm2 port.

Ps2	Session 1
-----	-----------

1	Comm1
---	-------

2	Comm2
---	-------

Comm1 is an RS232, 25-pin, serial port. Comm2 is an MMJ serial port.

DECSCPP—Select 80 or 132 Columns per Page

There are two control functions that can set the page width to 80 or 132 columns, DECSCPP and DECCOLM (DECCOLM—Select 80 or 132 Columns per Page).

Note

It is recommended that new applications use DECSCPP rather than DECCOLM. DECSCPP does not clear page memory or reset the scrolling regions as does DECCOLM. DECCOLM is provided mainly for compatibility with previous products.

Default: 80 columns

Format

Sequence	Sets Each Page to . . .
CSI \$ 9/11 2/4 7/12	80 columns.
CSI 0 \$ 9/11 3/0 2/4 7/12	80 columns.
CSI 8 0 \$ 9/11 3/8 3/0 2/4 7/12	80 columns.
CSI 1 3 2 \$ 9/11 3/1 3/3 3/2 2/4 7/12	132 columns.

Notes on DECSCPP

- DECSCPP changes the font to match the current number of columns on the page.
- DECSCPP does not move the cursor. If, however, the cursor is beyond the width of the new page when DECSCPP executes, then the cursor moves to the right column of the new page.
- If you switch from 132-column to 80-column pages, then you can lose data from page memory. Columns no longer present in page memory are lost. Make sure you set page columns *before* you enter data into page memory.

DECSCS—Select Communication Speed

Select a communication speed for a communication line.

Format

CSI *Ps1* ; *Ps2* * r
9/11 3/n 3/11 3/n 2/10 7/2

Parameters

Ps1

indicates the communication line.

Ps1	Comm Line Type
1, 0 or none	Host Transmit
2	Host Receive
3	Printer
4	Modem Hi
5	Modem Lo

Ps2

indicates the communication speed.

Ps2	Speed	Ps2	Speed
0 or none	default	6	9600
1	300	7	19200
2	600	8	38400
3	1200	9	57600
4	2400	10	76800
5	4800	11	115200

The default value depends on the type of communication line.

Communication Line	Default Communication Speed
Host Transmit	9600
Host Receive	Receive=transmit
Printer	4800
Modem Hi	Ignore
Modem Lo	Ignore

DECSCUSR—Set Cursor Style

Select the style of the cursor on the screen.

Format

CSI *Ps* SP q
9/11 3/*n* 2/0 7/1

Parameters

Ps indicates the style of the cursor.	
Ps	Cursor Style
0, 1 or none	Blink Block (Default)
2	Steady Block
3	Blink Underline
4	Steady Underline

This sequence causes the cursor to be displayed in a different style when the cursor is enabled.

Note on DECSCUSR

The escape sequence DECTCEM can enable or disable the cursor display.

DECSDDT—Select Disconnect Delay Time

Select an appropriate disconnect delay time.

Format

CSI *Ps* \$ q
9/11 3/*n* 2/4 7/1

Parameters

Ps
indicates the selection for the disconnect delay time.

Ps	Disconnect Delay Time
1	None
2	60 ms
3, 0 or none	2 seconds

Description

The disconnect delay feature determines the time allowed before the terminal disconnects from the communications line when the Received Line Signal Detect (RLSD) is lost.

All countries except the United Kingdom should use the 2 seconds delay. The 60 ms delay is for use in the United Kingdom.

If the terminal detects a loss of carrier and if the user has selected "no disconnect," then the terminal ignores RLSD (CD) after the beginning of the connection.

If the user tries to disconnect and reconnect the line, then the terminal checks if RLSD is asserted before granting the connection. Once it is connected, the terminal ignores the loss of carrier. In this case, the disconnect should be done on DSR going off.

DECSDPT—Select Digital Printed Data Type

Allow users to match the characters printed to the character sets the DEC ANSI conforming printer is capable of supporting.

Format

CSI *Ps*) p
9/11 3/*n* 2/9 7/0

Parameters

Ps
denotes the selection for printed data type.

Ps	Printed Data Type
1, 0 or none	Print National Only (Default)
2	National and Line Drawing
3	Print Multinational
4	Print All Characters

Description

This sequence takes effect only when a ProPrinter or DEC printer with ProPrinter emulation mode is attached.

DECSED—Selective Erase in Display

This control function erases some or all of the erasable characters in the display. DECSED can only erase characters defined as erasable by the DECSCA control function. DECSED works inside or outside the scrolling margins.

Available in: VT Level 4 mode only

Format

CSI ? *Ps* J
9/11 3/15 3/*n* 4/10

Parameters

Ps

represents the area of the display to erase, as follows:

Ps	Area Erased
0 (default)	From the cursor through the end of the display
1	From the beginning of the display through the cursor
2	The complete display

DECSEL—Selective Erase in Line

This control function erases some or all of the erasable characters in a single line of text. DECSEL erases only those characters defined as erasable by the DECSCA control function. DECSEL works inside or outside the scrolling margins.

Available in: VT Level 4 mode only

Format

CSI ? *Ps* K
9/11 3/15 3/*n* 4/11

Parameters

Ps

represents the section of the line to erase, as follows:

Ps	Section Erased
0 (default)	From the cursor through the end of the line
1	From the beginning of the line through the cursor
2	The complete line

DECSERA—Selective Erase Rectangular Area

This control function erases all erasable characters from a specified rectangular area in page memory. The select character protection attribute (DECSCA) control function defines whether or not DECSERA can erase characters.

When an area is erased, DECSERA replaces character positions with the space character (2/0). DECSERA does not change:

- Visual attributes set by the select graphic rendition (SGR) function
- Protection attributes set by DECSCA
- Line attributes

Available in: VT Level 4 mode only

Format

```
CSI  Pt  ;    Pl  ;    Pb  ;    Pr  $  {  
9/11 3/n 3/11 3/n 3/11 3/n 3/11 3/n 2/4 7/11
```

Parameters

Pt, Pl, Pb, and Pr

define the rectangular area to be selectively erased:

Pt is the top-line border. *Pt* must be less than or equal to *Pb*.

Default: *Pt* = 1.

Pl is the left-column border. *Pl* must be less than or equal to *Pr*.

Default: *Pl* = 1.

Pb is the bottom-line border.

Default: *Pb* = the last line of the active page.

Pr is the right-column border.

Default: *Pr* = the last column of the active page.

Notes on DECSERA

- The coordinates of the rectangular area are affected by the setting of origin mode (DECOM).
- DECSERA is not affected by the page margins.
- If the value of *Pt*, *Pl*, *Pb*, or *Pr* exceeds the width or height of the active page, then the value is treated as the width or height of that page.
- DECSERA does not change the active cursor position.

DECSFC—Select Flow Control

Select a flow control scheme for the communication port or printer port.

Format

CSI *Ps1* ; *Ps2* ; *Ps3* ; *Ps4* * s
9/11 3/n 3/11 3/n 3/11 3/n 3/11 3/n 2/10 7/3

Parameters

Ps1

determines which port is to be assigned the flow control scheme.

Ps1	Port Type
1, 0 or none	Comm Port
2	Printer Port

Ps2

indicates the flow method. On the VT510, the transmit side or the receive side of the communication line can select different or the same flow control methods.

Ps2	Transmit or Receive
1, 0 or none	Transmit
2	Receive
3	Transmit and Receive

Ps3

is for the flow control type. When the terminal is in PCTerm mode, XPC replaces XON/XOFF on the receive side.

Ps3	Flow Control Type
1, 0 or none	XON/XOFF or XPC (XPC is for receive only)
2	DTR
3	Both
4	None

Ps4

sets the flow control threshold.

Ps4	Flow Control Threshold
1, 0 or none	Low (64)
2	High (768)

The terminal uses "low" and "high" to denote the flow control threshold.

- If Low is selected, then the terminal sends an XOFF (if in software flow control) or lower DTR (in hardware flow control) to notify the host of holding transmission when the input buffer reaches 64 bytes.
- If High is selected, then 768 bytes is the flow control threshold.

"High" flow control threshold is not available for the printer port, which always uses low (64 bytes) as the flow control threshold.

DECSKCV—Set Key Click Volume

Set the keyclick volume.

Format

CSI *Ps* SP r
9/11 3/*n* 2/0 7/2

Parameters

Ps
indicates the selection of the click volume.

Ps	Key Click Volume
1	Off
2, 3, 4	Low
5, 6, 7, 8, 0 or none	High (Default)

DECSLCK—Set Lock Key Style

Allow user to select a Lock key style.

Format

CSI *Ps* SP v
9/11 3/*n* 2/0 7/6

Parameters

Ps
indicates the selection of the lock style.

Ps	Lock Style	Meaning
1, 0 or None	Cap Lock (Default)	Lock alpha keys on main keypad in uppercase.
2	Shift Lock	Lock alpha and numeric keys on main keypad in shifted setting.
3	Reverse Lock	Lock numeric keys in shifted setting, but lock alpha keys in lowercase setting.

DECSLPP—Set Lines Per Page

This control function sets the number of lines for each page in page memory.

Default: One session—6 pages of 24 lines

Format

CSI P_n t
9/11 3/ n 7/4

Parameters

P_n

Sets the lines per page according to the numeric parameter.

Description

If the number of lines specified is not one of the values supported by the implementation, then the next higher supported number of lines per page is assumed. If the number of lines specified exceeds the maximum number supported, then the maximum number supported is used. The valid P_n values are 24, 25, 36, 41, 42, 48, 52, 53, and 72.

The VT510 supports three pages of off-screen memory in the following configurations:

- 3 pages × 24 lines (default)
- 2 pages × 25 lines
- 2 pages × 36 lines
- 1 pages × 41 lines
- 1 pages × 42 lines
- 1 pages × 48 lines
- 1 pages × 52 lines
- 1 pages × 53 lines
- 1 pages × 72 lines

Notes on DECSLPP

- If you switch to a smaller page size, then data that was on the larger page may be split across the smaller pages. To avoid confusion, make sure you set the lines/page *before* you enter data into page memory.
- DECSLPP usually does not change the top and bottom scrolling margins. If, however, you change the page size so that the current scrolling margins exceed the new page size, then the terminal resets the margins to the page limits.

DECSLRM—Set Left and Right Margins

This control function sets the left and right margins to define the scrolling region. DECSLRM only works when vertical split screen mode (DECLRMM) is set.

Available in: VT Level 4 mode only

Default: Margins are at the left and right page borders.

Format

```
CSI  Pl  ;   Pr  s
9/11 3/n 3/11 3/n 7/3
```

Parameters

Pl

is the column number for the left margin.

Default: $Pl = 1$.

Pr

is the column number for the right margin.

Default: $Pr = 80$ or 132 (depending on the page width).

Notes on DECSLRM

- The value of the left margin (Pl) must be less than the right margin (Pr).
- The maximum size of the scrolling region is the page size, based on the setting of set columns per page (DECSCPP).
- The minimum size of the scrolling region is two columns.
- The terminal only recognizes this control function if vertical split screen mode (DECLRMM) is set.
- DECSLRM moves the cursor to column 1, line 1 of the page.
- If the left and right margins are set to columns other than 1 and 80 (or 132), then the terminal cannot scroll smoothly.

DECSMBV—Set Margin Bell Volume

Set the margin bell volume.

Format

CSI *Ps* SP u
9/11 3/*n* 2/0 7/5

Parameters

Ps
indicates the selection of the bell volume.

Ps	Key Click Volume
1, 0 or none	Off (default)
2, 3, 4	Low
5, 6, 7, 8	High

Description

This sequence is used to adjust the warning bell volume, which sounds as the current position nears the right margin.

DECSMKR—Select Modifier Key Reporting

Program the modifier keys to send extended keyboard reports when pressed or released. DECSMKR only works when key position mode (DECKPM) is set. Applications can use DECSMKR to gain more control over the keyboard by monitoring the action of each modifier key.

Format

```
CSI  Pm1  ;   Pf1 ... Pmn ;   Pfn +   r
9/11 3/n   3/11 3/n ... 3/n  3/11 3/n 2/11 7/2
```

Parameters

Pmn
is the key number that corresponds to the modifier key affected.
A *Pmn* value of 0 selects all modifier keys.

Pmn	Key	Default Function
1	left Shift	Local shift.
2	right Shift	Local shift.
3	Lock	Local lock.
4	Ctrl	Local control.
5	left Alt Function	Report up or down transitions.
6	right Alt Function	Report up or down transitions.
7	left Compose Character	Local compose.
8	right Compose Character	Local compose.

Pfn
controls the action of the modifier key specified by the corresponding Pmn value.

Pfn	Meaning
0	Selects the default function of the key.
1	The key performs its default modifier function.
2	The key sends an extended keyboard report when pressed or released. The key does not modify other keystrokes or perform its default function.
3	The key is disabled.

Description

DECSMKR can accept any number of parameter pairs. The terminal executes the parameters in the order in which it receives them. Applications cannot change the function of the F5 (Break) key. This restriction is required for secure systems to ensure that the operator has control of the communication line.

DECSNLS—Set Lines Per Screen

Set the maximum lines that can be displayed on the screen.

Format

```
CSI  Pn  *  |  
9/11 3/n 2/10 7/12
```

Description

The terminal supports three different font heights, which allows 26, 42, or 53 data lines to be displayed on the screen or 25, 41, or 52 data lines to be displayed on the screen, plus a status line. The different character fonts are:

	80/26	132/26	80/42	132/42	80/53	132/53
Cell Width	10	6	10	6	10	6
Cell Height	16	16	10	10	8	8

Notes on DECSNLS

- When 26 Line per Screen/72 Hz/Overscan is selected, the cell size changes from 10 × 16 (6 × 16) to 10 × 13 (6 × 13).
- When 42 lines or 53 lines are selected, users can only select either overscan feature or 72 Hz refresh rate, not both.

Valid Combinations

The terminal has the following valid combinations:

Lines Per Screen	Refresh Rate	Overscan
26	72 Hz	No
26	72 Hz	Yes
26	60 Hz	No
26	60 Hz	Yes
42	72 Hz	No
42	60 Hz	No
42	60 Hz	Yes
53	72 Hz	No
53	60 Hz	No
53	60 Hz	Yes

If an invalid combination is selected, the terminal automatically returns to a valid combination according to the following priority order:

- Lines Per Screen
- Refresh Rate
- Overscan

Examples

The current combination is 26 lines/72 Hz/Overscan. If you attempt to assign 42 lines per screen, then the terminal becomes 42 lines/72 Hz/No Overscan.

The current combination is 42 lines/60 Hz/Overscan. If you attempt to assign 72 Hz refresh rate, then the terminal becomes 42 lines/72 Hz/No Overscan.

The current combination is 53 lines/72 Hz/No Overscan. Any attempt to enable Overscan is invalid.

DECSP—Set Port Parameter

Configures communication parameters for either a communication port or a printer port.

Format

CSI *Ps1* ; *Ps2* ; *Ps3* ; *Ps4* + w
9/11 3/n 3/11 3/n 3/11 3/n 3/11 3/n 2/11 7/7

Parameters

Ps1

determines which port is to be configured.

Ps1	Port Selection
1, 0 or none	Comm Port
2	Printer Port

Ps2

determines the data bit.

Ps2	Data Bit
1, 0 or none	8 bits
2	7 bits

Ps3

determines the parity.

Ps3	Parity	Ps3	Parity
1, 0 or none	No Parity	5	Odd Parity, Unchecked
2	Even Parity	6	Mark Parity
3	Odd Parity	7	Space Parity
4	Even Parity, Unchecked		

Ps3 selections 4 and 5 are not available for the printer port.

Ps4

determines the stop bit.

Ps4	Stop Bit
1, 0 or none	1 bit
2	2 bits

DECSPPCS—Select ProPrinter Character Set

Allow users to match the characters printed to the character sets the IBM ProPrinter is capable of supporting.

Format

CSI *Pn* * p
9/11 3/*n* 2/10 7/0

Parameters

Pn
denotes the code page numbers for ProPrinter supported character sets.

Pn	Printer
210	PC Greek
220	PC Spanish
437	PC International
850	PC Multilingual
852	PC Slavic (Latin 2)
857	PC Turkish
860	PC Portuguese
862	PC Hebrew
863	PC French-Canadian
865	PC Danish
866	PC Cyrillic

Description

This sequence takes affect only when a ProPrinter or DEC printer with ProPrinter emulation mode is attached.

DECSPRTT—Select Printer Type

Allow users to select a printer type that matches their environment.

Format

CSI *Ps* \$ s
9/11 3/*n* 2/4 7/3

Parameters

Ps

indicates the selection of the printer type.

Ps	Printer Type
1, 0 or none	DEC ANSI (default)
2	IBM ProPrinter
3	DEC + IBM

Description

Printer type "DEC ANSI" indicates a printer type that is capable of recognizing the attached Digital ANSI conforming escape sequences.

"ProPrinter" indicates an attached printer type that recognizes the IBM ProPrinter escape sequences.

If the attached printer is an ANSI-compatible printer that has the IBM ProPrinter emulation feature, then select "DEC + IBM."

DECSR—Secure Reset

This control function sets the terminal to its power-up state. DECSR lets applications know the exact state of the terminal before any attempted operations. You can use DECSR to guarantee the terminal state for secure connections. DECSR is similar to RIS, but does not perform a disconnect from the host.

Format

CSI	<i>Pr</i>	+	<i>p</i>
9/11	3/ <i>n</i>	2/11	7/0

Parameters

Pr

is an optional random number from 0 to 16383. If *Pr* is included, then the terminal sends a confirmation (DECSRC) with this number.

DECSR Actions

- Clears user-defined keys.
- Clears soft character sets.
- Clears the screen and all off-screen page memory.
- Returns the cursor to the upper-left corner of the first page.
- Resets the scrolling margins.
- Resets origin mode (DECOM) to absolute.
- Sets visual character attributes to normal.
- Sets the selective erase (DECSCA) attribute to not selectively erasable.
- Sets all character sets to the defaults (G0, G1, G2, G3, GL, GR).
- Erases macro memory.
- Erases the paste buffer.
- Reinitializes the keyboard state by:
 - Clearing the keyboard input buffer
 - Canceling compose sequences
 - Resetting the keyboard to send its default codes (in NVR)
 - Unlocking the keyboard (resetting KAM)

DECSR replaces all set-up parameters to their saved settings or power-up default values. If you include the optional *Pr* parameter, then the terminal returns a secure reset report to the host.

Notes on DECSR

- DECSR works in all conformance levels (level 1 or higher).
- DECSR works in display controls mode (CRM) and causes the terminal to exit this mode.
- DECSR does not work in VT52 mode or printer controller mode. The host software must explicitly exit these modes. You can send DECSR to an attached printer if the terminal is in printer controller mode.
- The terminal buffers DECSR like any other incoming data.

DECSRC—Secure Reset Confirmation

The terminal returns this report to the host if the *Pr* parameter was included in the last secure reset (DECSR) control function. DECSRC report indicates that the secure reset operation was successful.

Format

CSI	<i>Pr</i>	*	q
9/11	3/ <i>n</i>	2/10	7/1

Parameters

Pr
is the number from 0 to 16383 that was included in the corresponding secure reset sequence.

DECSRFR—Select Refresh Rate

Selects a refresh rate.

Format

CSI *Ps* " t
9/11 3/*n* 2/2 7/4

Parameters

Ps
indicates the refresh rate selection. On the VT510, only 60 Hz and 72 Hz are supported.

Ps	Refresh Rate
1	50 Hz
2	60 Hz
3, 0 or none	Greater or equal to 70 Hz

Description

The terminal can be set to refresh its display at the power line frequency in electrically noisy environments. Normally this is unnecessary. The 72 Hz refresh rate is preferred to minimize flicker.

Because of a hardware limitation, overscan and 72 Hz refresh rate cannot co-exist when the lines per screen is set to 42 or 53. See DECSNLS for details. If a combination of lines per screen, refresh rate, and overscan become invalid, then the terminal returns to a valid combination in this order of priority:

1. Lines Per Screen
2. Refresh Rate
3. Overscan

DECSSCLS—Set Scroll Speed

Select how fast lines appear on the screen.

Format

CSI *Ps* SP *p*
9/11 3/*n* 2/0 7/0

Parameters

Ps
indicates the scroll speed and lines per second.

Ps	Scroll Speed	Lines per Second
0, 1, 2, 3 or none	Smooth 2 (default)	9
4, 5, 6, 7, 8	Smooth 4	18
9	Jump	As fast as received

Description

This is a user preference feature. If the User Features Locked is enabled, then the scroll speed cannot be changed by this control function.

The escape sequence DECSCLM can change the scroll speed between Smooth 2 and Jump scroll.

DECSSDT—Select Status Display (Line) Type

This control function lets the host select the type of status line displayed.

Available in: VT Level 4 mode only

Default: Indicator status display

Format

CSI *Ps* \$ ~
9/11 3/*n* 2/4 7/14

Parameters

Ps

indicates which status line the host selects:

Ps	Status Line Selected
0 (default)	No status line
1	Indicator status line
2	Host-writable status line

Notes on DECSSDT

- If you select no status line ($P_s = 0$), then the terminal uses the line as an additional user window line to display data.
- If you change from an indicator to a host-writable status line, then the new host-writable status line is empty.
- When you select the host-writable status line, most of the control functions that affect the main display also affect the status line. The following list describes the exceptions to the rule:

Control Function	Action
Select character set (SCS)	Both the main display and status line use the same character set.
Cursor position controls	Only the column parameters in cursor-positioning commands operate in the status line.
ANSI mode (DECANM)	Ignored if received in the status line.
Set conformance level (DECSCL)	Exits the status line.
Scrolling mode (DECSCLM)	Affects the main display and the status line.
Soft terminal reset (DECSTR)	Exits the status line.
Insert/replace mode (IRM)	Affects the main display and the status line.
Hard terminal reset (RIS)	Erases and exits the status line.
Tab stops	Affect the main display and status line.
Autowrap mode (DECAWM)	Affects the main display and status line.

- DECSSDT does not affect the status line type displayed in Set-Up. In Set-Up, the terminal always uses the indicator status line.

DECSSL—Select Set-Up Language

Select one of several languages for Set-Up display menus.

Format

CSI *Ps* p
9/11 3/*n* 5/0

Parameters

Ps indicates the set-up language.	
Ps	Set-Up Language
0, 1 or none	English
2	French
3	German
4	Spanish

DECST8C—Set Tab at Every 8 Columns

Set a tab stop at every eight columns starting with column 9.

Format

CSI	?	5	W
9/11	3/15	3/5	5/7

Description

Any tab stop setting before this command is executed is cleared automatically. Control function TBC clears the tab stops on the display; HTS sets a horizontal tab stop at the active column.

DECSTBM—Set Top and Bottom Margins

This control function sets the top and bottom margins for the current page. You cannot perform scrolling outside the margins.

Default: Margins are at the page limits.

Format

```
CSI  Pt  ;    Pb  r
9/11 3/n 3/11 3/n 7/2
```

Parameters

Pt

is the line number for the top margin.

Default: $Pt = 1$.

Pb

is the line number for the bottom margin.

Default: $Pb =$ current number of lines per screen.

Notes on DECSTBM

- The value of the top margin (Pt) must be less than the bottom margin (Pb).
- The maximum size of the scrolling region is the page size.
- DECSTBM moves the cursor to column 1, line 1 of the page.

DECSTR—Soft Terminal Reset

Perform a soft reset to the default values listed in [Table 5–9](#).

Format

```
CSI  !  p
9/11 2/1 7/0
```

Description

[Table 5–9](#) lists the terminal's default settings:

Table 5–9 Terminal's Default Settings

Mode	Mnemonic	State after DECSTR
Text cursor enable	DECTCEM	Cursor enabled.
Insert/replace	IRM	Replace mode.
Origin	DECOM	Absolute (cursor origin at upper-left of screen.)
Autowrap	DECAWM	No autowrap.
National replacement character set	DECNRCM	Multinational set.
Keyboard action	KAM	Unlocked.
Numeric keypad	DECNKM	Numeric characters.
Cursor keys	DECKKM	Normal (arrow keys).
Set top and bottom margins	DECSTBM	Top margin = 1; bottom margin = page length.
All character sets	G0, G1, G2, G3, GL, GR	Default settings.
Select graphic rendition	SGR	Normal rendition.
Select character attribute	DECSCA	Normal (erasable by DECSEL and DECSED).
Save cursor state	DECSC	Home position.
Assign user preference supplemental set	DECAUPSS	Set selected in Set-Up.
Select active status display	DECSASD	Main display.
Keyboard position mode	DECKPM	Character codes.
Cursor direction	DECRLM	Reset (Left-to-right), regardless of NVR setting.
PC Term mode	DECPCTERM	Always reset.

DECSTRl—Set Transmit Rate Limit

Limit the rate of character transmission according to the selected rate.

Format

CSI *Ps1* ; *Ps2* " *u*
9/11 3/*n* 3/11 3/*n* 2/2 7/5

Parameters

Ps1

indicates the key type.

Ps1	Key Type
1, 0 or none	All Keys
2	Graphic Key
3	Function Key

Ps2

indicates the characters per second.

Ps2	Characters Per Second
1, 0 or none	150 CPS
2	50 CPS
3	30 CPS

Description

When enabled, the terminal limits the rate at which it transmits the answerback, CPR, DA, TSI, and DSR messages, as well as the keyboard keys, and any other characters transmitted from the terminal. All messages and keystrokes are transmitted according to the selected "character per second" rate.

You can select different transmit rates for graphic keys and function keys, which usually transmit more than one byte at a time.

The transmit rate limit function can be enabled and disabled through the DECXRLM control function. When disabled, all messages and keystrokes are transmitted as fast as possible without any delay between characters.

DECSWBV—Set Warning Bell Volume

Set the warning bell volume.

Format

CSI *Ps* SP t
9/11 3/*n* 2/0 7/4

Parameters

Ps
indicates the selection of the click volume.

Ps	Bell Volume
1	Off
2, 3, 4	Low
5, 5, 7, 8, 0 or none	High (default)

Description

This sequence is used to adjust the warning bell volume (for errors and receipt of Ctrl G).

DECSWL—Single-Width, Single-Height Line

DECSWL makes the line with the cursor a single-width, single-height line. This line attribute is the standard for all new lines on the screen.

Format

ESC	#	5
1/11	2/3	3/5

DECTABSR—Tab Stop Report

The terminal sends this sequence to the host in response to a request presentation state report (DECRQPSR) sequence. DECTABSR informs the host of the terminal's current tab settings.

Programming Tip

Applications can use information in the tab stop report to save the current tab stops. Later, the application can restore the saved tab stops.

This operation is useful for applications that need to temporarily change the terminal's tab stops. When the application is finished, it can restore the tab stops that were in effect before the application changed them. You use the restore presentation state (DECRSPS) function to restore tab stops. Refer to DECRSPS—Restore Presentation State for additional information.

Format

```
DCS 2 $ u D ... D ST
9/0 3/2 2/4 7/5 ... 9/12
```

Parameters

D...D

is a data string indicating the column number location of each tab stop.

Example

The following is an example of a DECTABSR sequence:

```
DCS 2 $ u 9/ 17/ 25/ 33/ 41/ 49/ 57/ 65/ 73 ST
```

9, 17, 25, 33, 41, 49, 57, 65, and 73 are the column numbers for tab stops.

DECTCEM—Text Cursor Enable Mode

This control function makes the cursor visible or invisible.

Default: Visible

Format

CSI	?	2	5	h	Set: makes the cursor visible.
9/11	3/15	3/2	3/5	6/8	
CSI	?	2	5	l	Reset: makes the cursor invisible.
9/11	3/15	3/2	3/5	6/12	

DECTID—Select Terminal ID

Select a terminal ID.

Format

CSI *Ps* , q
9/11 3/*n* 2/12 7/1

Parameters

Ps
denotes the ID type.

Ps ID Type	
0	VT100
1	VT101
2	VT102
5	VT220
7	VT320
9	VT420
10	VT510

Description

This sequence selects the Primary DA Response ("Terminal ID") transmitted by the terminal. DECTID has no effect when the terminal is in VT52 mode.

The VT510 supports the following DA1 responses:

VT100 ID	ESC [? 1; 2 c
VT101 ID	ESC [? 1; 0 c
VT102 ID	ESC [? 6 c
VT220 ID (N.A.)	CSI ? 62; 1; 2; 7; 8 c
VT220 ID (Intl)	CSI ? 62; 1; 2; 7; 8; 9 c
VT320 ID (N.A.)	CSI ? 63; 1; 2; 7; 8 c
VT320 ID (Intl)	CSI ? 63; 1; 2; 7; 8; 9 c
VT420 ID (N.A.)	CSI ? 64; 1; 2; 7; 8; 9; 15; 18; 21 c
VT420 ID (Intl)	CSI ? 64; 1; 2; 7; 8; 9; 15; 18; 21 c
VT510 ID	Reference function DA1

The default ID is "VT510."

DECTME—Terminal Mode Emulation

Select a terminal emulation mode.

Format

CSI *Ps* SP ~
9/11 3/*n* 2/0 7/17

Parameters

Ps

The Ps parameter selects the terminal emulation mode as follows:

Table 5–10 Terminal Emulation Modes	
Ps	Terminal Mode
0, 1, or none	VT500 (VT Level 4)
2	VT100
3	VT52
4	VT420 PCTerm
5	WYSE 60/160
6	WYSE 60/160 PCTerm
7	WYSE 50+
8	WYSE 150/120
9	TVI 950
10	TVI 925
11	TVI 910+
12	ADDS A2
13	SCO Console

Description

This sequence allows you to select the terminal emulation mode. After any mode change is made, a soft reset is performed.

Notes on DECTME

- Printer operations are not affected or halted by a change in mode.
- A soft reset is always performed as a result of a mode change between VT modes.
Exception: Entering VT52 mode using DECANM does not cause a soft reset from VT100 mode.
- DECANM and DECSCL can also change the VT operating modes.

DECTSR—Terminal State Report—Terminal to Host

Requests and reports the state of the terminal.

Format

CSI *Ps* \$ u
9/11 3/*n* 2/4 7/5

Parameters

Ps
indicates the type of report the host requests from the terminal.

Ps	Report Requested
0 or none	Ignored. No report sent.
1	Terminal state report (DECTSR)

Response
DCS 1 \$ s D ... D ST
D ... D
See Device Attributes (DA1, DA2, DA3).

Description

DECRQTSR and DECTSR requests and reports the state of the terminal.

Errata

The printed document says that the report starts:
DCS 1 \$ S D ... D ST
This is incorrect. The sequence contains a lowercase ‘s’, not uppercase.

DECTST—Invoke Confidence Test

Select tests to be performed.

Format

CSI 4 ; Ps ... ; Ps y
9/11 3/4 3/11 3/n ... 3/11 3/n 7/9

Parameters

Ps
is the parameter indicating a test to be done.

Ps	Test
0	"All Tests" (1,2,3,6)
1	Power-Up Self Test
2	RS-232 Port Data Loopback Test
3	Printer Port Loopback Test
4	Speed Select and Speed Indicator Test
5	Reserved - No action
6	RS-232 Port Modem Control Line Loopback Test
7	EIA-423 Port Loopback Test
8	Parallel Port Loopback Test
9	Repeat (Loop On) Other Tests In Parameter String

Description

After the first parameter, "4", the parameters each select one test. Several tests may be invoked at once by chaining the parameters together separated by semicolons. The tests are not necessarily executed in the order in which they are entered in the parameter string.

"ESC # 8" invokes the Screen Alignment test for the VT510. Additionally, after executing the power-up selftest, the terminal displays either the diagnostic messages in the upper left corner of the screen or the "VT510 OK" message in the center of the screen and within a box. Upon receipt of any character except XON or if the user types a keystroke, the screen is cleared. If the terminal is in local mode, then characters from the host are ignored and the message remains visible even if characters are received from the host. DECTST causes a disconnect; therefore, it should not be used in conjunction with a modem.

DECUDK—User Defined Keys

DECUDK downloads definitions for user-defined keys.

Format

DCS	<i>Ps1 ; Ps2 ; Ps3</i>		<i>D . . . D</i>	ST
9/0			7/12	9/12
Device	Clear,	Final	Key Definition String	String
Control	Lock, and	Character		Terminator
String	Modifier			
Introducer	Parameters			

Parameters

Ps1	
Ps1	Meaning
0 (default) or none	Clear all keys before loading new values.
1	Load new UDK values; clear old values only when redefined.

When *Ps1* is 1, the terminal only clears the keys you are loading. By using a *Ps1* value of 1, you can redefine some keys without redefining them all.

Note

There are 804 bytes of memory for all user-defined keys. A key definition can only use the number of bytes available when that key is loaded.

Programming Tip

If *Ps1* is 1, a key load may fail because no memory space is available. The reason for this is as follows:

With *Ps1* set to 1, keys are cleared and loaded sequentially. If the new definition for a key is larger than the old one, then you may exceed the 804 byte limit.

For example, suppose **F6** contains 412 bytes, **F7** contains 366 bytes, and **F8** contains 20 bytes. You try to load **F8** with 40 bytes, **F6** with one byte, and **F7** with one byte, in that order. This works if all keys are cleared first (*Ps1* is 0), but not if keys are cleared one at a time (*Ps1* is 1). When you try to load **F8** with 40 bytes, the load fails because only 26 bytes are free at that time.

$$804 \text{ (maximum)} - 412 \text{ (F6)} - 366 \text{ (F7)} = 26$$

Ps2 is the *lock parameter*. *Pl* determines whether the key definitions are locked or unlocked after you load them.

Ps2	Action
0 or none	Lock the keys. If you want to load new values into the keys, then you must unlock the keys by using Set-Up.
1	Do not lock the keys against future redefinition.

If *Ps2* is 1 and the keys are already locked, then nothing happens.

The terminal uses a special lock to prevent or allow the programming of user-defined keys. You can turn on this lock from Set-Up or from the host (with a DECUDK device control string). The lock affects all programmable keys. When you use the lock, you should follow these guidelines:

- Unlock the keys to define them.
The keys must be unlocked before you can define them. You can only unlock the keys from set-up. If a key is locked and an application tries to redefine the key with a DECUDK sequence,

then the terminal ignores the sequence.

- Lock the keys to prevent redefinition.

You can lock the keys from Set-Up or from the host by sending a DECUDK sequence. New key definitions are locked by default.

Ps3

is the *modifier parameter*. Ps3 designates whether the key is shifted or unshifted.

Ps3	Action
0, 2, or none	Defines the shifted function key.
1	Defines the unshifted function key.
3	Defines the alternate unshifted function key.
4	Defines the alternate shifted function key.

|

is the *final character*. The vertical bar (7/12) identifies this control string as a DECUDK.

D ... D

are the *key definition strings* such as:

Key1/Mod1/Function #/UDS/UDS Direction;

Key2/Mod2/Function #/UDS/UDS Direction;.....;

You include these strings between the final character (|) and the string terminator (ST). Each string consists of a key selector number (*Kyn*), a string parameter (*Stn*), a slash (/) (2/15), and a direction (*Dir*). A semicolon (;) (3/11) separates different strings.

- Kyn is the key selector number of the key you are defining. Tables 5–11 and 5–12 list the definable keys and their identifying values. Table 5–13 compares the function keys on the VT keyboard to equivalent keys on the PC keyboard.
- Stn is the string parameters, which are the encoded definition of the keys, consisting of hex pairs in the following ranges:
3/0 through 3/9 (0 through 9)
4/1 through 4/6 (A through F)
6/1 through 6/6 (a through f)

When you combine these hex values, they represent an 8-bit quantity.

This method lets you use any of the 256 character codes in the key string. You can enter key definition strings in any order.

Default: Empty. The key is undefined.

- Dir is the user-defined sequence (UDS) direction.

Dir	Direction	UDS is sent to . . .
0, none	Normal	Host and/or the terminal, depending on the setting of the COMM mode parameter. The UDS is directed to: ◦ The host for full-duplex mode. ◦ The screen for half-duplex mode.
1	Local	Terminal screen only.
2	Remote	Host only.

Default: None (Normal)

Table 5–11 VT Keyboards

Pm = 1 Pm = none, 0, or 2		
Kyn	Key	Key
11	F1	Shift/F1
12	F2	Shift/F2
13	F3	Shift/F3
14	F4	Shift/F4
15	F5	Shift/F5
17	F6	Shift/F6
18	F7	Shift/F7
19	F8	Shift/F8
20	F9	Shift/F9
21	F10	Shift/F10
23	F11	Shift/F11
24	F12	Shift/F12
25	F13	Shift/F13
26	F14	Shift/F14
28	F15	Shift/F15
29	F16	Shift/F16
31	F17	Shift/F17
32	F18	Shift/F18
33	F19	Shift/F19
34	F20	Shift/F20

Note

Pm values of 3 and 4 are specific to the PC keyboard. They are ignored when you use a VT keyboard.

Table 5–12 PC Keyboards

Pm Value				
Kyn	1	None, 0 or 2	3	4
11	F1	Shift/F1	Alt/F1	Alt/Shift/F1
12	F2	Shift/F2	Alt/F2	Alt/Shift/F2
13	F3	Shift/F3	Alt/F3	Alt/Shift/F3
14	F4	Shift/F4	Alt/F4	Alt/Shift/F4
15	F5	Shift/F5	Alt/F5	Alt/Shift/F5
17	F6	Shift/F6	Alt/F6	Alt/Shift/F6
18	F7	Shift/F7	Alt/F7	Alt/Shift/F7
19	F8	Shift/F8	Alt/F8	Alt/Shift/F8
20	F9	Shift/F9	Alt/F9	Alt/Shift/F9
21	F10	Shift/F10	Alt/F10	Alt/Shift/F10
23	F11	Shift/F11	Alt/F11	Alt/Shift/F11
24	F12	Shift/F12	Alt/F12	Alt/Shift/F12

**Table 5–13 VT
and PC
Keyboard
Function Keys
Compared**

Keyboard		
Kyn	VT	PC
11	F1	F1
12	F2	F2
13	F3	F3
14	F4	F4
15	F5	F5
17	F6	F6
18	F7	F7
19	F8	F8
20	F9	F9
21	F10	F10
23	F11	F11
24	F12	F12
25	F13	–
26	F14	–
28	F15	–
29	F16	–
31	F17	–
32	F18	–
33	F19	–
34	F20	–

Description

DCS (9/0) indicates the beginning of a device control string. DCS is an 8-bit C1 character. You can use ESC P (1/11 5/0) for a 7-bit environment.

ST is the *string terminator*. ST (9/12) is a C1 8-bit character. You can use ESC \ (1/11, 5/12) for a 7-bit environment.

General Guidelines When Loading UDKs

- Clear UDK memory space before loading new definitions.

Use a DECUDK string to clear keys without locking them. Then you can use another DECUDK string to redefine the keys and lock them.

- If you redefine a key, then the old definition is lost. This clears space if the new definition uses less bytes than the old one.
- There is only one way to unlock UDKs.

To unlock a UDK, use Set-Up.

- When you clear a UDK, the key definition is empty.
- An invalid hex pair in a DECUDK string stops a UDK sequence.

When a load sequence stops (error or other cause), the terminal saves keys already loaded and sends the rest of DECUDK to the screen.

- You can disable the local function keys by redefining them with DECUDK.
- DECUDK works in all VT modes except VT52 mode.

Examples of DECUDK Device Control Strings

The following sequence clears unshifted UDKs:

DCS 0 ; 1 ; 1 | ST

The following sequence locks unshifted UDKs:

DCS 1 ; 0 ; 1 | ST

Suppose you want to define the unshifted F20 key to be "PRINT", without clearing or locking any other keys. The first part of your sequence would look like this:

DCS 1 ; 1 ; 1 | 3 4 /

34 is the code for the F20 key.

After the slash character (/) (2/15), you include the definition. The rest of the sequence after the slash character would look like this:

5 0 5 2 4 9 4 E 5 4 ST

The hex encoding for "PRINT" is as follows:

50 = P

52 = R

49 = I

4E = N

54 = T

The ST character (9/12) marks the end of the control string:

DCS 1 ; 1 ; 1 | 34 / 50 52 49 4E 54 ST

DECVCCM—Vertical Cursor-Coupling Mode

This control function determines whether the user window pans with the cursor when the cursor moves past the top or bottom border of the user window. DECVCCM is only useful when the height of the current user window is smaller than the page. The cursor must stay on the current page.

Default: Coupled

Format

CSI	?	6	1	h	Set: couples the cursor to the display.
9/11	3/15	3/6	3/1	6/8	
CSI	?	6	1	l	Reset: uncouples the cursor.
9/11	3/15	3/6	3/1	6/12	

Description

When DECVCCM is set, the cursor is coupled to the display for vertical movement. When the cursor moves past the top or bottom border of the user window, the window pans to keep the cursor in view.

If the cursor moves past the top of the display, the user window pans up. New lines appear at the top of the screen, while lines at the bottom move out of view.

When DECVCCM is reset, the cursor is uncoupled from the display. If the cursor moves past the top or bottom border of the user window, the cursor disappears.

DECXCPR—Extended Cursor Position

The host asks the terminal for the current cursor position, including the current page number.

Format

```
CSI  ?    6    n
9/11 3/15 3/6 6/14
```

Response

```
CSI  Pl ; Pc ; Pp R
```

Parameters

Pl; Pc; Pp

The terminal indicates that the cursor is currently at line *Pl*, column *Pc*, on page *Pp*.

DECXRLM—Transmit Rate Limiting

Applications can control the transmit rate limiting feature through the DECXRLM control function.

Default: Unlimited

Format

CSI ? 7 3 h Set: limited transmit rate.
9/11 3/15 3/7 3/3 6/8
CSI ? 7 3 I Reset: unlimited transmit rate.
9/11 3/15 3/7 3/3 6/12

Modem Connection and Disconnections

When the terminal makes a connection to the host system through a modem, the terminal performs the following operations to ensure it is ready to send and receive.

- Unlocks the keyboard (if it was locked).
- Clears any transmission in progress.
- Clears the keyboard buffer and all message buffers.
- Clears the input buffer.
- Clears XOFF sent and XOFF received.

Any of the following conditions will disconnect the connection to the host system in VT mode:

- You press **Shift** **F5** (**Shift** **Break**).
- You use the **Recall** or **Default** fields in the Set-Up Directory.
- *International model:* You change the host port you are using (from the RS-232 port to the EIS-423 port, or from the EIA-423 port to the RS-232 port).
- The terminal loses the data set read (DSR) signal.
- The terminal loses the received line signal detect (RLSD) signal for a period of time you defined in set-up. See the disconnect delay feature in the Communications Set-Up screen.
- The terminal does not receive an RLSD signal within 30 seconds after DSR.
- The terminal receives a self-test command from the host system.

The usual way to disconnect communications is to type **Shift** **F5** (**Shift** **Break**). The host system's response to the disconnect signal depends on the system and the software.

Terminal-to-Printer Data Flow Control

The terminal sends only data characters to the printer. The terminal does not send XON and XOFF characters. The terminal recognizes only XON and XOFF characters from the printer. Any other characters from the printer are ignored.

When the terminal receives XOFF from the printer, the terminal stops sending data. The terminal starts sending data again when it receives an XON, or when a clear comm operation is performed.

Using C1 Control Characters

Using an 8-bit setting for the printer port implies the use of 8-bit C1 control characters. Using a 7-bit setting implies the use of the 7-bit ESC form of C1 control characters. You select the setting by using the data bits/parity feature in the Printer Set-Up screen.

Note on DECXRLM

Older printers may not recognize the 8-bit form of C1 control characters. With these printers, you must set the printer port to 7-bits for correct operation.

DL—Delete Line

This control function deletes one or more lines in the scrolling region, starting with the line that has the cursor.

Format

CSI	<i>Pn</i>	M
9/11	3/ <i>n</i>	4/13

Parameters

Pn
is the number of lines to delete.
Default: *Pn* = 1.

Description

As lines are deleted, lines below the cursor and in the scrolling region move up. The terminal adds blank lines with no visual character attributes at the bottom of the scrolling region. If *Pn* is greater than the number of lines remaining on the page, DL deletes only the remaining lines. DL has no effect outside the scrolling margins.

DSR—Device Status Reports

The host computer and terminal exchange DSR sequences to provide the host with the operating status of the following features:

- | | |
|----------------------------|-----------------------|
| Operating status | User-defined keys |
| Keyboard status - language | Macro space report |
| Cursor position report | Memory checksum |
| Cursor position with page | Data integrity report |
| Printer port | |

DSR requests and reports follow one of two formats, ANSI or DEC format. The format for each is as follows:

Format

CSI	<i>Ps</i>	<i>n</i>	ANSI format	
9/11	3/ <i>n</i>	6/14		
CSI	?	<i>Ps</i>	<i>n</i>	DEC format
9/11	3/15	3/ <i>n</i>	6/14	

Parameters

Ps
indicates the type of DSR requested. See the following individual DSR reports for specific parameters within each report.

Description

There is a different DSR request for each feature. The following sections describe the possible DSR reports. If the terminal is in printer controller mode, then the printer receives the DSR request. The printer can respond through the bidirectional printer port.

DSR—Cursor Position Report (CPR)

The host asks the terminal for a cursor position report.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI 6 n	The host asks for a cursor position report (CPR).
CPR response (Terminal to host)	CSI <i>Pl</i> ; <i>Pc</i> R	The terminal indicates that the cursor is currently at line <i>Pl</i> , column <i>Pc</i> .

DSR—Data Integrity Report

The host asks for the status of the data integrity flag. The data integrity flag indicates the integrity of the data flow between the terminal and host. This flag is reset each time the terminal sends a data integrity report.

Programming Tip

Applications using the flag should always reset the flag before sending data to the terminal. You can reset the flag with a DSR request (CSI ? 75 n) sequence. After the application sends data, the application should send another request to find out if a parity error has occurred.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 75 n	The host asks for the status of the data integrity flag.
Possible responses (Terminal to host)	CSI ? 70 n	Ready, no communication errors or power-ups have occurred since last report.
	CSI ? 71 n	Malfunction, a communication error has occurred since the last report.
	CSI ? 73 n	The terminal has not reported data integrity since the last power-up or hard reset (RIS).

DSR—Extended Cursor Position Report (DECXCPR)

The host asks the terminal for the current cursor position, including the current page number.

Format

Exchange	Sequence	Meaning
Request (Host to Terminal)	CSI ? 6 n	The host asks for an extended cursor position report (DECXCPR).
DECXCPR response (Terminal to host)	CSI <i>Pl</i> ; <i>Pc</i> ; <i>Pp</i> R	The terminal indicates that the cursor is currently at line <i>Pl</i> , column <i>Pc</i> , on page <i>Pp</i> .

DSR—Keyboard

The host asks for the current keyboard language, operating status, and type.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 26 n	The host asks for the keyboard status.
Response (Terminal to host)	CSI ? 27; <i>Pn</i> ; <i>Pst</i> ; <i>Ptyp</i> n	The keyboard language is <i>Pn</i> , the keyboard status is <i>Pst</i> , and the keyboard type is <i>Ptyp</i> .

Parameters

Pn
[Table 5–14](#) is a complete list of keyboards and keyboard languages.

Table 5–14 Keyboard Languages

Pn	VT/PC Language
0	Keyboard language cannot be determined.
1	North American
2	British
3	Belgian (Flemish)
4	French Canadian
5	Danish
6	Finnish
7	German
8	Dutch
9	Italian
10	Swiss French
11	Swiss German
12	Swedish
13	Norwegian
14	French
15	Spanish
16	Portuguese
19	Hebrew
22	Greek
28	Canadian (English)
29	Turkish-Q/Turkish
30	Turkish-F/Turkish
31	Hungarian
33	Slovak
35	Polish
36	Romanian
38	SCS
39	Russian
40	Latin American

Pst

indicates the keyboard status.

Pst Keyboard Status‡

0	Keyboard Ready	The terminal sends typed characters to the current session.
3	No Keyboard	The terminal does not detect the keyboard.
8	Keyboard Busy	The other session is currently using the keyboard.

‡The terminal only sends Pst in VT level 4 mode.

Ptyp

indicates the keyboard type.

Ptyp Keyboard Type	
4	LK450
5	PCXAL

DSR—Macro Space Report

The host asks for the available macro space in the form of a macro space report (DECMSR).

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 62 n	The host asks for a macro space report (DECMSR).
DECMSR response (Terminal to host)	CSI <i>Pn</i> * {	The terminal indicates the number of bytes available for macro definitions.

Parameters

Pn

$$Pn = \frac{\text{number of bytes}}{16} \text{ rounded down.}$$

DSR—Memory Checksum (DECCKSR)

The host asks for a memory checksum report of current text macro definitions.

Programming Tip

Enter your macro definitions, then request and save a memory checksum. Later, you can request another checksum and see if it has changed.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 63; <i>Pid</i> n	The host asks for a memory checksum report of current text macro definitions (DECCKSR). <i>Pid</i> is an optional numeric parameter that provides a label to identify the particular checksum request. The checksum report returns the label. This label lets you differentiate between multiple checksum reports.
DECCKSR response (Terminal to host)	DCS <i>Pid</i> ! ~ <i>D..D</i> ST	<i>Pid</i> is a label indicating which DSR request the report is for. <i>D..D</i> is the data string consisting of four hexadecimal digits indicating the checksum. The digits can be in the range of 3/0 through 3/9 and 4/0 through 4/6.

DSR—Operating Status

The host requests the terminal's operating status.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI 5 n	The host requests the terminal's operating status. The host asks if the terminal is in good operating condition.
Responses (Terminal to host)	CSI 0 n or	The terminal indicates that it is in good operating condition.
	CSI 3 n	The terminal indicates that it has a malfunction.

DSR—Printer Port

The host asks for the status of the terminal's printer.

Note

Host software should check the printer status before entering any print mode or starting any printing function.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 15 n	The host asks for the current printer status.
Possible responses (Terminal to host)	CSI ? 13 n	No printer. The data terminal ready (DTR) signal has not been asserted on the printer port since the last power-up or reset.
	CSI ? 10 n	Printer ready. DTR is asserted on the printer port.
	CSI ? 11 n	Printer not ready. DTR is not currently asserted on the printer port.
	CSI ? 18 n	Printer busy. DTR is asserted on the printer port, but the other session is using the printer. In VT Level 1, the terminal sends the <i>printer not ready</i> sequence above.
	CSI ? 19 n	Printer assigned to other session. DTR is asserted on the printer port, but the printer is assigned to the other session. The printer is not available to this session. In VT100 mode, the terminal sends the <i>no printer</i> sequence above.

DSR—User-Defined Keys (VT Level 4 Only)

The host asks if the user-defined keys (UDKs) are locked or unlocked.

Format

Exchange	Sequence	Meaning
Request (Host to terminal)	CSI ? 25 n	The host asks if UDKs are locked or unlocked.
Possible responses (Terminal to host)	CSI ? 20 n	UDKs are unlocked.
	CSI ? 21 n	UDKs are locked.

ECH—Erase Character

This control function erases one or more characters, from the cursor position to the right. ECH clears character attributes from erased character positions. ECH works inside or outside the scrolling margins.

Available in: VT Level 4 mode only

Format

CSI	<i>Pn</i>	<i>X</i>
9/11	3/ <i>n</i>	5/8

Parameters

Pn

is the number of characters to erase. A *Pn* value of 0 or 1 erases one character.

Default: *Pn* = 1.

ED—Erase in Display

This control function erases characters from part or all of the display. When you erase complete lines, they become single-height, single-width lines, with all visual character attributes cleared. ED works inside or outside the scrolling margins.

Format

CSI *Ps* J
9/11 3/*n* 4/10

Parameters

Ps
represents the amount of the display to erase.

Ps	Area Erased
0 (default)	From the cursor through the end of the display
1	From the beginning of the display through the cursor
2	The complete display

Programming Tip

Use a *Ps* value of 2 to erase the complete display in a fast, efficient manner.

EL—Erase in Line

This control function erases characters on the line that has the cursor. EL clears all character attributes from erased character positions. EL works inside or outside the scrolling margins.

Format

CSI *Ps* K
9/11 3/*n* 4/11

Parameters

Ps

represents the section of the line to erase.

Ps	Section Erased
0 (default)	From the cursor through the end of the line
1	From the beginning of the line through the cursor
2	The complete line

HPA—Horizontal Position Absolute

Inquire as to the amount of free memory for programmable key operations.

Format

CSI *Pn* `
 9/11 3/*n* 6/0

Parameters

Pn
indicates horizontal position.

Description

HPA causes the active position to be moved to the n-th horizontal position of the active line. If an attempt is made to move the active position past the last position on the line, then the active position stops at the last position on the line.

HPR—Horizontal Position Relative

Inquire as to the amount of free memory for programmable key operations.

Format

CSI *Pn* a
 9/11 3/*n* 6/1

Parameters

Pn
 indicates horizontal position.

Description

HPR causes the active position to be moved to the n-th following horizontal position of the active line. If an attempt is made to move the active position past the last position on the line, then the active position stops at the last position on the line.

HTS—Horizontal Tab Set

HTS sets a horizontal tab stop at the column position indicated by the value of the active column when the terminal receives an HTS.

You can use either one of the following formats:

Format

HTS	or	ESC	H
8/8		1/11	4/8

Description

Executing an HTS does not effect the other horizontal tab stop settings.

HVP—Horizontal and Vertical Position

This control function works the same as the cursor position (CUP) function. New applications should use CUP instead of HVP. HVP is provided for compatibility with earlier VT products.

Format

CSI	f	Cursor moves to home position selected by DECOM			
9/11	6/6				
CSI	<i>Pl</i>	;	<i>Pc</i>	f	Moves cursor to line <i>Pl</i> , column <i>Pc</i>
9/11	3/ <i>n</i>	3/11	3/ <i>n</i>	6/6	

Parameters

Pl; Pc

If *Pl* or *Pc* is not selected or selected as 0, then the cursor moves to the first line or column, respectively. Origin mode (DECOM) selects line numbering and the ability to move the cursor into margins.

ICH—Insert Character

This control function inserts one or more space (SP) characters starting at the cursor position.

Available in: VT Level 4 mode only

Format

CSI	<i>Pn</i>	@
9/11	3/ <i>n</i>	4/0

Parameters

Pn

is the number of characters to insert.

Default: *Pn* = 1.

Description

The ICH sequence inserts *Pn* blank characters with the normal character attribute. The cursor remains at the beginning of the blank characters. Text between the cursor and right margin moves to the right. Characters scrolled past the right margin are lost. ICH has no effect outside the scrolling margins.

IL—Insert Line

This control function inserts one or more blank lines, starting at the cursor.

Format

CSI	<i>Pn</i>	L
9/11	3/ <i>n</i>	4/12

Parameters

Pn

is the number of lines to insert.

Default: *Pn* = 1.

Description

As lines are inserted, lines below the cursor and in the scrolling region move down. Lines scrolled off the page are lost. IL has no effect outside the page margins.

IND—Index

IND moves the cursor down one line in the same column. If the cursor is at the bottom margin, then the screen performs a scroll-up.

Format

ESC	D
1/11	4/4

IRM—Insert/Replace Mode

This control function selects how the terminal adds characters to page memory. The terminal always adds new characters at the cursor position.

Default: Replace.

Format

CSI	4	h	Set: insert mode.
9/11	3/4	6/8	
CSI	4	I	Reset: replace mode.
9/11	3/4	6/12	

Description

If IRM mode is set, then new characters move characters in page memory to the right. Characters moved past the page's right border are lost.

If IRM mode is reset, then new characters replace the character at the cursor position.

KAM—Keyboard Action Mode

This control function locks or unlocks the keyboard.

Default: Unlocked

Format

CSI	2	h	Set: locks the keyboard.
9/11	3/2	6/8	
CSI	2	l	Reset: unlocks the keyboard.
9/11	3/2	6/12	

Description

If KAM is set, then the keyboard cannot send characters to the host. The **Wait** indicator appears on the keyboard indicator line at the bottom of the screen. The terminal ignores all keystrokes that send characters to the host. KAM does not affect the **F3** (Set-Up) or **F4** (Session) keys.

If KAM is reset, then the terminal unlocks the keyboard. The keyboard can send characters to the host.

LNМ—Line Feed/New Line Mode

This control function selects the characters sent to the host when you press the Return key. LNM also controls how the terminal interprets line feed (LF), form feed (FF), and vertical tab (VT) characters.

Note

For compatibility with Digital's software, you should keep LNM reset (line feed).

Default: Line feed

Format

CSI	2	0	h	Set: new line.
9/11	3/2	3/0	6/8	
CSI	2	0	l	Reset: line feed.
9/11	3/2	3/0	6/12	

Description

If LNM is set, then the cursor moves to the first column on the next line when the terminal receives an LF, FF, or VT character. When you press Return, the terminal sends both a carriage return (CR) and line feed (LF).

If LNM is reset, then the cursor moves to the current column on the next line when the terminal receives an LF, FF, or VT character. When you press Return, the terminal sends only a carriage return (CR) character.

Note on LNM

When the auxiliary keypad is in keypad numeric mode (DECKPNM), the Enter key sends the same characters as the Return key.

LS—Locking Shift

When you use a locking shift, the character set remains in GL or GR until you use another locking shift. [Table 5–15](#) lists all locking shifts available.

Format

Table 5–15 Mapping Character Sets with Locking Shifts

Locking Shift	Code	Function
LS0 (locking shift 0)	SI 0/15	Map G0 into GL.(default)
LS1 (locking shift 1)	SO 0/14	Map G1 into GL.
The following locking shift functions are available only in VT Level 4 mode.		
LS1R (locking shift 1, right)	ESC ~ 1/11 7/14	Map G1 into GR.
LS2 (locking shift 2)	ESC n 1/11 6/14	Map G2 into GL.
LS2R (locking shift 2, right)	ESC } 1/11 7/13	Map G2 into GR.
LS3 (locking shift 3)	ESC o 1/11 6/15	Map G3 into GL.
LS3R (locking shift 3, right)	ESC 1/11 7/12	Map G3 into GR.

Examples

The following sequence designates the DEC Technical character set as G1, then maps G1 into GL:

ESC) > SO

Designate as G1. Map G1 into GL.

The following sequences designate the ISO Latin-1 supplemental character set as G2, then map G2 into GR:

ESC . A ESC }

Designate as G2. Map G2 into GR.

MC—Media Copy

MC is the control sequence that enables the terminal to control all print functions. There are two variations—ANSI standard and VT mode.

Format

CSI *Pn* i ANSI standard.

9/11 3/*n* 6/9

CSI ? *Pn* i VT mode.

9/11 3/15 3/*n* 6/9

Parameters

Pn (ANSI standard)

indicates the following ANSI standard print functions:

Pn	Action	Notes on Print Page
0	Prints the page that has the cursor.	- The terminal stores data from the keyboard until printing is complete. - If printer extent mode (DECPEX) is currently reset, then the print page function only prints the scrolling region.
2	Sends screen data through host port.	

Pn	Action	Notes on Printer Controller Mode
4	Turns off printer controller mode.	- Printer controller mode lets the host control printer operation. The terminal sends characters and control sequences directly to the printer, without displaying them on the screen. - The terminal sends all characters and control sequences except NUL, XON, XOFF, and the printer controller mode sequences. - Printer controller mode cancels autoprint mode. When the terminal leaves printer controller mode, it returns to the normal method for printing operations. - The printer's active column position should always be on the left margin before the terminal leaves printer controller mode.

Pn	Action
5	Turns on printer controller mode.
6	Disables a printer-to-host session.
7	Enables a printer-to-host session.

Pn (VT mode)

indicates the following VT mode print functions:

Pn	Action	Note on Print Cursor Line
1	Prints the line that has the cursor.	The cursor does not move.
Pn	Action	Notes on Autoprint Mode
4	Turns off autoprint mode.	
5	Turns on autoprint mode.	• The printer prints a line from the screen when you move the cursor off that line with an LF, FF, or VT character, or an autowrap occurs. • The printed line ends with a CR and the character (LF, FF, or VT) that moved the cursor off the previous line.
Pn	Action	Notes on Set/Reset Printer to Host
8	Disables communication from the printer port to the host.	CSI ? 8 i and CSI ? 9 i are media copy commands used in VT terminals to reset/set printer to host mode.
9	Enables communication from the printer port to the host.	
Pn	Action	Note on Print Screen Data
10	Prints the data on the screen.	Printer extent mode (DECPEX) does not affect this function.
Pn	Action	Notes on Print All Pages
11	Prints all pages in page memory.	• If the current page format is 3 pages of 24 lines each, the printer prints 3 pages of 24 lines. • The terminal stores new data from the keyboard until printing is complete. • If print form feed mode (DECPFF) is set, then the terminal sends a form feed (FF) to the printer after each page.

Note on MC Command

The ANSI Escape Sequences CSI 6 i and CSI 7 i to reset/set printer to host mode are functionally equivalent to CSI ? 8 i and CSI ? 9 i, respectively.

NEL—Next Line

Moves cursor to first position on next line. If cursor is at bottom margin, then screen performs a scroll-up.

Format

ESC	E
1/11	4/5

NP—Next Page

This control function moves the cursor forward to the home position on one of the following pages in page memory. If there is only one page, then the terminal ignores NP.

Default: Move to the next page.

Format

```
CSI  Pn  U
9/11 3/n 5/5
```

Parameters

Pn

indicates how many pages to move the cursor forward.

Default: $Pn = 0$.

If Pn is 0 or 1, then the cursor moves to the next page in page memory. If Pn tries to move the cursor past the last page in memory, then the cursor stops at the last page.

PP—Preceding Page

This control function moves the cursor backward to the home position on one of the preceding pages in page memory. If there is only one page, then the terminal ignores PP.

Default: Move to the preceding page.

Format

CSI	<i>Pn</i>	V
9/11	3/ <i>n</i>	5/6

Parameters

Pn

indicates how many pages to move the cursor backward.

Default: $Pn = 0$.

If Pn is 0 or 1, then the cursor moves to the preceding page. If Pn tries to move the cursor back farther than the first page in memory, then the cursor stops at the first page.

PPA—Page Position Absolute

This control function can move the cursor to the corresponding row and column on any page in page memory. You select the page by its number. If there is only one page, then the terminal ignores PPA.

Default: Move to page 1.

Format

CSI	<i>Pn</i>	SP	P
9/11	3/ <i>n</i>	2/0	5/0

Parameters

Pn

is the number of the page to move the cursor to. If *Pn* is greater than the number of the last page in memory, then the cursor stops at the last page. If *Pn* is less than the number of the first page, then the cursor stops at the first page.

PPB—Page Position Backward

This control function moves the cursor backward to the corresponding row and column on one of the preceding pages in page memory. If there is only one page, then the terminal ignores PPB.

Default: Move backward one page.

Format

CSI	<i>Pn</i>	SP	R
9/11	3/ <i>n</i>	2/0	5/2

Parameters

Pn

indicates the number of pages to move the cursor backward. If *Pn* tries to move the cursor back farther than the first page in memory, then the cursor stops at the first page.

PPR—Page Position Relative

This control function moves the cursor forward to the corresponding row and column on one of the following pages in page memory. If there is only one page, then the terminal ignores PPR.

Default: Move to the next page.

Format

CSI	<i>Pn</i>	SP	Q
9/11	3/ <i>n</i>	2/0	5/1

Parameters

Pn

indicates how many pages to move the cursor forward. If *Pn* tries to move the cursor beyond the last page in memory, then the cursor stops at the last page.

RIS—Reset to Initial State

This control function causes a nonvolatile memory (NVR) recall to occur. RIS replaces all set-up features with their saved settings.

The terminal stores these saved settings in NVR memory. The saved setting for a feature is the same as the factory-default setting, unless you saved a new setting.

Note

It is recommended that you not use RIS to reset the terminal. You should use a soft terminal reset (DECSTR) instead. RIS usually causes a communication line disconnect and may change the current baud rate settings. When performing a RIS, the terminal sends XOFF to the host to stop communication. When the RIS is complete, the terminal sends XON to resume communication.

Format

ESC c
1/11 6/3

RIS Actions

- Sets all features listed on set-up screens to their saved settings.
- Causes a communication line disconnect.
- Clears user-defined keys.
- Clears the screen and all off-screen page memory.
- Clears the soft character set.
- Clears page memory. All data stored in page memory is lost.
- Clears the screen.
- Returns the cursor to the upper-left corner of the screen.
- Sets the select graphic rendition (SGR) function to normal rendition.
- Selects the default character sets (ASCII in GL, and DEC Supplemental Graphic in GR).
- Clears all macro definitions.
- Erases the paste buffer.

RM—Reset Mode

There are two formats of this control function. You use the ANSI format to reset one or more ANSI modes. You use the DEC format to reset one or more DEC modes. You cannot reset ANSI and DEC modes with the same RM sequence.

Resetting DEC Modes

CSI	?	<i>Pd</i>	;	...	;	<i>Pd</i>	
9/11	3/15	3/ <i>n</i>	3/11	...	3/11	3/ <i>n</i>	6/12

Parameters

Pd

indicates a DEC mode to reset. [Table 5–8](#) lists the *Pd* values for DEC modes. You can use more than one *Pd* value in a sequence.

Examples

DEC Modes

The following sequence resets (Hebrew) keyboard mapping (DECHEBM) and Hebrew encoding mode (DECHEM):

CSI ? 34; 36 |

34 indicates (Hebrew) keyboard mapping.

36 indicates Hebrew encoding mode.

Programming Tip

Applications can use the SM and RM functions to restore any number of VT510 modes to a desired state. See the Report Mode (DECRPM) section in this chapter for details.

S7C1T—Send C1 Control Character to the Host

The VT510 can send C1 control characters to the host as single 8-bit characters or as 7-bit escape sequences. You should select the format that matches the operating level you are using.

The following sequence causes the terminal to send all C1 control characters as 7-bit escape sequences or single 8-bit characters:

Format

ESC SP F
1/11 2/0 4/7

Description

This sequence changes the terminal mode as follows:

Mode Before	Mode After
VT Level 4 mode, 8-bit controls	VT Level 4 mode, 7-bit controls.
VT Level 4 mode, 7-bit controls	Same. Terminal ignores sequence.
VT Level 1 or VT52 mode, 7-bit controls	Same. Terminal ignores sequence.

S8C1T—Send C1 Control Character to the Host

The following sequence causes the terminal to send C1 control characters to the host as single 8-bit characters:

Format

ESC	SP	G
1/11	2/0	4/6

Description

This sequence changes the terminal mode as follows:

Mode Before	Mode After
VT Level 4 mode, 8-bit controls	Same. Terminal ignores sequence.
VT Level 4 mode, 7-bit controls	VT Level 4 mode, 8-bit controls.
VT Level 1 mode	Same. Terminal ignores sequence.

SCODFK—Define Function Key

Programs function key to send a defined string in SCO Console mode.

Format

ESC Q Fn string
1/11 5/1

Parameters

Fn
is an ASCII code representing the function key being defined. F1, the first key, is represented by "0"; each following key has an ASCII value that increments 1 from the previous key's ASCII value as follows:

Key	Function Key	Fn ASCII Character
1 - 12	F1 through F12	0 - ;
13 - 24	Shift/F1 through Shift/F12	< - G
25 - 36	Ctrl/F1 through Ctrl/F12	H - S
37 - 48	Ctrl/Shift/F1 through Ctrl/Shift/F12	T - _
49	Kpd 7	`
50	Kpd 8	a
51	Kpd 9	b
52	Kpd -	c
53	Kpd 4	d
54	Kpd 5	e
55	Kpd 6	f
56	Kpd +	g
57	Kpd 1	h
58	Kpd 2	i
59	Kpd 3	j
60	Kpd 0	k

String
is the programmed key definition which is enveloped by two characters as the delimiters.

Description

The first character after the ASCII character representing a function key becomes the delimiter. It looks for an identical character as the second and final delimiter. If string is too long and no memory is available, then the command string is aborted and remaining characters are displayed on screen. The character becomes the final character. All the following characters are displayed on the screen.

Example:

Executing "ESC Q 0 'abcdefghijklmnpq" when 100 bytes of memory is available defines the string "abcdefghijkl" into F1 function key and displays "klmnpq" on the screen.

If you execute the same sequence when only a 4 byte string is allowed, then the whole sequence is aborted.

A "^" character causes any following character to subtract 32 from its ASCII value. For example, "^e" in the string sends "E" when the key is pressed.

Any code in the 00-7F region can be defined. An 8-bit code entered is stripped of its eighth bit and defined as a corresponding 7-bit code.

For example, executing ESCQ0'F1KEY' defines string F1KEY for function key F1; ESC1aF2KEYa defines string F2key for F2.

The maximum string length for a defined key is 255 characters.

SCORC—Restore Saved Cursor Position

Moves cursor to the position saved by save cursor command in SCO console mode.

Format

```
CSI  u  
9/11 7/5
```

Description

This command acts the same as the DECRC command, except that the cursor stays at the same page instead of moving to a page where the cursor was saved.

SCOSC—Save Current Cursor Position

Saves current cursor position for SCO console mode.

Format

```
CSI s  
9/11 7/3
```

Description

This command acts the same as the DECSC command, except that the page number where the cursor is located is not saved as a result of executing the command.

SCS—Select Character Set

Designate character sets to G-sets.

Format

ESC / *Dscs*
1/11

Parameters

I
is the intermediate character representing the G-set designator.

I 94-Character G-set

(G0

) G1

* G2

+ G3

I 96-Character G-set

- G1

. G2

/ G3

Dscs
represents a character set designator.

Dscs	Default 94-Character Set
% 5	DEC Supplemental
" ?	DEC Greek
" 4	DEC Hebrew
% 0	DEC Turkish
& 4	DEC Cyrillic
A	U.K. NRCS
R	French NRCS
9 or Q	French Canadian NRCS
`, E, or 6	Norwegian/Danish NRCS
5 or C	Finnish NRCS
K	German NRCS
Y	Italian NRCS
=	Swiss NRCS
7 or H	Swedish NRCS
Z	Spanish NRCS
% 6	Portuguese NRCS
" >	Greek NRCS
% =	Hebrew NRCS
% 2	Turkish NRCS
% 3	SCS NRCS
& 5	Russian NRCS
0	DEC Special Graphic
>	DEC Technical Character Set
<	User-preferred Supplemental
Dscs	Default 96-Character Set
A	ISO Latin-1 Supplemental
B	ISO Latin-2 Supplemental
F	ISO Greek Supplemental
H	ISO Hebrew Supplemental
M	ISO Latin-5 Supplemental
L	ISO Latin-Cyrillic
<	User-preferred Supplemental

SD—Pan Up

This control function moves the user window up a specified number of lines in page memory.

Format

CSI	<i>Pn</i>	T
9/11	3/ <i>n</i>	5/4

Parameters

Pn

is the number of lines to move the user window up in page memory. *Pn* new lines appear at the top of the display. *Pn* old lines disappear at the bottom of the display. You cannot pan past the top margin of the current page.

SGR—Select Graphic Rendition

This control function selects one or more character attributes at the same time.

Default: Clear all attributes.

Format

```
CSI  Ps  ;   Ps  ... m
9/11 3/n 3/11 3/n ... 6/13
```

Parameters

Ps

is a number representing a certain visual attribute. You can use more than one *Ps* value to select different character attributes. [Table 5–16](#) lists *Ps* values and the attributes they select.

Default: *Ps* = 0 (clears all attributes).

Table 5–16 Visual Character Attribute Values

Ps	Attribute
0	All attributes off
1	Bold
4	Underline
5	Blinking
7	Negative image
8	Invisible image
10	The ASCII character set is the current 7-bit display character set (default)— <i>SCO Console only</i> .
11	Map Hex 00-7F of the PC character set codes to the current 7-bit display character set— <i>SCO Console only</i> .
12	Map Hex 80-FF of the current character set to the current 7-bit display character set— <i>SCO Console only</i> .
22	Bold off
24	Underline off
25	Blinking off
27	Negative image off
28	Invisible image off

Examples

When you select more than one attribute in an SGR sequence, then they are executed in order. For example, you can use the following sequence to display text that is bold, blinking, and underlined:

```
CSI 0 ; 1 ; 5 ; 4 m
```

The following sequence displays the negative image of text:

```
CSI 7 m
```

Notes on SGR

- After you select an attribute, the terminal applies that attribute to all new characters received. If you move characters by scrolling, then the attributes move with the characters.

- If you display control characters, then the terminal ignores the bold attribute for displayed control characters.

When a PC character set is selected . . .	Displays . . .
Executing "CSI 10 m"	00-1F: control codes 20-7F: ASCII characters 80-FF: 8 bit PC characters
Executing "CSI 11 m"	00-1F: PC characters, except the following codes when XON/XOFF is enabled: 1B (ESC) 11 (DC1) 13 (DC3) 20-7F: ASCII characters (or PC 7 bit character) 80-FF: 8 bit PC characters
Executing "CSI 12 m"	00-1F: PC characters that are located in 80 - 9F in PC character set, except the following codes when XON/XOFF is enabled: 9B (ESC) 91 (DC1) 93 (DC3) 20-7F: PC characters located in A0-FF 80-FF: 8 bit PC characters.

When an ISO/ANSI character set is selected . . .	Displays . . .
Executing "CSI 10 m"	00-1F: control codes 20-7F: ASCII characters 80-9F: control code A0-FF: GR characters
Executing "CSI 12 m"	00-1F: control codes 20-7F: ISO/ANSI characters located in A0-FF 80-9F: control code A0-FF: GR characters

Commands "CSI 10-12 m" affect only the Hex 00-7F portion of the display character set. The hex 80-FF region of the display character set is left intact.

The ASCII character set with control codes residing in Hex 00 to 1F region is the default 7-bit display character set. This is true regardless of the ISO/ANSI or PC character set. When in an ISO/ANSI character set, issuing "CSI 10 m" is equivalent to "designating and invoking ASCII to G0 and GL."

When "ESC 11 m" is executed, the display character set is loaded with codes in the Hex 00 to 7F region of a PC character set. PC characters whose code values are less than 1F can be displayed through this sequence except 1B (ESC) and 11(DC1), 13(DC3) when XON/XOFF is enabled. 1B is always executed as an ESC to allow the application to execute the command to go back to the default character set. Hex 11 and 13 can be displayed only when XON/XOFF is disabled. This command does not work when the ISO/ANSI character set is selected.

Command "ESC 12 m" toggles the high bit of the current 8-bit character set. All the characters in Hex 80-FF region can be displayed as 7-bit codes except 9B (ESC) and 91(DC1), 93(DC3) when XON/XOFF is enabled. After the command is executed, 1B is executed as an ESC. Hex 11 and 13 can be displayed only when XON/XOFF is disabled.

When in an ISO/ANSI character set, issuing "CSI 12 m" is equivalent to "designating and invoking current 8 bit char set to G2 and GL."

SM—Set Mode

This control function has two formats. You use the ANSI format to set one or more ANSI modes. You use the DEC format to set one or more DEC modes. You *cannot* set ANSI and DEC modes with the same SM sequence.

Setting DEC Modes

CSI	?	<i>Pd</i>	;	...	<i>Pd</i>	h
9/11	3/15	3/ <i>n</i>	3/11	...	3/ <i>n</i>	6/8

Parameters

Pd

indicates a DEC mode to set. [Table 5–8](#) lists the *Pd* values for DEC modes. You can use more than one *Pd* value in a sequence.

Examples

DEC Modes

The following sequence sets (Hebrew) keyboard mapping (DECHEBM) and Hebrew encoding mode (DECHEM):

CSI ? 34; 36 h

34 indicates (Hebrew) keyboard mapping.

36 indicates Hebrew encoding mode.

Programming Tip

Applications can use the SM and RM functions to restore any number of VT510 modes to a desired state. See the Report Mode (DECRPM) section in this chapter for details.

SRM—Local Echo: Send/Receive Mode

This control function turns local echo on or off. When local echo is on, the terminal sends keyboard characters to the screen. The host does not have to send (echo) the characters back to the terminal display. When local echo is off, the terminal only sends characters to the host. It is up to the host to echo characters back to the screen.

Default: No local echo

Format

CSI	1	2	h	Set: local echo off.
9/11	3/1	3/2	6/8	
CSI	1	2	l	Reset: local echo on.
9/11	3/1	3/2	6/12	

Description

When the SRM function is set, the terminal sends keyboard characters to the host only. The host can echo the characters back to the screen.

When the SRM function is reset, the terminal sends keyboard characters to the host and to the screen. The host does have to echo characters back to the terminal.

SS—Single Shifts

You use a single shift when you want to display the next character from a different character set. A single shift maps the G2 or G3 set into GL. The character set is active for only one character, then the terminal returns to the previous character set in GL.

The terminal has two single-shift control functions available.

Format

Single-Shift Control	8-Bit Character	7-Bit Equivalent Sequence	Function
Single shift 2	SS2 8/14	ESC N 1/11 4/14	Maps G2 into GL for the next character.
Single shift 3	SS3 8/15	ESC O 1/11 4/15	Maps G3 into GL for the next character.

Example

Suppose the ASCII character set is in GL. You want to display the alpha character from the DEC Technical character set, already designated as G3. You do not want to replace the ASCII set just to display one character. Instead, you can use single shift 3 to temporarily map the DEC Technical set (G3) into GL.

SS3 a
single shift 3 alpha character

After displaying the alpha character, the terminal maps the ASCII set (G1) back into GL, replacing the DEC Technical set (G3).

SU—Pan Down

This control function moves the user window down a specified number of lines in page memory.

Format

CSI	<i>Pn</i>	S
9/11	3/ <i>n</i>	5/3

Parameters

Pn

is the number of lines to move the user window down in page memory. *Pn* new lines appear at the bottom of the display. *Pn* old lines disappear at the top of the display. You cannot pan past the bottom margin of the current page.

TBC—Tab Clear

This control function clears tab stops.

Format

CSI	<i>Ps</i>	<i>g</i>
9/11	3/ <i>n</i>	6/7

Parameters

Ps

indicates the tab stops to clear. There are only two values for *Ps*, 0 and 3.

0 or none (default)	The terminal only clears the tab stop at the cursor.
------------------------	--

3	The terminal clears all tab stops.
---	------------------------------------

VPA—Vertical Line Position Absolute

VPA inquires as to the amount of free memory for programmable key operations.

Format

CSI	<i>Pn</i>	d
9/11	3/ <i>n</i>	6/4

Parameters

Pn
is column number.

Description

VPA causes the active position to be moved to the corresponding horizontal position.

The default value is 1.

Move cursor to line *Pn*. VPA causes the active position to be moved to the corresponding horizontal position at vertical position *Pn*. If an attempt is made to move the active position below the last line, then the active position stops on the last line.

VPR—Vertical Position Relative

VPR inquires as to the amount of free memory for programmable key operations.

Format

CSI *Pn* e
9/11 3/*n* 6/5

Parameters

Pn
is column number.

Description

VPR causes the active position to be moved to the corresponding horizontal position.

This command causes the active position to be moved to the corresponding horizontal position at n-th following vertical position. If an attempt is made to move the active position below the last line, the active position stops at the last line.

SCO Console Emulation

This chapter describes the SCO Console mode. This chapter also describes the effects of switching from SCO Console to other emulation modes, including VT, PCTerm, and ASCII Terminal emulation.

6.1 SCO Console Emulation Mode

SCO Console mode is a terminal emulation mode. It uses ANSI sequences with minor modifications to run on a PC, which acts as a SCO UNIX operating system console.

Depending on the terminal status, SCO Console mode for the VT510 transmits ASCII characters (default) or scancode to the host when keys are pressed. Users can specify a keycode by issuing an escape sequence (DECPCTERM) or by keyboard encoding of Set-Up. When in SCO Console mode with the ASCII keycode selected, the terminal behavior is similar to VT mode. When in SCO Console mode with scancode selected, the terminal behavior is similar to VT420 PCTerm mode.

Table 6–1 shows the similarities and differences.

Table 6–1 SCO Console Mode with ASCII Keycodes and Scancodes

Default	SCO Console Mode with		VT Mode	PCTerm
	ASCII Keycodes	Scancodes		
Screen display	25×80	25×80	24×80	24×80
Autowrap	On	On	Off	Off
Autowrap behavior	Like WYSE	Like WYSE	Like VT	Like VT
Status line	None	None	Local	Local
Cursor position while switching pages	Saved	Saved	Not saved	Not saved
Character set	PCTerm (including PC, ISO/ANSI character sets)		VT character set (ISO/ANSI)	
CSI 10-12 m	Supported	Supported	Not supported	Not supported
SU	Acts like IND	Acts like IND	Acts like SU	Acts like SU
SD	Acts like RI	Acts like RI	Acts like SD	Acts like SD
CSI s and CSI u	Supported	Supported	Only DECSC and DECRC are supported.	
DECSC and DECRC	Supported	Supported		
CSI Pn z (switch pages)	Supported	Supported	Not supported	Not supported
ESC Q Fn String (Define Function)	Supported	Supported	Not supported	Not supported
Transmit flow control	None	None	XON/XOFF	XON/XOFF

6.1.1 Keyboard

SCO Console mode works with either a PC keyboard or a VT keyboard attached.

6.1.1.1 Local Function Keys in SCO Console Mode

The keys used to perform local functions in SCO Console mode are similar to those used for VT mode (see Chapter 8, [Keyboard Processing](#). [Table 6–2](#) shows the [exceptions](#)).

Table 6–2 Local Functions on VT and Enhanced PC Layouts

Fn#	Function	VT Layout	EPC Layout
1	Hold	Lock/F1	Scroll Lock
2	Print	Lock/F2	Print Screen
3	Set-Up	Lock/F3	Caps Lock/Print Screen, Alt/Print Screen
5	Break	Lock/F5	Caps Lock/Pause
7	Hard Reset	Lock/Ctrl/F3	Ctrl/Caps Lock/Print Screen
8	Toggle Autoprint	Lock/Ctrl/F2	Ctrl/Print Screen
9	Disconnect	Lock/Shift/F5	Shift/Caps Lock/Pause
10	Send Answerback	Lock/Ctrl/F5	Ctrl/Caps Lock/Pause
11	Print Composed Main Display	Not supported	
20	Pan Up	Not supported	
21	Pan Down	Not supported	
24	Pan Prev Page	Not supported	
25	Pan Next Page	Not supported	
30	Copy and Paste	Lock/F1 (hold down)	Scroll Lock
47	Primary KB language	Lock/Ctrl/Alt/F1	Caps Lock/Ctrl/Alt/F1
48	Secondary KB language	Lock/Ctrl/Alt/F2	Caps Lock/Ctrl/Alt/F2
53	±DEC/IBM PC Style	Not supported	

6.1.1.2 PC Keyboard

Tables 6–3 through 6–8 list the default codes sent by keys in ASCII character code mode.

The keys that can be affected by Caps Lock or Num Lock send a shifted value when in Lock state. If **Shift** is pressed when Caps Lock or Num Lock is in effect, the key sends its unshifted value. This "Reverse Lock" is the default for a Caps Lock function.

Alt has no effect on the key pressed unless otherwise indicated. A blank in any column means the key is not defined.

**Table 6–3 SCO Console Mode Function Key
Default Codes**

Key	Unmodified	Shift	Ctrl	Ctrl/Shift
F1	ESC[M	ESC[Y	ESC[k	ESC[w
F2	ESC[N	ESC[Z	ESC[l	ESC[x
F3	ESC[O	ESC[a	ESC[m	ESC[y
F4	ESC[P	ESC[b	ESC[n	ESC[z
F5	ESC[Q	ESC[c	ESC[o	ESC[@
F6	ESC[R	ESC[d	ESC[p	ESC[[
F7	ESC[S	ESC[e	ESC[q	ESC[<
F8	ESC[T	ESC[f	ESC[r	ESC[]
F9	ESC[U	ESC[g	ESC[s	ESC[^
F10	ESC[V	ESC[h	ESC[t	ESC[_
F11	ESC[W	ESC[i	ESC[u	ESC['
F12	ESC[X	ESC[j	ESC[v	ESC[{
Print Screen	Print Screen	ESC[0i		
Scroll Lock	Scroll Lock	Scroll Lock	N/A	N/A
Pause	Break	Break	N/A	N/A

**Table 6–4 SCO Console Mode Main Keypad
Default Codes**

Key	Unmodified	Shift	Ctrl	Ctrl/Shift
ESC	ESC	ESC	ESC	ESC
1	1	!	1	!
2	2	@	Nul	Nul
3	3	#	ESC	ESC
4	4	\$	FS	FS
5	5	%	GS	GS
6	6	^	RS	RS
7	7	&	US	US
8	8	*	DEL	DEL
9	9	(		
0	0	)		
-	-	_	US	US
=	=	+		
Backspace	BS	BS	DEL	DEL
Tab	HT	ESC[Z		
q	q	Q	DC1	DC1
w	w	W	ETB	ETB
e	e	E	ENQ	ENQ
r	r	R	DC2	DC2
t	t	T	DC4	DC4
y	y	Y	EM	EM
u	u	U	NAK	NAK
i	i	I	HT	HT
o	o	O	SI	SI
p	p	P	DLE	DLE
[	[	{	ESC	ESC
]	]	}	GS	GS
\	\		FS	FS
Return	CR	CR	LF	LF
Ctrl	Ctrl	Ctrl	Ctrl	Ctrl
a	a	A	SOH	SOH
s	s	S	DC3	DC3
d	d	D	EOT	EOT
f	f	F	ACK	ACK
g	g	G	BEL	BEL
h	h	H	BS	BS
j	j	J	LF	LF

Key	Unmodified	Shift	Ctrl	Ctrl/Shift
k	k	K	VT	VT
l	l	L	FF	FF
;	;	:		
'	'	"		
z	z	Z	SUB	SUB
x	x	X	CAN	CAN
c	c	C	ETX	ETX
v	v	V	SYN	SYN
b	b	B	STX	STX
n	n	N	SO	SO
m	m	M	CR	CR
,	,	<		
.	.	>		
/	/	?	US	US

Table 6–5 SCO Console Mode Editing Keypad Default Codes

Key	Unmodified	Shift	Ctrl	Ctrl/Shift
Insert	ESC[L	ESC[L	ESC[L	ESC[L
Home	ESC[H	ESC[H	ESC[H	ESC[H
Page Up	ESC[I	ESC[I	ESC[I	ESC[I
Delete	DEL	DEL	DEL	DEL
End	ESC[F	ESC[F	ESC[F	ESC[F
Page Down	ESC[G	ESC[G	ESC[G	ESC[G

Table 6–6 SCO Console Mode Cursor Keypad Default Codes

Key	Unmodified	Shift	Ctrl	Ctrl/Shift
↑	ESC[A	ESC[A	ESC[A	ESC[A
↓	ESC[B	ESC[B	ESC[B	ESC[B
⇒	ESC[C	ESC[C	ESC[C	ESC[C
⇐	ESC[D	ESC[D	ESC[D	ESC[D

Table 6–7 SCO Console Mode Numeric Keypad Default Codes, Num Lock Off

Keypad Function Key	Unmodified	Shift	Ctrl	Ctrl/Shift
Num Lock	Num Lock	Num Lock	DC3	DC3
/	/	/		
*	*	*		
7	ESC[H	7	ESC[H	7
8	ESC[A	8	ESC[A	8
9	ESC[I	9	ESC[I	9
-	-	-	-	-
4	ESC[D	4	ESC[D	4
5	ESC[E	5	ESC[E	5
6	ESC[C	6	ESC[C	6
+	+	+	+	+
1	ESC[F	1	ESC[F	1
2	ESC[B	2	ESC[B	2
3	ESC[G	3	ESC[G	3
0	ESC[L	0	ESC[L	0
.	DEL	.	DEL	DEL
Enter	CR	CR	LF	LF

Table 6–8 SCO Console Mode Numeric Keypad Default Codes, Num Lock On

Keypad Function Key	Unmodified	Shift	Ctrl	Ctrl/Shift
Numlock	Numlock	Numlock	DC3	DC3
/	/	/		
*	*	*		
7	7	ESC[H		
8	8	ESC[A		
9	9	ESC[I		
-	-	-	-	-
4	4	ESC[D		
5	5	ESC[E		
6	6	ESC[C		
+	+	+		
1	1	ESC[F		
2	2	ESC[B		
3	3	ESC[G		
0	0	ESC[L		
.	.	DEL		
Enter	CR	CR		

Note

Pressing **Alt/.** on the numeric keypad sends DEL, not a period.

6.1.1.3 VT Keyboard Key Mapping

More key mapping information is provided in Chapter 8, Keyboard Processing.

- On the main keypad, 8-bit codes are allowed for keyboard input.
- Function keys **F1** through **F12** send the same codes as **F1** through **F12** on a PC keyboard. Function keys **F13** through **F20** send the same codes as **F13** through **F20** in VT mode, except **Do** sends an ESC code.
- On the editing keypad, VT keyboard keys send the same sequences as the corresponding PC keyboard keys:

PC Keyboard	VT Keyboard
Insert	Insert Here
Home	Find
PageUp	Prev
Delete	Remove
End	Select
PageDown	Next

- For Autowrap, On is the default.
- On the cursor keypad, SCO Console mode sends the same sequence as VT510 mode.
- The cursor position is saved while switching pages.
- On the numeric keypad, SCO Console mode sends the same sequence as VT510 mode.
- SCO Console mode uses the PCTerm character set.

This chapter describes the character sets supported by the VT510 video terminal.

7.1 Overview

The VT510 video terminal supports most character sets for Eastern Europe; Cyrillic, Greek, Hebrew, and Turkish character sets. In addition to traditional DEC and ISO character sets, which conform to the structure and rules of ISO 2022, the VT510 supports a number of IBM PC *code pages* (page numbers in IBM's standard character set manual) in PCTerm mode to emulate the console terminal of industry-standard PCs.

Each code page defines both the exact repertoire (set of characters) and the assignment of code positions in the 0 - 255 range. Once configured, the code page is not usually changed within a single application.

7.2 Character Sets Supported

The following sections list the characters sets supported by the VT510 terminal.

7.2.1 VT Graphic Character Sets

The VT510 supports the following VT 8-bit multinational character sets:

- ISO Latin-1
- ISO Latin-2
- ISO Latin-Cyrillic
- ISO Latin-Greek
- ISO Latin-Hebrew
- ISO Latin-5
- KOI-8 Cyrillic
- DEC Multinational
- DEC Greek
- DEC Hebrew
- DEC Turkish

The VT510 also supports the following character sets:

- DEC Special Graphics (VT100 Line Drawing set)
- DEC Technical Character

When 7-bit NRCS Characters is selected, a corresponding 7-bit or NRC set is used depending on the keyboard language selected. (See [Chapter 8](#) for details.)

- U.K. NRC (ISO)
- French NRC (ISO)
- DEC French Canadian NRC
- Norwegian/Danish NRC (ISO)
- DEC Finnish NRC
- German NRC (ISO)

- Italian NRC (ISO)
- DEC Swiss NRC
- Swedish NRC (ISO)
- Spanish NRC (ISO)
- DEC Portuguese NRC
- SCS NRC
- Russian 7-bit (KOI-7)
- DEC 7-bit Greek
- DEC 7-bit Hebrew
- DEC 7-bit Turkish

7.2.2 PCTerm Graphic Character Sets

When PCTerm mode is selected, the terminal can still use all the VT 8-bit character sets listed in [Section 7.2.1](#). Additionally, the following PC code pages are supported:

- PC International (437)
- PC Multilingual (850)
- PC Slavic (852)
- PC Greek (210)
- PC Spanish (220)
- PC Turkish (857)
- PC Portuguese (860)
- PC Hebrew (862)
- PC French/Canadian (863)
- PC Danish/Norwegian (865)
- PC Cyrillic (866)

Note

7-bit NRC sets are not available in PCTerm mode.

7.2.3 Control Character Sets

The following control character sets are defined in the terminal. ASCII C0 and C1 control sets have corresponding fonts for graphic representation in control representation mode.

- ASCII C0 Control Set (32 characters)
- 8-bit ASCII C1 Control Set (32 characters)

The control representation mode font includes C0 and C1 control character set graphic representations plus DELETE (7/15), non-breaking space - NBSP (10/0), and 15/15 (total of 67 characters).

7.2.4 Character Set Structure

The VT 8-bit multinational character sets are formed from two 7-bit character set "halves." ASCII is used as the left half (GL), and the corresponding named supplemental set is used as the right half (GR). Although the character set "halves" can be designated and invoked independently, in practice, they are almost always used together. It is simpler and more convenient to think of them as a single 8-bit multinational set, although technically they are also separate 7-bit sets.

Many characters appear in more than one of the mentioned sets, so the total number of characters is far less than the sum of the individual totals. There is also a down-line loadable portion of

character RAM provided for a custom soft font, a Dynamically Redefinable Character Set (DRCS). This character RAM can contain up to 2×96 characters.

7.2.5 Characters Not in a Character Set.

The following characters are not technically part of any of character set. They are like "control characters," but they are not in the C0 or C1 control sets.

Name	Description	Code	Keystroke	Action on Receipt
SP	SPACE	32	Space Bar	Treated as space
DEL	DELETE	127	Delete Ctrl-8	Ignored (unless in CRM)
	Reserved	160	None	Error character generated
	Reserved	255	None	Ignored (unless in CRM)

Notes

NULL is the only fill character. (DEL and 255 are not usable as fill characters.)

If a 96-character character set is invoked into GL, then the receipt of codes 32 and 127 are part of that character set, and their graphic characters are displayed from the 96-character set. In this case, the codes 32 and 127 are not treated as SP or DEL.

If a 96-character character set is invoked into GR, the receipt of codes 160 and 255 are part of that character set, and their graphic characters are displayed from the 96-character character set. In this case, the codes 160 and 255 are not treated as an error condition or ignored, respectively.

7.3 VT Mode

All DEC character sets are supported in VT500 mode with either a VT keyboard or an attached Enhanced PC keyboard.

In VT100 mode, VT52 mode, or when 7-bit NRCS characters is selected (through Set-Up or DECNRCM), only ASCII, NRC sets, and DEC Special Graphic characters are available.

7.3.1 Selecting Character Sets

Character sets are designated with the select character set (SCS) control function. Refer to [Chapter 5](#) for more details.

7.3.2 Character Set Mappings

The following table lists the character set mappings.

Character Set	Mapped onto decimal . . .
C0 controls	0 to 31
Space	32
ASCII, all other NRC sets, Line Drawing, and DEC Technical set	33 to 126
DEL	127
C1 controls	128 to 159
10/0	160
DEC Multinational Graphic, DEC 7-Bit Greek, DEC 7-Bit Hebrew, or DEC 7-Bit Turkish	161 to 254
15/15	255
ISO Latin-1 Supplemental Graphic, ISO Latin-2 Supplemental Graphic, ISO Latin-Cyrillic, ISO Latin-Greek, ISO Latin-Hebrew, or ISO Latin-5.	160 to 255

These are typical assignments. By appropriate designating and invoking sequences, other mappings are possible for the graphic character sets.

7.4 PCTerm Mode

This section describes how to select character sets in PCTerm mode.

7.4.1 Selecting Character Sets

If a PC character set is selected, then all the SCS sequences used to designate and invoke character sets into G0, G1, G2, G3, GL, and GR (including single and locking shifts) are ignored. You select the character set through a Set-Up field or an escape sequence, DECPCTERM. You can use only one code page containing 256 characters at a time.

7.4.2 Character Set Mappings

In PC character sets, the conventional ISO C1 region, (80 hex through 9F hex) are all displayable character codes. The VT510 does not recognize control sequences that have 8-bit control codes; therefore, the 8-bit control mode becomes invalid.

C0 and C1 regions are all control codes. The C0 region (00 hex through 1F hex) in a PC character set is mixed with control characters and graphic characters. Among them, 11 locations can have both *control characters* and *displayable characters* in the same locations; other locations are occupied by displayable characters only. The 11 C0 locations include 0/5, 0/7, 0/8, 0/9, 0/A, 0/B, 0/C, 0/D, 1/1, 1/3, and 1/B.

Control codes are used for ENQ, BEL, BS, HT, LF, VT, FF, CR, DC1, DC3, and ESC operations.

You access the *displayable characters* by preceding the C0 control code with ESC to form a 2-byte pair (1Bh XXh). These 2-byte pairs, such as ESC 0/5, are only recognized in PCTerm mode and SCO console mode.

The C0 characters may also be displayed using the SGR sequence "CSI 11 m".

7.5 SCO Console Mode

SCO console mode can use both VT character sets and PC character sets. You select a VT character set through the PCTerm character set field in Set-Up, or you can use an SCS invocation sequence. You can select a PC character set through the same Set-Up field, or you can use the DECPCTERM escape sequence.

7.6 Soft Character Sets

The ANSI control string command to down-line load the soft character sets is DECDDL. The DECDDL command down-line loads one or more characters of a specified 94- or 96-character dynamically redefinable character set (DRCS) with a specified logical pixel pattern. The affected characters in the DRCS set also change the presentation of characters from the set currently being displayed on the screen. The DECDDL control does not clear the screen.

See [Chapter 5](#) for a description of the DECDDL—Dynamically Redefinable Character Sets command.

Keyboard Processing

This chapter describes the physical keyboards and keyboard-to-host interface supported by the VT510 video terminal.

8.1 Overview

The VT510 supports two keyboard layouts:

- VT keyboard (VT layout)
- Enhanced PC 101/102 keyboard (PC layout)

Any time the terminal is operating, a keyboard encoding is being used to translate graphic character keystrokes into character codes, called the keyboard character set. The Keyboard Character set is a function of the Keyboard Dialect, the default character set, the 7-bit or 8-bit characters setting, and any keyboard customization defined in terminal Set-Up.

When the graphic character assigned to a keystroke (which usually corresponds to the legend appearing on the keycap) has a corresponding character in the Keyboard Character set, the terminal transmits the corresponding character code for that keystroke. The code transmitted for any given keystroke can vary depending on the Keyboard Character set. If the graphic character assigned to a keystroke does not have a corresponding character in the Keyboard Character set, then that keystroke becomes dead.

The term keyboard layout describes the physical layout of a keyboard, specifically, the number of keys and their positions on the keyboard. The keys are marked with symbols which indicate their function. Graphic keys transmit codes that match the graphic character symbol on the key cap, whereas keys such as the `Tab` key transmit control codes. Modifier keys (`Shift`, `Ctrl` and `Alt`) change the behavior of other keys. Function keys transmit multi-byte sequences of control and character data; local function keys enable local terminal functions (For example, `Set-Up`, and `Print Screen`).

8.2 PC Keyboard—VT Style vs. PC Style

The VT510 is designed to operate with either a VT keyboard or an enhanced PC keyboard. These keyboards differ in the number and positioning of keys. This leads to different user expectations of where function and local function keys are positioned on the PC keyboard. For example, traditional VT keyboard users expect `F2` and `F3` to enable local functions for printing the screen and entering Set-Up. Traditional PC keyboard users expect that these same keys will send signals to the host to control the application. These keyboard layouts are called VT style and PC style layouts. The PC style layout is the factory default.

VT Style is an alternate way of mapping the PC keyboard to look like a VT keyboard. With this mapping, PC keys send the same function sequences as similarly named keys on the VT layout (Figure 8–1). Since the enhanced PC layout does not correspond key-for-key to the VT Layout, some adaptation is necessary. Some function keys, that are missing from the PC keyboard, can be created by pressing a Function Key in combination with `Caps Lock` (For example, press and hold `Caps Lock` while typing another key).

Pressing `Caps Lock/Num Lock` toggles between the VT Style and PC Style. When VT Style is selected, "VT" is displayed on the keyboard indicator line at the bottom of the screen. The style is saved in NVM when you select Save settings. VT Style and PC Style mappings apply to VT modes only.

Figure 8–1 VT Keyboard Layout

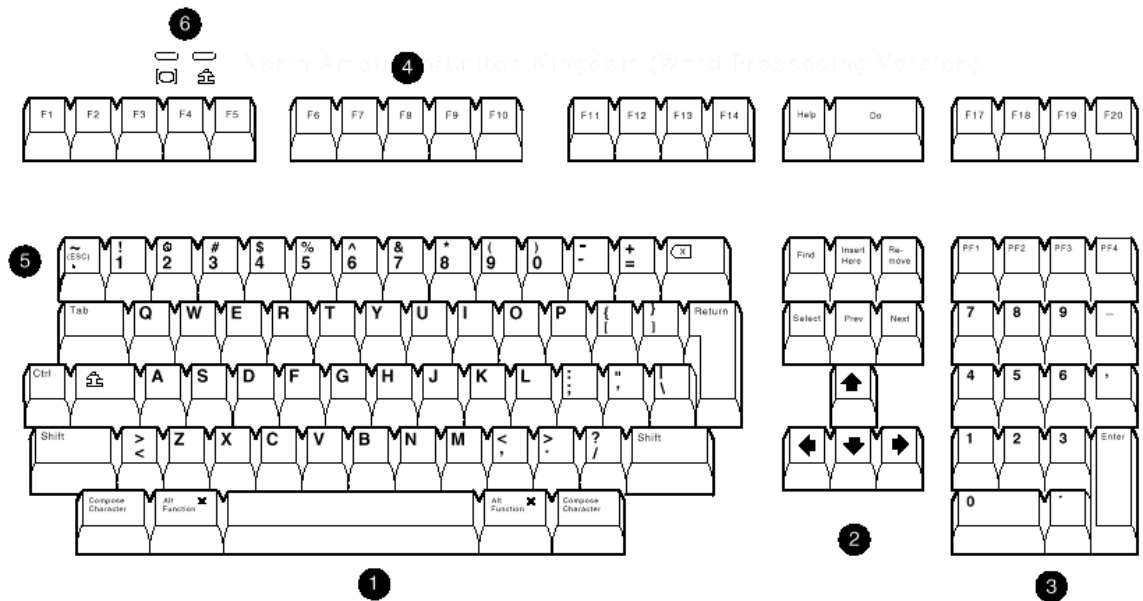
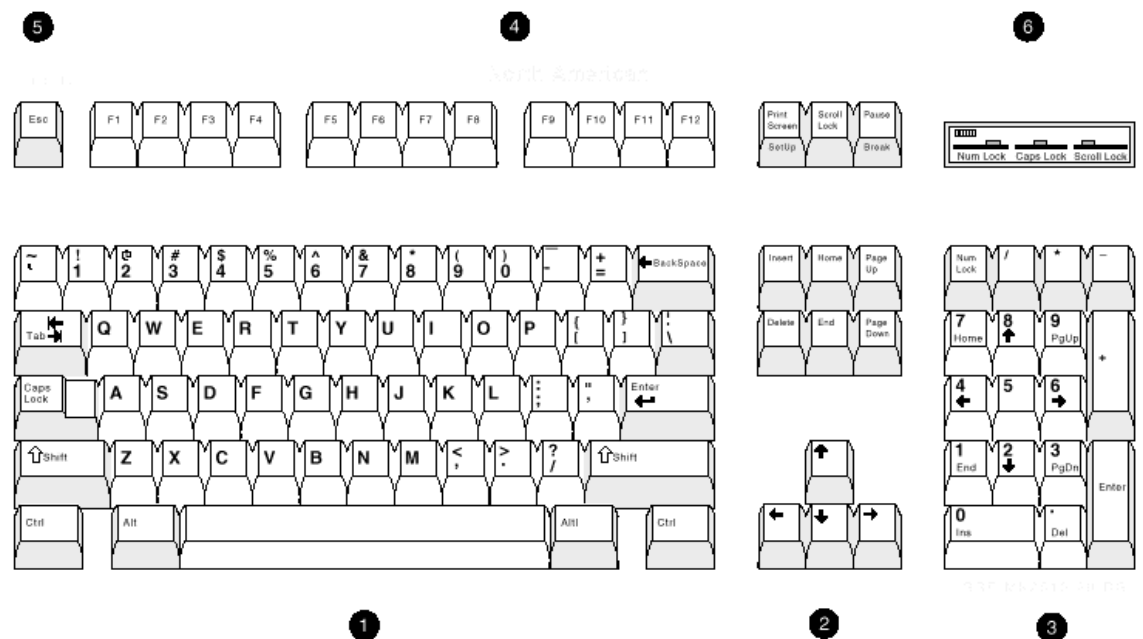


Figure 8–2 PC Keyboard Layout



- (1) Main keypad, (2) Editing keypad, (3) Numeric keypad,
(4) Function keys, (5) Escape key, (6) Indicator lights.

8.2.1 Differences—PC Style vs. VT Style

The differences between PC Style and VT Style are:

- Editing keypad keys **Home** and **Delete** send different sequences corresponding to their functions in PC Style and VT Style.
- Numeric keypad keys send different sequences when the keypad is in numeric mode to accommodate the **Num Lock** feature. With **Num Lock** off, the numeric keypad sends editing key sequences.
- Top row function keys **F1** through **F5** send function key sequences. Local terminal functions are relocated to the **Print Screen**, **Scroll Lock**, and **Pause** keys on the enhanced PC layout.
- **Print Screen**, **Scroll Lock**, and **Pause** are either local function keys or function keys in PC style, but are only function keys in VT Style.

- The keys to operate the Copy/Paste function are different between the two styles (See [Table 8–5](#)).
- Left `Alt/Space` works as compose in VT Style only.

8.3 Top Row Function Keys

There are 20 top row function keys on the VT layout and 12 on the enhanced PC layout.

`Caps Lock/F1` through `Caps Lock/F10` keys on the enhanced PC layout transmit the same sequences as `F11` through `F20` in the VT layout. Additional unique function sequences are transmitted by pressing function keys in combination with `Shift`, `Alt`, or `Ctrl` keys. See DECFNK and DECPAK for details.

In VT Style, the `F1` through `F5` keys operate the same as on the VT layout. In VT Style, `F13` (Print Screen), `F14` (Scroll Lock), and `F15` (Pause) send their corresponding function key sequences. In PC Style, the `F1` through `F5` keys send function sequences (unless re-programmed) while `Print Screen`, `Scroll Lock`, and `Pause` default to local functions.

8.4 Main Key Array

This section describes the main keys on the keyboard.

8.4.1 Character Sets

Only characters from the currently selected keyboard character set may be generated. All other graphic character keystrokes are ignored. The following keyboard character sets are supported: ISO Latin 1, ISO Latin 2, ISO Latin Cyrillic, ISO Latin Greek, ISO Latin Hebrew, ISO Latin 5 (Turkish), DEC Multinational, DEC Cyrillic, DEC Greek, DEC Hebrew and DEC Turkish. When 7-bit NRCS characters is selected, a corresponding 7-bit NRC set is supported based on the Keyboard Dialect.

PC character sets are not supported in VT modes since the terminal is being used as a conventional video terminal that happens to have a PC keyboard attached. New applications should be written to use ISO character sets. (PC character sets are available in PCTerm and SCO console mode.)

8.4.2 Caps Lock

The Lock function may operate in three ways:

Caps Lock Alphabetic keys send their shifted character.

Shift Lock All keys on the main key array send their shifted character.

Reverse Lock Pressing Shift temporarily reverses the effect of `Caps Lock`; that is, unshifted or lowercase alphabetic characters are transmitted.

Pressing and releasing the `Caps Lock` key toggles the lock state on or off.

The `Caps Lock` key is also used to generate keystrokes for keys that are not present on the keyboard. Known as keyboard extension, this is done by pressing and holding the `Caps Lock` key and then pressing the extension key. The lock state is not toggled when an extension keystroke is pressed. Extension keystrokes are shown in [Table 8–5, Local Functions](#).

8.4.3 Tab Key

With an enhanced PC keyboard, pressing `Shift/Tab` causes a CBT sequence, CSI Z, to be transmitted. A horizontal tabulation code (HT, 0/9) is transmitted when a VT keyboard is used.

8.4.4 Left Alt

Left `Alt` on the PC layout corresponds to the `Alt Function` on the VT layout. Depressing the left `Alt` key performs the following functions (unless re-programmed):

- Left `Alt/Space` acts as the compose key in VT Style.
- On the North American PC layout, left `Alt` is the same as `Alt Gr` when pressed and held in combination with keys from the numeric keypad. This feature keeps the two `Alt` keys equivalent to each other.

8.4.5 Right Alt

The right **Alt** key appears on the North American, Hebrew, and Greek PC layouts. On other country layouts, the right **Alt** key appears as **Alt Gr**. Depressing the right **Alt** key performs the following functions (unless re-programmed):

- Right **Alt/Space** acts as the compose key in VT Style. This feature keeps the two **Alt** keys equivalent.
- Right **Alt** key is the same as the **Alt Gr** key when pressed and held in combination with keys from the numeric keypad.
- The right **Alt** key is the same as the Right Alt function on the VT layout with the exceptions listed previously.

8.4.6 Alt Gr

The right **Alt** key is marked as **Alt Gr** on some Enhanced PC layouts. The words Alt Gr are short for alternate graphic and corresponds roughly to the function of the compose key.

- **Alt Gr** in combination with graphic character keys generate the character corresponding to the right side of the keyboard.
- **Alt Gr** is used for numeric keypad compose.

8.4.7 Modifier Keys

Table 8–1 shows the corresponding modifier keys for the VT and the enhanced PC layouts.

Table 8–1 VT Layout vs. Enhanced PC Layout Modifier Keys	
VT Keyboard	PC Keyboard
Left Shift	Left Shift
Right Shift	Right Shift
Lock	Caps Lock
Ctrl	Left Ctrl
none	Right Ctrl
Left Compose Character	Alt Gr
Right Compose Character	none
Left Alt	Left Alt
Right Alt	none or Right Alt

There is no **Ctrl** key on the right side of the VT keyboard. Left and right **Ctrl** keys on the enhanced PC layout generates the single control function on the VT Layout.

There is no key labeled "compose" on the enhanced PC layout. **Alt Gr** (Right **Alt**) accesses the supplemental characters on the top right half of the keyboard on some enhanced PC layout keyboards. **Alt Gr** can be used with the numeric keypad to enter character codes directly. Left **Alt/Space** can be used to initiate compose sequences when VT Style is selected.

8.5 Editing Keypad Keys

Figure 8–3 shows the layout for the DEC VT and enhanced PC editing keypads.

Figure 8–3 Editing Keypad Legends on DEC and PC Layouts

DEC VT Layout			Enhanced PC Layout		
Find	Insert Here	Re-Move	Insert	Home	Page Up
Select	Prev	Next	Delete	End	Page Down

When VT Style is selected, the enhanced PC layout editing keys transmit the same function sequences as the similarly legend editing keys in the VT layout. In PC Style, the **Home** key sends a Cursor Up (CUP) sequence to move the cursor home and the **Delete** key transmits a DEL (7/15).

Table 8–2 lists the key sequences for PC editing keypads.

Table 8–2 Editing Keypad Sequences for PC Layout

Enhanced PC Legend	VT Legend	VT Style Sequence	PC Style Sequence
Insert	Insert Here	CSI 2 ~	CSI 2 ~
Delete	Remove	CSI 3 ~	DEL
Home	Find	CSI 1 ~	CSI H
End	Select	CSI 4 ~	CSI 4 ~
Page Up	Prev Screen	CSI 5 ~	CSI 5 ~
Page Down	Next Screen	CSI 6 ~	CSI 6 ~

In addition to these unshifted sequences, the editing keys can send unique sequences when pressed in combination with **Shift**, **Alt**, or **Ctrl**. See DECFNK for details.

8.6 Cursor Keypad Keys

The cursor keypad keys send the same sequences as the VT layout. Additional unique sequences can be sent when pressed in combination with **Shift**, **Alt**, or **Ctrl**. See DECFNK for details.

8.7 Numeric Keypad Keys

The enhanced PC layout numeric keypad has three differences from the VT layout:

1. The four keys at the top of the keyboard are labeled **Num Lock**, **/**, *****, and **-** instead of **PF1** through **PF4**.
2. The **Num Lock** key toggles the keypad keys, sending either numerals or function sequences.

In VT Style, with Application Mode enabled, the numeric keypad keys send the same sequences as the corresponding keys on a VT layout. When the numeric keypad is in numeric mode, the four keys at the top of the keyboard still operates as **PF1** through **PF4**, and the **+** key sends an ASCII "+" character.

Table 8–3 shows the numeric keypad sequences in VT Style.

Table 8–3 PC Layout Numeric Keypad Sequences - VT Style

PC Key	DEC Key	Numeric Mode	Application Mode
		SS3 P	SS3 P
		SS3 Q	SS3 Q
		SS3 R	SS3 R
		SS3 S	SS3 S
			SS3 m
		"+"	SS3 l
		". "	SS3 n
		CR	SS3 M
		"0"	SS3 p
		"1"	SS3 q
		"2"	SS3 r
		"3"	SS3 s
		"4"	SS3 t
		"5"	SS3 u
		"6"	SS3 v
		"7"	SS3 w
		"8"	SS3 x
		"9"	SS3 y

In PC Style, when the numeric keypad is in Application Mode, the numeric keypad keys send the same sequences as the corresponding keys on the VT layout as shown in [Table 8–3](#).

In PC Style, when the numeric keypad is in Numeric Mode, the keypad keys send either editing keypad sequences corresponding to the gray legend in the lower part of the keyboard, or the numerals corresponding to the black legend in the upper part of the keyboard. The numerals are sent when or the has been depressed. The Num Lock state can be toggled on or off by pressing the key. "Num Lock" appears on the keyboard indicator line when this feature is activated. If Num Lock is on and "Reverse Lock" is activated, pressing temporarily reverses the effect of Num Lock.

[Table 8–4](#) lists the numeric keypad sequences in PC Style Numeric mode.

Table 8–4 PC Layout Numeric Keypad Sequences - PC Style, Numeric Mode

Keys	Num Lock Off unshifted, or Num Lock On with Shift	Num Lock On unshifted, or Num Lock Off with Shift
Num Lock		
/	/	/
*	*	*
-	-	-
+	+	+
. Del	DEL	.
Enter	CR	CR
0 Ins	CSI 2 ~	0
1 End	CSI 4 ~	1
2 ↓	CSI B	2
3 PgDn	CSI 6 ~	3
4 ←	CSI D	4
5		5
6 ⇒	CSI C	6
7 Home	CSI H	7
8 ↑	CSI A	8
9 PgUp	CSI 5 ~	9

8.8 Local Function Key Defaults

The keys used to perform local terminal functions differ between the VT keyboard, PC keyboard, and the mode selection. [Table 8–5](#) shows the corresponding keys for the default local functions. The function number in [Table 8–5](#) is used in the DECPFK host sequence or DECPAK's alternate function to specify a change to that local function key.

Table 8–5 Local Functions

Function Number	Function	VT Keyboard	SCO Console	PC Keyboard VT-Style	PC Keyboard PC-Style	PC Keyboard SCO
0	no function					
1	Hold	F1	Lock/F1	F1	Scroll Lock	
2	Print	F2	Lock/F2	F2	Print Screen	
3	Set-Up	F3	Lock/F3	F3	Lock/Print Screen	
					Lock/Sys Rq	
					or	
					Alt/Print Screen	
5	Break	F5	Lock/F5	F5	Lock/Pause	
7	Hard Reset	Ctrl/F3	Lock/Ctrl/F3	Ctrl/F3	Ctrl/Lock/Print Screen	
8	Toggle Autoprint	Ctrl/F2	Lock/Ctrl/F2	Ctrl/F2	Ctrl/Print Screen	
9	Disconnect	Shift/F5	Lock/Shift/F5	Shift/F5	Shift/Lock/Pause	
10	Send Answerback	Ctrl/F5	Lock/Ctrl/F5	Ctrl/F5	Ctrl/Lock/Pause	
11	Print Composed Main Display	Shift/F2	X ¹	Shift/F2	Shift/Print Screen	X
20	Pan Up	Ctrl/↑	X	Ctrl/↑	Ctrl/↑	X
21	Pan Down	Ctrl/↓	X	Ctrl/↓	Ctrl/↓	
24	Pan Prev Page	Ctrl/Prev	X	Ctrl/Page Up	Ctrl/Page Up	
25	Pan Next Page	Ctrl/Next	X	Ctrl/Page Down	Ctrl/Page Down	
30	Copy & Paste Mode	F1 (hold down)	Caps Lock/F1	F1	Scroll Lock	
31	C&P Cursor Left	←		←	←	
32	C&P Cursor Down	↓		↓	↓	
33	C&P Cursor Up	↑		↑	↑	
34	C&P Cursor Right	⇒		⇒	⇒	

¹ X means the function is not available in the default SCO state. If there is no "X" in the SCO Console column, then this means that the key sequence is the same as the VT Keyboard column. If there is no "X" in the PC Keyboard SCO column, then this means that the key sequence is the same as the PC Keyboard PC-Style column.

†The Shift2 Modifier is assigned to the **Alt Shift** key (German "right **Compose**") when it appears on the corresponding keyboard (German).

‡Toggle KB language is assigned to the named language key when it appears on the corresponding keyboard (Greek, Hebrew, and Russian).

± means the function switches alternately between on and off.

C&P means Copy and Paste.

Lock means the **Lock** key, **Caps Lock** key, or key with lock icon.

Function Number	Function	VT Keyboard	SCO Console	PC Keyboard VT-Style	PC Keyboard PC-Style	PC Keyboard SCO
35	C&P Start Selection	Select		Home	Home	
36	C&P Copy	Remove		End	End	
37	C&P Paste	Insert Here		Insert	Insert	
38	C&P ± Left-to-Right	Select		Home	Home	
41	Shift Modifier	Left or right Shift		Left or right Shift	Left or right Shift	
42	Control Modifier	Ctrl		Left or right Ctrl	Left or right Ctrl	
43	Alt Function Modifier	Left or right Alt		Left or right Alt	Left or right Alt	
44	Start Compose	Left or right Compose			Left Alt/Space	
45*	Group Shift Modifier	Group Shift		Alt/Gr	Alt/Gr	
46†	Shift2 Modifier	Alt/Shift				
47	Primary KB language	Ctrl/Alt/F1	Lock/Alt/Ctrl/F1	Ctrl/Alt/F1	Ctrl/Alt/F1	See SCO Console
48	Secondary KB language	Ctrl/Alt/F2	Lock/Alt/Ctrl/F2	Ctrl/Alt/F2	Ctrl/Alt/F2	See SCO Console
49‡	± KB language					
51	± Caps Lock State	Lock		Lock	Lock	
52	± Num Lock State			Num Lock	Num Lock	
53	± VT/IBM Style		X	Lock Num Lock	Lock Num Lock	X
54	Extend Kbd Modifier			Lock	Lock	
61	Screen saver					
62	Calculator	Lock/Alt/F12		Lock/Alt/F12	Lock/Alt/F12	

¹ X means the function is not available in the default SCO state. If there is no "X" in the SCO Console column, then this means that the key sequence is the same as the VT Keyboard column. If there is no "X" in the PC Keyboard SCO column, then this means that the key sequence is the same as the PC Keyboard PC-Style column.

†The Shift2 Modifier is assigned to the Alt Shift key (German "right Compose") when it appears on the corresponding keyboard (German).

‡Toggle KB language is assigned to the named language key when it appears on the corresponding keyboard (Greek, Hebrew, and Russian).

± means the function switches alternately between on and off.

C&P means Copy and Paste.

Lock means the Lock key, Caps Lock key, or key with lock icon.

Function Number	Function	VT Keyboard	SCO Console	PC Keyboard VT-Style	PC Keyboard PC-Style	PC Keyboard SCO
63	Clock	Lock/Alt/F11		Lock/Alt/F11	Lock/Alt/F11	
64	Character table	Lock/Alt/F10		Lock/Alt/F10	Lock/Alt/F10	
65	Transfer result	Shift/Enter		Shift/Enter	Shift/Enter	

¹ X means the function is not available in the default SCO state. If there is no "X" in the SCO Console column, then this means that the key sequence is the same as the VT Keyboard column. If there is no "X" in the PC Keyboard SCO column, then this means that the key sequence is the same as the PC Keyboard PC-Style column.

†The Shift2 Modifier is assigned to the Alt Shift key (German "right Compose") when it appears on the corresponding keyboard (German).

‡Toggle KB language is assigned to the named language key when it appears on the corresponding keyboard (Greek, Hebrew, and Russian).

± means the function switches alternately between on and off.

C&P means Copy and Paste.

Lock means the Lock key, Caps Lock key, or key with lock icon.

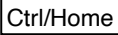
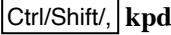
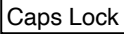
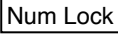
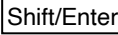
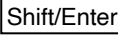
Note

See [Chapter 2](#) to re-define keys using the Define Key Editor.

Table 8–6 lists the local functions for ASCII emulations.

Table 8–6 Local Functions for ASCII emulations

Function	DEC VT Layout	EPC Layout
Hold Screen	F1	Scroll Lock
Print Page	Ctrl/Shift/. kpd	Ctrl/Shift/. kpd
Set-Up	F3	Alt/Print Screen
	Caps Lock/F3	Caps Lock/Print Screen
Break	F5	Ctrl/Pause
Hard Reset	Ctrl/F3 in Setup	
Soft Reset	Shift/F3	Alt/Shift/Print Screen
Autoprint Mode	Ctrl/F2	Ctrl/Print Screen
	Ctrl/Shift/F2	Ctrl/Shift/Print Screen
		Alt/Ctrl/Shift/Print Screen
Disconnect	Shift/F5	Pause
Send Answerback	Ctrl/F5	Shift/Pause
Display Next Page	Ctrl/Next	Ctrl/Page Down
	Shift/Ctrl/Next	Ctrl/Shift/Page Down
Active Other Window	Ctrl/Next or Ctrl/Prev	Ctrl/Page Down or Ctrl/Shift/Page Down or Ctrl/Page Up or Ctrl/Shift/Page Up
Display Prev. Page	Ctrl/Prev	Ctrl/Page Up
	Shift/Ctrl/Prev	Ctrl/Shift/Page Up
Display Page 0	Ctrl/0 kpd	Ctrl/0 kpd
Display Page 1	Ctrl/1 kpd	Ctrl/1 kpd
Display Page 2	Ctrl/2 kpd	Ctrl/2 kpd
Display Page 3	Ctrl/3 kpd	Ctrl/3 kpd
Display Page 4	Ctrl/4 kpd	Ctrl/4 kpd
Display Page 5	Ctrl/5 kpd	Ctrl/5 kpd
Block Mode	F4	Ctrl/Shift/Pause
	Ctrl/F4	
	Shift/Ctrl/F5	
Change Status Line	Ctrl/⇒	Ctrl/⇒
Display	Ctrl/Shift/⇒	Ctrl/Shift/⇒
Insert Mode	Ctrl/PF4	Ctrl/Insert
	Ctrl/Shift/PF4	Ctrl/Shift/Insert
Monitor Mode	Ctrl/Shift/1 kpd	Ctrl/Shift/1 kpd
Screen Saver	Ctrl/Shift/PF3	Ctrl/Shift/End
Speed Scroll Rate	Ctrl/Shift/↑↑	Ctrl/Shift/↑↑
Slow Scroll Rate	Ctrl/Shift/↓↓	Ctrl/Shift/↓↓
Home Cursor & Clear		Ctrl/Shift/Home
Display		
Roll Active Window Up	Ctrl/↑↑	Ctrl/↑↑
in Page		

Function	DEC VT Layout	EPC Layout
Roll Active Window Down In Page		
Toggle Split Screen	 kpd	 kpd
Raise Split Line	 kpd	 kpd
Lower Split Line	 kpd	N.A.
Adjust Window to Include Cursor		
Cursor Drag Mode	 kpd	N.A.
Caps Lock State		
Num Lock State	N.A.	
Keyclick State	 kpd	 kpd

8.8.1 Numeric Keypad Compose

Pressing and holding ,  or  while typing a decimal number on the numeric keypad sends the corresponding decimal character code when the compose key is released. If the  key on the keyboard is pressed while entering a number, then the value entered before pressing the  key is multiplied by 16 and added to the value entered next. This supports column/row entry commonly used in character coding tables.

When accessibility aids are enabled, remember to Lock and Unlock the compose key when entering codes.

8.8.2 Accessibility Aids

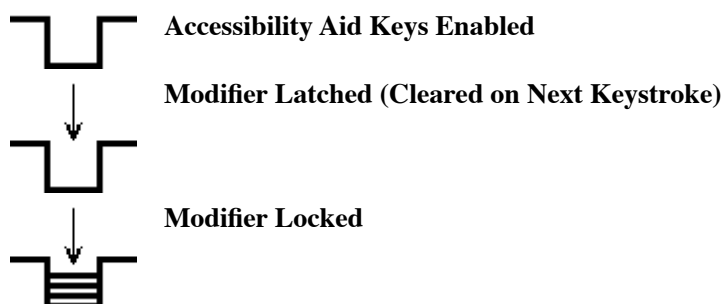
Accessibility aids allow the user with limited motor skills to use modifier key combinations in a sequential manner rather than a simultaneous manner (default). All modifier key combinations are supported.

The Accessibility aids option is enabled by depressing the  key five times in succession. This option can be disabled by pressing and holding a modifier key while pressing another key.

Once enabled, this option provides two levels of assistance known as Latch and Lock. The Latch state is achieved by pressing a modifier key once and affects the next key pressed. The Lock state is achieved by pressing a modifier key twice. All keys pressed are affected by the modifier until they are unlocked by either depressing the modifier key again or depressing another modifier key twice. Once any modifier is in the Locked state, pressing other modifier keys adds those modifiers to the Locked state.

A small icon appears on the Keyboard Indicator Line or Status Line to provide user feedback on the changing modifier state.

The icons for each state are as follows:



When Accessibility aid keys are disabled, no icon is displayed on the Keyboard Indicator Line or Status Line.

8.9 Controlling Keyboard LEDs

The following sequences allow the host to control keyboard modifier states for Caps Lock, Scroll Lock, and Num Lock or keyboard LEDs to indicate program status. They are as follows:

ESC [? 108 h – Set NumLock mode (DECNUMLK)

ESC [? 108 l – Reset NumLock mode

ESC [? 109 h – Set CapsLock mode (DECCAPSLK)

ESC [? 109 l – Reset CapsLock mode

"Scroll Lock" or "Hold Screen" cannot be controlled from the host because it suspends transmission from the host and there is no way to release it.

DECLL controls keyboard LEDs independently of any keyboard state. The use of LEDs for this purpose conflicts with their use as keyboard state indicators. The host control selects a mode of how the keyboard LEDs are to be used: as keyboard indicators; or host indicators. If host indicators is selected, then the DECLL sequence can be used to control the keyboard LEDs.

The following control sequences are used to control keyboard LEDs.

ESC [? 110 h – Set Keyboard LEDs Host Indicator Mode (DECKLHIM)

ESC [? 110 l – Reset Keyboard LEDs Host Indicator Mode (default)

Note

See [Chapter 2](#) to re-define keys using the Define Key Editor.

8.10 Keyboard Languages

[Table 8–7](#) lists the keyboard languages available for the VT terminal.

Table 8–7 VT Keyboard Layouts

Austrian/German	Greek	Russian
Belgian/French	Hebrew	SCS
British	Hungarian	Slovak
Canadian-French/English	Italian	Spanish
Czech	North American	Swedish
Danish	Norwegian	Swiss-French
Dutch	Polish	Swiss-German
Finnish	Portuguese	Turkish F
Flemish	Romanian	Turkish Q

[Table 8–8](#) lists the keyboard languages available for the PC terminal.

Table 8–8 Enhanced PC Keyboard Layouts

Belgian	Greek	Romanian
British	Hebrew	Russian
Czech	Hungarian	SCS
Danish	Italian	Slovak
Dutch	Latin American	Spanish
Finnish	North American	Swedish
French	Norwegian	Swiss-French
French-Canadian	Polish	Swiss-German
German	Portuguese	Turkish

8.11 Switching Between Keyboard Languages

The VT510 allows the user to easily switch between two different keyboard layouts for several languages (English and Hebrew, for example). This feature allows the VT510 to support both existing conventions and emerging standards for extending the graphic input repertoire and/or switching between languages in dual language environments.

The primary keyboard language corresponds to "Group 1" and generally references the legends on the left portion of the keyboard.

The secondary keyboard language corresponds to "Group 2" and generally references the legends on the right portion of the keyboard.

Unless otherwise overridden, **Ctrl/Alt/F1** activates the primary keyboard language (locking shift), and **Ctrl/Alt/F2** activates the secondary keyboard language. These factory defaults are standard on PCs.

Selecting a new Keyboard Dialect in Set-Up resets the keyboard group shift state to Group 1 and automatically initializes the keyboard character set and VT default character set as shown in table [Table 8–9](#). The keyboard character set is initialized to match the VT default character set.

Table 8–9 Default Character Set when changing Keyboard Language

Keyboard Language	VT Default Character Set	If desired, change to ...
Austrian/German, Belgian/French, British, Canadian-English, Canadian-French, Flemish, Danish, Dutch, Finnish, Italian, Portuguese, North American, Norwegian, Spanish, Swedish, Swiss-French, Swiss-German.	DEC MCS or ISO Latin-1	ISO Latin-1
Czech, Hungarian, Polish, Romanian, SCS, or Slovak	ISO Latin-2	ISO Latin-1
Greek	DEC Greek or ISO Latin-Greek	ISO Latin-Greek
Hebrew	DEC Hebrew or ISO Latin-Hebrew	ISO Latin-Hebrew
Russian	KOI-8 Cyrillic or ISO Latin-Cyrillic	ISO Latin-Cyrillic
Turkish	DEC Turkish or ISO Latin-Turkish	ISO Latin-Turkish

If 7-bit NRCS characters is selected, then the following corresponding NRC sets are available on the VT keyboards:

VT keyboards:

- Austrian/German - German NRC
- Belgian/French - French NRC
- British - U.K. NRC
- Canadian-English - French Canadian NRC
- Canadian-French - French Canadian NRC
- Danish - Norwegian/Danish NRC
- Finnish - Finnish NRC
- Flemish - French NRC

- **Greek - DEC Greek NRC**
- **Hebrew - DEC 7-bit Hebrew**
- **Italian - Italian NRC**
- **Norwegian - Norwegian/Danish NRC**
- **Portuguese - Portuguese NRC**
- **Russian - DEC Russian NRC**
- **SCS - SCS NRC**
- **Spanish - Spanish NRC**
- **Swedish - Swedish NRC**
- **Swiss-French - Swiss NRC**
- **Swiss-German - Swiss NRC**
- **Turkish F - DEC 7-bit Turkish**
- **Turkish Q - DEC 7-bit Turkish**

The 7-bit NRCS character sets are not available on the following keyboards:

- **Czech**
- **Dutch**
- **Hungarian**
- **North American**
- **Polish**
- **Romanian**
- **Slovak**

If 7-bit NRCS characters is selected, then the following corresponding NRC sets are available on the enhanced PC keyboards:

Enhanced PC Keyboards

- **Belgian - French NRC**
- **British - U.K. NRC**
- **Danish - Norwegian/Danish NRC**
- **Finnish - Finnish NRC (same KB as Sweden)**
- **French - French NRC**
- **French Canadian - French Canadian NRC**
- **German - Austrian/German NRC**
- **Greek - DEC Greek NRC**
- **Hebrew - DEC 7-bit Hebrew**
- **Italian - Italian NRC**
- **Latin American - Spanish NRC**
- **Norwegian - Norwegian/Danish NRC**
- **Portuguese - Portuguese NRC**
- **Russian - DEC Russian NRC**
- **SCS - SCS NRC**
- **Spanish - Spanish NRC**
- **Swedish - Swedish NRC (same KB as Finland)**
- **Swiss-French - Swiss NRC (Swiss Fr./Gr. is a single PC keyboard)**

- Swiss-German - Swiss NRC (Swiss Fr./Gr. is a single PC keyboard)
- Turkish - DEC Turkish NRC

The 7-bit NRCS character sets are not available on the following enhanced PC keyboards:

- Czech
- Dutch
- Hungarian
- North American
- Polish
- Romanian
- Slovak

8.11.1 Non-Spacing Accent Marks

Accented characters that do not appear directly on a keyboard can often be entered by typing a non-spacing diacritical mark followed by a base character.

Table 8–10 Non-Spacing Accent Marks

Symbol	Non-Spacing Accent Name
´	Acute accent
˘	Breve
ˇ	Hacek (caron)
¸	Cedilla
ˆ	Circumflex accent
¨	Diaeresis (umlaut)
˙	Dot above
˝	Double acute accent
`	Grave accent
ˊ	Greek Tonos
¯	Macron
˛	Ogonek
◊	Ring above
~	Tilde

Keystrokes assigned non-spacing accent marks do not send any character code immediately, but are used instead to modify the code transmitted by the subsequent keystroke. If a keystroke that has been assigned a non-spacing accent mark is followed immediately by a graphic character, then this forms a valid character in the keyboard character set, and the corresponding character code is transmitted.

When a non-spacing diacritical key is pressed, "Compose" appears on the keyboard indicator line until the next subsequent keystroke.

If the keystroke immediately following the non-spacing diacritical keystroke does not form a valid character in the keyboard character set, then the non-spacing diacritical mark is converted to the corresponding character, if any, and transmitted from the current keyboard character set. The second keystroke is then processed normally. This "error recovery" is different from previous VT auto-compose processing; however, this process follows the emerging standard used on personal computers.

8.11.2 German

For German keyboards, the **Group Shift** (left Compose) key functions as the keyboard repertoire extension mechanism. Group 1 and Group 2 are defined within the German keyboard dialect and are used whenever the keyboard language is set to German. **Group Shift** operates as a single shift prefix if pressed and released or as a modifier key if pressed and held.

8.11.3 Hebrew

When Hebrew is selected as the Keyboard Dialect, the **Ctrl/Hebrew** key toggles between the primary and secondary keyboard language.

When switching into 7-bit Hebrew, lowercase English letters are swapped with their corresponding Hebrew characters throughout the display. The **Shift/Hebrew** key also performs this swapping function at any time.

8.12 Keystation Numbers

Figure 8–4 shows the keystation numbers for a VT keyboard.

Figure 8–4 VT Layout - Keystation Numbers

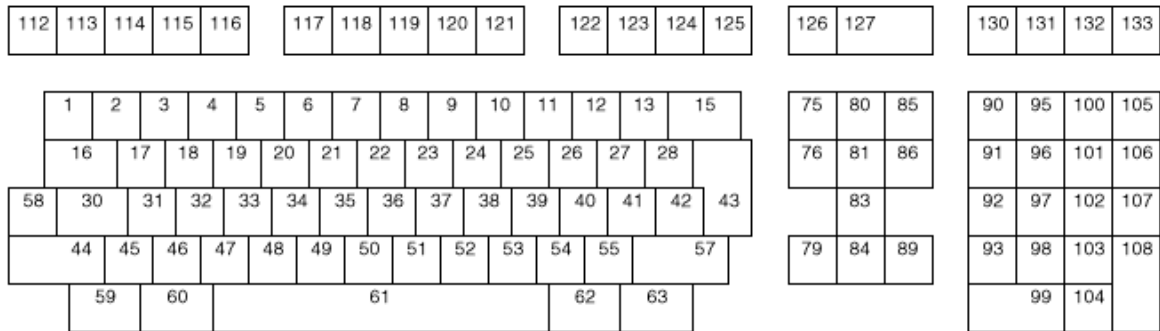


Figure 8–5 shows a typical legend for a North American VT keyboard.

Figure 8–5 VT Layout - Typical North American Key Legends

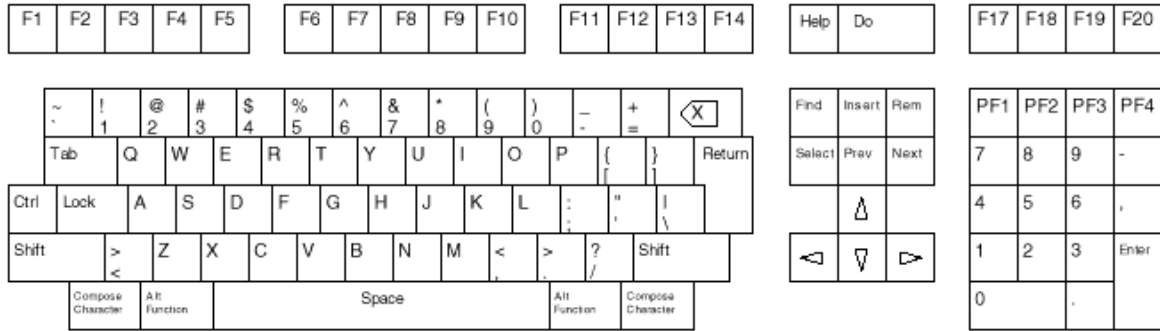


Figure 8–6 Enhanced PC Layout - Map Numbers

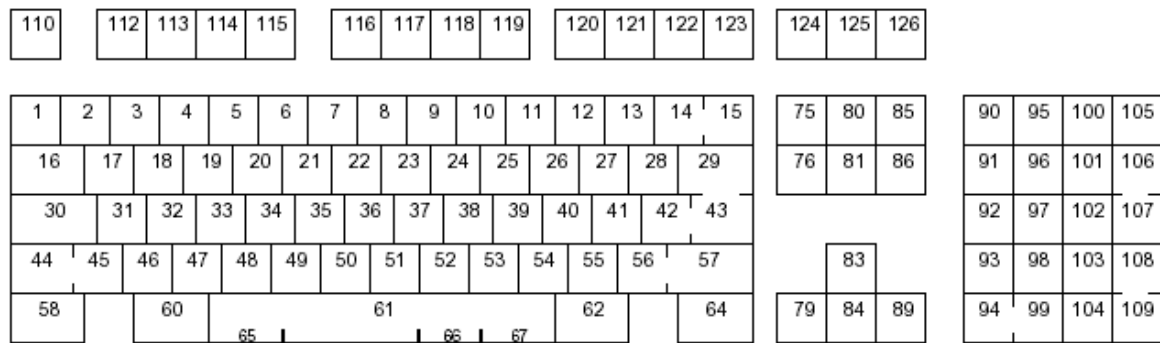
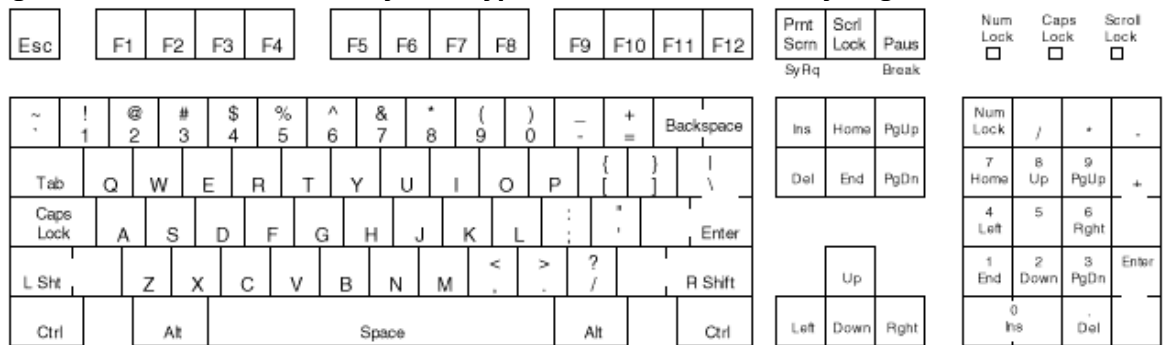


Figure 8–7 Enhanced PC Layout - Typical North American Key Legends



8.13 Scan Codes Sent in PCTerm Mode for EPC Keyboards

The following keys send the corresponding scan codes, regardless of any modifiers that may be in effect.

Key #	Legend (U.S.)	Scan Code Make/Break ¹
1	`	29/A9
2	1	02/82
3	2	03/83
4	3	04/84
5	4	05/85
6	5	06/86
7	6	07/87
8	7	08/88
9	8	09/89
10	9	0A/8A
11	0	0B/8B
12	-	0C/8C
13	=	0D/8D
14		7D/FD ²
15	Backspace	0E/8E
16	Tab	0F/8F
17	Q	10/90
18	W	11/91
19	E	12/92
20	R	13/93
21	T	14/94
22	Y	15/95
23	U	16/96
24	I	17/97
25	O	18/98
26	P	19/99
27	[	1A/9A
28	]	1B/9B
29	\	2B/AB [†]
30	Caps Lock	3A/BA
31	A	1E/9E
32	S	1F/9F
33	D	20/A0
34	F	21/A1

¹Make means when the key is pressed; Break means when the key is released.

²Not used on 101/102 keyboard layout.

[†]101 keyboard layout only

[‡]102 keyboard layout only

Key #	Legend (U.S.)	Scan Code Make/Break ¹
35	G	22/A2
36	H	23/A3
37	J	24/A4
38	K	25/A5
39	L	26/A6
40	, ,	27/A7
41	' `	28/A8
42		2B/AB‡
43	Return	1C/9C
44	Shift	2A/AA
45		56/D6‡
46	Z	2C/AC
47	X	2D/AD
48	C	2E/AE
49	V	2F/AF
50	B	30/B0
51	N	31/B1
52	M	32/B2
53	, ,	33/B3
54	. .	34/B4
55	/ _	35/B5
56		73/F3 ²
57	Shift	36/B6
58	Left Ctrl	1D/9D
60	Left Alt	38/B8
61	Space	39/B9
62	Right Alt	E0 38/E0 B8
64	Right Ctrl	E0 1D/E0 9D
90	Num Lock	45/C5
91	7 Home	47/C7
92	4 ←	4B/CB
93	1 End	4F/CF

¹Make means when the key is pressed; Break means when the key is released.

²Not used on 101/102 keyboard layout.

†101 keyboard layout only

‡102 keyboard layout only

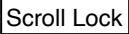
Key #	Legend (U.S.)	Scan Code Make/Break ¹
94		7C/FC ²
96		48/C8
97	Keypad 	4C/CC
98		50/D0
99		52/D2
100	Keypad 	37/B7
101		49/C9
102		4D/CD
103		51/D1
104		53/D3
105	Keypad 	4A/CA
106	Keypad 	4E/CE
107		7E/FE ²
108		E0 1C/E0 9C
109		78/F8 ²
110		01/81
112		3B/BB
113		3C/BC
114		3D/BD
115		3E/BE
116		3F/BF
117		40/C0
118		41/C1
119		42/C2
120		43/C3
121		44/C4
122		57/D7
123		58/D8

¹Make means when the key is pressed; Break means when the key is released.

²Not used on 101/102 keyboard layout.

†101 keyboard layout only

‡102 keyboard layout only

Key #	Legend (U.S.)	Scan Code Make/Break ¹
125		46/C6

¹Make means when the key is pressed; Break means when the key is released.

²Not used on 101/102 keyboard layout.

†101 keyboard layout only

‡102 keyboard layout only

The keys in the following table send different series of scan codes depending on the modifier state in effect (Shift, Control, Alt, and/or Num Lock). Many of them have a base scan code that is identical to another key. To make each key unique, some have an extra byte hex 'E0' or hex 'E1' added to their scan code sequence.

Key #	Legend (U.S.)	Base Case Make/Break	Shift Case Make/Break	Num Lock ON Make/Break
75		E0 52/E0 D2	E0 AA E0 52/E0 D2 E0 2A*	E0 2A E0 52/E0 D2 E0 AA
76		E0 53/E0 D3	E0 AA E0 53/E0 D3 E0 2A*	E0 2A E0 53/E0 D3 E0 AA
79		E0 4B/E0 CB	E0 AA E0 4B/E0 CB E0 2A*	E0 2A E0 4B/E0 CB E0 AA
80		E0 47/E0 C7	E0 AA E0 47/E0 C7 E0 2A*	E0 2A E0 47/E0 C7 E0 AA
81		E0 4F/E0 CF	E0 AA E0 4F/E0 CF E0 2A*	E0 2A E0 4F/E0 CF E0 AA
83		E0 48/E0 C8	E0 AA E0 48/E0 C8 E0 2A*	E0 2A E0 48/E0 C8 E0 AA
84		E0 50/E0 D0	E0 AA E0 50/E0 D0 E0 2A*	E0 2A E0 50/E0 D0 E0 AA
85		E0 49/E0 C9	E0 AA E0 49/E0 C9 E0 2A*	E0 2A E0 49/E0 C9 E0 AA
86		E0 51/E0 D1	E0 AA E0 51/E0 D1 E0 2A*	E0 2A E0 51/E0 D1 E0 AA
89		E0 4D/E0 CD	E0 AA E0 4D/E0 CD E0 2A*	E0 2A E0 4D/E0 CD E0 AA

*When the left  key is pressed, the AA-2A shift break and make should envelope the base codes as shown in the previous table. When the right  key is pressed, the code B6 (right shift break) is sent in place of AA and the code 36 (right shift make) is sent in place of 2A. If both right and left  keys are pressed, then both B6 and AA are sent where AA is shown, and both 36 and 2A are sent where 2A is shown.

Key #	Legend (U.S.)	Scan Code Make/Break	Shift Case Make/Break
95	Keypad 	E0 35/E0 B5	E0 AA E0 35/E0 B5 E0 2A*

The marker code Hex 'E0' precedes shift codes that are artificially produced by the keyboard logic. These artificially produced shift codes are not repeated during typamatic (auto repeat) operation of the corresponding key. Only the unique base code is repeated at the auto repeat rate. The terminal must ensure that both shift states are restored when all shift-dependent keys are released.

Key #	Legend (U.S.)	Scan Code Make/Break	Ctrl Case and Shift	
			Case Make/Break	Alt Case Make/Break
124	PrtSc	E0 2A E0 37/E0 B7 E0 AA	E0 37/E0 B7	
	SysRq			54/D4

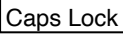
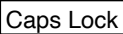
Notice there are two base scan codes associated with keystation 124. To ensure the make/break sequence, the code associated with the break must reflect the make code regardless of the Alt state at the time of the break.

Key #	Legend (U.S.)	Scan Code	Make	Ctrl down	Make
126	Pause	E1 1D 45	E1 9D	C5	
	Break		E0 46	E0	C6

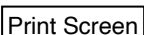
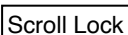
The Pause/Break key is a make-only key, meaning it generates a code when pressed, not when released. All scan codes or actions associated with this key occur with the make of the key.

8.14 Scan Codes Sent in PCTerm Mode for VT Keyboards

All keystrokes send scan codes except:

Key	Which sends . . .
 	Hold
 	Print
 	Enter Set-Up
 	Send Break

All keys including Shift, Alt, and Ctrl send make codes immediately upon being pressed, except the Caps Lock key. The make code for the Caps Lock key is sent after the Caps Lock key is released. The following keys send the corresponding scan codes, regardless of any modifiers that may be in effect.

Key #	Legend (U.S.)	Scan Code Make/Break
42		2B/AB
107	 ¹	7E/FE
124		E0 3D/E0 BD
125		E0 3E/E0 BE
126		E0 3F/E0 BF
127		01/81 (Like  key on PC keyboard)
130		E0 41/E0 C1
131		E0 2A E0 37/E0 B7 E0 AA Ctrl or Shift E0 37/E0 B7
		Alt 54/D4
132		46/C6
133	 ²	E1 1D 45 E1 9D C5
	 ²	Ctrl E0 46 E0 C6

¹On Numeric keypad.

²This is a make-only key; code is sent only when key is pressed.

This chapter describes the serial asynchronous communications through the host ports.

9.1 Overview

The VT510 is a single-session terminal. You have a choice of switching between Comm1, an EIA 232 25-pin port; and Comm2, an MMJ port, for host communications.

The 25-pin port uses either a male or a female connector but does not use both at the same time. In Set-Up, you can also designate this port as a printer port.

The VT510 has the following features:

- Baud rate up to 115.2 K
- 1024-byte receive buffer to prevent buffer overflow
- Flow control scheme that independently selects the transmit and the receive side of the communication line
- Expanded flow control options to include hardware DTR/DSR flow control
- A Hold state that causes an XOFF to be sent immediately
- Option to accept or ignore NUL characters
- Support for half-duplex mode

9.2 Introduction to Communications

This chapter describes the physical layer and data link layer. The physical layer is comprised of the mechanical, electrical, and functional arrangements necessary for a physical connection, including cables and telephone lines. The physical layer applies to both ANSI and ASCII terminal emulations.

The data link layer is the electronic protocol used to convey a unit of information from the VT510 to a host computer. This layer includes flow control, character conversions, and some forms of error control.

The asynchronous character format consists of a start bit (space), the data bits (1=mark, 0=space), the parity bit (if present), and one or two stop bits (mark). The data bits represent a character with the least significant bit leading. The parity bit may be enabled as either none, odd, even, space, or mark. Received parity may be checked or ignored. Only odd or even parity when received and must be the same as the parity transmitted. These parameters can be selected through Set-Up or by using the escape sequence, DECSPP.

For further information on the asynchronous character format, refer to ANSI X3.15-1976, *American National Standard for bit sequencing of the American National Standard Code for Information Interchange in Serial-By-Bit Data Transmission*.

The VT510 supports the following:

- Full-duplex mode
- Half-duplex mode

Full-duplex mode allows simultaneous two-way communication, while half-duplex mode limits communication to one way at any given time. The VT510 always supports full duplex (two wires) at the physical layer. Half-duplex mode is a simple data link layer protocol intended to support half-duplex modems. It is common to view the terminal and attached modem as a single system. Refer to [Section 9.3.4](#) for details on half-duplex mode. This feature can be selected from Set-Up or by the control function, DECHDPXM.

The VT510 does not support synchronous communications.

9.3 Physical Communications Link

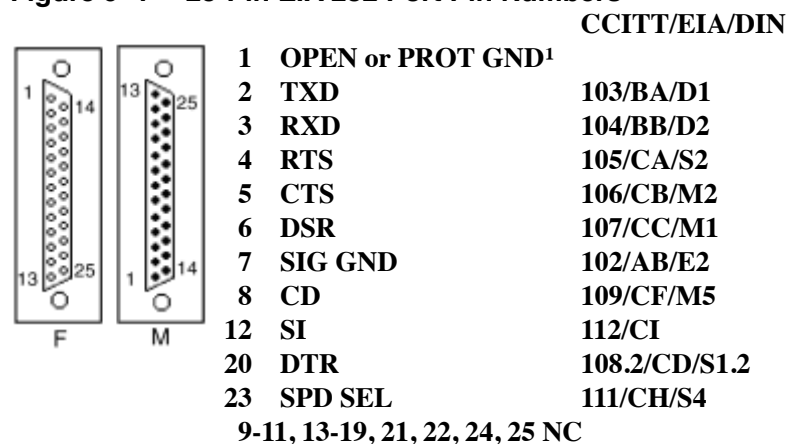
Communications lines may be connected to both Comm1 and Comm2 ports without interaction between the two lines. You can use the Communication Port select Set-Up menu to select the physical link to the cables. The port selection can also be programmed through the escape sequence, DECSCP.

When you change from EIA 232-E (25-pin port) to DEC 423 (MMJ port), the VT510 performs a disconnect on the EIA 232-E port. If the change is from DEC 423 to EIA 232-E, then the VT510 performs a disconnect on the EIA 423 port (deasserting DTR).

9.3.1 25-Pin Connectors

Figure 9–1 shows the 25-pin EIA 232 port.

Figure 9–1 25-Pin EIA 232 Port Pin Numbers



The DB-25P serial port accepts a variety of modems that meet national and international standards. Interface signals are labeled with both EIA and CCITT designations.

Table 9–1 shows the EIA interface signals and functions.

Table 9–1 EIA Interface Signals and Functions

Pin	Signal Name	Source	Function	CCITT/EIA/DIN
1			N.C. ¹	See ¹
2	TXD	Terminal	Transmitted Data	103/BA/D1
3	RXD	Modem	Received Data	104/BB/D2
4	RTS	Terminal	Request to Send	105/CA/S2
5	CTS	Modem	Clear to Send	106/CB/M2
6	DSR	Modem	Data Set Ready	107/CC/M1
7	SGND		Signal Ground	102/AB/E2
8	RLSD	Modem	Rec Line Signal Detector	109/CF/M5
9-11, 13-19, 21, 22, 24, 25			N.C.	
12	SPDI	Modem	Speed Mode Indication	112/CI/M4
20	DTR	Terminal	Data Term. Ready	108.2/CD/S1.2
23	SPDS	Terminal	Speed Select	111/CH/S4

¹ Pin 1 of the 25-pin connector normally will be open; however, the VT510 shall provide provisions for a 100-ohm resistor ground to be added to the VT510's printed circuit board. This may become necessary for products sold in Germany.

9.3.2 DEC Corporate Modular Jacks (MMJ)

The modular jacks provide DEC-423 compatible levels with the intent of allowing greater length between terminal and host. Limited modem support is also provided. The DTR output and DSR input are supported on this connector. Transmit ground for transmit data and DTR are isolated from receive ground used for receive data and DSR. [Figure 9–2](#) shows the MMJ port signals.

Figure 9–2 MMJ Modular Jack Pin Numbers

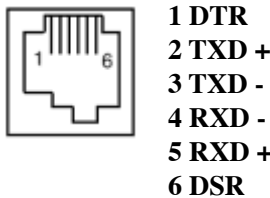


Table 9–2 describes the function of each MMJ interface signal.

Table 9–2 MMJ Interface Signals

Pin	Signal Name	Source	Function
1	DTR	Terminal	Data Terminal Ready*
2	TXD+	Terminal	Transmitted Data
3	TXD-	Terminal	Transmit Signal Ground
4	RXD-	Modem	Receive Signal Ground
5	RXD+	Modem	Received Data
6	DSR	Modem	Data Set Ready†

*Pin 1 is a terminal output signal and uses pin 3 as a reference.

†Pin 6 is a terminal input signal and uses pin 4 as a reference.

9.3.3 Connector Pins Description

This section describes the function of full-duplex mode connector pins. Half-duplex mode connector pins have different functions than the full-duplex mode connector pins as mentioned in [Section 9.3.4](#).

9.3.3.1 Transmitted Data–TXD

The TXD signal is supported on the 25-pin D-sub and the Corporate Modular Connector.

Data on this circuit represents the serially encoded characters that are transmitted from the VT510. This circuit is held at the mark state (-) during stop bits between characters and also at times when no data is being transmitted.

On the 25-pin connector when modem control is enabled, no data is transmitted unless Clear to Send is asserted from the modem. Assertion of DSR or RLSD from the modem is not required. This is specific to the 25-pin EIA connector and is intended to allow the VT510 to communicate with intelligent modems (like the DF224) before a connection has been established.

9.3.3.2 Received Data–RXD

The RXD signal is supported on the 25-pin D-sub as well as on the Corporate Modular Connector.

Data on this circuit represents the serially encoded characters to be received by the VT510.

When modem control is not enabled, received data is processed regardless of the state of the control lines.

On a 25-pin EIA connector, when modem control is enabled and a connection has been established (DSR from the modem is asserted), the received characters are ignored if RLSD is unasserted,

except when Disconnect Delay in Set-up is selected as No disconnect. This is an implementation in firmware of mark carrier clamping.

If a connection has not been established (DSR from the modem is not asserted), then received characters are processed even if RLSD is unasserted. This is an implementation that permits V.25 bis compatible autodial modems to be used without the user having to set "data leads only" to access the autodial functions.

9.3.3.3 Request to Send–RTS

The RTS signal is supported on the 25-pin D-sub but not on the Corporate Modular Connector.

For modems with this function, asserting RTS may put the modem in the transmit mode. When the modem is in the transmit mode, it then asserts CTS.

Note

For full-duplex modems without RTS inputs, CTS is asserted by the modem whenever it is capable of transmission.

9.3.3.4 Clear to Send–CTS

The CTS signal is supported on the 25-pin D-sub but not on the Corporate Modular Connector.

Assertion of CTS indicates that the modem is ready to receive data (that is, the terminal is clear to send data to the modem).

The data can be either a command to the modem if "off line" (DSR de-asserted) or transmitted data to the host if "on line" (DSR asserted).

9.3.3.5 Data Terminal Ready–DTR

The DTR signal is supported on the 25-pin D-sub as well as the Corporate Modular Connector.

The VT510 asserts DTR whenever it is ready to transmit and/or receive on the port to which it is associated with.

The VT510 de-asserts DTR when:

- Recalling saved settings
- Recalling factory defaults
- Pressing Shift/Break
- Powering up self-test
- Resetting to Initial State (RIS)–hard reset
- DTR is being used for hardware flow control and the receive buffer has reached the flow control threshold

The VT510 does not de-assert DTR when:

- Resetting session
- Clearing comm

9.3.3.6 Data Set Ready–DSR

The DSR signal is supported on the 25-pin D-sub as well as the Corporate Modular Connector.

When the DTR/DSR is not the selected flow control scheme, the assertion of DSR indicates the following:

- That the modem is in the data mode
- That the modem is connected to the communications channel
- The answer tone is finished
- The user is being charged by the PTT

- That the modem is ready to exchange the appropriate control signals to initiate data transmission/reception

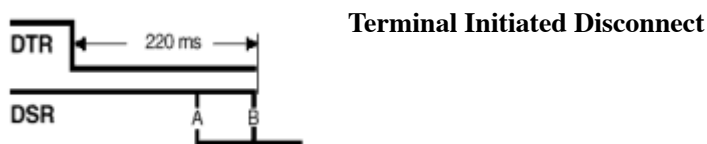
When DTR/DSR is the selected hardware flow control scheme only, refer to [Section 9.4.2.2.3](#).

9.3.3.6.1 Modem Initiated Disconnect

When DSR is unasserted during the progress of a call but before DTR is unasserted, the terminal interprets this condition as a lost or aborted call and disconnects. Any subsequent assertion of DSR is considered a new call. If connected to a modem and the modem's power is turned off or if DSR becomes an open circuit, then the VT510 interprets these conditions as a de-assertion. The state of the DSR input is shown on the indicator status line display.

9.3.3.6.2 Terminal Initiated Disconnect

The terminal disconnects itself from the modem by dropping DTR.



If DSR is de-asserted 220 ms after DTR goes de-asserted, then do nothing, since the modem has already done a disconnect. If DSR is still asserted 220 ms after DTR goes deasserted, then DTR must remain deasserted for greater than 2 seconds to assure that the modem does a disconnect.

9.3.3.6.3 Modem off line

When DSR is unasserted and modem control is enabled on the 25-pin EIA connector, the system is in a transitory "off line" state. In this state, the user may interact with a dial-up modem to enter numbers, change modem set-up, and so on.

When DSR is asserted and modem control is being used, the modem, terminal, and host system are ready for communication to a remote host.

9.3.3.7 Receive Line Signal Detector–RLSD

This signal is supported on the 25-pin D-sub but not on the Corporate Modular Connector.

RLSD is also called Carrier Detect. The modem asserts RLSD when the received signal is of sufficient amplitude or in the case of "training modems," when both modems have adjusted and equalized to the prevailing line conditions, to ensure demodulation of Received Data. The unasserted condition of RLSD indicates that there is neither a received signal nor that it is unsuitable for demodulation.

9.3.3.8 Speed Indicator–SPDI

The SPDI signal is supported on the 25-pin D-sub but not on the Corporate Modular Connector.

This signal comes from the modem. When Modem Control is enabled in Set-Up, the transmit and receive speed is set as follows:

State	Transmit and Receive Speed
Unasserted	Modem Low Speed
Asserted	Modem High Speed (default)

The user should set these Set-Up fields so that the fast speed (2400 baud, for example) is exactly twice the slow (1200 baud, in the following example) if V.25 bis compatibility is desired. Otherwise, the user may set the two speeds as applicable (1200/300 baud, in the following example).

	Assert	De-assert
Bell 212	1200	300
Bell 103	300	n/a
V.22	1200	600
V.22 bis	2400	1200
V.26 ter	2400	1200
V.32	4800	2400
V.32	9600	4800

9.3.3.9 Speed Select–SPDS

The SPDS signal is supported on the 25-pin D-sub but not on the Corporate Modular Connector.

If the speed is greater or equal to 1200 baud, then Speed Select will be high. If the speed is below 1200 baud, then Speed Select will be slow. This signal comes from the terminal.

9.3.4 Half-Duplex Mode

Half-duplex mode works on the EIA 232 line only. It uses the pin signals to control the data traffic to one direction at any given time.

In this mode, Data Terminal Ready (DTR) is always asserted. The terminal checks the assertion of Data Set Ready (DSR) constantly. The terminal disconnects once DSR is dropped. Ready to Send (RTS) is asserted only if the terminal intends to transmit.

Before transmitting, the terminal checks for the presence of Carrier Detect (CD). If CD is still detected for five seconds, then the terminal disconnects. Otherwise, the terminal asserts the Request to Send (RTS) signal and starts detecting the Clear to Send (CTS) signal. Transmission starts if CTS is detected within five seconds. If CTS is not monitored within five seconds, then the terminal disconnects. After transmission, RTS is de-asserted from the terminal. If CTS is still detected within the next five seconds, then the terminal disconnects.

9.3.5 Supported Modem Configurations

This section describes the supported modem configurations.

9.3.5.1 CCITT Public Switched Telephone Network (2-wire only)

V.21 Modem (V.24/V.28)
 Async, FDX, up to 200/300 bit/s
 V.22 Modem (V.24/V.28)
 Async, FDX, up to 1200 bit/s
 V.22 bis Modem (V.24/V.28)
 Async, FDX, up to 2400 bit/s
 V.23 Modem (V.24/V.28)
 Async, Asymmetric FDX
 Receive = 600 or 1200 bit/s,
 Transmit = Reverse channel at 75 bit/s
 V.25 bis Auto Dial Protocol Modem and Hayes

9.3.5.2 CCITT Dedicated Line (2-Wire / 4-Wire) Data Network

V.21 Modem (V.24/V.28, 2-W)
 Async, FDX, 200/300 B
 V.23 Modem (V.24/V.28)
 FDX, Async, 1200 B, 4-W
 V.25 bis compatible Modem
 Async, FDX, any standard speed from 300 baud up.
 X.20 bis Full-Duplex asynchronous 300 baud data service unit

9.3.5.3 European Public Circuit Switched Data Network

D1200S (Germany)
VC-DCE (Nordic Network)
VPC-DCE (Nordic Network)
X.20bis (V.21) Data Service Unit (V.28)
Async, FDX, up to 300 B

9.3.5.4 European Public Packet Switched Data Network (PSDN)

Germany: DATEX-P
Basic Datex P10-Service (X.25): Not supported
Additional Datex-P20 Service:
Fixed Connections to PAD facility with X.28 procedure:

Service	DCE
DATEX-P20	DFG300(X.20 bis), D300S (V.21) H300
DATEX-P20	D1200S12(V.22), D1200S (V.23,4-W) H1200
DATEX-P20	D1200S (V.23 with sec. channel, 75 baud Transmit / 1200 Baud Receive) H1200/75
DATEX-P20	H2400

Access to DATEX-P PAD facility from other public switched networks with X.28 procedure:

Service	Access	DCE
DATEX-P20	DATEX-L	DFG300 (X.20bis) L300
DATEX-P20	PSTN	D300S (V.21) F300
DATEX-P20	PSTN	D1200S12 (V.22) F1200
DATEX-P20	PSTN	D1200S (V.23 with sec channel, 75 baud Transmit, 1200 baud Receive) F1200/75
DATEX-P20		F2400

9.4 Data Link Layer

This section describes the data link layer.

9.4.1 7-Bit VT100 or 8-Bit VT510 Operation

The VT510 supports three broad classes of application requirements:

- Strict compatibility for existing VT100 applications:
 - No 8-bit control codes
 - No 8-bit data. Function keys are not limited to the VT100 set as in previous terminals.
- A migration path that allows most applications to run without modification:
 - No 8-bit control codes
 - 8-bit graphic data allowable
 - VT400 function keys are allowable (but not immediately useful) and can be integrated into the application over time.

- Full VT510 mode (VT level 4) functionality:
 - 8-bit data allowable
 - 8-bit controls allowable
 - All VT500 function and modifier keys

The VT510 application requirements are possible through the following VT terminal modes:

- VT52 Mode (7-bit characters, 7-bit controls)
- VT100 Mode (7-bit characters, 7-bit controls)
- VT400 Mode, 8-bit characters, 7-bit controls
- VT400 Mode, 8-bit characters, 8-bit controls
- VT400 Mode, 7-bit characters (Keyboard restricted to keys that generate 7-bit codes, 7-bit graphics, 7-bit controls, for NRC sets, worldwide version only).

These modes are the only supported modes. 8-bit characters are temporarily disabled when in VT100 mode.

The following are certain limitations within these modes:

- The terminal accepts both C0 and C1 controls when 8-bit characters are selected (7-bit NRCS Characters is not selected).
- The terminal accepts both GL and GR characters when 8-bit Characters are selected.
- When 7-bit characters are selected, the terminal sets the high bit of all received characters to zero. This action results in compatible character handling. The received characters, therefore, have only seven significant bits. This is applicable to both graphics characters and control characters.
- If the user needs to restrict the controls to 7 bits, but still needs 8-bit graphic characters and new function keys, then the user can select Transmit 7-bit controls with an 8-bit host line. To get full efficiency out of new applications, select 8-bit controls and an 8-bit host line. The 7-bit NRCS Characters selection is compatible with 7-bit NRC-based applications.
- C1 controls are never available directly from the default keyboard, although they may be available indirectly through the function keys or numeric compose.
- When the terminal is in printer controller mode and the 7-bit character mode, upon receiving 8-bit graphic characters, the terminal immediately retransmits them through the printer port to the printer. The highest bit is not stripped off.
- In printer controller mode when the terminal is receiving 8-bit characters and immediately retransmitting them on a 7-bit line, the high bit is stripped off.
- When the screen contains 8-bit (GR) characters and is being printed on a 7-bit printer, the characters are subject to constraints, which are described in [Chapter 10](#).
- Keyboard operation guidelines are described in [Chapter 8](#).

9.4.2 Flow Control

Flow control is a means to alternately suspend and resume the flow of characters over serial asynchronous point-to-point communication lines.

Depending on the type of flow control selected, the stop/resume transmission signal is different. Users can select a software flow control scheme, a hardware flow control scheme, or a combination software and hardware flow control scheme to meet their communication environment.

9.4.2.1 Receive Buffer

The VT510 operates at a transmission speed of up to 115.2K baud. To prevent loss of data, the receive buffer is 1024 byte. The substitute (SUB) character is substituted in the input buffer for any character with detected receive parity errors or framing errors. If the input buffer overflows, a SUB character is placed in the buffer in place of the lost characters. The SUB character is displayed as a reverse question mark.

9.4.2.2 Flow Control Selection

The VT510 supports bidirectional flow control. The flow control scheme can be selected independently for the transmit and receive direction. The user has the following choices for transmit data flow control:

- No Flow Control
- XON/XOFF
- DTR
- Both

The user has the following choices for receive flow control:

- No Flow Control
- XON/XOFF in VT mode or XPC in PC term mode
- DSR
- Both

You can select flow control methods from Set-Up or through the escape sequences, DECSFC.

9.4.2.2.1 No Flow Control

If No Flow Control is selected, then when the receive buffer is full, it does not send any signal; therefore, the **Hold** key is disabled. The **Ctrl/S** key stops the flow and the **Ctrl/Q** resumes the flow. It is left to the application and user to prevent buffer overflow.

Software that does not support flow control may use all the features of the VT510 by treating NUL as a fill character. If Ignore NUL is selected in Set-Up, then NUL is discarded upon receipt. All other received characters are stored in the input buffer for further processing. Ignore NUL or Accept NUL can be selected from Set-Up or through the escape sequence, DECNUML.

The VT510 can be used without flow control or fill characters, when the:

- Software never sends ESC sequences to the terminal
- Baud rate is limited to 19200 baud or less
- Software does not use smooth scrolling

9.4.2.2.2 XOFF/XON or XPC Flow Control

In the XOFF/XON flow control, two characters establish flow control. In VT mode, these two characters are DC1(11H) (XON) and DC3(13H) (XOFF). In VT mode, the VT510 sends XOFF to suspend the transmission of data and XON to resume the reception. Receipt of XOFF inhibits the VT510 from transmitting any code except XOFF and XON. If the keyboard data buffer overflows, then the keyboard locks and "wait" is displayed in the keyboard indicator line. Transmission resumes upon receipt of XON.

In PC Term mode, the terminal receives from the host 11H(DC1) and 13H(DC3) for flow control; but the terminal sends 65H and 67H to the host to resume/stop the communication, known as XPC flow control.

VT510 sends XOFF to indicate halt of transmission when:

1. The number of characters in the input buffer reaches the first XOFF point since the last XON was sent.
2. The number of characters in the input buffer reaches the second XOFF point since the last XON was sent.
3. The input buffer is full.
4. Entering a hold state.

Entering a Hold state causes an XOFF to be sent by the VT510. A Hold state can be generated by pressing the **Hold** key or entering **Ctrl/S**. When exiting from the hold state either by depressing the

Hold key or entering **Ctrl/Q**, the terminal sends XON to the host unless the input buffer is filled up to or beyond the first XOFF point (flow control threshold).

With the exception of exiting from the hold state, the VT510 sends an XON when the input buffer is emptied to the XON point and when XOFF was the last flow control sent.

Refer to the [Section 9.4.2.2.5](#) for more information on the thresholds.

The VT510 re-initializes the XON/XOFF(XPC) state under the following conditions:

1. The Clear Comm function is performed.
2. The completion of the NVR recall after the RECALL function is performed.
3. The completion of all tests in a self-test sequence is performed.
4. Successful completion of a connection - generally, DSR assertion.
5. The reception of RIS.

9.4.2.2.3 DSR/DTR Flow Control

When Hardware flow control is selected, DTR is asserted any time the terminal is not in self-test and the terminal's input buffer is below the flow control threshold; DTR is de-asserted when the input buffer is filled up to or beyond the threshold.

If the terminal is a sending device, then it polls DSR before transmitting each character. If DSR is not being asserted by the receiving device, then transmission of the character is suspended until DSR is asserted.

As in the XON/XOFF software flow control, VT510 deasserts DTR when:

1. The number of characters in the input buffer is greater than or equal to the flow control threshold (first XOFF point).
2. Entering the Hold state.

As in the XON/XOFF flow control, entering a Hold state causes a deassertion of DTR. When exiting from the hold state either by depressing the **Hold** key or by entering **Ctrl/Q**, the terminal asserts DTR unless the input buffer is filled up to or beyond the flow control threshold (first XOFF point).

With the exception of exiting from the hold state, the VT510 reasserts DTR when the input buffer is emptied to the "resume transmission" (XON) point.

DSR/DTR can be used for hardware flow control when they are not needed for modem control functions. When Modem Control is enabled, DSR/DTR should not be selectable.

9.4.2.2.4 Software and Hardware Combined Flow Control

For software and hardware flow control, both DTR/DSR and XON/XOFF(XPC) are used to control the data flow. When the receive input buffer reaches the flow control threshold, XOFF is sent and DTR is deasserted. When transmitting, the terminal must be in the XON state with DSR asserted to transmit the next character, except in the case of (XON/XOFF).

9.4.2.2.5 Flow Control Threshold

[Table 9–3](#) shows the flow control threshold points available for the VT510.

Table 9–3 Flow Control Threshold Points

1st Flow Control Threshold	Resume Transmission Point	2nd Flow Control Threshold
64	32	896
768	128	896

9.4.2.3 Show Control Character Mode

While in show control character mode, all characters are temporarily stored in the input buffer and eventually displayed on the screen in the CRM font. XON and XOFF can serve both as data

(to be displayed) and flow control. Note that this circumstance is unique to show control character mode.

9.4.2.4 Set-Up on Flow Control Handling

Entering Set-Up leaves the received character processing in the state it was in before entering Set-Up. Characters continue to fill the silo, depending on the flow control type selected. A signal (XOFF or lower DTR) will be sent when the silo gets to the first flow control threshold.

9.4.2.5 When Disconnect Occurs

If a disconnect occurs while the VT510's silo contains characters that have not been displayed or printed, then the characters may be discarded in order to prepare for the next call. XON is sent when a new connection is established.

9.5 Features that Affect Communications

The following features described in [Chapter 2](#) affect communication:

- Clear communications
- On-Line/Local
- Transmit speed
- Receive speed
- Transmit rate limiting
- Word size
- Parity
- Stop bits
- Modem control
- Modem high speed
- Modem low speed
- Disconnect delay
- Local echo

9.5.1 Clear Communications

If the terminal appears to be "locked" in Set-Up, then select Clear communications to clear the current session. This action:

1. Aborts any print operation in progress if the printer is allocated to the active session.
2. Aborts any escape sequence, control sequence, or device control string (DCS) processing.
3. Clears the keyboard buffer.
4. Clears the receive buffer.
5. Clears the transmit buffer.
6. Takes the terminal out of printer controller mode.
7. Sends XON to the host port if software flow control is in use. Asserts DTR if hardware flow control is in use.
8. Resets XOFF received flags on the printer and host ports if software flow control is in use.

The communications had been cleared when "Done" appears on the terminal status line.

Note

Clear communications does not cause a disconnect and does not clear the screen.

9.5.2 Connection

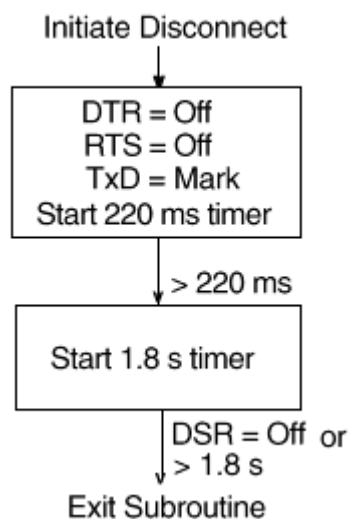
If XON/XOFF flow control is enabled, then a single XON will be transmitted after the connection is made. When a connection is made, the VT510 performs the following operations to ensure that it is ready to send and receive:

1. Unlock the keyboard, if it is locked.
2. Clear any transmit in progress on this port.
3. Clear all buffers associated with this port.
4. Clear the "XOFF sent" and "XOFF received" state of this port.
5. At the reception of the carrier tone, the modem asserts CD, and data transmission/reception may begin.

9.5.3 Line Disconnection

A disconnect sequence occurs on the following conditions:

- Operator initiates a disconnect by pressing Shift/Break (note that the BREAK can be disabled through Keyboard Set-Up).
- Doing a RECALL, or DEFAULT.
- Loss of DSR.
- Switching to the EIA 232-E port from the DEC423 port or to the DEC423 port from the EIA232-E port.
- With the 25-pin connector and modem control enabled, after a connection has been established, loss of RLSD for greater than 2 seconds (or 60 ms) on host line 1.
- Receipt of a DECTST or RIS sequence (Except the keyboard test in DECTST that leaves the terminal on line).
- Turning off the terminal.



Normally DTR and RTS are asserted by the VT510 whenever power is applied except:

- During a disconnect sequence
- When hardware flow control DTR/DSR is being used and the receive buffer reaches 64 bytes threshold

9.5.4 Break

The Break key operates as described in [Chapter 8](#).

A break consists of a 275 ± 25 ms space condition on the transmit data line if transmission is enabled. The break (space) condition is separated from any subsequent mark-to-space transition by at least 100 ms.

9.5.4.1 Received Break

If a break is detected on the received data circuit, then it is ignored. One or more error characters (reverse question mark) may be displayed because the framing errors were generated.

This chapter describes the features of the VT510 printer ports.

10.1 Overview

The VT510 expands the traditional printer port extension to include support for IBM ProPrinter and compatible low-cost parallel interface printers. In Set-Up, you specify the type of printer attached and the terminal automatically selects the correct printing commands and character sets.

The VT510 printer port has the following features:

- Centronics parallel port (25-pin, female, D-type connector)
- IBM ProPrinter mode
- Ability to transparently pass all characters (including the NUL character) through to the printer in printer controller mode

10.2 Serial Ports

The VT510 has two bi-directional serial communication/printer ports. Comm1 has full modem support at 300 to 115.2K baud:

EIA 232	EIA 423
Comm 1:	Comm 2:
Two 25-pin D-sub male/female	6-pin MMJ
(use one or the other)	

You can connect both a serial and parallel printer to the VT510 at the same time. In this case, the printer port (serial or parallel) selected in Set-Up is used and the other port is inactive.

The interface signals are the same as those previously described in [Chapter 9](#).

10.2.1 Serial Port Flow Control

The VT510 serial printer port is fully bi-directional when Printer To Host communications is enabled in Set-Up or by a Media Copy (MC) control sequence. Transmit and receive flow control protocols can be set independently to one of four choices through Set-Up or by the control sequence, DECSFC.

- None
- XON/XOFF
- DTR/DSR
- Both

The printer port has two buffers: one is for output, which is 16 bytes long; and the other is for input, which is 254 bytes long. XON/XOFF: For the 254-character receive input buffer, the XON point is 32. The first XOFF point is 64 characters, and the second is 220 characters.

DTR/DSR: If enabled, the terminal polls DSR before transmitting each character. The terminal asserts DTR until the buffer fills up to or beyond the first XOFF point. When the buffer empties to the XON point, the terminal again asserts DTR. When not used for hardware flow control, DTR is asserted whenever the firmware is active and not in its self-test.

When the input buffer is full and receives another character byte, it overflows. In the event of an input buffer overflow, the last byte in the buffer is replaced with the SUB character.

10.3 Parallel Port

The Centronics parallel interface is considered to be industry standard and is not explained in this chapter. This port uses a 25-pin D-type connector with pin assignments shown in [Figure 10–1](#) and listed in [Table 10–1](#).

Figure 10–1 Parallel Printer Port

1 STROBE L	10 ACKNLG L
2 DAT <0>	11 BUSY
3 DAT <1>	12 PE
4 DAT <2>	13 SLCT
5 DAT <3>	14 AUTO FEED XT L
6 DAT <4>	15 ERROR L
7 DAT <5>	16 INIT L
8 DAT <6>	17 SLCT IN L
9 DAT <7>	18 – 25 GND

Table 10–1 Parallel Connector Signals

Pin	Signal Name	Source	Function
1	STROBE L	Terminal	Strobe
2	DATA0	Terminal	Data Bit 1
3	DATA1	Terminal	Data Bit 2
4	DATA2	Terminal	Data Bit 3
5	DATA3	Terminal	Data Bit 4
6	DATA4	Terminal	Data Bit 5
7	DATA5	Terminal	Data Bit 6
8	DATA6	Terminal	Data Bit 7
9	DATA7	Terminal	Data Bit 8
10	ACKNLG L	Printer	Acknowledge
11	BUSY	Printer	Printer cannot receive data.
12	PE	Printer	Printer is out of paper.
13	SLCT	Printer	Printer is on-line.
14	AUTO FEED XT L	Terminal	Feed 1 line automatically after printing.
15	ERROR L	Printer	Paper end, printer off-line, or printer error
16	INIT L	Terminal	Resets the printer to power-on defaults
17	SLCT IN L	Terminal	The VT510 monitors SLCT to determine if a parallel printer is attached and on-line.
18 – 25	GROUND	N/A	

The VT510 uses the eight data lines along with control signals STROBE, ACKNLG, and BUSY to transfer data to the printer. All other interface signals are ignored.

10.3.1 Parallel Port Flow Control

Characters are transferred to the printer in 8-bit bytes. A parallel printer port is not duplex; data can only go from the terminal to the printer.

The terminal and the printer use interface signals STROBE, ACKNLG, and BUSY to synchronize the data transfer as follows:

1. When the terminal is ready to send data to the printer, the terminal puts the data on the data lines and then generates a STROBE pulse to the printer.
2. When the printer senses the STROBE, it responds with a BUSY signal.
3. When printer receives the data, it pulses ACKNLG to signal the terminal that data has been read into memory.
4. If the printer buffer is full, the printer keeps the BUSY signal asserted.

The parallel printer port does not recognize XON/XOFF flow control.

10.4 Print Modes

There are five modes that affect the control of printing operations:

- Printer Controller mode
- Local Controller mode
- Auto Print mode
- Print Form Feed mode
- Print Extent mode

These print modes follow standard terminal transmission except for Printer Controller mode.

10.4.1 Printer Controller Mode

When printer controller mode is enabled, all data received at the host port is passed through to the printer port, except for:

- XON/XOFF when XON/XOFF flow control is enabled at either the host port or the printer port
- NUL when Ignore NUL characters is selected in the communication setup.

If printer-to-host communication is enabled, then all data received at the printer port is passed through to the host port with the exceptions listed previously.

10.5 Print Operation

The VT510 supports the following print operations:

- Print Line
- Print Page or Scrolling Region
- Print Screen

Print Screen prints the screen as it appears, including the status lines if enabled. Print Screen, however, does not print the keyboard indicator line.

10.6 Transmitting Print Data

Users have three choices for a printer type:

- DEC ANSI
- IBM ProPrinter
- DEC + IBM emulation mode

Depending on the printer type selected, the terminal transmits different control functions to the printer. You can select the printer type in Set-Up or by using the escape sequence, DECSPRTT.

10.6.1 DEC ANSI

The DEC ANSI protocol supports the following printed data types:

- When in 8-bit mode:
 - ASCII only

- ASCII + Line Drawing
- When in 7-bit mode:
 - National only
 - National + Line Drawing

You select a printed data type based on the 7- or 8-bit printer port setting and the character set capability of the attached printer. The VT510 sends initialization, transmission, and closing sequences according to the printed data type selected.

If IBM PC characters are displayed, then the terminal translates the corresponding characters in the ISO Latin-1 character set and transmits them based on the current printed data type selected. If there is no corresponding character in ISO Latin-1, then the Space character is used instead.

10.6.2 IBM ProPrinter or Compatible Printer

When an IBM ProPrinter is attached, the terminal sends IBM ProPrinter sequences. You can neither designate, nor invoke the ANSI sequence or any other operating sequences in this mode.

The VT510 supports the following printer character sets for the IBM printer: PC Multilingual, PC Latin 2 (PC Slavic), PC International, PC Spanish, PC Portuguese, PC Danish, PC French Canadian, PC Hebrew, PC Greek, PC Turkish, and PC Cyrillic.

During initialization, the VT510 sends a control sequence to the printer to select the same code page as the printer character set selected in the terminal. You must select the appropriate printer character set for the correct data to be printed either in Set-Up or by the escape sequence, DECSPPCS.

In an IBM ProPrinter, each code page has the following character sets:

- Character Set 1
- Character Set 2
- All Characters

The three character sets are identical, except for the interpretation of C0 (0/0 through 1/15) and C1 (8/0 through 9/15) characters.

In Character Set 1, the controls in C0 are identical to those in C1, except that 80 (hex) is ignored instead of being interpreted as NUL.

Character Set 2 uses the same controls as C0, except that some printable characters are added to unused character code spaces. The C1 region is all printable characters.

The "All Characters" set only has printable characters. No control codes are allocated to this character set.

The VT510 initializes the printer to use Character Set 2 at the start of a print screen operation. When a C0 printable character is encountered, the VT510 sends the appropriate sequence to print from the "All Characters" set.

10.6.2.1 Data Transmission

The printer is either an IBM personal printer or a compatible printer. It recognizes the IBM ProPrinter mode escape sequences and supports the terminal's code pages. The printer should support a software-selectable code page switching sequence. If it does not, then you need to take appropriate action to set the correct code page before a printing operation takes place.

Characters are sent to the printer along with their display attributes. If a C0 printable character is encountered, the VT510 prints from the "All Characters" set. If a character that is not in the selected code page is encountered, then a Space character is sent instead.

The VT510 supports bold, underline, and double-height and width display attributes.

If the terminal has DEC Supplemental or ISO Latin 1 characters on the screen, then the VT510 translates the characters to corresponding characters in the code page indicated by the printer character set before the transmission takes place. If there is no corresponding character in the selected IBM code page, then the Space character is used instead.

10.6.3 DEC ANSI with IBM ProPrinter Emulation

You can specify both DEC ANSI printed data types and IBM ProPrinter PC character sets in Set-Up when this type of printer is selected. If the terminal is set to display DEC or ISO character sets, then the VT510 treats the printer as a DEC ANSI printer and transmits data to the attached printer using ANSI protocols.

If the terminal is set to display PC character sets, then the VT510 transmits data to the attached printer using the IBM ProPrinter protocol. During initialization, the VT510 sends a forces the printer into IBM ProPrinter emulation mode. The terminal sends all other sequences just as in the IBM ProPrinter mode.

10.6.3.1 Data Transmission

If the printer is a DEC ANSI printer with IBM ProPrinter emulation, the printer is in its default ANSI mode state. The printer recognizes IBM ProPrinter mode escape sequences and supports the IBM code pages. The printer supports the software selectable code page switching sequence. If it does not, then you must take appropriate action to set the correct code page before a printing operation takes place.

10.6.3.2 Printer Status

The Session Management Extension is not supported, so the terminal does not respond with "Printer Busy" and "Printer Not Assigned."

If the printer stalls, the word "Printer" flashes continuously in the Indicator Status Line.

10.7 Suspending Print Operations

Entering Set-Up can suspend, but not abort, printing operations. Upon exit from Set-Up, all unfinished printing operations resume unless an explicit action has been taken in Set-Up to abort the print operation, such as Clear communications, Reset terminal, or Restore factory defaults.

10.8 Aborting Print Operations

The Clear communications function from Set-up, aborts the current print operation, clears the output silo, and re-initializes the printer port UART. This function may be used to get out of a "hung" state.

Note

After a Clear communications command, printing may not stop immediately because characters may still be in the printer's input buffer.

Part III

VT510 ASCII Emulation

This part contains the following chapters:

- Chapter 11, [ASCII Emulations and Control Functions](#)
- Chapter 12, [ASCII Escape Sequences](#)
- Chapter 13, [Defining and Loading ASCII Character Fonts](#)

ASCII Emulations and Control Functions

This chapter describes the ASCII emulations and control functions.

11.1 Overview

The VT510 video terminal includes emulations for several popular ASCII video terminals. ASCII terminals represent the other major class of serial asynchronous video terminals in widespread use today. Like the VT100, these terminals communicate with a host computer by transmitting and receiving a data stream of 7-bit or 8-bit coded characters. To communicate successfully, the computer and the terminal use a convention for interpreting this data stream. This convention specifies how to recognize characters in the data stream as well as how to recognize control information, such as formatting or editing commands. ASCII terminals support the ASCII character set, *American Standard Code for Information Interchange*, ANSI X3.4; however, they do not follow other ANSI standards that define controls for use with the ASCII character set. The original ASCII terminal protocol pre-dates the ANSI standards. These early protocols have since been widely copied and extended to form the ASCII terminal market.

11.2 ASCII Emulations

The VT510 includes a protocol emulation for the WYSE 160 Native mode, WYSE 60 Native mode, and many of its popular emulations of other ASCII terminals:

- WYSE 160/60 Native mode
- WYSE 160/60 PCTerm
- WYSE 50+
- WYSE 150/120
- Televideo 950
- Televideo 925
- Televideo 910+
- ADDS A2

Emulation means that virtually all host application programs designed for these terminals will function correctly.

Note that the VT510 video terminal cannot be identical to all of these terminals simultaneously, and the VT510 includes enhancements in its user interface and character handling ability. The VT510 can handle additional languages and keyboards that extend beyond the original terminals being emulated.

The VT510 does not emulate the graphics modes of the WYSE 160, such as the TEK 4010/4014 personality.

11.2.1 Enhanced Mode

WYSE 160 Native mode includes function sequences for almost all features supported by the WYSE 160 terminal, including those supported by individual sub-emulations, such as Televideo or ADDS. When one of these sub-emulations is selected, WYSE Enhanced mode (EnH) allows WYSE Native mode sequences to be used from within a sub-emulation to access features that are not normally supported by the physical terminal being emulated. This capability allows existing applications, for one of these emulated terminals, to be enhanced to take advantage of WYSE Native mode features. This enhanced mode does not apply to the VT or ANSI modes, which follow the corresponding ANSI standards.

11.3 Coding Notation

Since ASCII control functions do not follow a structured syntax, the notation used to describe function sequences and parameters is important to avoid confusion. Escape sequences are shown with a space between each character to make them easier to read. These spaces are not part of the Escape sequence.

Words in all caps: Strings of two or more capital letters, such as ESC or SPACE, are used as symbols, usually to represent a single character within an Escape sequence.

Words in italics: Words or abbreviations of two or more characters in italics represent sequence parameters that are described subsequently.

Single characters: Single characters, such as "E" or "8", represent the corresponding ASCII character in the sequence.

Regular text (upper- and lowercase): Regular text is sometimes used to describe sequences that are similar or related to other sequences.

Table 11–1 lists the symbols used in the ASCII control functions.

Table 11–1 Symbols Used in ASCII Terminal Escape Sequences

Symbol Meaning		Symbol Meaning	
Ctrl-	Ctrl immediately followed by a single character represents a control character in the range of 0/0 to 1/15.	EnH	Enhance Mode: This is the same as the WYSE native code when Enhance Mode is on.
Same	Same as the WYSE native code (code is native to this emulation).	Wyse	Same as the WYSE native code (code is <i>not</i> native to this emulation, but is a WYSE enhancement).
n/a	Not available in this emulation.		
Symbol	Code	Symbol	Code
NUL	0/0	DLE	1/0
SOH	0/1	DC1 (XON)	1/1
STX	0/2	DC2	1/2
ETX	0/3	DC3 (XOFF)	1/3
EOT	0/4	DC4	1/4
ENQ	0/5	NAK	1/5
ACK	0/6	SYN	1/6
BEL	0/7	ETB	1/7
BS	0/8	CAN	1/8
HT	0/9	EM	1/9
LF	0/10	SUB	1/10
VT	0/11	ESC	1/11
FF	0/12	FS	1/12
CR	0/13	GS	1/13
SO	0/14	RS	1/14
SI	0/15	US	1/15
SPACE	4/0		

11.3.1 Communication Flow Control with PCTerm Mode

XON/XOFF transmit flow control is disabled while in PCTerm mode. This means XON and XOFF characters received from the host are ignored or displayed as characters from a PC character set. In PCTerm mode, the keyboard generates scan codes. There are no multi-character reports (escape sequences or paste data) sent from the terminal. The terminal can still send XPC as receive flow control.

When switching out of PCTerm mode, the XON/XOFF state of the communications link is unknown. If XON/XOFF flow control is enabled, then the terminal shall transmit a single XON and clear the XOFF received state.

11.3.2 Protecting Data

Turning on Write-Protect mode does not start protecting the data. It only defines the range of data that needs to be protected. Not until Protect mode is turned on does the data become protected. Changing emulation does not change the Write-Protect mode. Characters range from 1/0 to 1/15, and embedded attributes are automatically write-protected even when Write-Protect mode is off.

To protect a certain area of data in the screen, users should:

1. Turn on Write-Protect mode.
2. Turn off Write-Protect mode, if the protected area has been defined.
3. Turn on Protect mode.

Protect Mode On: The protect condition usually is valid for the current page only. The exception is when Autopage mode is on, if a cursor movement command causes the cursor to move from one protected page to an unprotected page, the unprotected page becomes protected. If a cursor movement command causes the cursor to move from an unprotected page to a protected page, then the protected page becomes unprotected. [Table 11–2](#) lists these commands.

Table 11–2 Commands that may Carry Protect Mode Features through Pages

Command	Sequence
Cursor left (backspace)	Ctrl/H
Cursor left (delete)	DEL
Cursor right	Ctrl/L
Cursor up; Scroll	ESC j
Cursor down; Scroll (linefeed)	Ctrl/J
Cursor to start of next line	Ctrl/_
Tabulate cursor	ESC i or Ctrl/I
Backtab	ESC I

- Turning on Protect mode enables Line Edit mode. The terminal ignores the command to enable Page Edit mode. Editing command activities are limited to the unprotected data. Any protected line can neither be inserted nor be deleted.
- No cursor movement commands can cause data, protected or unprotected, to scroll off the page.
- The tabulation cursor becomes the field tab.
- Changing emulation turns off Protect mode, except when changing the terminal to a TVI emulation; in which case, the Protect mode does not change.
- Some cursor commands can move the cursor into the protected area, some cannot.

>[Table 11–3](#) lists the commands that can move the cursor to the protected area. Even if the cursor can be moved into the protected area by executing these commands, no data can be

written into the area. The cursor jumps to the first unprotected position when data started writing into the screen.

Table 11–4 lists the commands that cannot move the cursor to the protected area. Any attempt to move the cursor to the protected area by executing these commands, moves the cursor up, down, left, or right into the first unprotected position.

Table 11–3 Commands that Can Move the Cursor to Protected Area

Move Cursor . . .	Sequence
Up; scroll (reverse linefeed)	ESC j
Down; no scroll	Ctrl/V
To specific line	ESC [<i>line</i>
To a specific column	ESC _ <i>column</i>
In current page	ESC = <i>line col</i>
To a specific page	ESC h <i>page</i>
In specific 80-column page	ESC w @ <i>page line col</i>
In specific 80-column window	ESC - <i>wnd/page line col</i>
In 80/132-column current page	ESC a ll R <i>ccc C</i>

Table 11–4 Commands that Cannot Move Cursor into Protected Area

Move Cursor . . .	Sequence
Left (backspace)	Ctrl/H †
Left (delete)	DEL
Right	Ctrl/L
Up; No scroll	Ctrl/K †‡
Down; Scroll (linefeed)	Ctrl/J
To start of line	Ctrl/M
To start of next line	Ctrl/_
Home	ESC { or Ctrl/^

†The cursor can be moved to a protected field under certain conditions. Refer to the specific cursor command section for details.

‡Cursor can be moved to a protected field for TVI personalities.

ASCII Escape Sequences

This chapter summarizes the ASCII escape and control Sequences that the VT510 recognizes when it is operating in its ASCII emulation modes.

12.1 ASCII Communications

Table 12–1 lists the serial port control functions for each mode. The word "same" in **Table 12–1** signifies the same control sequence as the WYSE 160/60 column.

Table 12–1 Serial Port Control Functions

ACK Mode	WYSE			TVI				ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
ACK Mode On (default)	ESC e 7	n/a	same	same	EnH	EnH	EnH	EnH
ACK Mode Off	ESC e 6	n/a	same	same	EnH	EnH	EnH	EnH
Answerback Mode								
Answerback Off (default)	ESC e Sp	n/a		same	n/a	n/a	n/a	EnH
Answerback On	ESC e !	n/a	same	same	n/a	n/a	n/a	n/a
Answerback Message	ESC c ; <i>answer</i> Ctrl/Y ESC] <i>answer</i> Ctrl/M							
			same	same	n/a	n/a	n/a	EnH
Send Answerback message – Format 1:	ESC c <	n/a	same	same	n/a	n/a	n/a	EnH
Format 2:	Ctrl/E	same	same	same	same	same	same	EnH
Conceal Answerback	ESC c =	n/a	same	same	n/a	n/a	n/a	EnH
Assign Serial Ports								
S1=comm1 (host)	ESC e 8	n/a	same	same	EnH	EnH	EnH	EnH
S1=comm2 (host)	ESC e 9	n/a	same	same	EnH	EnH	EnH	EnH
Monitor Mode								
Monitor Mode Off (default)	ESC u or ESC X	same	same	same	same	same	same	EnH on or off
Monitor Mode On	ESC U	same	same	same	same	same	same	EnH
Receive Null Character Mode								
Ignore Null Characters from Host (default)	ESC e R	n/a	same	same	EnH	EnH	EnH	EnH
Accept Null Characters from Host	ESC e S	n/a	same	same	EnH	EnH	EnH	EnH
Set Serial Port Parameters								
Set Comm1 parameters	ESC c 0 <i>Baud Stop Parity Word</i>				ESC { <i>Baud Stop Parity Word</i>			
		n/a	same	same		n/a	n/a	EnH
Set Comm2 parameters	ESC c 1 <i>Baud Stop Parity Word</i>				ESC } <i>Baud Stop Parity Word</i>			
		n/a	same	same		n/a	n/a	EnH
	<u>Baud (Comm1)</u>		<u>Baud (Comm2)</u>		<u>Description</u>			
	D		N/A		115200			
	B		N/A		76800			
	C		N/A		56700			
	0		@ (4/0)		38400			
	1		0 (3/0)		19200			
	2		1 (3/1)		9600			
	3		3 (3/3)		4800			
	4		5 (3/5)		2400			

	WYSE			TVI			ADDS	
ACK Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
	7		8 (3/8)			1200		
	8		9 (3/9)			600		
	9		: (3/10)			300		
	<u>Stop</u>		<u>Description</u>					
	0 (3/0)		1 Stop Bit (default)					
	1 (3/1)		2 Stop Bits					
	<u>Word</u>		<u>Description</u>					
	0 (3/0)		7 bits					
	1 (3/1)		8 bits (default)					
	<u>Parity</u>		<u>Description</u>					
	0 (3/0)		No Parity (default)					
	1 (3/1)		Odd Parity					
	2 (3/2)		Mark Parity					
	3 (3/3)		Even Parity					
	4 (3/4)		Space Parity					
Serial Port Handshaking								
Set Comm1 Receive	ESC c 2 Handshaking							
	n/a	same	same	n/a	n/a	n/a		EnH
Set Comm2 Receive	ESC c 3 Handshaking							
	n/a	same	same	n/a	n/a	n/a		EnH
Set Comm1 Transmit	ESC c 4 Handshaking							
	n/a	same	same	n/a	n/a	n/a		EnH
Set Comm2 Transmit	ESC c 5 Handshaking							
	n/a	same	same	n/a	n/a	n/a		EnH
	<u>Handshaking</u>		<u>Receive Flow Control</u>			<u>Transmit Flow Control</u>		
	0		None			None (default)		
	1		XON-XOFF/XPC (default)			XON/XOFF		
	2		DTR			DSR		
	3		XON-XOFF/XPC and DTR			XON/XOFF and DSR		
Set Host port Receive to DTR	n/a	Ctrl/N	n/a	n/a	Ctrl/N	Ctrl/N	Ctrl/N	n/a
Set Host port Receive to XON/XOFF or XPC	n/a	Ctrl/O	n/a	n/a	Ctrl/O	Ctrl/O	Ctrl/O	n/a
Set Duplex Mode								
Full-Duplex On (default)	ESC D F	ESC }	same	same	same	same	same	EnH
Half-Duplex On	ESC D H	ESC {	same	same	same	same	same	EnH
Set Block Mode								

	WYSE			TVI			ADDS	
ACK Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Block Mode Off (default)	ESC C	same	same	same	Wyse	Wyse	Wyse	EnH
Block Mode On	ESC B	same	same	same	same	same or ESC c	same or ESC c	EnH
Set Local Mode								
Set Local Mode	n/a	n/a	n/a	n/a	ESC c	n/a	n/a	n/a
Set Maximum Transmit Rate Limit								
Transmit Speed	ESC c 6 <i>Max</i>	n/a	same	same	n/a	n/a	n/a	EnH
Function Keys	ESC c 7 <i>Max</i>	n/a	same	same	n/a	n/a	n/a	EnH
	<u>Max</u>		<u>Characters Per Second</u>					
	0		30					
	1		50					
	2		No Limit (default)					
	3		150					

12.2 Terminal Management

Table 12–2 lists the terminal management control functions. The word "same" in **Table 12–2** signifies the same control sequence as in the WYSE 160/60 column.

Table 12–2 Terminal Management Control Functions

Select Terminal Personality	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
ADDS A2	ESC ~ %	ESC v %	same as 160/60	same	Wyse	Wyse	Wyse	Wyse
TVI 950	ESC ~ (	ESC v (	same	same	Wyse	Wyse	Wyse	Wyse
TVI 925	ESC ~ \$	ESC v \$	same	same	Wyse	Wyse	Wyse	Wyse
TVI 910	ESC ~ #	ESC v #	same	same	Wyse	Wyse	Wyse	Wyse
VT510, 7-bit	ESC ~ <	ESC v <	same	same	Wyse	Wyse	Wyse	Wyse
VT510, 8-bit	ESC ~ =	ESC v =	same	same	Wyse	Wyse	Wyse	Wyse
VT100	ESC ~ ;	ESC v ;	same	same	Wyse	Wyse	Wyse	Wyse
VT52	ESC ~ 6	ESC v 6	same	same	Wyse	Wyse	Wyse	Wyse
WYSE 160/60	ESC ~ 4	ESC v 4	same	same	Wyse	Wyse	Wyse	Wyse
WYSE 160/60 PCTerm	ESC ~ 5	ESC v 5	same	same	Wyse	Wyse	Wyse	Wyse
WYSE 150/120	ESC ~ 3	ESC v 3	same	same	Wyse	Wyse	Wyse	Wyse
WYSE 50+	ESC ~ "	ESC v "	same	same	Wyse	Wyse	Wyse	Wyse
Send Terminal ID								
Inquire emulation ID	ESC Sp	n/a	same	same	ESC M	ESC M	ESC M	EnH
Set Enhanced Mode								
Enhanced Mode On (EnH) (default)	ESC ~ !	ESC v !	same	same	Wyse	Wyse	Wyse	Wyse
Enhanced Mode Off	ESC ~ Sp	ESC v Sp	same	same	Wyse	Wyse	Wyse	Wyse
Save Set-Up Modes								
To NVM (except Tabs)	ESC c V	n/a	same	same	n/a	n/a	n/a	EnH
To NVM with Tabs	ESC c W	n/a	same	same	n/a	n/a	n/a	EnH
Select Terminal Defaults								
PCTerm Mode Group ¹	n/a	ESC m	n/a	n/a	n/a	n/a	n/a	n/a
All Modes ²	ESC c T	n/a	same	same	n/a	n/a	n/a	EnH
Power On Reset	ESC c X	n/a	same	same	n/a	n/a	n/a	EnH
Load/Send Time of Day								

¹Sets the following parameters for PCTerm mode:

- Autoscroll = On
- Received CR = CR
- Communication Mode = Full Duplex
- Enhanced Mode = Off
- Wrap EOL = Off
- Attribute = Char

²Sets the parameters shown in [Table 11-2](#).

³*Hour* ranges from 00 to 23; *Minute* ranges from 00 to 59 decimal. Entering "00" in hour is displayed as 12 AM. The a/p is a single character—a = AM, p = PM.

Select Terminal Personality	WYSE			TVI			ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Load Time of Day ³	ESC c 8 Hr Min							
	ESC Sp 1 a/p Hr Min							
			same	same	ESC Sp 1 a/p Hr Min			EnH
Send Time of Day	n/a	ESC Sp 2	n/a	n/a	ESC Sp 2	ESC Sp 2	ESC Sp 2	n/a

¹Sets the following parameters for PCTerm mode:

- Autoscroll = On
- Received CR = CR
- Communication Mode = Full Duplex
- Enhanced Mode = Off
- Wrap EOL = Off
- Attribute = Char

²Sets the parameters shown in [Table 11-2](#).

³*Hour* ranges from 00 to 23; *Minute* ranges from 00 to 59 decimal. Entering "00" in hour is displayed as 12 AM. The a/p is a single character — a = AM, p = PM.

12.3 Keyboard Processing

[Table 12–3](#) lists the keyboard processing control functions. The word "same" in [Table 12–3](#) signifies the same control sequences as in the WYSE 160/60 column.

Table 12–3 Keyboard Processing Control Functions

	WYSE				TVI			ADDS
Locking the Keyboard	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Unlock keyboard (default)	Ctrl/N or ESC "	ESC "	same	same	ESC "	ESC "	ESC "	Ctrl/B ESC 6
Lock keyboard	Ctrl/O or ESC #	ESC #	same	same	ESC #	ESC #	ESC #	Ctrl/D ESC 5
Enable/Disable Key Repeat								
Key Repeat On (default)	ESC e -	n/a	same	same	EnH	EnH	EnH	EnH
Key Repeat Off	ESC e ,	n/a	same	same	EnH	EnH	EnH	EnH
Caps Lock								
Caps Lock off	ESC e '	n/a	same	same	EnH	EnH	EnH	EnH
Caps Lock on	ESC e &	n/a	same	same	EnH	EnH	EnH	EnH
Lock key=Caps Lock (default)	ESC e T	n/a	same	same	EnH	EnH	EnH	EnH
Lock key=Shift Lock	ESC e V	n/a	same	same	EnH	EnH	EnH	EnH
Lock key=Reverse Lock	ESC e U	n/a	same	same	EnH	EnH	EnH	EnH
Keycode Mode								
ASCII Key Codes ¹	ESC e H	ESC c	same	same	EnH	EnH	EnH	EnH
PC Scan Codes	ESC e I	ESC H	same	same	EnH	EnH	EnH	EnH
Keyclick								
On (default)	ESC e %	ESC >	same	same	ESC >	ESC >	ESC >	EnH
Off	ESC e \$	ESC <	same	same	ESC <	ESC <	ESC <	EnH
Margin Bell								
Sound Bell	Ctrl/G	same	same	same	same	same	same	same
Select Bell tone	ESC c \ <i>Tone</i>	n/a	same	Wyse	n/a	n/a	n/a	EnH
	<u>Tone</u>	<u>Description</u>						
	0	Off						
	1	Low pitch						
	2	High pitch						
	3	High pitch						
Margin Bell off (default)	ESC e L	ESC n	same	same	EnH	EnH	EnH	EnH
Margin Bell on ²	ESC e M	ESC o	same	same	EnH	EnH	EnH	EnH
Set Margin Bell at cursor	ESC ` J	n/a	same	same	n/a	n/a	n/a	EnH
Set Margin Bell on and set position	n/a	ESC o	n/a	n/a	n/a	n/a	n/a	n/a

¹Default for all but PCTerm mode.²Default = 72 for 80-column mode; 124 for 132-column mode.³ See Table 12–4 for redefinable keys.⁴*Seq* is an ASCII character data string sequence. Any 7-bit character except "DEL" (Hex 7F) is legal for the sequence.

	WYSE	TVI					ADDS	
Locking the Keyboard	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Modifier Keys								
Request Status Inquiry	n/a	ESC [	n/a	n/a	n/a	n/a	n/a	n/a
Report: 81(Hex) + <i>a</i> where <i>a</i> indicates the modifier key being pressed as follows:								
		<u>Bit</u>	<u>Modifier Key</u>					
		0	Caps lock					
		1	Always 0					
		2	Always 0					
		3	Scroll lock On					
		4	Alt					
		5	Ctrl					
		6	Num lock On					
		7	Shift key					

Redefining Keys³

Program Function keys ⁴	ESC z <i>Fkey Seq</i> DEL						
	n/a	same	same	Wyse	Wyse	Wyse	EnH
	<u>Fkey Unshifted</u>	<u>Fkey Shifted</u>		<u>VT Keyboard</u>		<u>Enhanced PC Keyboard</u>	
	@	`		F1		F1	
	A	a		F2		F2	
	B	b		F3		F3	
	C	c		F4		F4	
	D	d		F5		F5	
	E	e		F6		F6	
	F	f		F7		F7	
	G	g		F8		F8	
	H	h		F9		F9	
	I	i		F10		F10	
	J	j		F11		F11	
	K	k		F12		F12	
	L	l		F13			
	M	m		F14			
	N	n		F15			

¹Default for all but PCTerm mode.

²Default = 72 for 80-column mode; 124 for 132-column mode.

³ See [Table 12–4](#) for redefinable keys.

⁴*Seq* is an ASCII character data string sequence. Any 7-bit character except "DEL" (Hex 7F) is legal for the sequence.

	WYSE			TVI			ADDS	
Locking the Keyboard	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
	O		o		F16			
Program Key Direction	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Program Key Direction	ESC Z <i>Dir Key Seq</i> DEL		same	same		Wyse	Wyse	EnH
	ESC <i>FKey Dir Seq</i> Ctrl-Y				ESC <i>FKey Dir Seq</i> Ctrl-Y			
Key-Sent direction	Communication Mode			Key Definition is Sent to . . .				
	Local			Terminal screen only				
	Remote			Host only				
	Normal			Depending on Comm mode:				
	Full-duplex			Host only				
	Half-duplex			Both the host and the terminal screen				
	Block			Terminal screen only				
	Half Block			Both the host and the terminal screen				
	<u>Dir</u>	<u>Key Transmission Direction</u>						
	0	Normal						
	1	Remote (default)						
	2	Local						
	3	Normal (TVI950 and WYSE 160/60 PCTerm only)						
Inquire Key Direction	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Inquiry:	ESC Z ~ <i>Key</i>	n/a	same	same	n/a	Wyse	Wyse	EnH
Response:	<i>Dir Key Sequence</i> DEL							
Clear Key Definitions								
Clear Function key direction and definition	ESC z <i>Fkey</i> DEL	n/a	same	same	Wyse	Wyse	Wyse	EnH
Set key direction and Clear key definition	ESC Z <i>Dir Key</i> DEL							
		n/a	same	same	n/a	Wyse	Wyse	EnH
Clear all redefinable key definitions	ESC c U	n/a	same	same	n/a	n/a	n/a	EnH

¹Default for all but PCTerm mode.

²Default = 72 for 80-column mode; 124 for 132-column mode.

³ See [Table 12–4](#) for redefinable keys.

⁴*Seq* is an ASCII character data string sequence. Any 7-bit character except "DEL" (Hex 7F) is legal for the sequence.

[Table 12–4](#) lists the redefinable keys.

Table 12–4 Redefinable Keys

Emulation	Redefinable Keys			
WYSE 160/60, WYSE 150/120, WYSE 50+, TVI925, and TVI 910	<u>Unshifted</u>	<u>Shifted</u>	<u>VT Keyboard</u>	<u>EPC Keyboard</u>
	SP	%		ESC
	!	&	Tab	Tab
	"	'	Backspace	Backspace
	#	(	Remove	
	\$	)	Return	Enter
	*	/	Select	Home
	+	0	↑	↑
	,	1	↓	↓
	-	2	←	←
	.	3	⇒	⇒
	s	4	Enter	Enter (keypad)
	q	p	PF4	Insert
	r	w		Page Down
	}	z	PF3	
	5	6	PF2	Delete
	7	8	PF1	
	R	X		Print Screen
	:	;		Page Up
	\	]		End
	Q	W	F17	
	S	Y	F18	
	T	Z	F19	
	P	V	F20	
TVI 950	<u>Fkey</u>	<u>Fkey</u>	<u>VT Keyboard</u>	<u>Enhanced PC Keyboard</u>
	1	<	F1	F1
	2	=	F2	F2
	3	>	F3	F3
	4	?	F4	F4
	5	@	F5	F5
	6	A	F6	F6
	7	B	F7	F7
	8	C	F8	F8
	9	D	F9	F9
	:	E	F10	F10
	;	F	F11	F11
	G	L	F12	F12

Emulation	Redefinable Keys		
	H	M	F13
	I	N	F14
	J	O	F15
	K	P	F16
WYSE 160/60 PCTerm	<u>Fkey Unshifted</u>	<u>VT Keyboard</u>	<u>EPC Keyboard</u>
	1	!	F1
	2	@	F2
	3	#	F3
	4	\$	F4
	5	%	F5
	6	^	F6
	7	&	F7
	8	*	F8
	9	(	F9
	0	)	F10

12.4 Screen Processing

[Table 12–5](#) lists the screen processing control functions. The word "same" in [Table 12–5](#) signifies the same control sequences as in the WYSE 160/60 column.

Table 12–5 Screen Processing Control Functions

Video Display Features	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Border Display (Overscan On/Off)	ESC c] <i>Intensity</i>							
		n/a	same	same	n/a	n/a	n/a	EnH
Parameters:	<u>Intensity</u>	<u>Overscan</u>						
	0	Off (default)						
	1	On						
Screen Display On (default)	ESC ` 9	ESC N	same	same	ESC n	ESC n	ESC n	EnH
Screen Display Off	ESC ` 8	ESC O	same	same	ESC o	ESC o	ESC o	EnH
Normal background ¹	ESC ^ 0	ESC d	same	same	ESC d	ESC d	ESC d	EnH
Reverse background	ESC ^ 1	ESC b	same	same	ESC b	ESC b	ESC b	EnH
Screen Saver on (default)	ESC e Q	n/a	same	same	EnH	EnH	EnH	EnH
Screen Saver off	ESC e P	n/a	same	same	EnH	EnH	EnH	EnH
Scroll Type								
WYSE Mode	ESC ` <i>Scroll</i>	n/a	same	same	n/a	n/a	n/a	EnH
Parameters:	<u>Scroll</u>	<u>Scrolling Type</u> ²						
	@	Jump Scroll (default)						
	<, =	Slow Smooth Scroll						
	>, ?	Fast Smooth Scroll						
TVI - Smooth scroll off	n/a	n/a	n/a	n/a	ESC 9	ESC 9	ESC 9	n/a
TVI - Smooth scroll on	n/a	n/a	n/a	n/a	ESC 8	ESC 8	ESC 8	n/a
Character Cell								
Select 10 × 16 size (def)	ESC e >	n/a	same	Wyse	EnH	EnH	EnH	EnH
Select 10 × 13 size	ESC e ?	n/a	same	Wyse	EnH	EnH	EnH	EnH
Column Display								
Select 80-column display (default)	ESC ` :	n/a	same	same	n/a	n/a	n/a	EnH
Select 132-column display	ESC ` ;	n/a	same	same	n/a	n/a	n/a	EnH
Display Data Lines								
24 data lines (default)	ESC e (	n/a	same	same	EnH	EnH	EnH	EnH
25 data lines	ESC e)	ESC ^	same as 160/60	same as 160/60	EnH	EnH	EnH	EnH
42 data lines	ESC e *	n/a	same	same	EnH	EnH	EnH	EnH
43 data lines	ESC e +	ESC _	same as 160/60	same as 160/60	EnH	EnH	EnH	EnH

Video Display Features	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
In ASCII emulation mode, the VT510 displays four types of data lines:								
• 24 data lines, 1 status line, and 2 label lines • 25 data lines, 1 status line, and 1 label line • 42 data lines, 1 status line, and 2 label lines⁴ • 43 data lines, 1 status line, and 1 label line								
Fill Page with "H"s in PCTerm Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Fill page with "H"s	n/a	ESC F	n/a	n/a	n/a	n/a	n/a	n/a
Width-Change-Clear Mode								
Width-change-clear off	ESC e .	n/a	same	same	EnH	EnH	EnH	EnH
Width-change-clear on (default)	ESC e /	n/a	same	same	EnH	EnH	EnH	EnH
Page Size								
Set Page Size ⁵	ESC w <i>Length</i>		same	same	ESC \ <i>Length</i> ⁶			EnH
WYSE 160/60, WYSE 50+ Parameters:		ESC \ <i>Length</i> ⁶						
	<u><i>Length</i></u>	<u>Size of a Page</u>		<u>Description</u>				
	G	1*lines		Page number = current screen data lines				
	H	2*lines		Page number = current screen data lines * 2				
	I	4*lines		Page number = current screen data lines * 4				
WYSE 160/60 PCTerm and TVI Parameters:	J	*		One page number = screen data lines; the other contains rest of lines in memory.				
	<u><i>Length</i></u>	<u>Size of a Page</u>		<u>Description</u>				
	1	1*lines		Page number = current screen data lines				
	2	2*lines		Page number = current screen data lines * 2				
	3	4*lines		Page number = current screen data lines * 4				
Display Page	4	*		One page number = screen data lines; the other contains rest of lines in memory.				
Display Previous Page ⁷	ESC w B or ESC J	n/a	same	same	ESC J	ESC J	ESC J	EnH
Display Next Page ⁸	ESC w C or ESC K	n/a	same	same	ESC K	ESC K	ESC K	n/a
Display Specific Page	ESC w <i>Page</i>	n/a	same	same	n/a	n/a	n/a	EnH
Split Screen	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Split screen horizontally	ESC x A <i>Line</i>	n/a	same	same	n/a	n/a	n/a	n/a
Adjustable horiz split	ESC x C <i>Line</i>	n/a	same	same	n/a	n/a	n/a	n/a

Video Display Features	WYSE				TVI			ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2	
Split screen horizontally and Clear data	ESC x 1 <i>Line</i>	n/a	same	same	n/a	n/a	n/a	n/a	
Adjustable horiz split screen and Clear data	ESC x 3 <i>Line</i>	n/a	same	same	n/a	n/a	n/a	n/a	
Lower horiz split	ESC x P	n/a	same	same	n/a	n/a	n/a	n/a	
Raise horiz split	ESC x R	n/a	same	same	n/a	n/a	n/a	n/a	
Windows									
Activate upper window	ESC]	n/a	same	same	n/a	n/a	n/a	n/a	
Activate lower window	ESC }	n/a	same	same	n/a	n/a	n/a	n/a	
Activate other window or page	ESC J or ESC K	n/a	same	same	n/a	n/a	n/a	n/a	
Roll window up in page	ESC w E	n/a	same	same	n/a	n/a	n/a	n/a	
Roll window down in page	ESC w F	n/a	same	same	n/a	n/a	n/a	n/a	
Redefine screen as one window	ESC x @	n/a	same	same	n/a	n/a	n/a	n/a	
Redefine screen as one window and Clear page	ESC x 0	n/a	same	same	n/a	n/a	n/a	n/a	
Cursor Drag Mode									
Cursor drag mode on (default)	ESC d)	n/a	same	same	n/a	n/a	n/a	n/a	
Cursor drag mode off	ESC d (	n/a	same	same	n/a	n/a	n/a	n/a	
Status Line Display									
Editing Status line on	ESC ` a	n/a	same	same	n/a	n/a	n/a	EnH	
Standard Status line on (default)	ESC ` b	n/a	same	same	n/a	n/a	n/a	EnH	
Status line off	ESC ` c	n/a	same	same	n/a	n/a	n/a	EnH	
	<u>Starts at Column</u>		<u>Standard Status Line</u>			<u>Editing Status Line</u>			
	26		HH:MM AM (or PM)			PROT (Protect mode is on)			
	31					WRPT (Write protect is on)			
	35		Column-Row (Cursor)						
	36					INS (Insert mode is on)			
Messages:	<u>Message</u>		<u>Description</u>						
	LOCK		Keyboard locked						
	COMP		In compose process						
	NUM		NUM LOCK state						
	CAPS		The Caps Lock key is on						

Video Display Features	WYSE				TVI			ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2	
	1...		Page indication (No message displayed for page 0)						
	*		Monitor mode state						
	FDX		Full-duplex mode						
	HDX		Half-duplex mode						
	BLK		Block mode						
	HBLK		Half-duplex block mode						
	LCL		Local mode						
	HLD		Hold state						
	%		Page Printing						
	>		Autoprint mode						
	=		Controller print mode						
	PBSY		Printer is busy or no printer is attached.						
	PTR		Printer is ready.						
	<AUX		Terminal is in auxiliary receive mode.						
Computer Message	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2	
Program/Display computer message	ESC F <i>Message</i> CR								
		n/a	same	same	Wyse	Wyse	Wyse	EnH	
Clear computer message	ESC F CR	n/a	same	same	same	same	same	EnH	
Function Key Label Display									
Display F key label	ESC z <i>Field Label</i> CR								
		n/a	same	same	Wyse ⁹	Wyse	Wyse	EnH	
Clear F key label	ESC z <i>Field</i> CR								
		n/a	same	same	Wyse	Wyse	Wyse	EnH	
Field Parameters:			Field			Field			
	<u>Key</u>	<u>Unshifted</u>		<u>Shifted</u>	<u>Key</u>	<u>Unshifted</u>		<u>Shifted</u>	
	F1	0		P	F9	8		X	
	F2	1		Q	F10	9		Y	
	F3	2		R	F11	:		Z	
	F4	3		S	F12	;		[	
	F5	4		T	F13	<		\	
	F6	5		U	F14	=		]	
	F7	6		V	F15	>		^	
	F8	7		W	F16	?		_	
Label Parameters:			<u>Emulation Mode</u>		<u>Column Mode</u>		<u>Maximum Length</u>		
	Hidden		80		9				
	Hidden		132		7				
	Embedded		80		8				

Video Display Features	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
	Embedded		132		7			
Label Line Display	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Program/Display computer message on unshifted Label line.	ESC z (<i>Text</i> CR		same	same as 160/60	ESC f <i>text</i> CR or ESC f <i>text</i> Ctrl-M			EnH
Program/Display computer message on shifted Label line.	ESC z) <i>text</i> CR		n/a	same	n/a	n/a	n/a	EnH
Clear unshifted Label line.	ESC z (CR	ESC f CR or ESC f Ctrl-M	same	same as 160/60	ESC f CR or ESC f Ctrl-M			EnH
Clear shifted Label line.	ESC z) CR	n/a	same	same	n/a	n/a	n/a	EnH
Disable shifted Label line.	ESC z DEL	n/a	same	same	n/a	n/a	n/a	EnH
WYSE and ADDS Enhanced Modes:	There are two distinct label lines when the terminal is using 24 or 42 data lines: • The upper one is for shifted label line (shifted function key label line). • The lower one is for unshifted label line (unshifted function key label line). When using the 25 or 43 data line configuration mode, only one label line is available at a time. By default, it displays the unshifted label line (or unshifted function key label line). This line becomes a shifted label line (or shifted function key label) whenever the Shift key is pressed. Releasing the Shift key returns the line to be the unshifted display line.							
TVI and PCTerm Modes:	Only an unshifted label line (unshifted function label line) is available. To be visible, the label line must be explicitly turned on by the command "Turn on User Line." For TVI personalities, the label line is displayed only when data line is set to 24 or 42 data lines. For PCTerm mode, turning on the display line forces the terminal to 24 or 42 data lines. If the cursor is at the 25th or 43rd line before the line is set, then the cursor moves to the same column at line 24 or 42, and the entire data area scrolls up one line. The label line is displayed at the 25th or 43rd line, respectively.							
Text Parameter:	The parameter <i>text</i> is a character string to be entered in the label line. The allowable size of <i>text</i> varies among emulation modes and column modes. In embedded emulation mode, the first character shown on the user line is an attribute space.							
	<u>Emulation Mode</u>		<u>Column Mode</u>		<u>Maximum Number</u>		<u>Column Position</u>	
	Hidden		80		80		1–80	
	Hidden		132		132		1–132	
	Embedded		80		79		2–80	
	Embedded		132		131		2–132	
User Label Line Display for TVI Mode	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Turn off User line (default)	n/a	ESC e or ESC h	same	n/a	ESC h	ESC h	ESC h	n/a
Turn on User line	n/a	ESC g	n/a	n/a	ESC g	ESC g	ESC g	n/a

Video Display Features	WYSE				TVI			ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2	
Character Attributes									
Assign character attribute (main display area)	ESC G <i>Attr</i>	ESC G <i>Attribute</i>	same	same	same	same	same	EnH	
Parameters:	<i>Attr</i>	Display Attributes						TVI <i>Attr</i>	
Wyse 160/60, 150/120, 50+, and ADDS	SP	Space character						SP	
	0	Normal						0, p	
	1	Invisible						1, q	
	2	Blink						2, r	
	3	Invisible and blink						3, s	
	4	Reverse						4, t	
	5	Reverse and invisible						5, u	
	6	Reverse and blink						6, v	
	7	Reverse, blink, invisible						7, w	
	8	Underline						8, x	
	9	Underline and invisible						9, y	
	:	Underline and blink						:, z	
	;	Underline, blink, invisible						;, {	
	<	Underline and reverse						<, l	
	=	Underline, reverse, invisible						=, }	
	>	Underline, reverse, blink						>, ~	
	?	Underline, reverse, blink, invisible						?, DEL	
	p	Dim							
	q	Dim and invisible							
	r	Dim and blink							
	s	Dim, blink, invisible							
	t	Dim and reverse							
	u	Dim, reverse, invisible							
	v	Dim, reverse, blink							
	w	Dim, reverse, blink, invisible							
	x	Dim and underline							
	y	Dim, underline, invisible							
	z	Dim, underline, blink							
	{	Dim, underline, blink, invisible							
		Dim, underline, reverse							
	}	Dim, underline, reverse, invisible							
	~	Dim, underline, reverse, blink							
	DEL	Dim, underline, reverse, blink, invisible							

Video Display Features	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
PCTerm Mode	<u>Attribute</u>		<u>Display Attributes</u>					
	0, @, P, `, p		Normal					
	1, A, Q, a, q		Invisible					
	2, B, R, b, r		Blink					
	3, C, S, c, s		Invisible and blink					
	4, D, T, d, t		Reverse					
	5, E, U, e, u		Reverse and invisible					
	6, F, V, f, v		Reverse and blink					
	7, G, W, g, w		Reverse, blink, invisible					
	8, H, X, h, x		Underline					
	9, I, Y, i, y		Underline and invisible					
	:, J, Z, j, z		Underline and blink					
	;, K, [, k, {		Underline, blink, invisible					
	<, L \, l, l		Underline and reverse					
	=, M,], }		Underline, reverse, invisible					
	>, N, ^, n, ~		Underline, reverse, blink					
	?, O, _, o, DEL		Underline, reverse, blink, invisible					
Selecting Display Attribute Mode	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Char attribute mode on ¹⁰	ESC e 1	n/a	same	n/a	n/a	n/a	n/a	n/a
Char attribute mode off	ESC e 0	n/a	same	n/a	n/a	n/a	n/a	n/a
Page attribute mode on ¹¹	ESC e 2	n/a	same	same	EnH	EnH	EnH	EnH
Line attribute mode on	ESC e 3	n/a	same	same	EnH	EnH	EnH	EnH
Assign display attribute to a message field	ESC A <i>Area Attr</i>	n/a	same	same	ESC \ <i>Area Attr</i> (EnH)			EnH
Parameters:	<u>Area</u>	<u>Description</u>						
	0	Data Area						
	1	Label Line						
	2	Terminal message field on Status Line						
	3	Computer message field on Status Line						
Display Attributes								
Assign write-protect character attribute	ESC ` <i>Wpca</i>	n/a	same	same	n/a	n/a	n/a	EnH or ESC 0 <i>Spca</i>
Parameters – WYSE 160/60, WYSE 50+,	<u>Wpca</u>	<u>Description</u>	Parameters – ADDS mode†			<u>Spca</u>	<u>Description</u>	

Video Display Features	WYSE				TVI		ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
and ADDS using WYSE sequences: ¹²	6	Reverse				0	Dim	
	7	Dim (default)				1	Blink	
	A	Normal				2	Invisible	
	B	Blink				3	Normal	
	C	Invisible				4	Reverse	
	E	Underline				5	Underline	
	F	Reverse						
	G	Dim						
Clear unprotected page to display attribute	n/a	n/a	n/a	ESC ! <i>Attr</i>	n/a	Wyse	Wyse	EnH
Request attribute under cursor	n/a	ESC D	n/a	n/a	n/a	n/a	n/a	n/a
Response to Display Attributes								
Response:‡	Two bytes—the Bit for attribute under the cursor, then "a".							
	<u>Bit</u>	<u>Attribute</u>		<u>Bit</u>	<u>Attribute</u>			
	0	Underline		4	Always 0			
	1	Reverse		5	Always 0			
	2	Blink		6	Always 0			
	3	Invisible		7	Dim			
Assign Line Attributes								
Assign line attributes	ESC G <i>LineAttr</i>	n/a	same	same	n/a	n/a	n/a	EnH
Parameters:	<u><i>LineAttr</i></u>		<u>Line Attribute</u>					
	@		Single height, single width (default)					
	A		Single height, double width					
	B		Top-half of double height, single width					
	C		Bottom-half of double height, single width					
	D		Top-half of double height, double width					
	E		Bottom-half of double height, double width					

Video Display Features	WYSE				TVI		ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2

¹"WYSE 160/60 PCTerm" and TVI should have Enhanced Mode off.

²Users can change the scrolling rate by entering keys from the keyboard. Pressing Ctrl/Shift/↑ to increase the rate and Ctrl/Shift/↓ to decrease it.

⁴Because of hardware limitations, selecting 42 or 43 data lines forces the screen size to be 53 lines.

⁵Upon receipt of the Set Page Size command, the terminal:

- Clears the display memory to NUL (0/0) characters.
- Displays first page "page 0."
- Moves the cursor to its home position.
- Changes the split screen to a full screen format.
- Turns off the Protect mode.
- Unlocks the locked line.

⁶Enhanced Mode off.

⁷In WYSE mode, command "ESC J" works only when the screen is not split. If the screen is split, refer to "Activate Other Window or Page" command. Command "ESC w B" in WYSE modes works when the screen is not split or the split screen is caused by an adjustable split. It does not work when the split screen is a result of a simple split. Users can use local function keys to display the previous page. On a VT keyboard, press Ctrl/Prev; on an Enhanced PC keyboard, press Ctrl/Page Up.

⁸Users can use a local function key to display the next page. Press Ctrl/N, where N is a number on the numeric keypad key equivalent to the page number, on both VT and Enhanced PC keyboard.

⁹Unshifted key label only; the Shifted Key label is not available for TVI modes.

¹²Assigning parameters "6," "7" or "A" clears previously defined write-protected attributes; "B" through "G" add on top of previous attribute(s) and display a combination of new and old display attributes.

† All the 7-bit ASCII codes are legal in this sequence. Example: "A" turns on the dim attribute, "T" turns on the reverse attribute, "S" turns on dim, blink and reverse attributes.

‡ Example: Dim + Reverse + Invisible reports VTS (8AH) + a.

Table 12–6 lists the data protection control functions. The word "same" in **Table 12–6** signifies the same control sequences as in the WYSE 160/60 column.

Table 12–6 Data Protection Control Functions

Write-Protect Mode	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Write-protect mode off (default)	ESC (	same	same	same	same	same	same	EnH or Ctrl/O
Write-protect mode on ¹	ESC)	same	same	same	same	same	same	EnH or Ctrl/N
Protect Mode								
Protect mode off (default)	ESC '	same	same	same	same	same	same	EnH
Protect mode on	ESC &	same	same	same	same	same	Wyse	EnH

¹Turning on Write-Protect mode does not start protecting the data. It only defines the range of data that needs to be protected. Not until Protect mode is turned on does the data become protected. Changing personality does not change the Write-Protect mode. Graphic characters range from 1/0 to 1/15, and embedded attributes are automatically write-protected even when Write-Protect mode is off.

Table 12–7 lists the cursor control functions. The word "same" in **Table 12–7** signifies the same control sequences as in the WYSE 160/60 column.

Table 12–7 Cursor Control Functions

	WYSE				TVI			ADDS
Autopage Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
On	ESC d +	n/a	same	same	ESC v	ESC v	ESC v	EnH
Off (default)	ESC d *	n/a	same	same	ESC w	ESC w	ESC w	EnH
Autoscrolling Mode								
On (default)	ESC O	n/a	same	same	n/a	n/a	n/a	EnH
Off	ESC N	n/a	same	same	n/a	n/a	n/a	EnH
Cursor Display Feature								
Set cursor appearance	ESC ` <i>Cursor</i>	ESC . <i>Cursor</i>	same as 160/60	same as 160/60	ESC . <i>Cursor</i>			EnH ¹
Parameters:	<u><i>Cursor</i></u>	<u>Description</u>						
	0	Cursor Display Off						
	1	Cursor Display On						
	2	Steady Block Cursor						
	3	Blinking Line Cursor						
	4	Steady Line Cursor						
	5	Blinking Block Cursor						
End-of-Line Wrap Mode								
On (default)	ESC d /	ESC ~	same as 160/60	same as 160/60	n/a	n/a	n/a	EnH
Off	ESC d .	ESC 0	same as 160/60	same as 160/60	n/a	n/a	n/a	EnH
When the cursor position is located at the bottom and right margin position of the current page with wrap flag on, any displayable character received has the following effect:								
	<u>Autopage</u>	<u>Autoscroll</u>	<u>Page Mode</u>		<u>Cursor Moves to . . .</u>			
	On	–	Single		Home position of the active page.			
			Multiple		Home position of the next page.			
	Off	On	Multiple		Left margin of bottom line after a scroll is performed.			
	Off	Off	Multiple		Home position of the active page.			
Received CR Mode								
CR (default)	ESC e 4	ESC 9	same as 160/60	same as 160/60	EnH	EnH	EnH	EnH

¹Ctrl/W = Cursor off; Ctrl/X = Cursor on.

²Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is on.

³Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is off.

⁴Either Autopage mode or Single Page mode is enabled.

⁵Enhanced Mode off.

⁶When screen is split, "0" = upper window; "1" = lower window.

	WYSE				TVI			ADDS
Autopage Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
CRLF	ESC e 5	ESC 8	EnH	same as 160/60	EnH	EnH	EnH	EnH
Lock Cursor Line Mode								
On	ESC ` H	n/a	same	same	ESC ! 1	n/a	n/a	EnH
Off (default)	ESC ` I	n/a	same	same	ESC ! 2	n/a	n/a	EnH
Recognize DEL Mode								
On (default)	ESC e E	EnH	same	same	EnH	EnH	EnH	EnH
Off	ESC e D	EnH	same	same	EnH	EnH	EnH	EnH
Cursor Movement								
Cursor Left (Back Space)	Ctrl/H	same	same	same	same	same	same	Wyse or Ctrl/U
	<u>Autopage</u>	<u>Page Mode</u>		<u>Cursor Moves to . . .</u>				
	On	Multiple		Bottom right margin of the preceding page (if any) even though the bottom right margin is protected. Under this command, the Protect mode feature carries through all pages. Refer to Protect Mode for details.				
	Off	Single		Bottom right margin of the active page even though the right margin is protected.				
Cursor Left (Delete)	DEL	same	same	same	same	same	same	same
Cursor Right	Ctrl/L	same	same	same	same	same	same	Ctrl/F
	<u>Autopage</u>	<u>Autoscroll</u>	<u>Page</u>	<u>Cursor Moves to . . .</u>				
	On	–	Multiple	Home position or the first unprotected position (when Protect mode is on) of the next page (if any). Under this command, the Protect mode feature carries through all pages. Refer to Protect mode for details.				
	Off	On	Single ²	Left margin of the bottom margin line after a scroll is performed.				
	Off	Off	Single ³	Home position of the active page.				
Cursor Up								
Cursor Up—No Scroll	Ctrl/K	same	same	same	same	same	same	Ctrl/Z
Cursor Up—Scroll (Reverse Linefeed)	ESC j	same	same	same	same	same	Wyse	EnH

This command moves the cursor as follows:

¹ Ctrl/W = Cursor off; Ctrl/X = Cursor on.

² Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is on.

³ Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is off.

⁴ Either Autopage mode or Single Page mode is enabled.

⁵ Enhanced Mode off.

⁶ When screen is split, "0" = upper window; "1" = lower window.

Autopage Mode	WYSE				TVI		ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
	<u>Autopage</u>	<u>Autoscroll</u>	<u>Page</u>	<u>Protect</u>	<u>Cursor Moves to . . .</u>			
	–	–	Single	–	(Command is ignored.)			
	On	–	Multiple	On or Off	Bottom line of the previous page. Under this command, the Protect mode feature carries through all pages.			
	Off	On	Multiple	Off	Top margin of the active page after a reverse scroll is performed.			
	Off	On	Multiple	On	(Command is ignored.)			
	Off	Off	Multiple	–	(Command is ignored.)			

Cursor Down

Cursor Down—No Scroll	Ctrl/V	same	same	same	same	same	same	same
Cursor Down—Scroll (Linefeed)	Ctrl/J	same	same	same	same	same	same	same
	<u>Autopage</u>	<u>Autoscroll</u>	<u>Page Mode</u>	<u>Cursor Moves to . . .</u>				
	Off	Off	–	Top margin line of the active page. If the destination position is protected, the cursor moves right to the first unprotected position.				
	On	–	Single	Top margin line of the active page. If the destination position is protected, the cursor moves right or down to the first unprotected position.				
	On	–	Multiple	Top margin line of the next page. Under this command, the Protect mode feature carries through all pages. Refer to Protect Mode for details.				
	Off	On	Multiple	Bottom line of the active page after a scroll is performed.				

Cursor to Start

Cursor to start of line	Ctrl/M	same	same	same	same	same	same	same
Cursor to start of next line	Ctrl/_	same	same	same	same	same	same	EnH
	<u>Autopage</u>	<u>Page</u>	<u>Cursor Moves to . . .</u>					
	Off	Single ⁴	Home position of the current page or the first unprotected position of the current page if Protect mode is on.					
	On	Multiple	Home position of the next page or the first unprotected position of the next page if Protect mode is on. Under this command, the Protect mode					

¹Ctrl/W = Cursor off; Ctrl/X = Cursor on.

²Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is on.

³Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is off.

⁴Either Autopage mode or Single Page mode is enabled.

⁵Enhanced Mode off.

⁶When screen is split, "0" = upper window; "1" = lower window.

	WYSE				TVI			ADDS
Autopage Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
feature carries through all pages. Refer to Protect Mode for details.								
Address Cursor	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Cursor to Home	Ctrl/^ or ESC {	Ctrl/^	same	same	Ctrl/^	same	same	EnH or Ctrl/A
Address Cursor to a specific column (<i>Col</i>)	ESC _ <i>Col</i>	n/a	same	same	ESC] <i>Col</i>			EnH or Ctrl/P <i>Col</i>
Address Cursor to a specific line	ESC [<i>Line</i>	n/a	same	same	same	same	same	EnH or Ctrl/K <i>Line</i>
Address Cursor in Current Page	ESC = <i>Line Col</i>	same	same	same	same	same	same	EnH or ESC Y <i>Line Col</i>
Address Cursor in specific page	ESC h <i>page</i>	n/a	same	same	n/a	n/a	n/a	EnH
Address Cursor in specific page, line, and column	ESC w @ <i>Page Line Col</i>	n/a	same	same	ESC - <i>Page Line Col</i> ⁵			EnH
Address Cursor in specific Window/Page	ESC - <i>Window/Page Line Col</i> ⁶	same	same	same	EnH	EnH	EnH	EnH
Address Cursor in 80/132 column current Page	ESC a <i>Line R Col C</i>	n/a	same	same	ESC d <i>Line R Col C</i>			EnH
Read Cursor								
Read Cursor address in current Page	ESC ?	same	same	same	same	same	same	EnH
Read Window, Page no., and Cursor address	ESC /	same	same	same	same	same	same	EnH
Read Cursor address in Current 80/132-col page	ESC b	n/a	same	same	n/a	n/a	n/a	EnH
Read Page number and Cursor address	ESC w ` ESC w . ESC w , ESC w '	n/a	same	same	n/a	n/a	n/a	EnH

¹Ctrl/W = Cursor off; Ctrl/X = Cursor on.

²Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is on.

³Autopage mode is either set to off, or the terminal is in Single Page mode and Autoscroll mode is off.

⁴Either Autopage mode or Single Page mode is enabled.

⁵Enhanced Mode off.

⁶When screen is split, "0" = upper window; "1" = lower window.

Note

The terminal responds to the Read Cursor command in the following format: `lll R ccc C`

The two parameter values, `lll` and `ccc`, are the 3-byte decimal values for the line/column location of the cursor. For example, `"019R010C"` indicates the cursor is at line 19, column 10. The sequence with parameters `"001"` (3/0)(3/0)(3/1) and `"001"` (3/0)(3/0) (3/1) indicates the cursor is at the home position.

Table 12–8 lists the ASCII editing control functions. The word "same" in **Table 12–8**, **Table 12–10**, and **Table 12–11** signify the same control sequences as in the WYSE 160/60 column.

Table 12–8 ASCII Editing Control Functions

	WYSE			TVI			ADDS	
Local Edit Mode	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Off ¹ (default)	ESC l	n/a	same	same	same	same	same	n/a
On ²	ESC k	n/a	same	same	same	same	same	n/a

¹Duplex Edit Mode On.

²Duplex Edit Mode Off.

In local edit mode, instead of sending keycodes to the host, some function keys send codes directly to the terminal to speed up the editing process. **Table 12–9** lists these local edit functions and their operating keys for both the VT keyboard and the Enhanced PC keyboard.

Table 12–9 Local Edit Mode Function Keys

Function	VT Keyboard	EPC Keyboard	Sequence
Insert Character	PF1		ESC Q
Insert Line	Shift PF1		ESC E
Delete Character	PF2	Delete	ESC W
Delete Line	Shift PF2	Shift Delete	ESC R
Clear Line	PF3	End	ESC T
Clear Screen	Shift PF3	Shift End	ESC Y
Insert		Insert	ESC q
Replace		Shift Insert	ESC r
Next Page		Page Down	ESC K
Previous Page		Page Up	ESC J
Send			ESC 7
Print		Print	ESC P
Home		Shift Home	ESC {
Tab		Shift Tab	ESC I

Table 12–10 lists the tab stop control functions.

Table 12–10 Tab Stop Control Functions

Tab Stop Function	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Clear All Tab Stops	ESC 0	ESC 3	same as 160/60	same as 160/60	ESC 3	ESC 3	ESC 3	n/a
Clear Tab Stop	ESC 2	same	same	same	same	same	same	EnH
Set Tab Stop	ESC 1	same	same	same	same	same	same	EnH
Tabulate Cursor or Field Tab ¹	ESC i or Ctrl/I	same	same	same	same	same	same	EnH
The cursor moves as follows:								
<u>Autopage</u>		<u>Autoscroll</u>	<u>Page Mode</u>		<u>Cursor Moves to . . .</u>			
On		–	Single		Top margin line of the active page			
On		–	Multiple		Top margin line of the other page.			
Off		On	Multiple		Bottom margin of the active page after a scroll is performed.			
Off		Off	Multiple		Top margin line of the active page.			
Backtab or Field Tab ²	ESC I	same	same	same	same	same	same	EnH
The cursor moves as follows:								
<u>Autopage</u>		<u>Autoscroll</u>	<u>Page Mode</u>		<u>Cursor Moves to . . .</u>			
On		–	Single		Home			
On		–	Multiple		Bottom margin line of the previous page if any.			
Off		On	Multiple		Bottom margin line of the active page.			
Off		Off	Multiple		Bottom margin line of the active page.			

¹Executes "tabulate cursor" when protect mode is off.

²If protect mode is on, this command executes a "field tab" to move the cursor left to the first unprotected character position after the nearest protect field. The tab stop setting is completely ignored. The cursor moves backward from the first position of an unprotected area to the first position of another unprotected area.

Executing a Backtab from an unprotected page to a previous protected page changes its protect mode to "unprotected."

Table 12–11 lists the page edit control functions.

Table 12–11 Page Edit Control Functions

Page Edit Mode	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Off (default)	ESC e "	n/a	same	same	ESC O	ESC O	ESC O	EnH
On	ESC e #	n/a	same	same	ESC N	ESC N	ESC N	EnH
Insert/Replace Mode								
Replace mode On (default)	ESC r	same	same	same	same	Wyse	Wyse	EnH
Insert mode On	ESC q	ESC z	same as 160/60	same as 160/60	same as 160/60	Wyse	Wyse	EnH
Insert Characters								
Insert Space Character	ESC Q	same	same	same	same	same	same	EnH
Insert Line of Spaces	ESC E	same	same	same	same	same	same	EnH or ESC M
Insert Column of NULs	ESC c M	n/a	same	same	n/a	n/a	n/a	EnH
Deleting Data on Current Page Only								
Delete Cursor Character	ESC W	same	same	same	same	same	same	EnH
Delete Cursor Line	ESC R	same	same	same	same	same	same	EnH
Delete Cursor Column	ESC c J	n/a	same	same	n/a	n/a	n/a	EnH
Erasing Data on Page Function								
Clear Page to NULs	ESC *	same	same	same	same	same	same	EnH
Clear Page to spaces	ESC +	n/a	same	same	n/a	n/a	n/a	EnH or Ctrl/L
Clear page to write-protected spaces	ESC ,	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected page to write-protected spaces	n/a	ESC ,	n/a	n/a	ESC ,	ESC ,	ESC ,	n/a
Clear unprotected page to NULs	ESC :	same	same	same	same	same	same	ESC : (EnH)
Clear unprotected page to spaces	ESC ; or Ctrl/Z	same or ESC +	same as 160/60	same as 160/60	same or ESC +	same or ESC +	same or ESC +	EnH
Clear unprotected page to a specific character ¹	ESC . <i>Char</i>	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected page to NULs from cursor	ESC y	same	same	same	same	same	same	EnH
Clear unprotected page to spaces from cursor	ESC Y	same	same	same	same	same	same	ESC k
Clear unprotected page foreground to NULs from cursor position	ESC c Q	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected page foreground to spaces from cursor position	ESC c P	n/a	same	same	n/a	n/a	n/a	EnH

¹"Char" is an ASCII character. Any 8-bit code is treated as a 7-bit code since the most significant bit of the code is stripped off.

Page Edit Mode	WYSE		TVI				ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Clear unprotected area to end of line with NULs	ESC c L	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected area to end of line with spaces	ESC c O	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected line from cursor position to NULs	ESC t	same	same	same	same	same	same	EnH
Clear unprotected line from cursor position to spaces	ESC T	same	same	same	n/a	n/a	n/a	ESC K
Clear unprotected line foreground from cursor position to NULs	ESC c S	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected line foreground from cursor position to spaces	ESC c R	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected position to NULs	ESC c K	n/a	same	same	n/a	n/a	n/a	EnH
Clear unprotected column to a specific character ¹	ESC c I <i>Char</i>	n/a	same	same	n/a	n/a	n/a	EnH
Clear column to write-protected space	ESC V	same	same	same	same	same	same	EnH

¹"*Char*" is an ASCII character. Any 8-bit code is treated as a 7-bit code since the most significant bit of the code is stripped off.

Table 12–12 lists the rectangle area control functions.

Table 12–12 Rectangle Area Control Functions

Draw Rectangle in 80/132 column Current Page	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Format 1	ESC c G <i>Line Col</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Format 2 ¹	ESC c G <i>Line ~ Col</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Fill Rectangle with a Specific Character in 80/132-column Cursor Page								
Format 1	ESC c H <i>Line Col Char</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Format 2 ¹	ESC c H <i>Line ~ Col Char</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Fill unprotected Rectangle with a specific character								
Format 1	ESC c F <i>Line Col Char</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Format 2 ¹	ESC c F <i>Line ~ Col Char</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Draw Rectangle ²								
Draw rectangle area to right and below cursor.	ESC c N <i>Width Height</i>							
	n/a	same	same	same	n/a	n/a	n/a	EnH
Swap/Copy/Move	WYSE 160/60							
Swap, copy, or move the data in a rectangle to another rectangle.	ESC c ^ <i>Action S_page P S_n1 R S_n2 C S_n3 R S_n4 C D_page P D_n1 R D_n2 C</i>							
Parameters:	<u>Action</u>	<u>Description</u>						
	0	Swaps data between the source and the destination rectangles.						
	1	Copies the content of the source rectangle to a destination rectangle, preserving the contents of the source rectangle.						
	2	Moves the contents of the source rectangle to destination rectangle, clearing the source rectangle.						
	<u>Parameter</u>	<u>Description</u>						
	<i>S_page</i>	Page number of the source rectangle						
	<i>S_n1</i>	Starting line number of the source rectangle (top left corner)						
	<i>S_n2</i>	Starting column number of the source rectangle (top left corner)						
	<i>S_n3</i>	Ending line number of the source rectangle (bottom right corner)						
	<i>S_n4</i>	Ending column number of the source rectangle (bottom right corner)						
	<i>D_page</i>	Page number of the destination rectangle						
	<i>D_n1</i>	Starting line number of the destination rectangle (top left corner)						

¹Not recommended for 80-column mode.²Parameters Line and Column are ASCII characters whose decimal values are the desired numbers plus 31.

Draw Rectangle in 80/132 column Current Page	WYSE		TVI				ADDS	
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
	<i>D_n2</i>	Starting column number of the destination rectangle (top left corner)						
Swap/Copy/Move		PCTerm	150/120	WY 50+	950	925	910+	A2
Swap, copy, or move the data in a rectangle.		n/a	same as 160/60	same as 160/60	n/a	n/a	n/a	EnH

¹Not recommended for 80-column mode.

²Parameters Line and Column are ASCII characters whose decimal values are the desired numbers plus 31.

12.5 Printing/Sending Data

This section describes commands to print and send data.

12.5.1 Sending Data

This section describes commands to send a defined area of data on the screen through the main communication port using ASCII control functions. Individual commands address the specific data to be sent.

In send operations, delimiter characters are sent along with data to mark a line end, transfer complete, or the protect data and its area. [Table 12–13](#) lists the possible delimiter types. Depending on the scope of data (line, page) and the type of data (protected, unprotected) that has been sent, different delimiter characters are enclosed. The individual commands explain which delimiter should be sent with each command.

[Table 12–13](#) lists the types of delimiters to send data.

Table 12–13 Send Data Delimiters

Delimiter Type	Sent . . .
End of the line	At the end of each line but not at the end of the page (block).
Transfer complete	After each Send Data action, except for a "Send the Cursor Character" command.
Field separator	To take the position of the protected field data when "Send Unprotected" command is executed and Protect mode is on.
Start protect field and End protect field	To mark the beginning and the ending of a write-protected field of data when Protect mode is on.

All the send data actions in this section, except "Send the Cursor Character" command, use the following rules:

- The attributes are not sent along with their data. But a Space character (Hex 20) is inserted between two distinct attribute data to indicate the attribute difference.
- During the process of sending characters, the cursor moves to the location of the character being sent.
 - Protect mode on: The cursor does not move.
 - Protect mode off: The cursor moves to the left margin of the next line.
- Null characters on the screen are not sent.
- Delimiter characters:
 - TVI modes and WYSE 160/60 PCTerm modes: Defined by command "Define Delimiters" (Refer to the command for details.)
 - Other personalities: See [Table 12–14](#).

Table 12–14 Delimiters for Other Personalities

Delimiter	Terminal sends...
End of line	Either "US" or "CRLF" depending on the Set-Up Block End selection.
End of transmission	Either "CR" or "ETX" depending on the Set-Up Block End selection.
Protected field	A single FS (Hex 1C).
Protected field separator	Native mode: "ESC)" and "ESC (" when the protect field attribute is Dim or Normal; "ESC)sp" and "ESC (sp" when the protect field attribute is any other combination. Other modes: "ESC)" and "ESC ("

- If any line on the screen is locked, then the line is not sent; however, the end of line delimiter is still sent to indicate a line has been skipped.
- If the first character sent is a protected character, then the ESC) sequence is not sent before the first character; if the last character sent is a protected character, then the ESC (sequence is not sent before the terminator.
- If the cursor position is located at the right margin of the bottom margin line and the Protect mode is off:

Autopage	Autoscroll	Page Mode	Cursor moves to . . .
On	–	Single	Home position of the active page.
On	–	Multiple	Home position of the other page.
Off	On	Multiple	Left margin of the bottom line after a scroll is performed.
Off	Off	Multiple	Home position of the active page.

Table 12–15 lists the print/send control functions.

Table 12–15 Print/Send Control Functions

	WYSE				TVI			ADDS
Function	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Define delimiters	n/a	ESC x <i>Dlmtr Char1 Char2</i>						
			n/a	n/a	ESC x <i>Dlmtr Char1 Char2</i>			n/a
Parameters:	<u><i>Dlmtr</i></u>		<u>Description</u>					
	0	Protected Field separator						
	1	Start protect field						
	2	End protect field						
	3	End of the line						
	4	Termination						
	Parameters <i>Char1</i> and <i>Char2</i> are ASCII characters.							
	<u>Delimiter</u>		<u>Default ASCII Character</u>					
	Field separator		FS NUL (1/12 0/0)					
	Start protect field		ESC) (1/11 2/9)					
	End protect field		ESC ((1/11 2/8)					
	End of the line		US NUL (1/15 0/0)					
	Termination		CR NUL (1/13 0/0)					
	A NUL character that is defined in a delimiter is not sent out.							
Begin Print/Send								
Begin Print/Send at Top of Page	ESC d `	n/a	same	same	n/a	n/a	n/a	EnH
Begin Print/Send at Top of Screen	ESC d &	n/a	same	same	n/a	n/a	n/a	EnH
Sending a Character or Line								
Send the Cursor Character	ESC M	n/a	same	same	n/a	n/a	n/a	n/a
Send the Cursor Line	ESC 6	same	same	same	same	same	same	n/a
Send unprotected Cursor Line	ESC 4	same	same	same	same	same	same	n/a
Sending a Page								
Send Page	ESC 7	same	same	same	same	same	same	EnH
Send an unprotected Page	ESC 5	same	same	same	same	same	same	n/a
Sending a Block								
Mark Block Beginning	ESC 8	n/a	same	same	n/a	n/a	n/a	EnH
Mark Block Ending ¹	ESC 9	n/a	same	same	n/a	n/a	n/a	EnH
Send Block ²	ESC s	same	same	same	same	same	same	EnH
Send an unprotected Character in a Block	ESC S	same	same	same	same	same	same	EnH

¹See Note 1.²See Note 2.

ASCII emulations do not send:

- Initialization codes
- Ending sequences to designated character sets and display attributes
- "Fallbacks" to represent data that cannot be printed by the attached printer

The only factors that affect printing data are data and parity settings.

Data/Parity Setting Characters are printed . . .

8-bit	As is
7-bit	After stripping off the most significant bit

Notes

1. Note on Mark Block Ending:

- **Replace mode on:** "STX" or "ETX" replaces the existing character at the active position and moves the cursor right one position.
- **Insert mode on:** Writing "STX" or "ETX" moves all characters and their attributes one position to the right of the cursor position (inclusive) in the current line without wraparound.

2. Note on Send Block:

- **Protect mode on:** Unprotected data is sent as is; protected data, enveloped between "start of protect field" and "end of protect field" delimiters, is sent.
 - **Protect mode off:** All the data in the range, protected or unprotected, is sent.
-

Table 12–16 lists the printing data control functions.

Table 12–16 Printing Data Control Functions

	WYSE				TVI			ADDS
Autoprint Mode ¹	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Off (default)	Ctrl/T	ESC A	Ctrl/T	Ctrl/T	ESC A or Ctrl/T			Ctrl/T
On	Ctrl/R	ESC @	Ctrl/R	Ctrl/R	ESC @ or Ctrl/R			Ctrl/R
Control Print Mode ³								
Off	Ctrl/T	ESC a	Ctrl/T	Ctrl/T	ESC a	ESC a	ESC a	ESC 4
On	ESC d # or Ctrl/X ²	ESC '	same as 160/60	same as 160/60	ESC '	ESC '	ESC '	ESC 3 ESC d # (EnH)
Aux Receive Mode								
Off (default)	ESC d Sp	n/a	same	same	n/a	n/a	n/a	EnH
On	ESC d !	n/a	same	same	n/a	n/a	n/a	EnH
Bidirectional Mode ⁴								
Off (default)	ESC d \$	Ctrl/T	same as 160/60	same as 160/60	Ctrl/T	Ctrl/T	Ctrl/T	EnH
On	ESC d %	Ctrl/R	same as 160/60	same as 160/60	Ctrl/R	Ctrl/R	Ctrl/R	EnH
Print Page to the Cursor Position								
Formatted Page ⁵	ESC P	same	same	same	same	same	Wyse	EnH
Formatted Page (unprotected)	ESC @	n/a	same	same	n/a	n/a	n/a	EnH
Unformatted Page ⁵	ESC p or ESC L	n/a	same	same	ESC L	n/a	n/a	ESC p
Print a Page with Time Message ⁵								
	n/a	ESC L	n/a	n/a	n/a	ESC L	ESC L	n/a
Pass a Character to the Printer Port								
	Ctrl/P Char	Wyse	same	same	Wyse	Wyse	Wyse	n/a
Define the Page Print Terminator								
	n/a	ESC p Ps Char						
			n/a	n/a	ESC p Char			n/a

¹The symbol ">" is displayed in the status line when in Autoprint mode.

²Enhanced mode is off.

³The symbol "=" is displayed in the status line when in Autoprint mode.

⁴Turning off Bidirectional mode turns off Aux Receive mode and forces a normal print mode.

⁵The time prints out in a format such as "TIME AM/PM 00:00." At the end of the time message, the following is sent to the printer port: NUL Ctrl/F NUL. The page prints as formatted data. The terminal sends an "ACK" to the host at the end of a print operation if ACK mode is on.

12.5.2 Graphics Characters

Table 12–17 lists the graphic characters.

Table 12–17 Graphic Characters

Graphics Mode	WYSE				TVI			ADDS
	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
On	ESC H or Ctrl/B	ESC \$	same as 160/60	same as 160/60	ESC \$ or Wyse	ESC \$ or Wyse	ESC \$ or Wyse	EnH
Off (default)	ESC H or Ctrl/C	ESC %	same as 160/60	same as 160/60	ESC % or Wyse	ESC % or Wyse	ESC % or Wyse	EnH
Display Graphic Character								
Display character from the first 32 character set. ¹	ESC H <i>Char</i>	n/a	same	same	n/a	n/a	n/a	EnH

¹The parameter Char is an ASCII character in the range of Hex 20 to Hex 7E.

With Graphics mode on, all received codes in the range of 00 to 7E are displayed as characters in the range of Hex 00 to Hex 1F in the same character set.

This is done by stripping off the highest 3 bits of the code. For example:

Receiving . . .	Displays . . .
21, 41, or 61 (Hex)	01 (Hex) character
2C, 4C, or 6C (Hex)	0C (Hex) character
31, 51, or 71 (Hex)	11 (Hex) character

Most character sets in ASCII emulation contain characters in the range of Hex 00 and Hex 1F. For example, the Native character set contains 16 line-drawing graphics characters from Hex 10 to Hex 1F and 16 CRM characters from Hex 00 to 0F. But in most of the cases, Hex codes 00 to 1F are treated as control codes. To display these graphic characters, Graphic mode has to be turned on.

When the Graphic mode is on, new characters entered have normal attributes regardless of the state of the Write-Protect mode attribute. With Protect mode on, all the entered characters, except CRM characters, are protected.

To design a character font, see [Chapter 13](#).

Defining and Loading ASCII Character Fonts

This chapter defines the ASCII character fonts and describes how to load the fonts.

13.1 Character Sets

In WYSE mode and ADDS Enhanced mode, desired characters can be acquired by loading them from various built-in character sets or by designing them using a pixel pattern^{*}.

13.1.1 PCTerm on US Keyboard

The default and only character set supported for PCTerm on the US Keyboard is the PC International character set. This is a 256-character set, of which the first 128 characters form the PC Standard character set^{*}.

^{*}Refer to the IBM Code Page 437.

13.1.2 WYSE and ADDS Enhanced Personalities

WYSE mode and ADDS enhanced mode support ten built-in ROM character sets and four *font banks*, which are physical locations in memory. Each character set has 128 characters (Hex 00 to Hex 7F), thus each font bank has the same space allocation.

The character set must be loaded into a font bank to become an "active" character set. Characters cannot be displayed directly by calling their physical locations; they can only be displayed through a series of steps to make them active. First, the primary and secondary logical character sets are defined by pointing to the desired font banks. Then, the active character set is defined for the terminal to display and receive codes. See [Table 13–1](#) on how to display and load character sets.

Table 13–1 Display and Load Character Sets

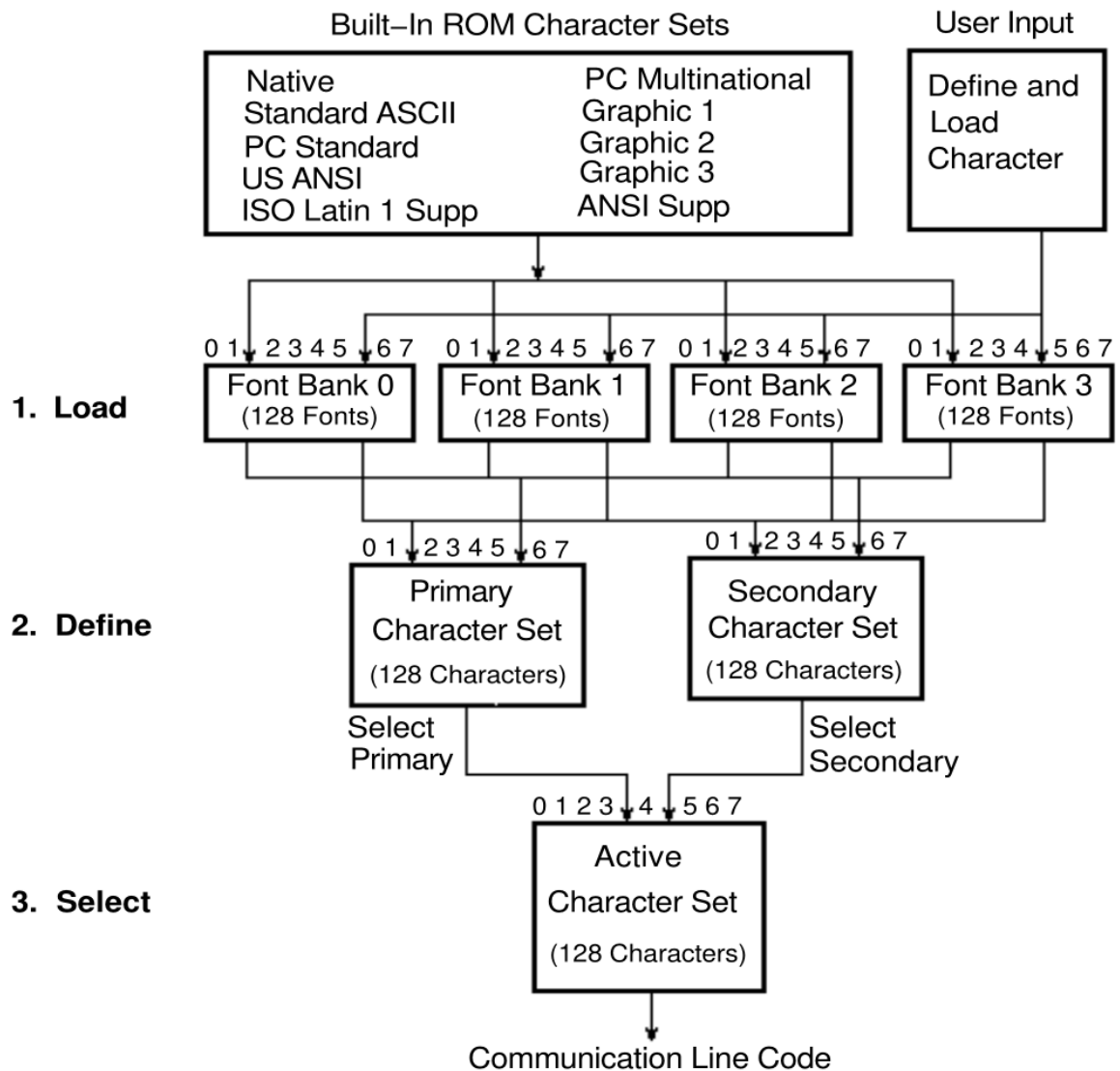
Step	Description
1. Load font bank with built-in character set	Load a target character set from one of the ten built-in character sets into a font bank.
or Define and load character	Design a character and load it into a font bank.
2. Define logical character set	Define primary or secondary character set to point to the font bank.
3. Select active character set	Set active character set to point to either a primary or a secondary character set.

Notes

- WYSE 160: You can display all 512 characters in the four font banks on the screen at the same time.
- WYSE 50+: You can only display one font bank on the screen at a time.

[Figure 13–1](#) shows the process for selecting character sets.

Figure 13–1 Process of Selecting Character Sets



The VT510 selects a character set from one of the built-in character sets by the following process. Refer to [Table 13–1](#) for details.

1. Load the character sets into the font banks.
2. Define the primary and secondary character set from the font buffers.
3. Select which character set will be the active characters set.

Note

Changing a character set in a font bank may change the active character set if the active character set and primary/secondary character set happen to point to the same font bank.

13.2 Loading, Defining, and Selecting Character Sets

[Table 13–2](#) shows how to load, define, and select the character sets. The word "same" in [Table 13–2](#) signifies the same control sequences as in the WYSE 160/60 column.

Table 13–2 Loading Font Bank

	WYSE			TVI				ADDS
Load Font Bank ¹	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Load ROM-resident character set into one of the four font banks. ²	ESC c @ <i>Bank Charset</i>							
		n/a	same	same	n/a	n/a	n/a	EnH
Parameters:	<u>Bank</u>	<u>Description</u>						
	0	Font Bank 0						
	1	Font Bank 1						
	2	Font Bank 2						
	3	Font Bank 3						
	<u>Charset</u>	<u>Character Set Selected</u>						
	@ (4/0) or ` (6/0)	Native						
	A (4/1) or a (6/1)	PC Multinational						
	B (4/2) or c (6/3) or d (6/4)	Standard ASCII						
	C (4/3)	Graphics 1						
	D (4/4) or b (6/2)	PC standard						
	E (4/5)	Graphics 2						
	F (4/6)	Graphics 3						
	G (4/7) or H (4/8) or g (6/7)	US ANSI						
	N (4/14) or j (6/10)	ISO Latin-1 Supplemental						
	I (4/9) or e (6/5)	ANSI Supplemental						
Define Logical Character Set								
Primary Character Set	ESC c B	n/a	same	same	n/a	n/a	n/a	EnH
	<i>Bank</i>							
Secondary Character Set	ESC c C	n/a	same	same	n/a	n/a	n/a	EnH
	<i>Bank</i>							
Select Active Character Set ³								
Primary Character Set (default)	ESC c D	n/a	same	same	n/a	n/a	n/a	EnH
Secondary Character Set	ESC c E	n/a	same	same ⁴	n/a	n/a	n/a	EnH
Automatic Font Loading								
On (default)	ESC e O	n/a	same	same	n/a	n/a	n/a	n/a
Off	ESC e N	n/a	same	same	n/a	n/a	n/a	n/a

¹If not redefined, the default primary character set is Native/Wyse 50+ in Font Bank 0; the secondary character set is PC Multinational in Font Bank 1.

²Example: To load Bank 1 with the Multinational character set, enter: ESC c @ 1 A
This sequence is sent to the terminal. If ACK mode is on, the terminal returns an ACK to indicate that the font has loaded successfully.

³The default active character set, including that for TVI mode, is Native/Wyse 50+.

⁴See Note on the following page.

	WYSE				TVI			ADDS
Load Font Bank ¹	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Clear Font Bank								
Erase a character set from one of the four font banks. ¹	ESC c ? <i>Bank</i>	n/a	same	same	n/a	n/a	n/a	EnH

¹If not redefined, the default primary character set is Native/Wyse 50+ in Font Bank 0; the secondary character set is PC Multinational in Font Bank 1.

²Example: To load Bank 1 with the Multinational character set, enter: ESC c @ 1 A
This sequence is sent to the terminal. If ACK mode is on, the terminal returns an ACK to indicate that the font has loaded successfully.

³The default active character set, including that for TVI mode, is Native/Wyse 50+.

⁴See Note on the following page.

13.3 Designing a Character Font

Table 13–3 shows how to design a character font.

Table 13–3 Design a Character Font

	WYSE				TVI			ADDS
Define and Load a Character	160/60	PCTerm	150/120	WY 50+	950	925	910+	A2
Design a character font	ESC c A <i>Bank Pp bb..bb</i> Ctrl/Y							
		n/a	same	same	n/a	n/a	n/a	EnH

The sequence in Table 13–3 defines the *bb..bb* string and loads it in the *Pp* position of the *Bank* font bank.

Bank is the font bank that holds a defined character set. Refer to Table 13–2.

Pp is a 2-byte Hex code in the range of 00 to 7F specifying a position in that font bank.

bb...bb is a Hex code string defining the bit pattern of the defined character. The string contains one of the following:

- 16 2-byte Hex codes for a 10 × 16 cell size character
- 13 2-byte Hex codes for a 10 × 13 cell size character
- 10 2-byte Hex codes for a 10 × 10 cell size character

The command aborts if Ctrl/Y in the sequence is received before the expected number of Hex codes in *bb...bb* is received.

Perform the following steps to draw a graphic character in a 10 × 16 cell:

1. Divide the cell into 10 × 16 pixels and draw the desired font pattern. Observe the following rules:
 1. Start the pattern from the left column in the cell.
 2. For an alphanumeric character, the pattern should occupy only the left seven columns; for other graphic characters, the pattern can extend to the eighth column. (The ninth and tenth column follow whatever you place in the eighth column.)
 3. If possible, leave some space at both the top and bottom pixels to allow for spacing.
2. Turn the font pattern into a value represented as 1s and 0s; where "1" represents a filled pixel, and "0" represents an empty pixel.
3. Encode the bit pattern, 4 bits (a *nibble*) at a time, row by row, into hexadecimal numbers 0 to F. (Do not encode the right two columns.)

4. Form the two nibbles per row into a 2-byte Hex code.

5. Feed the 16 Hex codes into the *bb...bb* string. See the following example.

Load the following character into position 31 (Hex) in Font Bank 3.

	MSB				LSB				
Bit -->	7	6	5	4	3	2	1	0	A B Hex Code
1								00	
2								00	
3	X	X	X	X	X	X	X	X	FE
4	X					X			82
5	X					X			82
6	X					X			82
7	X					X			82
8	X	X	X	X	X	X	X	X	FE
9	X					X			82
10	X					X			82
11	X					X			82
12	X					X			82
13	X	X	X	X	X	X	X	X	FE
14								00	
15								00	
16								00	
17								00	

Use the following sequence to load this graphic character:

ESC c A 3 31 00 00 FE 82 82 82 82 FE 82 82 82 82 FE 00 00 00 00 Ctrl/Y

Once loaded, the font remains in the font bank until:

- The terminal is powered off
- This font bank is explicitly re-loaded or cleared
- This font bank is implicitly re-loaded by changing the terminal's personality or the terminal changes between a 24-line and a 44-line display while Automatic Font Load is on. Refer to [Table 13–2](#).